

The coupled Boussinesq control system with physical viscosity retained

8 September 2026

This standalone calculation reconstructs the released Alpöge–Buckmaster controls, proves the complete first controlled stage in the original physical coordinates, and gives compact fields and explicit smooth compactly supported forces realizing it with arbitrary prescribed thermal diffusivity and physical viscosity. It then proves a modified two-layer stage with an explicit positive equal-diffusion range, both original temperature thresholds, the changing parent, the second selected vorticity return, and a finite hold, together with all original-profile compact fields and forces. The next complete calculation follows the actual third growth and its full transition: both parent temperatures evolve, the second vorticity revives during transition, and the original third target is reached with an explicit positive physical diffusion range. A compact continuation argument then proves the complete third shooting family and selected third return on a further definite positive equal-diffusion interval. Every phase, deformation, feedback, scalar force, vector force and diffusion cost is retained. It also derives the exact original third-steering map, proves an initial steering interval with complete compact fields and forces, and completes the finite third shooting return without assuming root uniqueness. The reader additionally derives the evolving parent feedback, exact profile diffusion, and the pressure-independent obstruction to retaining the released infinite state with a smooth viscous force. The exact two-parameter similarity then transfers the complete finite third return to every spatial radius at one fixed positive physical diffusion. Its full force-jet weights prove a raw-copy term obstruction and define the force-flat packet space required by a corrected infinite accumulation. It proves that complete packets cannot be reset at consecutive scales because their endpoint curls do not match, then derives the exact based-increment residual, every space-time residual derivative, and the vector-and-matrix state interface that a return-renormalization map must satisfy. The infinite viscous singularity construction remains open. Original A_0, λ_0 , spatial scaling μ , and the paper’s time-scaling $\nu = \sqrt{A_0}$ are retained. The appendix sections use explicitly declared alternative notation for physical viscosity to avoid confusing it with this ν .

Primary source: Levent Alpöge and Tristan Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, 76-page released PDF, <https://cims.nyu.edu/~tristanb/boussinesq.pdf>. The exact downloaded edition is identified by the SHA-256 in `checks/replay_exact.json`. Printed-page and equation citations accompany the calculations. Source infinite existence results are attributed as such; the analytic finite-stage and operator proofs are written here in full.

Contents

1	Physical coordinates and exact finite viscous realization	3
2	The actual first controlled stage in the original units	7
3	Solving the next growth interval with a diffusing parent	14
4	Reaching both original growth targets with retained diffusion	18
5	A direct two-layer return theorem with positive physical diffusivity	23

6	Compact spatial realization of the changed two-stage trajectory	29
7	Exact third growth on the moving second holding state	37
8	The actual third growth target on the moving second hold	42
9	The actual third transition and its revived second parent	47
10	Compact spatial equations through the third threshold	53
11	Full compact spatial realization through the third transition	59
12	The exact third steering map with two evolving parents	69
13	Persistence and completion of the third shooting return	75
14	Compact spatial realization of the actual local third steering	80
15	Exact coupled feedback and the scope of common damping	86
16	The exact cost along the released infinite trajectory	89
17	Fixed-diffusion similarity and the force-flat cascade space	91
18	The based cascade: exact residual array and state interface	95
19	Signed residuals, the exact returned core, and a compact correction	100
A	Full source controls and aggregate background identities	106
B	Exact periodic profile evolution and the affine region	115

1 Physical coordinates and exact finite viscous realization

We use the orientation

$$x = (x_1, x_2) \in \mathbb{R}^2, \quad e_2 = (0, 1), \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \nabla^\perp = J\nabla, \quad \text{curl } u = \partial_1 u_2 - \partial_2 u_1.$$

The physical equations throughout this note are

$$\theta_t + u \cdot \nabla \theta = \kappa \Delta \theta + f_\theta, \tag{1}$$

$$u_t + (u \cdot \nabla)u + \nabla p = \varepsilon \Delta u + \theta e_2 + f_u, \quad \text{div } u = 0. \tag{2}$$

Here $\kappa, \varepsilon \geq 0$ are respectively thermal diffusivity and physical viscosity. In particular ε is not the source's time-scaling parameter ν and is not its schedule error tolerance. The datum is retained exactly:

$$A_0 \geq 1, \quad \lambda_0 = \lceil \sqrt{A_0}/2 \rceil, \quad \theta_{\text{in}}(x) = -\frac{A_0}{\lambda_0} \sin(\lambda_0 x_2) \chi_0(x), \quad u_{\text{in}}(x) = 0.$$

The original radial cutoff $\chi_0 \in C_c^\infty(B_{R_0})$, $R_0 > 2$, equals one on B_2 and lies in $[0, 1]$. Define

$$\mu = \lambda_0, \quad \nu = \sqrt{A_0}, \quad (X, T) = (\mu x, \nu t).$$

These are explicit maps between the source coordinates and the physical coordinates, rather than replacements of A_0 or λ_0 by one. For source fields decorated with a tilde their pullback is

$$\begin{aligned} \theta(x, t) &= \frac{\nu^2}{\mu} \tilde{\theta}(\mu x, \nu t), & u(x, t) &= \frac{\nu}{\mu} \tilde{u}(\mu x, \nu t), \\ p(x, t) &= \frac{\nu^2}{\mu^2} \tilde{p}(\mu x, \nu t), & \psi(x, t) &= \frac{\nu}{\mu^2} \tilde{\psi}(\mu x, \nu t). \end{aligned} \tag{3}$$

Their forces pull back with factors ν^3/μ and ν^2/μ , respectively. Their source diffusion coefficients must be

$$\tilde{\kappa} = \frac{\mu^2}{\nu} \kappa, \quad \tilde{\varepsilon} = \frac{\mu^2}{\nu} \varepsilon. \tag{4}$$

Indeed each term on the left of the scalar equation acquires ν^3/μ , whereas $\kappa \Delta_x \theta$ acquires $\kappa \nu^2 \mu$. Each momentum and buoyancy term acquires ν^2/μ , whereas $\varepsilon \Delta_x u$ acquires $\varepsilon \nu \mu$. Division gives exactly (4). Also $\nabla_x^\perp \psi = u$, $\Delta_x \psi = \nu \tilde{\omega}$, and $\text{div}_x u = \nu \text{div}_X \tilde{u}$. The inverse map substitutes $(x, t) = (X/\mu, T/\nu)$ and reciprocates the displayed factors, proving a bijection for corresponding forces, coefficients, and initial values. The cutoff is $\tilde{\chi}_0(X) = \chi_0(X/\mu)$, so the pulled-back initial temperature is exactly the displayed θ_{in} . These computations extend source Lemma 1.2, equations (1.6)–(1.7), pp. 6–7, by retaining both diffusion coefficients.

1.1 The released profile, the initial interpolation, and the base

Fix $0 < s_0 < 1$ and an even smooth cutoff χ supported in $(-\pi, \pi)$, equal to one on $[-s_0, s_0]$. On $[-\pi, \pi]$ put

$$F(s) = \sin s + \chi(s)(s - \sin s)$$

and extend periodically. Let

$$P(s) = \int_0^s F(r) dr - \frac{1}{2\pi} \int_{-\pi}^\pi \int_0^y F(r) dr dy.$$

Oddness and zero period integral of F show that P is even, periodic, has zero mean, and $P' = F$. Near zero, $F(s) = s$ and $P(s) = P(0) + s^2/2$. No sinusoidal identity is substituted for this profile.

Take a smooth nondecreasing step η that is zero for $t \leq 0$ and one for $t \geq 1$. Every derivative of positive order vanishes at its endpoints: smoothness and its constant extension prove this.

Choose $R_H > R_0 + 1$ and radial $H \in C_c^\infty(B_{R_H})$ equal to $|x|^2/2$ on B_{R_0+1} . Let $\alpha = 0$ on $[0, 1/\nu]$ and $e_0 = R_\alpha e_2$. Define

$$\begin{aligned} \theta_0 &= -\frac{A_0}{\mu} F(\mu e_0 \cdot x) \chi_0(x), & \theta_{\text{ref}} &= -\frac{A_0}{\mu} F(\mu x_2) \chi_0(x), \\ \theta_b &= \theta_0 + (1 - \eta(\nu t))(\theta_{\text{in}} - \theta_{\text{ref}}), & \psi_b &= \dot{\alpha} H, \quad u_b = \dot{\alpha} J \nabla H. \end{aligned} \quad (5)$$

For $t \leq 1/\nu$, $u_b = 0$ and $\partial_t \theta_b = \nu \eta'(\nu t)(\theta_{\text{ref}} - \theta_{\text{in}})$. For later times, $u_b = \dot{\alpha} J x$ on the support of θ_0 . Since $\dot{e}_0 = \dot{\alpha} J e_0$,

$$(\partial_t + \dot{\alpha} J x \cdot \nabla)(\mu e_0 \cdot x) = \mu \dot{\alpha} (J e_0 \cdot x + e_0 \cdot J x) = 0.$$

Radiality gives $J x \cdot \nabla \chi_0 = 0$. Therefore the exact zero-pressure base forces are

$$\begin{aligned} f_{\theta,b}^0 &= \nu \eta'(\nu t)(\theta_{\text{ref}} - \theta_{\text{in}}), \\ f_{u,b}^0 &= \ddot{\alpha} J \nabla H + \dot{\alpha}^2 (J \nabla H \cdot \nabla)(J \nabla H) - \theta_b e_2. \end{aligned} \quad (6)$$

On $|\mu e_0 \cdot x| < s_0$, $|x| < 2$, and $t \geq 1/\nu$, the base has the exact values

$$\theta_b = -A_0 e_0 \cdot x, \quad u_b = \dot{\alpha} J x, \quad G_b = -A_0 e_0, \quad D_b = \dot{\alpha} J.$$

This is source (4.9)–(4.10), pp. 38–39, in the original coordinates.

1.2 A compact realization of the complete first controlled stage

Use the first-stage coefficients constructed and proved below, with physical frequency $K_1 = \mu \lambda_1$, phase $\zeta = R_\alpha e(s_*)$, $|\zeta| = 1$, and signed amplitudes Θ, Ω satisfying

$$\dot{\Theta} = -\frac{J \zeta \cdot G_b}{K_1} \Omega, \quad \dot{\Omega} = K_1 \zeta_1 \Theta. \quad (7)$$

Here $e(\phi) = (\sin \phi, \cos \phi)$ and $J \zeta \cdot G_b = -A_0 \sin s_*$, a rotationally invariant scalar. Choose $C_\ell \geq 4$, set $\ell = s_0/(C_\ell \mu)$, and take radial $f \in C_c^\infty(B_1)$ equal to one on $B_{1/2}$. Put

$$g(x) = f(x/\ell), \quad s = K_1 \zeta \cdot x, \quad w = \eta(\Gamma(t - 1/\nu)), \quad \Gamma = \nu \sin s_*, \quad T = w \Theta, \quad Q = \frac{w \Omega}{K_1^2}.$$

Although g is written in laboratory coordinates, it is exactly the paper's material envelope: $M = R_\alpha$, so $f(M^{-1}x/\ell) = f(x/\ell)$ by radiality. Set

$$\begin{aligned} V &= T F(s) g, & \Psi &= Q P(s) g, \\ v &= Q \{K_1 J \zeta F(s) g + P(s) J \nabla g\}, & \theta &= \theta_b + V, \quad u = u_b + v, \quad p = 0. \end{aligned} \quad (8)$$

The new fields are extended by zero before $t = 1/\nu$. They are smooth across that time because w is flat there. Their support is in $B_\ell \subset B_{s_0/\mu} \cap B_2$, where the base is affine for every angle. The phase and envelope satisfy

$$(\partial_t + D_b x \cdot \nabla) s = 0, \quad (\partial_t + D_b x \cdot \nabla) g = 0.$$

For the phase this follows from $\dot{\zeta} = \dot{\alpha} J \zeta = -D_b^T \zeta$; for the envelope it follows from radiality.

Write $\partial_\perp g = J \zeta \cdot \nabla g$. Direct expansion gives the following complete added forces at zero diffusion:

$$E_\theta^0 = \dot{w} \Theta F g + Q P (J \nabla g \cdot G_b) + K_1 Q T (F^2 - P F') g \partial_\perp g, \quad (9)$$

$$E_u^0 = 2 D_b v + \dot{Q} \{K_1 J \zeta F g + P J \nabla g\} + (v \cdot \nabla) v - T F g e_2. \quad (10)$$

Every function of s in these equations is evaluated at the phase in (8). The vector derivative in the last formula can, equivalently, be evaluated by the explicitly specified matrix

$$\nabla v = QJ [K_1^2 \zeta \otimes \zeta F'g + K_1 F(\zeta \otimes \nabla g + \nabla g \otimes \zeta) + P \nabla^2 g], \quad (v \cdot \nabla)v = (\nabla v)v.$$

Thus (10) is a direct component formula, without solving a pressure equation or an inverse-curl problem.

Here is a proof of both residual identities. Since $J\zeta \cdot \zeta = 0$,

$$v \cdot \nabla V = K_1 QT(F^2 - PF')g(J\zeta \cdot \nabla g).$$

Also $v \cdot G_b = QK_1(J\zeta \cdot G_b)Fg + QP(J\nabla g \cdot G_b)$ and the material derivative of V is $\dot{T}Fg$. Equation (7) implies $\dot{T} + QK_1 J\zeta \cdot G_b = \dot{w}\Theta$, proving (9). For any trace-free matrix D , direct multiplication proves $DJ + JD^T = 0$. Consequently

$$(\partial_t + Dx \cdot \nabla)J\nabla\Psi = DJ\nabla\Psi + J\nabla[(\partial_t + Dx \cdot \nabla)\Psi].$$

The material derivative of Ψ is $\dot{Q}Pg$, and $(v \cdot \nabla)u_b = D_b v$ on the new support. Adding these two terms, the self-advection, and the negative buoyancy increment proves (10). Away from the new support every increment vanishes, so this verification is global.

For every prescribed $\kappa, \varepsilon \geq 0$ the *same* controlled fields now solve (1)–(2) with the explicit forces

$$f_\theta^{\kappa, \varepsilon} = f_{\theta, b}^0 + E_\theta^0 - \kappa \Delta \theta_b - \kappa \Delta V, \quad (11)$$

$$f_u^{\kappa, \varepsilon} = f_{u, b}^0 + E_u^0 - \varepsilon \Delta u_b - \varepsilon \Delta v. \quad (12)$$

Indeed substituting these right-hand sides cancels the displayed diffusion terms exactly and leaves the two residual identities already proved. This proves a concrete finite controlled viscous stage, for arbitrary physical coefficients, retaining the paper's actual profile and motion. Its costs are calculated next; no claim that these are small or uniform over infinitely many stages is part of this construction.

All fields and forces are supported in the fixed spatial ball B_{R_H} . They are smooth on the complete closed finite interval, including interpolation, activation, growth, transition, steering, and any specified finite holding duration. The pressure $p = 0$ satisfies $p(0, t) = 0$. Since $\psi_b + \Psi$ is compactly supported, $u = J\nabla(\psi_b + \Psi)$ is divergence free and has finite spatial energy. It also has exactly the Biot–Savart relation $u = K\omega$: if $N = (2\pi)^{-1} \log|x|$, integration by parts against the compact streamfunction gives $N * \Delta(\psi_b + \Psi) = (\Delta N) * (\psi_b + \Psi) = \psi_b + \Psi$; applying $J\nabla$ proves the assertion in distributions and hence for these smooth fields. Odd F , even P , and radial cutoffs give odd θ, u, f_θ, f_u and even ω . The finite-slab forces can also be specified with compact temporal support. Extend the state by its stationary initial values to the left and its stationary held values to the right; flatness of the interpolation and the selected pulse proves smoothness of these extensions. For any fixed $\tau_{\text{ext}} > 0$, multiply the resulting force formulas by

$$\eta(1 + t/\tau_{\text{ext}})\eta(1 + (T_f - t)/\tau_{\text{ext}}).$$

Here T_f denotes the finite final time of the prescribed hold. This multiplier is one on $[0, T_f]$ and zero outside $(-\tau_{\text{ext}}, T_f + \tau_{\text{ext}})$. Thus the same solution on its specified slab has forces in $C_c^\infty(\mathbb{R}^2 \times \mathbb{R})$, with explicitly specified spatial and temporal support.

On the smaller open set

$$|x| < \ell/2, \quad |K_1 \zeta \cdot x| < s_0,$$

the exact fields are

$$\theta = (-A_0 e_0 + w K_1 \Theta \zeta) \cdot x, \quad u = (\dot{\alpha} J + w \Omega J \zeta \otimes \zeta)x. \quad (13)$$

Their Laplacians vanish there, and so do the additional diffusive forces *and all their spatial derivatives* there. Therefore the evolving parent background for a next layer inside this open set is exactly the coupled gradient and shear in (13), with the actual time-dependent $\Theta, \Omega, \alpha, w$. It has not been replaced by prescribed unchanging D, G . The finite-stage force supplies the compensating diffusion outside this affine region.

1.3 Every spatial diffusion cost, including the cutoff terms

Put $k = K_1 \zeta$, so $|k| = K_1$. Direct differentiation gives

$$\Delta V = T\{K_1^2 F'' g + 2F' k \cdot \nabla g + F \Delta g\}, \quad (14)$$

$$\begin{aligned} \Delta v = QJ\{K_1^2 k F'' g + K_1^2 F' \nabla g + 2F' k(k \cdot \nabla g) \\ + 2F \nabla(k \cdot \nabla g) + F k \Delta g + P \nabla \Delta g\}. \end{aligned} \quad (15)$$

For the second formula first differentiate $\Delta \Psi = Q\{K_1^2 F' g + 2F k \cdot \nabla g + P \Delta g\}$, and apply J . These are exact differential expressions on all of $\mathbb{R}^2$ and retain the phase, mixed, and envelope terms. For example, their zeroth-order norms satisfy

$$\begin{aligned} \|\Delta V\|_\infty &\leq |T|\{K_1^2 \|F''\|_\infty \|g\|_\infty + 2K_1 \|F'\|_\infty \|\nabla g\|_\infty + \|F\|_\infty \|\Delta g\|_\infty\}, \\ \|\Delta v\|_\infty &\leq |Q|\{K_1^3 \|F''\|_\infty \|g\|_\infty + 3K_1^2 \|F'\|_\infty \|\nabla g\|_\infty \\ &\quad + 2K_1 \|F\|_\infty \|\nabla^2 g\|_\infty + K_1 \|F\|_\infty \|\Delta g\|_\infty + \|P\|_\infty \|\nabla \Delta g\|_\infty\}. \end{aligned} \quad (16)$$

The Hessian norm is its operator norm, and vector norms are Euclidean. Thus no cutoff diffusion term has been absorbed into a frequency-only estimate. Here $\|\partial^\beta g\|_\infty = \ell^{-|\beta|} \|\partial^\beta f\|_\infty$, including every original s_0, C_ℓ, μ through $\ell = s_0/(C_\ell \mu)$.

For an arbitrary nonnegative integer m put

$$E_m = \max_{|\beta|=m} \|\partial^\beta g\|_\infty, \quad B_m(W) = \sum_{j=0}^m \binom{m}{j} K_1^j \|W^{(j)}\|_\infty E_{m-j}.$$

Leibniz' rule, $\partial^\beta W(k \cdot x) = k^\beta W^{(|\beta|)}(k \cdot x)$, and $\sum_{|\beta|=j, \beta \leq \gamma} \binom{\gamma}{\beta} = \binom{m}{j}$ for $|\gamma| = m$ prove

$$\begin{aligned} \max_{|\gamma|=m} \|\partial^\gamma \Delta V\|_\infty &\leq 2|T| B_{m+2}(F), \\ \max_{|\gamma|=m} \|\partial^\gamma \Delta v\|_\infty &\leq 2\sqrt{2}|Q| B_{m+3}(P). \end{aligned} \quad (17)$$

The two in each line counts the two coordinate Laplacian terms; $\sqrt{2}$ counts the two components of $J\nabla$.

Let $S = |\Theta^{\text{seed}}|$, let L be the prescribed growth gain, let Λ_1 be the paper's pulse compression, and let $H_t \geq 0$ be the specified physical holding duration. The proved selected stage below has durations $L/\Gamma, 1/\Gamma, (1 + \Lambda_1^{-1})/\Gamma$, and H_t , respectively. On its growth and transition $|\Theta| = S \exp(\Gamma(t - t_{\text{in}}))$ and $|\Omega| = (K_1/\nu)|\Theta|$. On its selected steering,

$$|\Theta| \leq S e^{L+2+\Lambda_1^{-1}}, \quad |\Omega| \leq \frac{K_1}{\nu} |\Theta|.$$

On its hold $\Omega = 0$ and Θ is constant. Consequently the explicit costs

$$\begin{aligned} I_\Theta &= S \frac{e^{L+1} - 1}{\Gamma} + S e^{L+2+\Lambda_1^{-1}} \left(\frac{1 + \Lambda_1^{-1}}{\Gamma} + H_t \right), \\ I_\Omega &= \frac{K_1}{\nu} S \left\{ \frac{e^{L+1} - 1}{\Gamma} + e^{L+2+\Lambda_1^{-1}} \frac{1 + \Lambda_1^{-1}}{\Gamma} \right\} \end{aligned} \quad (18)$$

satisfy $\int |T| dt \leq I_\Theta$ and $\int |w\Omega| dt \leq I_\Omega$ over the complete physical stage. This follows by integrating the exponential exactly, using $0 \leq w \leq 1$, and then using the stated bounds on the remaining intervals. Combining with (17) gives

$$\int \max_{|\gamma|=m} \|\partial^\gamma (-\kappa \Delta V)\|_\infty dt \leq 2\kappa B_{m+2}(F) I_\Theta,$$

$$\int \max_{|\gamma|=m} \|\partial^\gamma(-\varepsilon \Delta v)\|_\infty dt \leq \frac{2\sqrt{2}\varepsilon}{K_1^2} B_{m+3}(P) I_\Omega. \quad (19)$$

In particular the holding scalar cost is proportional to H_t ; there is no assertion that a forced hold has zero scalar diffusion cost.

The base costs are equally explicit. Write

$$C_m = \max_{|\beta|=m} \|\partial^\beta \chi_0\|_\infty, \quad \mathcal{B}_m(W) = \sum_{j=0}^m \binom{m}{j} \mu^j \|W^{(j)}\|_\infty C_{m-j}.$$

On the initial interpolation $\theta_b = (1 - \eta(\nu t))\theta_{\text{in}} + \eta(\nu t)\theta_{\text{ref}}$; afterwards it is a rotated F -profile with the same radial cutoff. Thus, writing $T_f = 1/\nu + (L + 2 + \Lambda_1^{-1})/\Gamma + H_t$,

$$\begin{aligned} \int_0^{T_f} \max_{|\gamma|=m} \|\partial^\gamma(\kappa \Delta \theta_b)\|_\infty dt &\leq \frac{2\kappa A_0}{\mu} \left\{ \frac{\max(\mathcal{B}_{m+2}(\sin), \mathcal{B}_{m+2}(F))}{\nu} + (T_f - 1/\nu) \mathcal{B}_{m+2}(F) \right\}, \\ \int_0^{T_f} \max_{|\gamma|=m} \|\partial^\gamma(\varepsilon \Delta u_b)\|_\infty dt &\leq \varepsilon \max_{|\gamma|=m} \|\partial^\gamma J \nabla \Delta H\|_\infty \int_0^{T_f} |\dot{\alpha}| dt. \end{aligned} \quad (20)$$

The last angle variation is exactly the finite integral of the explicit control and has the following fixed upper bound. For $z(\tau)$ in the stage proof and its selected pulse multiplier m_* , $|Z| \leq 1/4$, and

$$\int |\dot{\alpha}| dt \leq \frac{4}{\sqrt{15}} \int |\dot{Z}| dt \leq \frac{4 \sin s_*}{\sqrt{15}} \left(1 + m_* \Lambda_1 \int_0^1 |\eta'_2(y)| dy \right).$$

This uses $\alpha = s_* - \arcsin Z$, $\int_0^1 \eta' = 1$, and the change of variables $y = \Lambda_1(\tau - 1)$ on the pulse.

For completeness all mixed derivatives also have an exact finite replay formula; spatial bounds alone are not being called bounds for time derivatives. For $A = T, W = F$, or $A = Q, W = P$, and any multi-index γ , Leibniz gives

$$\partial_t^b \partial_x^\gamma [AW(k \cdot x)g] = \sum_{\beta \leq \gamma} \binom{\gamma}{\beta} (\partial^\beta g) \sum_{a+c+d=b} \frac{b!}{a!c!d!} A^{(a)}(k^{\gamma-\beta})^{(c)} \partial_t^d W^{(|\gamma-\beta|)}(k \cdot x). \quad (21)$$

For $d > 0$ the remaining derivative is the finite sum

$$\partial_t^d W^{(n)}(k \cdot x) = \sum_{\substack{m_1, \dots, m_d \geq 0 \\ \sum j m_j = d}} \frac{d!}{\prod_{j=1}^d m_j! (j!)^{m_j}} W^{(n+\sum m_j)}(k \cdot x) \prod_{j=1}^d (k^{(j)} \cdot x)^{m_j}.$$

For $d = 0$ it is $W^{(n)}(k \cdot x)$. This identity follows by induction: differentiation either increases the derivative order on W and creates $k' \cdot x$, or differentiates one of the displayed factors; grouping the resulting multiplicities gives exactly the factorial coefficients at order $d + 1$. On the support use $|k^{(j)} \cdot x| \leq \ell |k^{(j)}|$ and $|\partial^\beta g| \leq E_{|\beta|}$ to obtain an explicit upper bound from each term. Sum the two coordinate second derivatives in (21) for ΔV and then differentiate once more spatially for Δv . The same formula with $k = \mu e_0$, $g = \chi_0$ handles θ_0 ; the initial interpolation additionally uses the exact derivatives $\partial_t^a \eta(\nu t) = \nu^a \eta^{(a)}(\nu t)$. For u_b , mixed diffusion derivatives are exactly $\varepsilon \alpha^{(b+1)} \partial^\gamma J \nabla \Delta H$. All coefficient derivatives are defined by the explicit smooth stage equations on a compact interval and the finitely many fixed profile derivatives, so these formulas quantify every specified mixed order without an asymptotic omission.

2 The actual first controlled stage in the original units

The source for this section is Alpöge–Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, <https://cims.nyu.edu/~tristanb/boussinesq.pdf>, released PDF, 76 pages. The source locations used here are Lemma 1.2 and (1.6), pp. 6–7; (3.1)–(3.12), pp. 8–10; the complete proofs of Lemmas 3.7–3.8 and (3.24)–(3.34), pp. 15–18; (3.58) and the following second-stage calculation, p. 27; and the first holding integral on p. 36. The proofs below are given in full. The source's unit-scale convention is not imposed on the fields in this section.

2.1 Parameters, coordinates, and the original coupled equations

Let $x = (x_1, x_2) \in \mathbb{R}^2$, $e_2 = (0, 1)$, and

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_\alpha = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}, \quad e(s) = (\sin s, \cos s).$$

In particular $R_\alpha e(s) = e(s - \alpha)$ and $\text{curl } u = \partial_1 u_2 - \partial_2 u_1$. Retain $A_0 > 0$, $\lambda_0 \geq 1$, and the source's two scaling parameters

$$\mu = \lambda_0, \quad \nu = \sqrt{A_0}, \quad K_1 = \mu \lambda_1.$$

Here ν is the source's *time-scaling parameter*. Physical momentum viscosity below is denoted $\nu_{\text{phys}} \geq 0$; thermal diffusivity is $\kappa_{\text{phys}} \geq 0$. The first pulse compression is the source's Λ_1 , which is different from the physical spatial frequency K_1 .

Fix a smooth nondecreasing step $\eta : \mathbb{R} \rightarrow [0, 1]$ equal to zero on $(-\infty, 0]$ and one on $[1, \infty)$, and $h \in C_c^\infty((0, 1))$ with $h \geq 0$ and $\int_0^1 h = 1$. Write $H = \|h\|_\infty$. The source's choice is $h = \eta_2$, $c_p = 3$; the following calculation retains any

$$c_p > 1, \quad \Lambda_1 \geq 2c_p H, \quad 0 < s = s_* \leq \frac{1}{8}, \quad c_p \Lambda_1 H \sin s \leq \frac{1}{4}. \quad (22)$$

Since $\int h = 1$, $H \geq 1$ and $\Lambda_1 > 2$. Let $k_1 \geq 0$ be an integer, $\lambda_1 > 1$, and set

$$S = \frac{A_0}{\mu} \lambda_1^{-k_1-6}, \quad L = (k_1 + 5 + \frac{1}{8}) \log \lambda_1 \geq 1, \quad P_* = S e^L = \frac{A_0}{\mu} \lambda_1^{-7/8}, \quad \Gamma = \nu \sin s, \quad t_0 = \nu^{-1}. \quad (23)$$

The condition $L \geq 1$ is ensured explicitly by $\lambda_1 \geq \exp((k_1 + 41/8)^{-1})$.

The first layer's complete preceding affine background, phase transport, and signed amplitude pair are

$$D_{<1} = \dot{\alpha} J, \quad G_{<1} = -A_0 R_\alpha e_2, \quad \dot{M}_1 = D_{<1} M_1, \quad M_1(t_0) = I, \quad (24)$$

$$\zeta_1 = M_1^{-T} e(s), \quad \dot{\Theta}_1 = a_1 \Omega_1, \quad \dot{\Omega}_1 = b_1 \Theta_1, \quad a_1 = -\frac{J \zeta_1 \cdot G_{<1}}{K_1 |\zeta_1|^2}, \quad b_1 = K_1 \zeta_{1,1}. \quad (25)$$

The entry data are

$$\alpha(t_0) = 0, \quad \Theta_1(t_0) = -S, \quad \Omega_1(t_0) = -\frac{K_1}{\nu} S, \quad w_1(t) = \eta(\Gamma(t - t_0)).$$

The activation factor multiplies the physical increment, not the homogeneous amplitude equation. It is flat at t_0 and equals one by $t_0 + \Gamma^{-1}$.

For this actual background, $M_1 = R_\alpha$: differentiation gives $\dot{R}_\alpha = \dot{\alpha} J R_\alpha$ and the initial values agree. Uniqueness follows, for example, by differentiating $R_\alpha^{-1} M_1$. Consequently

$$\zeta_1 = R_\alpha e(s), \quad |\zeta_1| = 1, \quad J \zeta_1 \cdot G_{<1} = -A_0 \sin s, \quad a_1 = \frac{A_0 \sin s}{K_1}, \quad b_1 = K_1 \sin(s - \alpha). \quad (26)$$

The determinant identity used here is explicit: $J e(s) = (-\cos s, \sin s)$ has scalar product $\sin s$ with e_2 , and simultaneous rotation preserves that scalar product. In particular a_1 is constant because both the parent gradient and the new phase rotate; it has not been frozen independently of the evolving parent.

2.2 Exact growth and transition

For the first stage the feedback sums F_0 and F_1 in source (3.4) are empty and equal zero. Thus both the source's growth control $\dot{\alpha} = F_0$ and its transition control $(1 - \eta_b)F_0 + \eta_b F_1$ give $\alpha = 0$. Set

$$t_a = t_0 + \frac{L}{\Gamma}, \quad t_1 = t_a + \Gamma^{-1}, \quad t_b = t_1 + \Gamma^{-1}(1 + \Lambda_1^{-1}).$$

On $[t_0, t_1]$ the exact solution of the original pair is

$$P_1(t) := -\Theta_1(t) = S e^{\Gamma(t-t_0)}, \quad \Omega_1(t) = -\frac{K_1}{\nu} P_1(t). \quad (27)$$

Indeed $a_1 K_1 / \nu = \Gamma$ and $\Gamma K_1 / \nu = K_1 \sin s = b_1$. This proves both differential equations and both initial data. Uniqueness follows from the linear integral equation: on a bounded interval with coefficient norm at most B , the n -fold iterated integral of the difference of two solutions is bounded by its supremum times $(B\ell)^n / n!$, which tends to zero. The same convergent integral series supplies existence on every finite interval. Since P_1 is strictly increasing, t_a is exactly the first crossing of P_* , and the transition contributes exactly one to $\log P_1$. The steering entry amplitude is therefore $P_{\text{ent}} = e P_*$.

2.3 The trial steering family and proof of its unique return

Use $m \in [0, c_p]$ for the pulse trial parameter, to distinguish it from the unchanged spatial parameter $\mu = \lambda_0$. With $\tau = \Gamma(t - t_1)$ and $\tau_b = 1 + \Lambda_1^{-1}$ set

$$z(\tau; m) = \begin{cases} 1 - \eta(\tau), & 0 \leq \tau \leq 1, \\ -m \Lambda_1 h(\Lambda_1(\tau - 1)), & 1 \leq \tau \leq \tau_b, \end{cases} \quad Z = \sin s z, \quad \alpha = s - \arcsin Z. \quad (28)$$

Extend $Z = \sin s$ before t_1 and $Z = 0$ after t_b . The two smooth pieces and extensions agree to every order, since the step is flat and h is supported in the interior. The source's selected pulse height in (3.12) is precisely $h_1 = m \Lambda_1 \sin s$; its allowed interval is $[0, c_p \Lambda_1 \sin s]$. The design gives

$$|Z| \leq \frac{1}{4}, \quad \zeta_{1,1} = Z, \quad \zeta_{1,2} = \sqrt{1 - Z^2} \geq \frac{\sqrt{15}}{4}, \quad \dot{\alpha} = -\frac{\dot{Z}}{\zeta_{1,2}}, \quad 0 \leq \alpha < \frac{1}{2}. \quad (29)$$

For the last bound, $Z \leq \sin s$ implies $\alpha \geq 0$, and $\arcsin(1/4) \leq (1/4)/\sqrt{1 - 1/16} < 1/3$ gives $\alpha \leq 1/8 + 1/3 < 1/2$. These are the source's actual feedback and the original laboratory phase component; gravity remains e_2 .

Define $\mathcal{P}$ first as the solution of the linear equation

$$\mathcal{P}_{\tau\tau} = z\mathcal{P}, \quad \mathcal{P}(0) = \mathcal{P}_\tau(0) = 1. \quad (30)$$

The same linear integral series proves existence, uniqueness, and smoothness for every trial on the full finite interval. Moreover the exact inverse map to the original amplitude pair is

$$\Theta_1(t) = -P_{\text{ent}} \mathcal{P}(\tau), \quad \Omega_1(t) = -\frac{K_1}{\nu} P_{\text{ent}} \mathcal{P}_\tau(\tau). \quad (31)$$

To check it, its component derivatives are exactly

$$-\Gamma P_{\text{ent}} \mathcal{P}_\tau = a_1 \Omega_1, \quad -\frac{K_1}{\nu} \Gamma P_{\text{ent}} z \mathcal{P} = K_1 \sin s z \Theta_1 = b_1 \Theta_1.$$

The data at t_1 match (27). Conversely the first amplitude equation with constant a_1 and $a_1 b_1 = \Gamma^2 z$ gives (30) for $-\Theta_1 / P_{\text{ent}}$, with the stated initial derivative. Thus the auxiliary equation and original pair are explicitly inverse descriptions of the same dynamics, not a substituted oscillator.

On $0 \leq \tau \leq 1$, $z \geq 0$. Before any first zero of either $\mathcal{P}$ or $\mathcal{P}_\tau$, one has $\mathcal{P}_{\tau\tau} \geq 0$ and therefore $\mathcal{P}_\tau \geq 1$, $\mathcal{P} \geq 1 + \tau$. Continuity excludes that first zero and proves these inequalities through $\tau = 1$. Only now define $v = \mathcal{P}_\tau / \mathcal{P}$. It satisfies

$$v_\tau = 1 - \eta - v^2, \quad v(0) = 1.$$

Since $(v - 1)_\tau = -(v + 1)(v - 1) - \eta$, the integrating-factor formula gives $v \leq 1$. Since $v > 0$ and $v_\tau \geq -v^2$, $(1/v)_\tau \leq 1$, so

$$\frac{1}{1 + \tau} \leq v(\tau) \leq 1, \quad \log 2 \leq \int_0^1 v d\tau \leq 1, \quad v_d := v(1) \in [1/2, 1], \quad 2 \leq \mathcal{P}(1) \leq e. \quad (32)$$

On the negative pulse set $y = \Lambda_1(\tau - 1)$ and $\chi(y) = \mathcal{P}(1 + y/\Lambda_1)/\mathcal{P}(1)$. There is no division by an unknown evolving amplitude in this definition. Its exact equation and data are

$$\chi_{yy} + \frac{mh(y)}{\Lambda_1}\chi = 0, \quad \chi(0) = 1, \quad \chi_y(0) = v_d/\Lambda_1. \quad (33)$$

Before a possible first zero, χ is concave, so $\chi \leq 1 + y/\Lambda_1 \leq 2$. Since $mh/\Lambda_1 \leq c_p H/\Lambda_1 \leq 1/2$, one has $\chi_{yy} \geq -1$, hence

$$\chi(y) \geq 1 + \frac{v_d}{\Lambda_1}y - \frac{y^2}{2} \geq \frac{1}{2} \quad (0 \leq y \leq 1).$$

The strict positive lower bound excludes a first zero. Thus every trial satisfies on the entire pulse

$$\frac{1}{2} \leq \chi(y) \leq 1 + y/\Lambda_1, \quad \chi_y = \frac{v_d}{\Lambda_1} - \frac{m}{\Lambda_1} \int_0^y h(r)\chi(r) dr \geq -\frac{2c_p}{\Lambda_1}. \quad (34)$$

Consequently the original temperature is strictly negative for every trial, and the quotient is well defined. Writing $V(y; m) = v(1 + y/\Lambda_1; m) = \Lambda_1 \chi_y / \chi$ gives

$$V_y = -mh(y) - \frac{V^2}{\Lambda_1}, \quad V(0; m) = v_d, \quad -4c_p \leq V(y; m) \leq v_d \leq 1. \quad (35)$$

Here is the parameter dependence needed to select the return. In the first-order integral system for (χ, χ_y) the coefficient matrix depends affinely on m . On the compact parameter interval, its iterated integrals converge uniformly with factorial denominators. Differentiating r times in m introduces at most a degree- r polynomial in the iteration index, still dominated by that factorial. Thus this series and each parameter derivative converge uniformly. The lower bound $\chi \geq 1/2$ then gives smooth parameter dependence of V . Differentiating its exact equation and using an integrating factor gives

$$\partial_m V(1; m) = - \int_0^1 h(y) \exp\left(-\frac{2}{\Lambda_1} \int_y^1 V(r; m) dr\right) dy < 0. \quad (36)$$

The strict sign follows because $h \geq 0$ has mass one and the exponential is positive. Indeed the explicit bounds also give $e^{-2/\Lambda_1} \leq -\partial_m V(1; m) \leq e^{8c_p/\Lambda_1}$. At $m = 0$, direct solution of $V_y = -V^2/\Lambda_1$ gives $V(1; 0) = v_d/(1 + v_d/\Lambda_1) > 0$. At $m = c_p$, integration gives $V(1; c_p) \leq v_d - c_p < 0$. Continuity and strict monotonicity therefore give exactly one $m_* \in (0, c_p)$ with $V(1; m_*) = 0$. This is exactly the source's selected return: by (31) and positivity, it is equivalent to $\Omega_1(t_b) = 0$.

For this selected trial V is nonincreasing and ends at zero, so $0 \leq V \leq v_d$. Integration of (35) now gives the exact identity and bounds

$$m_* = v_d - \frac{1}{\Lambda_1} \int_0^1 V(y; m_*)^2 dy, \quad \frac{1}{2} - \frac{1}{4\Lambda_1} \leq m_* \leq 1. \quad (37)$$

For the lower bound, $v_d - v_d^2/\Lambda_1$ is increasing for $v_d \in [1/2, 1]$, because its derivative is $1 - 2v_d/\Lambda_1 > 0$.

For every trial and every steering time,

$$0 \leq \log \mathcal{P}(\tau) \leq 1 + \Lambda_1^{-1}.$$

On the ramp this follows from (32); on the pulse it follows from $\mathcal{P}(1) \geq 2$, $\chi \geq 1/2$, and $\log \chi \leq \log(1 + \Lambda_1^{-1}) \leq \Lambda_1^{-1}$. For the selected trial the pulse logarithmic contribution is nonnegative, so the endpoint obeys

$$\log 2 \leq \log \mathcal{P}(\tau_b) \leq 1 + \Lambda_1^{-1}, \quad 2eP_* \leq P_b := P_1(t_b) \leq e^{2+\Lambda_1^{-1}} P_*. \quad (38)$$

The total duration from entry to return is exactly

$$t_b - t_0 = \frac{L + 2 + \Lambda_1^{-1}}{\nu \sin s}. \quad (39)$$

Throughout the selected stage P_1 is nondecreasing, $S \leq P_1 \leq P_b$, and $-(K_1/\nu)P_1 \leq \Omega_1 \leq 0$. For every unselected trial the valid bound instead is $|\Omega_1| \leq 4c_p(K_1/\nu)e^{2+\Lambda_1^{-1}}P_*$; its sign need not stay negative.

Near t_b , $h = 0$, so $Z = 0$ and $\alpha = s$. The equation $\mathcal{P}_{\tau\tau} = 0$, together with $\mathcal{P}_\tau(\tau_b) = 0$, implies that $\mathcal{P}$ is constant on that whole neighborhood. Thus α , Θ_1 , and Ω_1 match their stationary continuations to all orders. The endpoint state is

$$\alpha = s, \quad M_1 = R_s, \quad \zeta_1 = e_2, \quad w_1 = 1, \quad \Theta_1 = -P_b, \quad \Omega_1 = 0. \quad (40)$$

2.4 Control bounds and exact whole-stage integrals

For $j = 1, 2$ define the explicit finite profile quantities

$$B_j = \Gamma^j \sin s (\|\eta^{(j)}\|_\infty + c_p \Lambda_1^{j+1} \|h^{(j)}\|_\infty).$$

From (28), $|\partial_t^j Z| \leq B_j$. Differentiating the original control, including its denominator, gives

$$\dot{\alpha} = -\dot{Z}(1 - Z^2)^{-1/2}, \quad (41)$$

$$\ddot{\alpha} = -\ddot{Z}(1 - Z^2)^{-1/2} - Z\dot{Z}^2(1 - Z^2)^{-3/2},$$

$$|\dot{\alpha}| \leq \frac{4}{\sqrt{15}}B_1, \quad |\ddot{\alpha}| \leq \frac{4}{\sqrt{15}}B_2 + \frac{16}{15\sqrt{15}}B_1^2. \quad (42)$$

Higher derivatives exist because the displayed smooth composition has argument in $[-1/4, 1/4]$; no endpoint differentiability is omitted.

For exact coefficient costs use the invertible fixed map $Y = (\Theta_1, \nu\Omega_1/K_1)^T$. Its inverse is $(Y_1, K_1 Y_2/\nu)^T$, and direct differentiation gives

$$\dot{Y} = \widehat{B}Y, \quad \widehat{B} = \begin{pmatrix} 0 & \Gamma \\ \nu Z & 0 \end{pmatrix}.$$

Let $J_\eta = \int_0^1 (1 - \eta(r)) dr \in [0, 1]$. On a hold of physical duration $T_h \geq 0$, the following are exact:

interval	duration	$\int \widehat{B}_{12} dt$	$\int \widehat{B}_{21} dt$
growth	L/Γ	L	L
transition	$1/\Gamma$	1	1
steering ramp	$1/\Gamma$	1	J_η
negative pulse	$1/(\Lambda_1 \Gamma)$	$1/\Lambda_1$	m_*
holding	T_h	ΓT_h	0

For example, on the pulse substitution $y = \Lambda_1 \Gamma(t - t_1 - \Gamma^{-1})$ gives $\int \nu|Z| dt = m_* \int_0^1 h = m_*$. Thus the two transition-and-steering costs are exactly $2 + \Lambda_1^{-1}$ and $1 + J_\eta + m_*$; their sum is at most $4 + c_p + \Lambda_1^{-1}$.

The activation and amplitude integrals needed by force estimates can also be retained without an asymptotic replacement. Set

$$J_a = \int_0^1 \eta'(r)e^r dr \in [1, e], \quad J_g = \int_0^1 \eta(r)e^r dr = e - J_a.$$

For the selected stage including the hold, direct change of variable and integration by parts give

$$\begin{aligned} \int_{t_0}^{t_b} |\dot{w}_1 \Theta_1| dt &= S J_a, & \int_{t_0}^{t_b} |\dot{w}_1 \Omega_1| dt &= \frac{K_1}{\nu} S J_a, \\ I_P := \int_{t_0}^{t_b+T_h} w_1 P_1 dt &= \frac{S}{\Gamma} (J_g + e^{L+1} - e) + P_b T_h \end{aligned} \quad (43)$$

$$+ \frac{eP_*}{\Gamma} \int_0^{1+\Lambda_1^{-1}} \mathcal{P}(\tau; m_*) d\tau, \quad (44)$$

$$I_\Omega := \int_{t_0}^{t_b+T_h} w_1 |\Omega_1| dt = \frac{P_b - SJ_a}{a_1}. \quad (45)$$

For the last identity, $P'_1 = -a_1 \Omega_1 = a_1 |\Omega_1|$, $w_1(t_0) = 0$, and $w_1(t_b + T_h) = 1$, hence $\int w_1 P'_1 = P_b - \int \dot{w}_1 P_1 = P_b - SJ_a$. In particular

$$I_P \leq \frac{S}{\Gamma} (J_g + e^{L+1} - e) + \frac{eP_*}{\Gamma} (1 + \Lambda_1^{-1}) e^{1+\Lambda_1^{-1}} + P_b T_h.$$

These are finite specified integrals of the prescribed profiles and the uniquely selected exact linear solution.

2.5 The evolving background inherited by layer two

On the affine part of the first wave, the complete background for the next layer is

$$G_{<2} = -A_0 R_\alpha e_2 + w_1 K_1 \Theta_1 \zeta_1, \quad D_{<2} = \dot{\alpha} J + w_1 \Omega_1 J \zeta_1 \otimes \zeta_1. \quad (46)$$

These are source (3.3) with $A_1 = -K_1 \Theta_1 |\zeta_1|$ and the physical units retained. Their complete derivatives are

$$\dot{G}_{<2} = \dot{\alpha} J G_{<2} + (K_1 \dot{w}_1 \Theta_1 + w_1 A_0 \sin s \Omega_1) \zeta_1, \quad (47)$$

$$\begin{aligned} \dot{D}_{<2} &= \ddot{\alpha} J + (\dot{w}_1 \Omega_1 + w_1 K_1 \zeta_{1,1} \Theta_1) J \zeta_1 \otimes \zeta_1 \\ &\quad + w_1 \Omega_1 \dot{\alpha} (-\zeta_1 \otimes \zeta_1 + J \zeta_1 \otimes J \zeta_1). \end{aligned} \quad (48)$$

Indeed $\dot{\zeta}_1 = \dot{\alpha} J \zeta_1$, $J \dot{\zeta}_1 = -\dot{\alpha} \zeta_1$, and the product rule gives each displayed term. In particular, using $J \zeta_1 \cdot G_{<2} = -A_0 \sin s$ gives the exact identity

$$\dot{G}_{<2} + D_{<2}^T G_{<2} = K_1 \dot{w}_1 \Theta_1 \zeta_1. \quad (49)$$

Thus the inherited gradient satisfies its true affine transport equation after activation, even while the first stage rotates it. The affine vorticity is $2\dot{\alpha} + w_1 \Omega_1$ and its scalar source is $G_{<2,1} = A_0 \sin \alpha + w_1 K_1 \Theta_1 \zeta_{1,1}$; therefore its exact vorticity residual is

$$(2\dot{\alpha} + w_1 \Omega_1)' - G_{<2,1} = 2\ddot{\alpha} - A_0 \sin \alpha + \dot{w}_1 \Omega_1. \quad (50)$$

At the selected endpoint define $A_1 = K_1 P_b > 0$. Then

$$G_{<2}(t_b) = (A_0 \sin s, -A_0 \cos s - A_1), \quad D_{<2}(t_b) = 0, \quad \sigma_1^2 = |G_{<2}(t_b)| = \sqrt{A_0^2 + A_1^2 + 2A_0 A_1 \cos s}. \quad (51)$$

Consequently

$$2eA_0 \lambda_1^{1/8} \leq A_1 \leq e^{2+\Lambda_1^{-1}} A_0 \lambda_1^{1/8}, \quad A_1 + A_0 \cos s \leq \sigma_1^2 \leq A_1 + A_0.$$

The lower norm bound is its positive vertical component in absolute value; the upper bound is the triangle inequality, which also follows by squaring the displayed expression. No horizontal component of the parent gradient has been discarded.

On the first hold the actual feedback is $F_1 = 0$. The endpoint state (40) is stationary under the original equations: $b_1 = K_1 \zeta_{1,1} = 0$ gives $\dot{\Omega}_1 = 0$, then $\dot{\Theta}_1 = a_1 \Omega_1 = 0$, and $\dot{\alpha} = 0$ gives $\dot{\zeta}_1 = 0$. Thus (51) remains the actual inherited background throughout this hold. For an arbitrary perturbation of the weighted amplitude pair its propagator over length T_h is exactly

$$\exp\left(T_h \begin{pmatrix} 0 & \Gamma \\ 0 & 0 \end{pmatrix}\right) = \begin{pmatrix} 1 & \Gamma T_h \\ 0 & 1 \end{pmatrix}. \quad (52)$$

This follows by squaring the matrix and obtaining zero, and retains the nonzero first holding coefficient cost.

For clarity one can construct the next growth interval itself rather than prescribing an unspecified holding time. Retain $\lambda_2 = \lambda_1^{Q_2}$ with $Q_2 \geq 202$, an integer $k_2 \geq 0$, $L_2 = (k_2 + 41/8) \log \lambda_2$, and the source's

$$s_2 = L_2 \nu / \sigma_1, \quad K_2 = \mu \lambda_2, \quad \Gamma_2 = \sigma_1 \sin s_2.$$

An explicit sufficient finite choice ensuring $L_2 \geq 2$ and $0 < s_2 \leq 1/8$ is $\log \lambda_1 \geq 4096(k_2 + 41/8)Q_2$. To prove this, put $B = (k_2 + 41/8)Q_2$ and $x = \log \lambda_1$. The gain implies $\sigma_1 \geq \sqrt{2e\nu}e^{x/16}$; the exponential series gives $e^{x/16} \geq x^2/512$. Hence $s_2 \leq 512B/(\sqrt{2e}x) \leq 1/8$. Also $L_2 = Bx \geq 4096B^2 > 2$.

During the second growth, $\alpha = s$, $\Omega_1 = 0$, and $D_{<2} = 0$, so $M_2 = I$, $\zeta_2 = e(s_2)$, and the original coefficient formulas yield exactly

$$a_2 = \frac{A_1 \sin s_2 + A_0 \sin(s + s_2)}{K_2}, \quad b_2 = K_2 \sin s_2, \quad \gamma_2 = \sqrt{[A_1 \sin s_2 + A_0 \sin(s + s_2)] \sin s_2} > 0. \quad (53)$$

Both signs are positive because $0 < s + s_2 \leq 1/4$. For its actual seed $S_2 = (A_0/\mu)\lambda_2^{-k_2-6}$ and entry vorticity $-\sqrt{b_2/a_2}S_2$, the exact solution is $P_2 = S_2 e^{\gamma_2(t-t_b)}$, $\Omega_2 = -\sqrt{b_2/a_2}P_2$. Substitution verifies both equations; the first threshold time is $t_{a,2} = t_b + L_2/\gamma_2$. Its activation is the source's $\eta(\Gamma_2(t - t_b))$. To see that activation completes before threshold, note

$$R := \frac{\gamma_2^2}{\Gamma_2^2} = \frac{A_1 + A_0 \cos s + A_0 \sin s \cot s_2}{\sigma_1^2}, \quad \cos s \leq R \leq 1 + \frac{\nu \sin s}{L_2 \sigma_1}. \quad (54)$$

For the lower bound, discard the positive cotangent term, use $\sigma_1^2 \leq A_1 + A_0$, and then $(A_1 + A_0 \cos s)/(A_1 + A_0) \geq \cos s$. For the upper bound use $A_1 + A_0 \cos s \leq \sigma_1^2$ and $\cot s_2 \leq 1/s_2$, the latter following from $(\sin r - r \cos r)' = r \sin r \geq 0$. Since $\sigma_1 \geq \nu$ and $L_2 \geq 2$, $R \leq 1 + \sin s/2 < 2$, and $\Gamma_2 L_2 / \gamma_2 = L_2 / \sqrt{R} > 1$. The exact first holding integral therefore is

$$H_1 = \Gamma \frac{L_2}{\gamma_2} = \sin s \frac{s_2}{\sin s_2} \frac{1}{\sqrt{R}}, \quad \frac{\sin s}{\sqrt{1 + \nu \sin s / (L_2 \sigma_1)}} \leq H_1 \leq \frac{\sin s}{\sqrt{\cos s (1 - s_2^2/6)}}. \quad (55)$$

Here $\sin r \leq r$ follows by integrating $\cos r \leq 1$; $\sin r \geq r - r^3/6$ follows by integrating $\cos r \geq 1 - r^2/2$, itself obtained from $1 - \cos r = \int_0^r \sin u \, du \leq r^2/2$. Thus no small-angle replacement is used in the bounds. The second transition also has $F_2 = 0$, because its only shear coefficient is $\Omega_1 = 0$; it prolongs the first pair's stationarity by exactly Γ_2^{-1} . The source counts that transition in the first layer's tail, not in H_1 .

2.6 Exact uncut forced fields and their retained diffusion cost

This subsection gives an explicit PDE realization of the just-proved finite stage on $\mathbb{R}^2$. Its fields have the spatial domain $\mathbb{R}^2$ and need not decay; compact spatial localization is a separate construction. Let F be any smooth odd 2π -periodic function. Define $Q_0(r) = \int_0^r F(v) \, dv$ and $\mathcal{Q}(r) = Q_0(r) - (2\pi)^{-1} \int_{-\pi}^{\pi} Q_0(v) \, dv$. Oddness gives zero period integral for F , so Q_0 is periodic; substitution $v = -u$ gives $Q_0(-r) = Q_0(r)$. Thus $\mathcal{Q}$ is even and mean-zero, and $\mathcal{Q}' = F$. Put $q(x, t) = K_1 \zeta_1(t) \cdot x$ and

$$\begin{aligned} \theta &= G_{<1} \cdot x + w_1 \Theta_1 F(q), \\ u &= \dot{\alpha} Jx + \frac{w_1 \Omega_1}{K_1} J \zeta_1 F(q), \end{aligned} \quad (56)$$

$$p = \frac{\dot{\alpha}^2}{2} |x|^2 + \frac{1}{2} x_2 (G_{<1} \cdot x) + \left(\frac{2w_1 \Omega_1 \dot{\alpha}}{K_1^2} + \frac{w_1 \Theta_1 \zeta_{1,2}}{K_1} \right) \mathcal{Q}(q). \quad (57)$$

An additive function of time can be subtracted if $p(0, t) = 0$ is desired. The exact inviscid forces are

$$f_\theta^0 = \dot{w}_1 \Theta_1 F(q), \quad f_u^0 = \left(\ddot{\alpha} - \frac{A_0}{2} \sin \alpha \right) Jx + \frac{\dot{w}_1 \Omega_1}{K_1} J\zeta_1 F(q). \quad (58)$$

We verify every component. The background derivative satisfies $\dot{G}_{<1} = \dot{\alpha} J G_{<1}$, hence its scalar transport residual vanishes. The phase satisfies $(\partial_t + \dot{\alpha} Jx \cdot \nabla)q = 0$. The wave velocity has scalar product zero with ∇q , so it advects neither its own temperature nor its own velocity profile. Its advection of the background temperature equals $(w_1 \Omega_1 / K_1)(J\zeta_1 \cdot G_{<1})F(q)$, which cancels $w_1 \dot{\Theta}_1 F(q)$ by (25). The remaining scalar residual is exactly $\dot{w}_1 \Theta_1 F(q)$.

For momentum, write $C = w_1 \Omega_1 / K_1$ and $v = C J\zeta_1 F(q)$. Differentiating v and adding its transport of $\dot{\alpha} Jx$ gives, after subtracting the wave buoyancy,

$$\left[\frac{\dot{w}_1 \Omega_1}{K_1} J\zeta_1 + \frac{w_1 \dot{\Omega}_1}{K_1} J\zeta_1 - 2C\dot{\alpha}\zeta_1 - w_1 \Theta_1 e_2 \right] F(q).$$

The last three vector terms have zero scalar product with $J\zeta_1$: this is $w_1 \dot{\Omega}_1 / K_1 - w_1 \Theta_1 \zeta_{1,1} = 0$. Their scalar product with ζ_1 is $-2C\dot{\alpha} - w_1 \Theta_1 \zeta_{1,2}$. Since $(\zeta_1, J\zeta_1)$ is an orthonormal basis, the wave pressure in (57) cancels precisely these three terms, leaving the displayed activation force. The base momentum residual before pressure is $(\ddot{\alpha} J - \dot{\alpha}^2 I - e_2 \otimes G_{<1})x$. The gradient of the first two pressure terms removes its symmetric part. Its antisymmetric part is $(\ddot{\alpha} - G_{<1,1}/2)Jx = (\ddot{\alpha} - A_0 \sin \alpha / 2)Jx$, as claimed. Finally $\text{div } u = 0$ since $\text{tr } J = 0$ and $J\zeta_1 \cdot \zeta_1 = 0$.

For the retained-viscosity equations

$$\partial_t \theta + u \cdot \nabla \theta = \kappa_{\text{phys}} \Delta \theta + f_\theta, \quad \partial_t u + (u \cdot \nabla)u + \nabla p = \nu_{\text{phys}} \Delta u + \theta e_2 + f_u, \quad \nabla \cdot u = 0,$$

the very same state and controls solve them with the explicitly specified compensating forces

$$\begin{aligned} f_\theta &= f_\theta^0 - \kappa_{\text{phys}} w_1 \Theta_1 K_1^2 F''(q), \\ f_u &= f_u^0 - \nu_{\text{phys}} w_1 \Omega_1 K_1 J\zeta_1 F''(q). \end{aligned} \quad (59)$$

Indeed the affine Laplacians vanish, while two spatial derivatives of the profile multiply by $K_1^2 |\zeta_1|^2 = K_1^2$. This proves the asserted equations by substitution, without changing either ν or ν_{phys} and without assuming damping leaves the parent coefficients unchanged. The exact cumulative L_x^∞ norms of the two added forces through the entire stage and hold are

$$\kappa_{\text{phys}} K_1^2 \|F''\|_\infty I_P, \quad \nu_{\text{phys}} K_1 \|F''\|_\infty I_\Omega. \quad (60)$$

Equality holds because $x \mapsto K_1 \zeta_1 \cdot x$ is surjective onto $\mathbb{R}$ and $|J\zeta_1| = 1$; this verifies the norm statement as well as an upper bound. The scalar holding contribution is exactly $\kappa_{\text{phys}} K_1^2 \|F''\|_\infty P_b T_h$; the velocity holding contribution is zero. These finite costs are not asserted to be summable over the source's infinite frequency sequence.

The construction proves a finite controlled stage, its actual evolving affine background, its return, and its next growth holding interval. It does not establish an infinite viscous cascade or a smooth terminal-time viscous force. In particular, simply replacing the first temperature by an exponentially damped amplitude during holding would replace A_1 in (51) by a time-dependent quantity and would change both coefficients and the rate in (53). The compensating forces in (59) give an exact specified relation that retains the original coupled finite trajectory; their costs have been retained explicitly rather than claimed negligible.

3 Solving the next growth interval with a diffusing parent

This section continues an uncompensated equal-diffusion variant of the first stage through the next growth interval. It keeps the laboratory direction and the original coefficients. The exact amplitude system is realized below both on a forced affine parent and on an actual two-wave state with an explicitly computed interaction force. The latter force specifies the difference between the affine-gradient calculation and the full spatial equations.

3.1 The inherited gradient and the original second pair

Retain the preceding section's $A_0, \mu, \nu, K_1, s, \lambda_1$, and let $\nu_{\text{phys}} = \kappa_{\text{phys}} = d > 0$. For a sine first wave on its rotating affine base, the exact amplitude equations with retained diffusion are

$$\dot{\Theta}_{1,d} = a_1 \Omega_{1,d} - dK_1^2 \Theta_{1,d}, \quad \dot{\Omega}_{1,d} = b_1 \Theta_{1,d} - dK_1^2 \Omega_{1,d}.$$

If (Θ_1, Ω_1) denotes the first inviscid pair of the preceding section on its specified control and times, direct differentiation proves

$$(\Theta_{1,d}, \Omega_{1,d})(t) = e^{-dK_1^2(t-t_0)}(\Theta_1, \Omega_1)(t).$$

Both the forward map and its inverse multiplication by the positive exponential satisfy the indicated systems, so they are a bijection of solution spaces. In particular the selected vorticity still returns to zero at the specified t_b . This uses the inviscid times and does not assert the damped temperature crosses the same growth threshold. Its exact log-gain from the original seed is

$$\log(P_b/S) - \frac{dK_1^2}{\nu \sin s}(L + 2 + \Lambda_1^{-1}).$$

Set $P_{d,b} = e^{-dK_1^2(t_b-t_0)}P_b > 0$, $A_{1,d} = K_1 P_{d,b}$, and $r = t - t_b \geq 0$. During the fixed first holding control $\alpha = s$ the exact first pair is

$$\Theta_{1,d}(r) = -P_{d,b}e^{-qr}, \quad \Omega_{1,d} = 0, \quad q = dK_1^2, \quad G_d(r) = (A_0 \sin s, -A_0 \cos s - A_{1,d}e^{-qr}).$$

The gradient therefore changes at the rate $G'_d(r) = (0, qA_{1,d}e^{-qr})$. Let the new laboratory phase be $\zeta_2 = e(s_2)$ with $0 < s_2 \leq 1/8$, and let $K_2 = \mu\lambda_2 > 0$. The actual first hold has zero velocity gradient, so $\zeta'_2 = 0$. The second affine-wave coefficients are exactly

$$a_{2,d}(r) = -\frac{J\zeta_2 \cdot G_d(r)}{K_2} = \frac{A_{1,d}e^{-qr} \sin s_2 + A_0 \sin(s + s_2)}{K_2}, \quad b_2 = K_2 \sin s_2, \quad \delta_2 = dK_2^2. \quad (61)$$

Their signs are positive because $0 < s + s_2 \leq 1/4$. The exact second sine amplitude system is

$$\Theta'_2 = a_{2,d}(r)\Omega_2 - \delta_2\Theta_2, \quad \Omega'_2 = b_2\Theta_2 - \delta_2\Omega_2. \quad (62)$$

In particular the changing inherited gradient has not been replaced by its endpoint value. An affine realization alone would require the scalar force $G'_d(r) \cdot x$: the affine Laplacian is zero, so that force cannot be inferred to vanish from the old sine-wave diffusion.

3.2 Exact operator change and convergent fundamental solutions

Define

$$c_0 = A_0 \sin(s + s_2) \sin s_2 > 0, \quad c_1 = A_{1,d} \sin^2 s_2 > 0, \quad W(r) = e^{\delta_2 r} \Omega_2(r).$$

Because b_2 is constant, differentiating the second original equation gives the exact system equivalence

$$W'' = (c_0 + c_1 e^{-qr})W, \quad \Omega_2 = e^{-\delta_2 r} W, \quad \Theta_2 = \frac{e^{-\delta_2 r}}{b_2} W'. \quad (63)$$

Conversely these last two formulas give $\Omega'_2 = b_2\Theta_2 - \delta_2\Omega_2$ and $\Theta'_2 = e^{-\delta_2 r} W''/b_2 - \delta_2\Theta_2 = a_{2,d}\Omega_2 - \delta_2\Theta_2$, since $b_2 a_{2,d} = c_0 + c_1 e^{-qr}$. Thus the initial-data map is the invertible map $(\Theta_2(0), \Omega_2(0)) \mapsto (W(0), W'(0)) = (\Omega_2(0), b_2\Theta_2(0))$.

Put

$$x(r) = x_0 e^{-qr/2}, \quad x_0 = \frac{2\sqrt{c_1}}{q}, \quad \vartheta = \frac{2\sqrt{c_0}}{q} > 0, \quad W(r) = \mathcal{W}(x(r)).$$

Here $x(r)$ is a scalar change of independent variable and is unrelated to either physical spatial coordinate x_1, x_2 , which remain fixed in the PDE below. Since $x' = -qx/2$ and $x'' = q^2x/4$, one has

$$W'' - (c_0 + c_1 e^{-qr})W = \frac{q^2}{4} [x^2 \mathcal{W}_{xx} + x \mathcal{W}_x - (x^2 + \vartheta^2) \mathcal{W}].$$

This proves the operator transformation including its nonzero factor. Rather than invoking a special-function name, define on $x > 0$

$$f(x) = x^\vartheta \sum_{n=0}^{\infty} a_n x^{2n}, \quad a_0 = 1, \quad a_n = \frac{a_{n-1}}{4n(n+\vartheta)}, \quad g(x) = f(x) \int_{x_0}^x \frac{d\xi}{\xi f(\xi)^2}. \quad (64)$$

For every compact interval in $x > 0$, the consecutive-term absolute ratio $x^2/[4n(n+\vartheta)]$ tends uniformly to zero. Every fixed derivative multiplies these terms by a polynomial in n and a bounded power of x^{-1} on that compact interval; its series therefore converges uniformly as well. Termwise substitution is valid. The coefficient of $x^{\vartheta+2n}$ in $x^2 f'' + x f' - (x^2 + \vartheta^2) f$ is $4n(n+\vartheta)a_n - a_{n-1} = 0$ for $n \geq 1$; the $n = 0$ coefficient is also zero. All coefficients are positive, so $f > 0$. The defining integral for g consequently exists on every compact subinterval of $x > 0$. If $I' = 1/(xf^2)$, differentiation yields

$$f(fI)' - f'(fI) = f^2 I' = 1/x.$$

Also $I''/I' = -1/x - 2f'/f$; substituting $g = fI$ into $g'' + g'/x - (1 + \vartheta^2/x^2)g$ leaves $fI'' + (2f' + f/x)I' = 0$. Thus g solves the same equation and its Wronskian with f is the nonvanishing function $1/x$. For completeness this proves completeness of the two solutions: at any $x > 0$ their value-and-derivative matrix is invertible, so one linear combination matches arbitrary Cauchy data; linear uniqueness then identifies that combination with any solution on overlapping compact subintervals of $(0, \infty)$.

For the original data let $W_0 = \Omega_2(0)$ and $W_1 = b_2 \Theta_2(0)$. The exact solution is

$$W(r) = c_f f(x(r)) + c_g g(x(r)), \quad c_f = \frac{W_0}{f(x_0)}, \quad c_g = -\frac{2f(x_0)}{q} \left(W_1 + \frac{qx_0}{2} c_f f'(x_0) \right). \quad (65)$$

Indeed $g(x_0) = 0$, $g'(x_0) = 1/[x_0 f(x_0)]$, and $x'(0) = -qx_0/2$ verify both initial conditions directly. For every finite r , $x(r) > 0$, so the displayed functions and their derivatives are finite there. If $d = 0$, the preceding substitutions dividing by q are not used: the original equation is $W'' = (c_0 + c_1)W$, with the exact solution $W = W_0 \cosh(\sqrt{c_0 + c_1} r) + W_1 \sinh(\sqrt{c_0 + c_1} r)/\sqrt{c_0 + c_1}$.

3.3 Signs and quantitative physical gain with this feedback

Let $\gamma_0 = \sqrt{c_0}$ and $\gamma_i = \sqrt{c_0 + c_1}$. Use the original instantaneous growing eigenline at entry: $\Theta_2(0) < 0$ and $\Omega_2(0) = \sqrt{b_2/a_{2,d}(0)} \Theta_2(0) < 0$. Then $Y = -W$ has $Y(0) = Y_0 > 0$ and $Y'(0) = \gamma_i Y_0 > 0$. As long as $Y > 0$, its equation gives $Y'' > 0$ and $Y' \geq Y'(0) > 0$; this excludes a first zero by continuity. Define

$$Y_- = Y_0 \left(\cosh(\gamma_0 r) + \frac{\gamma_i}{\gamma_0} \sinh(\gamma_0 r) \right), \quad Y_+ = Y_0 e^{\gamma_i r}.$$

The differences have zero initial value and derivative and satisfy

$$\begin{aligned} (Y - Y_-)'' - \gamma_0^2 (Y - Y_-) &= c_1 e^{-qr} Y, \\ (Y_+ - Y)'' - \gamma_i^2 (Y_+ - Y) &= c_1 (1 - e^{-qr}) Y. \end{aligned}$$

For $a > 0$, the formula $Z(r) = \int_0^r \sinh(a(r-u)) S(u)/a \, du$ has $Z(0) = Z'(0) = 0$ and $Z'' - a^2 Z = S$, by two differentiations. Uniqueness proves it is the solution of that initial-value problem. Both sources above are nonnegative; both this kernel and its first r derivative are nonnegative for $0 \leq$

$u \leq r$. Therefore $Y_- \leq Y \leq Y_+$ and $Y'_- \leq Y' \leq Y'_+$ for every finite $r \geq 0$. The inverse physical map now gives both amplitudes negative and the precise temperature-gain bounds

$$e^{-\delta_2 r} \left(\cosh(\gamma_0 r) + \frac{\gamma_0}{\gamma_i} \sinh(\gamma_0 r) \right) \leq \frac{-\Theta_2(r)}{-\Theta_2(0)} \leq e^{(\gamma_i - \delta_2)r}. \quad (66)$$

In particular, if the actual numerical coefficients satisfy $\delta_2 \geq \gamma_i$, this particular growth control never raises the second temperature above its entry amplitude. This is a proved obstruction to that specified uncompensated control, not a statement about other controls or the existence of an infinite viscous sequence.

3.4 The exact map to spatial fields and its interaction force

The affine parent $\theta_b = G_d(r) \cdot x$, $u_b = 0$ is a solution with pressure $p_b = x_2 G_d(r) \cdot x/2$ and forces $f_{\theta,b} = G'_d(r) \cdot x$ and $f_{u,b} = -(A_0 \sin s/2)Jx$. The scalar statement follows from $\Delta\theta_b = 0$. For momentum, direct differentiation gives $\nabla p_b - \theta_b e_2 = (G_d x_2 - e_2(G_d \cdot x))/2 = -G_{d,1}Jx/2$. An added sine wave whose amplitudes solve (62) has zero additional PDE residual except for its activation, by the same explicit phase, transverse-vector, and pressure calculations below. Thus the solved system has an exact forced affine-background realization.

There is also a full two-wave realization specifying the original spatial discrepancy. Let F_1, F_2 be smooth odd periodic profiles, each with mean-zero primitive $Q'_j = F_j$, and assume $F'_1(0) = 1$. This includes both sine and the released profile which is linear near zero. Set

$$G_0 = (A_0 \sin s, -A_0 \cos s), \quad T(r) = -P_{d,b}e^{-qr}, \quad q_1 = K_1 x_2, \quad q_2 = K_2 \zeta_2 \cdot x,$$

and let $w_2(r)$ be any specified smooth activation, for example the source's $\eta(\Gamma_2 r)$. Define on $\mathbb{R}^2$

$$\begin{aligned} \theta &= G_0 \cdot x + T F_1(q_1) + w_2 \Theta_2 F_2(q_2), \\ u &= \frac{w_2 \Omega_2}{K_2} J \zeta_2 F_2(q_2), \\ p &= \frac{1}{2} x_2 (G_0 \cdot x) + \frac{T}{K_1} Q_1(q_1) + \frac{w_2 \Theta_2 \zeta_{2,2}}{K_2} Q_2(q_2). \end{aligned} \quad (67)$$

Their complete retained-diffusion forces are

$$\begin{aligned} f_\theta &= -dK_1^2 T [F_1(q_1) + F_1''(q_1)] + w'_2 \Theta_2 F_2(q_2) \\ &\quad - dK_2^2 w_2 \Theta_2 [F_2(q_2) + F_2''(q_2)] + f_{\text{cross}}, \end{aligned} \quad (68)$$

$$f_{\text{cross}} = \frac{w_2 \Omega_2}{K_2} \sin s_2 K_1 T [F'_1(q_1) - 1] F_2(q_2), \quad (69)$$

$$f_u = -\frac{A_0 \sin s}{2} Jx + \frac{w'_2 \Omega_2}{K_2} J \zeta_2 F_2(q_2) - dK_2 w_2 \Omega_2 J \zeta_2 [F_2(q_2) + F_2''(q_2)]. \quad (70)$$

Here is a direct verification. The old wave has derivative $T' = -dK_1^2 T$ and Laplacian $K_1^2 T F_1''$, giving the first scalar force. The old temperature gradient differs from the coefficient gradient $G_d = G_0 + K_1 T e_2$ by precisely $K_1 T [F'_1(q_1) - 1] e_2$. Its advection by u is the displayed f_{cross} because $J \zeta_2 \cdot e_2 = \sin s_2$. The remainder of that old-temperature advection is $(w_2 \Omega_2 / K_2) J \zeta_2 \cdot G_d F_2 = -w_2 a_{2,d} \Omega_2 F_2$; it cancels the coupling part of $w_2 \Theta'_2 F_2$. The new wave advects its own temperature and velocity by zero since $J \zeta_2 \cdot \zeta_2 = 0$. Its amplitude damping and its Laplacian give the two new-profile terms in the forces. The old wave pressure has gradient $T F_1(q_1) e_2$, so it cancels old wave buoyancy exactly. The new pressure has gradient $w_2 \Theta_2 \zeta_{2,2} \zeta_2 F_2$. The new momentum coupling vector before that pressure is $w_2 \Theta_2 (\zeta_{2,1} J \zeta_2 - e_2) F_2 = -w_2 \Theta_2 \zeta_{2,2} \zeta_2 F_2$, by resolving e_2 in the orthonormal basis $(\zeta_2, J \zeta_2)$. Hence it cancels as well. The base pressure supplies exactly the first vector-force term, as proved above, and divergence is zero by transversality. This verifies both full equations including all interactions.

For the sine profiles, $F_j + F_j'' = 0$ identically. The remaining scalar interaction force is then exactly

$$f_{\text{cross}} = -\frac{A_{1,d}e^{-qr}}{K_2}w_2\Omega_2 \sin s_2 [\cos(K_1x_2) - 1] \sin(q_2), \quad \|f_{\text{cross}}(r)\|_\infty \leq \frac{2A_{1,d}e^{-qr}}{K_2}|w_2\Omega_2| \sin s_2.$$

The bound follows from $|\cos u - 1| \leq 2$ and $|\sin v| \leq 1$. For the released linear-core profile the same explicit formulas hold, and $F_1' - 1 = 0$ on its original linear interval, so the interaction force vanishes there. Its old-profile diffusion defect is instead $-dK_1^2Tq_1$ on that interval, since $F_1(q_1) = q_1$ and $F_1''(q_1) = 0$. Its gradient is exactly $G_d'(r)$. Thus exponential damping of that released linear core requires the stated core force, and is not an unforced consequence of its zero Laplacian. The explicit forces, inverse solution maps, and convergent formulas above carry the feedback calculation to an actual spatial solution without conflating the sine state, its affine gradient, and the released profile. No compact-support or infinite-iteration claim is made in this section.

4 Reaching both original growth targets with retained diffusion

We now join the exact evolving-parent growth calculation to the direct second return theorem. Both stopping events are the original temperature thresholds. This changes the first growth time from the earlier fixed-inviscid-time variant in Section 3. The construction below specifies the frequencies and a nonempty interval of positive physical diffusivities, and proves that their actual entry states satisfy every inequality in Theorem 2. The spatial fields and their full forces are supplied by the compact construction following that theorem. In particular the amplitude damping on the released linear profile has its explicitly stated core force.

4.1 A finite choice within the original frequency schedule

Retain the original physical data

$$A_0 \geq 1, \quad \nu = \sqrt{A_0}, \quad \mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil,$$

and all fixed source profiles, including $s_*, s_0, C_\ell, \eta, \eta_2, \Lambda_1$. Thus $0 < s_* \leq 1/8$, $C_\ell \geq 4$, $H = \|\eta_2\|_\infty \geq 1$, $\Lambda_1 \geq 6H$, and $3\Lambda_1 H \sin s_* \leq 1/4$. The first section's λ_1 denoted a source frequency; in this section we write that same quantity as $\hat{\lambda}_1$ and retain the physical frequencies separately as $K_j = \mu\hat{\lambda}_j$. Keep the source's $Q_* \geq 200$, $Q_j = Q_* + j$, its specified finite constants $K(k)$, and the exact choices

$$\begin{aligned} \hat{\lambda}_2 &= \hat{\lambda}_1^{Q_2}, & k_j &= \max\{k \in \mathbb{N}_0 : 120k \leq Q_j, K(k) \leq \log \hat{\lambda}_j\}, \\ L_j &= (k_j + 41/8) \log \hat{\lambda}_j, & S_j &= \frac{A_0}{\mu} \hat{\lambda}_j^{-k_j-6}, \\ P_j^* &= \frac{A_0}{\mu} \hat{\lambda}_j^{-7/8} = S_j e^{L_j}. \end{aligned} \tag{71}$$

Here $K(k)$ is the source's prescribed parameter sequence, not the physical frequency K_j . The maximum is a nonempty finite set once $\log \hat{\lambda}_1 \geq K(0)$: $k = 0$ belongs to it and $k \leq \lfloor Q_j/120 \rfloor$. No smooth dependence of k_j on the frequencies is required.

The following fixed constants give one explicit sufficient frequency choice; their large numerical size is immaterial to finite existence:

$$\begin{aligned} C_A &= e^{2+\Lambda_1^{-1}}, & C_\sigma &= \sqrt{1+C_A}, & C_N &= 1 + 3C_\sigma/32, \\ C_L &= (\lfloor Q_2/120 \rfloor + 41/8)Q_2, \\ Y_0 &= 1 + \max \left\{ \begin{array}{l} 64, K(0), Q_2, 72 \cdot 16^3 H C_L^2 / \sqrt{e}, \quad 8 \log(4/e), \\ (16/47) \log(4C_\ell C_N / s_0), \quad (16/31) \log(2C_N / s_0) \end{array} \right\}. \end{aligned} \tag{72}$$

Choose any $\widehat{\lambda}_1 \geq e^{Y_0}$ and then use (71) literally. Write $L = L_2$, $\Gamma_1 = \nu \sin s_*$ and $C_{\text{post}} = 2 + \Lambda_1^{-1}$. Define the explicit positive number

$$d_{\max} = \min \left\{ \frac{\Gamma_1}{2K_1^2}, \frac{\Gamma_1 \log 2}{K_1^2 C_{\text{post}}}, \frac{\nu \log(3/2)}{16K_1^2}, \frac{L\nu \log 3}{12800K_2^2} \right\}. \quad (73)$$

Every entry in this minimum uses only the fixed original data and the chosen frequencies. In particular it does not require solving a circular inequality involving a yet unknown diffusive growth scale. All subsequent statements hold for each $0 < d \leq d_{\max}$, with physical viscosity and thermal diffusivity both equal to d .

4.2 First actual threshold and the changed first endpoint

Put $q = dK_1^2$ and $t_1^{\text{in}} = 1/\nu$. Growth and transition still have $\alpha = 0$. The signed first pair is exactly

$$\Theta_1(t) = -S_1 e^{(\Gamma_1 - q)(t - t_1^{\text{in}})}, \quad \Omega_1(t) = \frac{K_1}{\nu} \Theta_1(t). \quad (74)$$

Indeed its temperature coupling is $a_1 = A_0 \sin s_*/K_1$, its vorticity coupling is $b_1 = K_1 \sin s_*$, and $a_1 K_1/\nu = \Gamma_1$, $b_1 \nu/K_1 = \Gamma_1$. Differentiation verifies both damped equations and both original seed values. The linear integral equation proves uniqueness. Since $q \leq \Gamma_1/2$, the original first threshold is attained uniquely and transversely at

$$t_1^a = t_1^{\text{in}} + \frac{L_1}{\Gamma_1 - q}, \quad t_1^1 = t_1^a + \Gamma_1^{-1}, \quad t_1^b = t_1^1 + \Gamma_1^{-1}(1 + \Lambda_1^{-1}). \quad (75)$$

The first activation $\eta(\Gamma_1(t - t_1^{\text{in}}))$ is already one at t_1^a , since $L_1 \geq 1$.

Starting at t_1^a , take the original first transition and steering control with their displayed durations, translated to these actual times. The inverse map

$$(\Theta_1, \Omega_1)(t) = e^{-q(t - t_1^a)} (\Theta_{1,E}, \Omega_{1,E})(t)$$

relates their damped solution to the inviscid controlled solution with data

$$(\Theta_{1,E}, \Omega_{1,E})(t_1^a) = (-P_1^*, -K_1 P_1^*/\nu).$$

Differentiation gives the two damped equations, and multiplication by $e^{q(t - t_1^a)}$ gives the inverse equations and identical data. Both factors are positive. Thus the quotient, its entire trial family, and its unique first return root proved in Section 2 are unchanged on these translated intervals. At the selected endpoint

$$\alpha = s_*, \quad \zeta_1 = e_2, \quad \Omega_1 = 0, \quad A_d := -K_1 \Theta_1(t_1^b) = A_E e^{-qC_{\text{post}}/\Gamma_1}, \quad (76)$$

where the proved inviscid endpoint satisfies

$$2eA_0 \widehat{\lambda}_1^{1/8} \leq A_E \leq C_A A_0 \widehat{\lambda}_1^{1/8}. \quad (77)$$

This expression does not damp the inviscid growth duration: that duration has been replaced by the actual threshold time in (75). The first three diffusion exponents are exactly $qL_1/(\Gamma_1 - q)$ in growth, q/Γ_1 in transition, and $q(1 + \Lambda_1^{-1})/\Gamma_1$ in steering. The physical gains are respectively L_1 , $1 - q/\Gamma_1$, and the original selected steering gain minus the last exponent.

4.3 Actual second scale and all fixed design inequalities

The true inherited gradient at $t_2^{\text{in}} = t_1^b$ is $G_d = (A_0 \sin s_*, -A_0 \cos s_* - A_d)$. Evaluate the source scale at this changed endpoint, as required by its original rule:

$$\sigma^2 = |G_d| = \sqrt{A_0^2 + A_d^2 + 2A_0 A_d \cos s_*}, \quad s = s_2 = \frac{L\nu}{\sigma}, \quad a = \sin s, \quad \Gamma = \Gamma_2 = \sigma a. \quad (78)$$

Since $qC_{\text{post}}/\Gamma_1 \leq \log 2$, equations (77)–(78) imply

$$eA_0\hat{\lambda}_1^{1/8} \leq A_d \leq C_A A_0\hat{\lambda}_1^{1/8}, \quad \sqrt{e}\nu\hat{\lambda}_1^{1/16} \leq \sigma \leq C_\sigma\nu\hat{\lambda}_1^{1/16}. \quad (79)$$

For the lower scale bound use $\sigma^2 \geq A_d$, since the square of its left side differs from A_d^2 by a positive quantity. For the upper bound use the triangle inequality $\sigma^2 \leq A_0 + A_d \leq (1 + C_A)A_0\hat{\lambda}_1^{1/8}$. The choice $Y_0 > 8 \log(4/e)$ also gives $A_d > 4A_0$.

Let $y = \log \hat{\lambda}_1 \geq Y_0$. Then $64 < L \leq C_L y$. The exponential series with nonnegative terms proves $e^{y/16} \geq y^3/(6 \cdot 16^3)$; by (72) its right side is strictly greater than $12HC_L^2 y^2/\sqrt{e}$. Consequently

$$0 < s \leq \frac{L}{\sqrt{e} e^{y/16}} < \frac{1}{12HL} \leq \frac{1}{8}, \quad 0 < a \leq \frac{1}{8}, \quad 3LHa < \frac{1}{4}. \quad (80)$$

For $0 \leq s \leq 1/8$, integrating $\cos r \geq 1 - r^2/2 \geq 1/2$ on $[0, s]$ gives $s/2 \leq \sin s \leq s$. Thus

$$\frac{L\nu}{2} \leq \Gamma \leq L\nu. \quad (81)$$

The fixed base coefficients in the second-return coordinates obey

$$0 < C_0 = \frac{A_0 \cos s_*}{\sigma^2} < \frac{1}{4}, \quad 0 < C_1 = \frac{A_0 \sin s_*}{\sigma^2 a} \leq \frac{2\nu \sin s_*}{\sigma L} < \frac{1}{32}. \quad (82)$$

The last inequality uses $\sigma \geq \nu$, $L \geq 64$ and $\sin s_* < 1$; it has no missing comparison constant.

The same frequency choice proves the source support nesting for the entire second return and its finite hold. Set $N = 1 + (3 - \cos s)/a$. The exact matrix reconstruction in Theorem 2 gives both $\|M_2\|$ and $\|M_2^{-1}\|$ at most N , by the triangle inequality for its displayed shear. Equations (79) and $a \geq s/2$ imply $1/a \leq 2C_\sigma\hat{\lambda}_1^{1/16}/L$, whence $N \leq C_N\hat{\lambda}_1^{1/16}$. The final two logarithmic bounds in (72) give, strictly,

$$N\hat{\lambda}_1^{-3} < \frac{s_0}{2C_\ell}, \quad N\hat{\lambda}_1^{-2} < s_0. \quad (83)$$

Indeed their respective left sides are at most $C_N e^{-47y/16}$ and $C_N e^{-31y/16}$, and our choices bound these by $s_0/(4C_\ell)$ and $s_0/2$. For $\ell_1 = s_0/(C_\ell \mu)$ and $\ell_2 = \hat{\lambda}_1^{-3}/\mu$, this places the entire second support inside the first envelope's constant region and its exact linear profile region: $|x| \leq N\ell_2 < \ell_1/2$ and $|K_1 \zeta_1 \cdot x| \leq K_1 N\ell_2 < s_0$. Every original factor of μ is present in these identities.

4.4 Second growth with the actual diffusing parent

Write $r = t - t_2^{\text{in}}$. The second growth and transition have $\alpha = s_*$, $\Omega_1 = 0$, $M_2 = I$, $\zeta_2 = e(s)$ and $A_1(r) = A_d e^{-qr}$. These claims follow from the closed subsystem: $\zeta_{1,1} = 0$ makes $\dot{\Omega}_1 = -q\Omega_1 = 0$; both controls F_1^{ctl} and F_2^{ctl} are then zero, and the phase matrix equation is $\dot{M}_2 = 0$. The first temperature equation is $\dot{A}_1 = -qA_1$. The second pair has the exact coefficients

$$a_2(r) = \frac{A_d e^{-qr} \sin s + A_0 \sin(s_* + s)}{K_2}, \quad b_2 = K_2 \sin s, \quad d_2^{\text{phys}} = dK_2^2, \quad (84)$$

and the exact original entry data $\Theta_2(0) = -S_2$, $\Omega_2(0) = \sqrt{b_2/a_2(0)}\Theta_2(0)$. This is precisely the system solved by the convergent fundamental functions in Section 3; the substitution here uses its displayed changed coefficients, not frozen values.

Put $a_i = a_2(0)$, $u_i = \sqrt{b_2/a_i}$, $\gamma_i = \sqrt{a_i b_2}$ and

$$n_i = \frac{\gamma_i^2}{\Gamma^2} = \frac{A_d + A_0 \cos s_*}{\sigma^2} + C_1 \cos s.$$

One has

$$\frac{4}{5} < n_i < \frac{33}{32} < 2. \quad (85)$$

The lower bound follows from $A_d/\sigma^2 \geq A_d/(A_d + A_0) > 4/5$. For the upper bound, square $\sigma^2 = \sqrt{(A_d + A_0 \cos s_*)^2 + A_0^2 \sin^2 s_*}$ to get $(A_d + A_0 \cos s_*)/\sigma^2 \leq 1$, then use (82). In particular $\gamma_i \geq \Gamma/\sqrt{2} \geq L\nu/(2\sqrt{2})$.

On the explicitly fixed horizon $0 \leq r \leq 16/\nu$, the third bound in (73) gives $qr \leq \log(3/2)$. Since both terms in a_2 are positive, it follows that

$$\frac{1}{2}a_i \leq a_2(r) \leq a_i. \quad (86)$$

To prove all signs and ratio estimates without assuming them, set $X = -e^{d_2^{\text{phys}} r} \Theta_2$ and $Y = -e^{d_2^{\text{phys}} r} \Omega_2$. They solve $X' = a_2 Y$, $Y' = b_2 X$ with positive initial values. Before any first zero, both derivatives are positive, excluding that zero by continuity. The linear integral equation, whose coefficient matrix is bounded on this horizon, gives existence there by its uniformly convergent iterated-integral series. Therefore $u = Y/X = \Omega_2/\Theta_2$ exists and satisfies $u' = b_2 - a_2 u^2$. The constant functions u_i and $\sqrt{2}u_i$ are respectively subsolution and supersolution on this horizon, because $b_2 - a_2 u_i^2 \geq 0$ and $b_2 - 2a_2 u_i^2 \leq 0$. Subtracting either comparison equation from the equation for u gives the linear difference coefficient $-a_2(u + u_{\text{comparison}})$; its integrating factor preserves the stated ordering. Hence

$$u_i \leq u \leq \sqrt{2}u_i, \quad \frac{1}{\sqrt{n_i}} \leq v := \frac{\sigma \Omega_2}{K_2 \Theta_2} \leq \sqrt{\frac{2}{n_i}}, \quad \frac{1}{2} < v < 2. \quad (87)$$

Here $\sigma u_i/K_2 = \Gamma/\gamma_i = 1/\sqrt{n_i}$; the equality follows by substituting the original a_i, b_2, K_2 .

For $P_2 = -\Theta_2$, the exact physical logarithmic derivative is

$$(\log P_2)' = a_2 u - d_2^{\text{phys}}.$$

The last bound in (73) gives $d_2^{\text{phys}} \leq L\nu \log 3/12800 < L\nu/16 \leq \gamma_i/4$. Thus throughout the fixed horizon

$$\frac{\gamma_i}{4} \leq (\log P_2)' \leq \sqrt{2}\gamma_i \leq 2\Gamma. \quad (88)$$

Indeed $a_2 u \geq a_i u_i/2 = \gamma_i/2$ and $a_2 u \leq \sqrt{2}a_i u_i = \sqrt{2}\gamma_i$. Consequently the original second target $P_2^* = S_2 e^L$ is reached at exactly one $r = r_a$, with nonzero derivative, and

$$\frac{L}{2\Gamma} \leq r_a \leq \frac{4L}{\gamma_i} \leq \frac{8\sqrt{2}}{\nu}, \quad r_1 = r_a + \Gamma^{-1} < \frac{16}{\nu}, \quad t_2^a = t_2^{\text{in}} + r_a, \quad t_2^1 = t_2^{\text{in}} + r_1. \quad (89)$$

For existence of the crossing, the upper candidate time is inside the fixed horizon; integrating the lower rate to that time gives at least L logarithmic gain. Strict monotonicity and the intermediate value theorem give existence and uniqueness. The transition length is at most $2/(L\nu) \leq 1/(32\nu)$, so the entire transition is also inside that already justified horizon. The lower bound on r_a ensures the second activation $\eta(\Gamma r)$ equals one before the growth threshold, since $L \geq 64$.

4.5 Closing every entry bound and the complete finite statement

At the actual steering entry t_2^1 , take $P_{1\text{in}} = A_d e^{-qr_1}/K_1$ and $P_{2\text{in}} = P_2(r_1)$. Equations (87) and (89) give the initial quotient interval in Theorem 2. Also

$$\frac{8}{15} < B_{\text{in}} = \frac{A_d e^{-qr_1}}{\sigma^2} \leq 1, \quad 0 < \delta_2 = \frac{dK_2^2}{\Gamma} \leq \frac{\log 3}{6400}, \quad 0 < \delta_1 = \frac{dK_1^2}{\Gamma} \leq \delta_2 < \frac{1}{16}. \quad (90)$$

The lower parent bound is $(4/5)(2/3)$; the upper is $A_d \leq \sigma^2$. The diffusion bounds follow from (73), (81) and $K_1 < K_2$. Equations (80), (82), (87), and (90) prove every entry/design hypothesis of the direct return theorem from the original seeds and the explicit choices (72)–(73).

Theorem 1. *For each fixed original source datum and profiles above, choose $\hat{\lambda}_1$ by (72), retain the source recursion (71), and choose any $0 < d \leq d_{\text{max}}$ in (73). The complete modified two-layer coefficient system exists through both original growth targets, both prescribed transitions, both selected vorticity returns, and a second holding interval of physical length $1/(32\Gamma_2)$. Its times, amplitudes, original coordinates, controls, and spatial support inequalities are the exact formulas above and the inverse formulas of Theorem 2. At the second return the new vorticity amplitude and horizontal phase component are zero, the first parent vorticity is strictly positive, and*

$$\log \frac{-\Theta_2(t)}{S_2} \geq L_2 + \frac{\gamma_i}{4\Gamma_2} + \frac{\log 3}{320} \quad \text{for } t_2^b \leq t \leq t_2^b + (32\Gamma_2)^{-1}. \quad (91)$$

The inherited parent is evolved throughout; no inviscid estimate is assumed to persist under diffusion. With the full profile/localization forces below these coefficients determine compact spatial solutions from the original initial datum and zero initial velocity.

Proof. The first stage was proved by the exact map at its actual threshold. The second growth and transition, their signs and all entry bounds were proved on a common explicit physical horizon. Theorem 2 therefore applies, constructs its entire trial family before taking the least root, and proves the return and nonstationary parent hold. The second growth contributes exactly L_2 to logarithmic gain. The transition contributes at least $\gamma_i/(4\Gamma_2)$ by (88). The steering and entire hold contribute at least $\log 3/320$ by that theorem and (90). Adding these three actual integrals proves (91). All finite join controls are smooth by flatness of the original step and interior support of its pulse; the solution of the smooth ODE therefore matches to every order at those joins. The spatial identities below prove the stated compact realization with every required force. $\square$

For use in those spatial bounds, throughout both stages and their hold one may use the following proved amplitude upper bounds:

$$\begin{aligned} \sup |\Theta_1| &\leq \max\{C_A P_1^*, 2\sigma^2/K_1\}, & \sup |\Omega_1| &\leq \max\{(K_1/\nu)C_A P_1^*, 9\sigma\}, \\ \sup |\Theta_2| &\leq e^{18} P_2^*, & \sup |\Omega_2| &\leq (3K_2/\sigma)e^{18} P_2^*. \end{aligned} \quad (92)$$

For layer one, use its first-stage positive multiplier bounds and the direct second-stage bounds $B \leq 2, W < 9$. For layer two, growth never exceeds its threshold; the transition has dimensionless coefficient $a_2 K_2/(\sigma\Gamma) = a_2 b_2/\Gamma^2 \leq n_i < 2$ and $v < 2$, so its logarithmic gain is at most 4. The selected steering gain is at most $12(1 + L^{-1}) < 14$, and holding only decreases temperature. These observations prove $e^{18} P_2^*$ as well as the vorticity bound from $0 \leq v \leq 3$ on the selected steering, $v < 2$ before it, and $v = 0$ on holding.

The exact second-stage diffusion exponents before steering are $dK_2^2 r_a$ and dK_2^2/Γ ; the first parent's exponent over these same intervals is $q(r_a + 1/\Gamma)$. The steering and holding exponents retain the evolving $|\zeta_2|^2$ inside the integrals in Theorem 2; together these are the complete two-stage amplitude diffusion costs.

Finally the proved viscosity range is explicitly frequency dependent. For fixed original data and Q_2 ,

$$0 < d_{\text{max}} \leq \frac{\nu C_L \log 3}{12800\mu^2} (\log \hat{\lambda}_1) \hat{\lambda}_1^{-2Q_2} \longrightarrow 0 \quad (\hat{\lambda}_1 \longrightarrow \infty). \quad (93)$$

To verify the limit, set $y = \log \hat{\lambda}_1$ and use $e^{2Q_2 y} \geq (2Q_2 y)^2/2$, giving $ye^{-2Q_2 y} \leq 1/(2Q_2^2 y) \rightarrow 0$. This is the dependence of the range proved here, not an assertion that every larger diffusivity fails or that a modified infinite construction is impossible. The actual finite result preserves both original thresholds and controls. A fixed-positive-diffusivity iteration would require further estimates or a changed scale design; it does not follow by iterating a range whose upper bound tends to zero.

4.6 An exact scaling restriction for this growth rule

The original datum ties the physical spatial scale to the initial amplitude: $\mu = \lceil \nu/2 \rceil$ with $\nu = \sqrt{A_0} \geq 1$. Consequently

$$\frac{\mu^2}{\nu} \geq \frac{1}{2}, \quad \hat{d} = d \frac{\mu^2}{\nu} \geq \frac{d}{2}, \quad (94)$$

where $\hat{d}$ is exactly the transformed diffusion coefficient in the original map $(X, T) = (\mu x, \nu t)$. To prove the first inequality, if $1 \leq \nu \leq 2$ then $\mu = 1$ and $\mu^2/\nu = 1/\nu \geq 1/2$. If $\nu > 2$, then $\mu \geq \nu/2$, whence $\mu^2/\nu \geq \nu/4 > 1/2$. Equality occurs at the original datum value $A_0 = 4$, $\nu = 2$, $\mu = 1$; this identifies the equality case and does not change the parameters of any constructed trajectory. The second inequality follows by multiplication by $d > 0$, using the exact diffusion operator map.

For the first modified growth rule proved here, attaining $P_1^* = S_1 e^{L_1} > S_1$ requires and is equivalent to $dK_1^2 < \Gamma_1$. This follows directly from (74): at equality the temperature amplitude is constant, and above it the amplitude strictly decreases for all growth times. Therefore every instance of this same growth rule must satisfy

$$\hat{d} \hat{\lambda}_1^2 < \sin s_*, \quad d \lambda_1^2 < 2 \sin s_*. \quad (95)$$

The first inequality is the identical physical inequality divided by ν , since $K_1 = \mu \hat{\lambda}_1$; the second follows from (94). Thus increasing A_0 while preserving the original spatial scaling cannot make $\hat{d}$ tend to zero at a fixed positive physical d . This necessary first-growth restriction is a result about the specified modified amplitude equation and original growth control. Any route past it must change an identified part of that dynamics or scale design and verify the corresponding full spatial forces. It is not an obstruction to all forced viscous flows.

5 A direct two-layer return theorem with positive physical diffusivity

This section proves a finite controlled return for the actual two-layer coefficient system, including the evolving first parent. It uses the released Alpöge–Buckmaster steering profile from Section 3.6.3, pp. 18–22, and its identification in Section 3.7.6, pp. 31–32. The estimates below are proved directly for the modified equations; no estimate for an inviscid trajectory is transferred to a diffusive one. The spatial realization of these amplitude equations requires the explicit profile and localization forces calculated in the companion sections. This section makes no assertion about an infinite sequence.

5.1 Original parameters, entry state, and control

Keep $A_0 \geq 1$, $\mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil$, and the paper's time-scaling parameter $\nu_{AB} = \sqrt{A_0}$. The physical frequencies are $K_i = \mu \hat{\lambda}_i$, $i = 1, 2$. The prescribed source parameters s_* , $L = L_2$ and the source profiles η, η_2 are retained. Thus $0 < s_* \leq 1/8$, η is smooth nondecreasing, zero on $(-\infty, 0]$ and one on $[1, \infty)$, and $\eta_2 \in C_c^\infty((0, 1))$ is nonnegative with $\int_0^1 \eta_2 = 1$. Put $H = \|\eta_2\|_\infty$. Physical viscosity and thermal diffusivity are both $d \geq 0$. They are distinct from ν_{AB} , μ , and the shooting parameter $m \in [0, 3]$.

Let t_1 be the actual second-layer steering-entry time. Its data are

$$\begin{aligned}\alpha(t_1) &= s_*, \quad \zeta_1(t_1) = e_2, \quad \Omega_1(t_1) = 0, \quad \Theta_1(t_1) = -P_{1\text{in}} < 0, \\ M_2(t_1) &= I, \quad \zeta_2(t_1) = e(s), \quad \Theta_2(t_1) = -P_{2\text{in}} < 0, \\ \Omega_2(t_1) &= -\frac{K_2}{\sigma} P_{2\text{in}} v_{\text{in}}.\end{aligned}\tag{96}$$

Here $e(s) = (\sin s, \cos s)$, $\sigma > 0$ is the actual inherited scale, $s = L\nu_{\text{AB}}/\sigma$ is the actual source second angle, and $\Gamma = \sigma \sin s$. In particular σ, s, Γ are not reset to fixed values if the preceding diffusive stage changes them. The data $P_{1\text{in}}, P_{2\text{in}}, v_{\text{in}}$ are the actual values after growth and transition. Theorem 1 proves from both original seeds and both actual threshold crossings that these values satisfy the explicit entry margins below, for its displayed nonempty range of positive d . No coefficient comparison during steering is assumed.

For compact notation define the following exact coordinate change:

$$\begin{aligned}a &= \sin s, \quad h_0 = \cos s = \sqrt{1 - a^2}, \quad B_{\text{in}} = \frac{K_1 P_{1\text{in}}}{\sigma^2}, \\ C_0 &= \frac{A_0 \cos s_*}{\sigma^2}, \quad C_1 = \frac{A_0 \sin s_*}{\sigma^2 a}, \\ \delta_1 &= \frac{dK_1^2}{\Gamma}, \quad \delta_2 = \frac{dK_2^2}{\Gamma}, \quad \tau = \Gamma(t - t_1).\end{aligned}\tag{97}$$

Every factor used as a divisor in this change is displayed and positive; physical time is recovered by $t = t_1 + \tau/\Gamma$. Set $T = 1 + L^{-1}$, $T_H = T + 1/32$, and use the exact control

$$z_m(\tau) = \begin{cases} 1 - \eta(\tau), & 0 \leq \tau \leq 1, \\ -mL\eta_2(L(\tau - 1)), & 1 \leq \tau \leq T, \\ 0, & T \leq \tau \leq T_H. \end{cases} \quad Z_m(t) = az_m(\Gamma(t - t_1)).\tag{98}$$

Flatness and interior support make these definitions smooth at both joins. This is exactly the source pulse, with source pulse amplitude $mL \sin s$, entry first component a , and entry phase length one.

5.2 Exact closed equations and their inverse map

Write $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. The two phase gradients have the exact signed determinant $\det(\zeta_1, \zeta_2) = -a$. Since $|\zeta_1| = 1$, put

$$\begin{aligned}h &= \zeta_1 \cdot \zeta_2, \quad D_* = h^2 + a^2 = |\zeta_2|^2, \quad Y = \sqrt{D_* - a^2 z_m^2}, \\ B &= \frac{-K_1 \Theta_1}{\sigma^2}, \quad W = \frac{\Omega_1}{\sigma}, \quad n = B + C_0 + C_1 h.\end{aligned}$$

The symbol D_* is a positive scalar, not the parent velocity matrix. The square root selects the positive second component of ζ_2 . The complete older/geometry system in τ is

$$\begin{aligned}h' &= W, & h(0) &= h_0, \\ W' &= B \frac{Y - h z_m}{D_*} - \delta_1 W, & W(0) &= 0, \\ B' &= -C_1 W - \delta_1 B, & B(0) &= B_{\text{in}}, \\ n' &= -\delta_1 B, & n(0) &= B_{\text{in}} + C_0 + C_1 h_0.\end{aligned}\tag{99}$$

The last equation is a proved consequence of the preceding ones, not an additional independent constraint. Define $k = n/D_*$. The newest pair is recovered from

$$\begin{aligned}v' &= z_m - kv^2, & v(0) &= v_{\text{in}}, \\ P_2 &= P_{2\text{in}} \exp\left(\int_0^\tau (kv - \delta_2 D_*) dr\right), \\ \Theta_2 &= -P_2, & \Omega_2 &= -\frac{K_2}{\sigma} P_2 v.\end{aligned}\tag{100}$$

Here P_2 is recovered only on an interval where v has been proved finite. The proof below constructs that interval before using this map.

For clarity, the full geometric inverse and the older physical state are

$$\begin{aligned}\zeta_2 &= (az_m, Y), & \zeta_1 &= \left(\frac{haz_m - aY}{D_*}, \frac{hY + a^2z_m}{D_*} \right), \\ \phi_2 &= \arcsin(az_m/\sqrt{D_*}), & \phi_1 &= \phi_2 - \arctan(a/h), & \alpha &= s_* - \phi_1, \\ M_1 &= R_\alpha, & M_2 &= R_{\alpha-s_*} \begin{pmatrix} 1 & -(h-h_0)/a \\ 0 & 1 \end{pmatrix}, & \Theta_1 &= -\sigma^2 B/K_1, & \Omega_1 &= \sigma W.\end{aligned}\tag{101}$$

The expression for M_1 uses its original first-activation datum $M_1 = I$ when $\alpha = 0$ and $p_1 = e(s_*)$; it is not an artificial reset of the first deformation at t_1 .

Proof of the exact equations and inverse. The actual parent is

$$G = -A_0 R_\alpha e_2 - K_1 P_1 \zeta_1, \quad D_{<2} = \dot{\alpha} J + \Omega_1 J \zeta_1 \otimes \zeta_1, \quad P_1 = -\Theta_1.$$

The first coefficient is $a_1 = A_0 \sin s_*/K_1$, because $\phi_1 + \alpha = s_*$. Therefore the diffusive first pair gives $\dot{A}_1 = -A_0 \sin s_* \Omega_1 - dK_1^2 A_1$ and $\dot{\Omega}_1 = -A_1 \sin \phi_1 - dK_1^2 \Omega_1$, where $A_1 = K_1 P_1$. The phase equation gives $\dot{h} = a\Omega_1$. The determinant representation is $\zeta_2 = h\zeta_1 - aJ\zeta_1$; solving its two components gives (101) and $-\sin \phi_1 = a(Y - hz_m)/D_*$. Substitution and division by the nonzero physical factors in (97) prove the first three equations in (99). Differentiating $B + C_0 + C_1 h$ proves the exact cancellation $n' = -\delta_1 B$.

The full second coefficient retains the original base term:

$$a_2 = \frac{aA_1 + A_0(a \cos s_* + h \sin s_*)}{K_2 D_*} = \frac{\sigma^2 a n}{K_2 D_*}, \quad b_2 = K_2 a z_m.$$

In particular the numerator changes when the first temperature diffuses; it has not been held fixed. The equations $\dot{\Theta}_2 = a_2 \Omega_2 - dK_2^2 D_* \Theta_2$ and $\dot{\Omega}_2 = b_2 \Theta_2 - dK_2^2 D_* \Omega_2$ give exactly (100) by differentiation. Conversely differentiating its exponential and product proves these two physical equations, so the coordinate map and its inverse preserve the complete dynamics.

For the matrices, write $S = \begin{pmatrix} 1 & -(h-h_0)/a \\ 0 & 1 \end{pmatrix}$. Then $\dot{S} = \Omega_1 (J e_2 \otimes e_2) S$ because $\dot{h} = a\Omega_1$. Differentiating $R_{\alpha-s_*} S$ proves $\dot{M}_2 = D_{<2} M_2$ and $M_2(t_1) = I$. Furthermore $S^{-T}(a, h_0) = (a, h)$, proving $M_2^{-T} e(s) = \zeta_2$ with its original oriented datum. Both matrices have determinant one. Differentiating $\zeta_{2,1} = Z_m$ in its phase equation proves the exact control

$$\dot{\alpha} = F_2^{\text{ctl}} - \frac{\dot{Z}_m}{Y}, \quad F_2^{\text{ctl}} = -\frac{\Omega_1 \zeta_{1,1} (J \zeta_1 \cdot \zeta_2)}{Y} = \frac{a \Omega_1 \zeta_{1,1}}{Y}.$$

Thus no extra angular control has been substituted for the source feedback. All divisions and inverse trigonometric branches are justified by the bounds proved next. $\square$

5.3 An explicit finite parameter range

Theorem 2. *Suppose the numerical entry and design parameters in (96)–(97) satisfy*

$$\begin{aligned}0 < s &\leq \arcsin(1/8), & 0 < a &\leq \frac{1}{8}, & L &\geq 64, & 3LHa &\leq \frac{1}{4}, \\ \frac{1}{2} &\leq B_{\text{in}} \leq 2, & 0 < C_0 &\leq \frac{1}{4}, & 0 < C_1 &\leq \frac{1}{32}, \\ \frac{1}{2} &\leq v_{\text{in}} \leq 2, & 0 &\leq \delta_1 \leq \frac{1}{16}.\end{aligned}\tag{102}$$

These are explicit inequalities on actual entry data and fixed design numbers, not hypotheses on a subsequently unknown trajectory. The displayed range of s fixes the original positive-cosine branch;

the inequality on $\sin s$ alone would not fix that branch. Then the entire family $m \in [0, 3]$ exists through $\tau = T$, with $P_2 > 0$, and its endpoint map $m \mapsto v(T; m)$ is continuous. It satisfies the quantitative endpoint margins

$$v(T; 0) \geq \frac{1}{7}, \quad v(T; 3) \leq -\frac{1}{480}. \quad (103)$$

The zero set is a nonempty compact subset of $(0, 3)$ and hence has a least element m_* . Every zero satisfies

$$\frac{13}{96} \leq m_* \leq 3 - \frac{1}{480}, \quad W(T; m_*) \geq \frac{13h_0}{3840} \exp(-\delta_1/L) > 0. \quad (104)$$

No monotonicity of the endpoint map or uniqueness of the root is claimed.

The selected state continues through the nonzero holding interval $[T, T_H]$, with $\zeta_{2,1} = \Omega_2 = 0$ and with the first parent still evolving according to (99). On the whole selected interval $[0, T_H]$,

$$\begin{aligned} h_0 \leq h < 3, \quad 0 \leq W < 9, \quad \frac{191}{512} < B \leq 2, \quad \frac{1}{4} < n < 3, \\ \frac{63}{64} \leq D_* < 10, \quad Y \geq \frac{\sqrt{59}}{8}, \quad \zeta_{1,2} > \frac{1}{16}, \quad \frac{1}{40} < k < 4. \end{aligned} \quad (105)$$

The same geometry bounds hold throughout every trial through T . Every trial has $|v| < 6$ through T , so its linear amplitude pair never loses the negative sign of Θ_2 .

The exact selected logarithmic gain at every $\tau_f \in [T, T_H]$ is

$$\log \frac{P_2(\tau_f)}{P_{2\text{in}}} = \int_0^T kv \, d\tau - \delta_2 \int_0^{\tau_f} D_* \, d\tau, \quad \frac{\log 3}{160} - 10\delta_2\tau_f \leq \log \frac{P_2(\tau_f)}{P_{2\text{in}}} \leq 12T. \quad (106)$$

In particular the strictly positive physical range

$$0 < d \leq \min \left\{ \frac{\Gamma}{16K_1^2}, \frac{\Gamma \log 3}{6400K_2^2} \right\} \quad (107)$$

gives $\log(P_2(\tau_f)/P_{2\text{in}}) \geq \log 3/320 > 0$. The interval is expressed in actual physical units. Theorem 1 supplies an explicit positive $d_{\max}$ using only the original data and chosen frequencies, and proves (102) and (107) for every $0 < d \leq d_{\max}$ while retaining the dependence of the actual entry data and Γ on d .

Proof. First construct geometry without using the newest amplitude pair. On an initial interval where $B > 0$, the bracket $Y - hz_m$ is nonnegative. If $0 \leq z_m \leq 1$, this follows by squaring: $Y^2 - h^2 z_m^2 = D_*(1 - z_m^2) \geq 0$; if $z_m \leq 0$, it follows from $Y, h > 0$. Consequently the integrating-factor formula for W gives $W \geq 0$. Therefore $h \geq h_0$ and $B' = -C_1 W - \delta_1 B \leq 0$, giving $B \leq B_{\text{in}}$. The control bound is $|az_m| \leq 1/4$, so

$$Y^2 \geq h_0^2 - \frac{1}{16} \geq \frac{59}{64} > 0.$$

For $0 \leq \tau \leq 1$, $Y - hz_m \leq \sqrt{D_*}$, hence $W' \leq B_{\text{in}}/h_0$ and $h \leq h_0 + B_{\text{in}}\tau^2/(2h_0)$. On the negative pulse the same argument gives

$$W' \leq \frac{B_{\text{in}}}{h_0} \{1 + mL\eta_2(L(\tau - 1))\}.$$

During holding its right side is B_{in}/h_0 . Integration on any interval up to T_H therefore yields

$$W \leq \frac{B_{\text{in}}}{h_0} (T_H + 3) \leq \frac{26}{3} < 9,$$

because $h_0 \geq 15/16$, $T_H \leq 17/16$, and $B_{\text{in}} \leq 2$. The duration after the first part is at most $L^{-1} + 1/32 \leq 3/64$, so

$$h \leq h_0 + \frac{B_{\text{in}}}{2h_0} + \frac{B_{\text{in}}}{h_0}(T_H + 3)\frac{3}{64} \leq 1 + \frac{16}{15} + \frac{13}{32} = \frac{1187}{480} < 3.$$

No quotient variable has entered these estimates.

The exact integrating-factor identity for the first temperature is

$$B(\tau) = e^{-\delta_1 \tau} B_{\text{in}} - C_1 \int_0^\tau e^{-\delta_1(\tau-r)} W(r) dr.$$

Since $W \geq 0$, $h' = W$, $\delta_1 T_H \leq 17/256$, and $e^{-x} \geq 1 - x$ for $x \geq 0$, it follows that

$$B(\tau) \geq \frac{1}{2} \left(1 - \frac{17}{256} \right) - \frac{1}{32}(h(\tau) - h_0) > \frac{239}{512} - \frac{3}{32} = \frac{191}{512} > \frac{1}{4}.$$

This strict estimate excludes a first exit through $B = 0$. All square-root denominators stay uniformly positive and h, W, B remain bounded. The smooth finite-dimensional vector field is therefore bounded with bounded first derivatives on a compact neighborhood of every such solution. Local existence, uniqueness and the integral equation extend the solution through T_H ; the same neighborhood can be chosen uniformly for $m \in [0, 3]$. Differentiating that integral equation and applying the elementary Gronwall estimate proves continuous dependence on m on this interval. This constructs the entire older/geometry trial family before the newest pair.

Now $n = B + C_0 + C_1 h > 1/4$, and $n' = -\delta_1 B \leq 0$, so $n \leq B_{\text{in}} + C_0 + C_1 h_0 \leq 73/32 < 3$. The asserted D_* and k bounds follow. For the first phase, $|az_m| \leq 1/4$, $a \leq 1/8$, $h \geq 15/16$ and $Y > 7/8$ give

$$\zeta_{1,2} = \frac{hY + a^2 z_m}{D_*} > \frac{(15/16)(7/8) - 1/32}{10} = \frac{101}{1280} > \frac{1}{16}.$$

Thus the branches and control denominators in (101) stay in their specified domains. This proves all geometry claims.

Construct the Riccati variable next. On $[0, 1]$, its forcing $z_m = 1 - \eta$ is nonnegative. The reciprocal differential inequality $(1/v)' \leq 4$, initially while $v > 0$, gives

$$\frac{v_{\text{in}}}{1 + 4v_{\text{in}}\tau} \leq v(\tau) \leq v_{\text{in}} + \tau \leq 3. \quad (108)$$

The positive lower bound rules out a first zero, and the upper bound follows by integrating $v' \leq 1$. These bounds also exclude finite escape. Put $v_d = v(1)$. In particular $v_d \geq 1/6$. Moreover

$$\int_0^1 k v^2 d\tau \geq \frac{1}{40} \int_0^1 \frac{v_{\text{in}}^2}{(1 + 4v_{\text{in}}\tau)^2} d\tau = \frac{v_{\text{in}}^2}{40(1 + 4v_{\text{in}})} \geq \frac{1}{480}.$$

The last inequality follows because $x^2/(1 + 4x)$ is increasing for $x > 0$, as its derivative is $(2x + 4x^2)/(1 + 4x)^2 > 0$. Integrating the Riccati equation proves

$$\frac{1}{6} \leq v_d \leq 3 - \frac{1}{480}. \quad (109)$$

On the pulse put $V(y) = v(1 + y/L)$ and $M_2(y) = \int_0^y \eta_2(r) dr$. Its exact integral equation is

$$V(y) = v_d - m M_2(y) - \frac{1}{L} \int_0^y k(1 + r/L) V(r)^2 dr.$$

Under the temporary bound $|V| \leq 6$, it gives $V \leq v_d < 3$ and $V \geq -3 - 144/L \geq -21/4 > -6$. The bound is strict initially and strictly improves itself at every possible first exit, so $|V| < 6$ on the full pulse. The quotient has therefore been constructed with a uniform finite bound, and

the exponential formula (100) constructs a negative Θ_2 solving the entire original linear pair. Its uniqueness follows by subtracting two solutions of that linear pair and applying the integral Gronwall inequality to their zero initial difference. In particular the construction agrees with the unique linear-pair solution and has not assumed absence of a temperature zero.

For $m = 0$, the pulse solves $v' = -kv^2$, so $v(T; 0) \geq v_d/(1 + 4v_d/L) \geq 1/(6 + 4/L) > 1/7$. For $m = 3$, the integral equation and (109) give $v(T; 3) \leq v_d - 3 \leq -1/480$. The uniformly finite quotient and geometry imply continuous dependence of $v(T; m)$ on m by their integral equations. The intermediate value theorem proves existence of a zero. Its zero set is closed in $[0, 3]$ and excludes both endpoints, proving the stated compactness and least-root selection.

At any selected zero, $v' = -mL\eta_2 - kv^2 \leq 0$ throughout the pulse. Thus $0 \leq v \leq v_d$ there, and

$$m = v_d - \int_1^T kv^2 d\tau \in \left[v_d - \frac{4v_d^2}{L}, v_d \right] \subseteq \left[\frac{13}{96}, 3 - \frac{1}{480} \right].$$

For the lower inclusion use $v_d \leq 3$, $L \geq 64$ and $v_d \geq 1/6$. On the pulse,

$$W' + \delta_1 W = B \frac{Y + hmL\eta_2}{D_*} \geq \frac{h_0}{40} mL\eta_2 (L(\tau - 1)).$$

Its integrating-factor formula, $W(1) \geq 0$ and the unit integral of η_2 prove (104). On holding, $z_m = 0$ and $v(T) = 0$, so uniqueness gives $v = 0$. The older geometry continues by the already established bounds, and (100) gives the exact remaining temperature decay.

Finally, (108) gives

$$\int_0^1 kv d\tau \geq \frac{1}{40} \int_0^1 \frac{v_{\text{in}}}{1 + 4v_{\text{in}}\tau} d\tau = \frac{1}{160} \log(1 + 4v_{\text{in}}) \geq \frac{\log 3}{160}.$$

The selected pulse adds nonnegative gain, the hold adds zero gain, and $v \leq 3$, $k < 4$ bound the gain above by $12T$. The exact diffusion exponent is at most $10\delta_2\tau_f$. This proves (106); substitution of (107) gives its positive lower bound. $\square$

5.4 Returned state, holding dynamics, and diffusion costs

At the selected physical return time $t_b = t_1 + T/\Gamma$ one has

$$\zeta_2(t_b) = (0, \sqrt{D_*(T)}), \quad \Omega_2(t_b) = 0, \quad \zeta_1(t_b) = \frac{(-a, h(T))}{\sqrt{D_*(T)}}, \quad \alpha(t_b) = s_* + \arctan(a/h(T)),$$

and $\Theta_1(t_b) = -\sigma^2 B(T)/K_1 < 0$, $\Omega_1(t_b) = \sigma W(T) > 0$. The first parent is therefore neither frozen nor returned to zero. Its full motion creates the specified returned state of the second wave. The controlled holding interval has physical duration $1/(32\Gamma)$. On it the complete system is

$$h' = W, \quad W' = B/\sqrt{h^2 + a^2} - \delta_1 W, \quad B' = -C_1 W - \delta_1 B, \\ P_2(\tau) = P_2(T) \exp\left(-\delta_2 \int_T^\tau (h(r)^2 + a^2) dr\right).$$

These formulas retain the evolving phase length inside the diffusion integral. They also prove the exact stationarity of $\zeta_{2,1}, \Omega_2$ throughout this finite hold.

The returned parent vorticity also gives positive scalar curl of the full local velocity at the origin. Indeed $z_m = z'_m = 0$ on holding, so the proved control and inverse give the exact physical formulas

$$\dot{\alpha} = -\frac{a^2 \Omega_1}{h^2 + a^2}, \quad \omega(0, t) = 2\dot{\alpha} + \Omega_1 = \Omega_1 \left(1 - \frac{2a^2}{h^2 + a^2}\right) \geq \frac{31}{32} \Omega_1 > 0. \quad (110)$$

For the inequality use $h \geq h_0$, $h_0^2 + a^2 = 1$ and $a^2 \leq 1/64$. The positive forcing in the holding equation for W gives $W(\tau) \geq W(T)e^{-\delta_1(\tau-T)} > 0$ by its integrating factor. These statements

include the return endpoint because the original pulse is identically zero near that endpoint. The compact spatial construction has constant envelopes and exact linear profiles near the origin, so its curl there is precisely the quantity in (110); no cutoff derivative enters it.

The total first-wave damping exponents on the ramp, pulse and hold are δ_1 , δ_1/L and $\delta_1/32$, respectively. The corresponding second-wave exponents are exactly

$$\delta_2 \int_0^1 D_* d\tau, \quad \delta_2 \int_1^T D_* d\tau, \quad \delta_2 \int_T^{T_H} D_* d\tau.$$

Each exponent lies between $(63/64)\delta_2$ and $10\delta_2$ times its displayed interval length. In original physical coordinates these equal $dK_2^2 \int |\zeta_2(t)|^2 dt$, with $K_2 = \mu \hat{\lambda}_2$ and $dt = d\tau/\Gamma$; none of the factors μ , ν_{AB} , K_i or Γ has been dropped. The exact first-amplitude identity in the proof also records the additional parent-response term $-C_1 \int e^{-\delta_1(\tau-r)} W(r) dr$; a pure common-damping multiplier would miss that term.

For the parent affine fields the altered gradient is not independently prescribed. With $B_1 = -\sigma^2 B_{\zeta_1}$, direct differentiation gives

$$\dot{G} + D_{<2}^T G = -dK_1^2 B_1, \quad C = 2\dot{\alpha} + \Omega_1, \quad \dot{C} - G_1 = 2\ddot{\alpha} - A_0 \sin \alpha - dK_1^2 \Omega_1.$$

Thus $G \cdot x$, $D_{<2} x$ have scalar force $(-dK_1^2 B_1) \cdot x$ and full vector force $\frac{1}{2}(2\ddot{\alpha} - A_0 \sin \alpha - dK_1^2 \Omega_1) Jx$ with pressure $-\frac{1}{2}x^T \text{sym}(\dot{D}_{<2} + D_{<2}^2 - e_2 \otimes G)x$. Their Laplacians vanish. These affine formulas are local exact fields; the compact original-profile realization and its additional spatial forces are supplied by the accompanying finite-profile construction. They must not be discarded when interpreting the amplitude result as a statement about the full spatial equations.

Here is a direct verification of the affine force and pressure, including their signs. Put $D = D_{<2}$, $u(x, t) = D(t)x$ and $\theta(x, t) = G(t) \cdot x$. Both summands in D have zero trace, so $\text{div } u = 0$ and $D^2 = -\det(D)I$ by multiplication of a general trace-free 2×2 matrix. The scalar residual is exactly $(\dot{G} + D^T G) \cdot x$. In that derivative the term $K_1 \dot{\Theta}_1 \zeta_1$ cancels $(\Omega_1 J \zeta_1 \otimes \zeta_1)^T (-A_0 R_\alpha e_2)$, apart from $-dK_1^2 B_1$; the angular terms cancel because $\dot{\zeta}_1 = \dot{\alpha} J \zeta_1$ and $\partial_t(R_\alpha e_2) = \dot{\alpha} J R_\alpha e_2$. Also $\zeta_1 \cdot J \zeta_1 = 0$ eliminates the self-interaction. For momentum let $\mathcal{A} = \dot{D} + D^2 - e_2 \otimes G$. Then $\partial_t u + (u \cdot \nabla)u - \theta e_2 = \mathcal{A}x$, whereas the displayed pressure contributes $-\text{sym}(\mathcal{A})x$. Their sum is $\text{skew}(\mathcal{A})x$. The scalar curl of Dx is $C = D_{21} - D_{12}$; since D^2 is scalar and the curl of $(G \cdot x)e_2$ is G_1 ,

$$\text{skew}(\mathcal{A}) = \frac{\dot{C} - G_1}{2} J.$$

Finally $\dot{\Omega}_1 = -K_1 P_1 \zeta_{1,1} - dK_1^2 \Omega_1$ and $G_1 = A_0 \sin \alpha - K_1 P_1 \zeta_{1,1}$ give precisely the force stated above. Both affine Laplacians are zero by direct second differentiation. Thus these are the full vector equations with the specified pressure, not merely a vorticity residual.

6 Compact spatial realization of the changed two-stage trajectory

We now realize the actual damped two-stage trajectory, including its changed first threshold, inherited scale, second threshold, selected second return, and evolving finite hold, in the full spatial equations. The coefficients used here are those constructed in the direct return theorem and Theorem 1. The present calculation supplies their entire spatial realization; in particular it does not interpret a modal damping equation as an unforced equation for the released profile. The physical equations and orientation are

$$\begin{aligned} \theta_t + u \cdot \nabla \theta &= d\Delta \theta + f_\theta, \\ u_t + (u \cdot \nabla)u &= d\Delta u + \theta e_2 + f_u, & \text{div } u &= 0, & p &= 0, \\ J &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, & e_2 &= (0, 1), & x &\in \mathbb{R}^2. \end{aligned}$$

Both physical coefficients equal the explicitly selected $d > 0$. The original parameters remain

$$A_0 \geq 1, \quad \mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil, \quad \nu = \sqrt{A_0}, \quad K_j = \mu \lambda_j \quad (j = 1, 2).$$

Thus d is not the source time-scaling parameter ν . Here λ_j denotes exactly the source frequency written $\hat{\lambda}_j$ in the preceding entry calculation.

6.1 The actual coefficients and the nested material regions

Write $e(s) = (\sin s, \cos s)$, $e_0 = R_\alpha e_2$, $p_1 = e(s_*)$, and $p_2 = e(s)$, where $s = L_2 \nu / \sigma$, $a = \sin s$, $h_0 = \cos s$, and $\Gamma = \sigma a$ use the actual diffusive inherited scale σ . The activation functions are

$$w_1 = \eta(\Gamma_1(t - t_{01})), \quad w_2 = \eta(\Gamma(t - t_{02})), \quad \Gamma_1 = \nu \sin s_*, \quad t_{01} = 1/\nu.$$

The source step η is smooth nondecreasing, zero on $(-\infty, 0]$ and one on $[1, \infty)$. All inactive increments below are defined to be zero. The finite coefficient construction has the exact equations

$$G_{<j} = -A_0 e_0 + \sum_{i < j} w_i K_i \Theta_i \zeta_i, \quad D_{<j} = \dot{\alpha} J + \sum_{i < j} \frac{w_i \Omega_i}{r_i^2} J \zeta_i \otimes \zeta_i, \quad (111)$$

$$\begin{aligned} \dot{M}_j &= D_{<j} M_j, & M_j(t_{0j}) &= I, & \zeta_j &= M_j^{-T} p_j, \quad r_j^2 = |\zeta_j|^2, \\ \dot{\Theta}_j &= -\frac{J \zeta_j \cdot G_{<j}}{K_j r_j^2} \Omega_j - d K_j^2 r_j^2 \Theta_j, & \dot{\Omega}_j &= K_j \zeta_{j,1} \Theta_j - d K_j^2 r_j^2 \Omega_j. \end{aligned} \quad (112)$$

The common angle is the selected original first control followed by the exact second feedback control. In particular the first threshold time is $t_{01} + L_1/(\Gamma_1 - d K_1^2)$, not the inviscid threshold time. The source first transition and pulse are translated to that actual threshold. During second growth and transition, $\alpha = s_*$, $M_2 = I$, $\Omega_1 = 0$, and Θ_1 diffuses. The second activation time is exactly $t_{02} = t_{b1}$, the actual first return time, with no intervening gap. During second steering and holding the exact inverse is

$$\begin{aligned} M_1 &= R_\alpha, \quad M_2 = R_{\alpha-s_*} \begin{pmatrix} 1 & -(h - h_0)/a \\ 0 & 1 \end{pmatrix}, \quad \Omega_1 = \sigma W, & \Theta_1 &= -\sigma^2 B / K_1, \quad (113) \\ h' &= W, \quad W' = B \frac{Y - hz}{h^2 + a^2} - \delta_1 W, & B' &= -C_1 W - \delta_1 B, \quad Y = \sqrt{h^2 + a^2 - a^2 z^2}, \end{aligned}$$

where $\tau = \Gamma(t - t_1)$, a prime denotes $d/d\tau$, $\delta_1 = d K_1^2 / \Gamma$, $C_1 = A_0 \sin s_*/(\sigma^2 a)$, and z is the selected source ramp, pulse, and zero hold. The proved bounds give

$$h_0 \leq h < 3, \quad 0 \leq W < 9, \quad 1/4 < B \leq 2, \quad h_0 \geq \sqrt{63}/8, \quad r_1 = 1, \quad h_0 \leq r_2 < \sqrt{10}.$$

These are actual changing coefficients, not independently chosen affine backgrounds. Every $D_{<j}$ has trace zero: the rotation has trace zero and $\text{tr}(J \zeta_i \otimes \zeta_i) = \zeta_i \cdot J \zeta_i = 0$. Jacobi's determinant formula therefore gives $\det M_j = 1$. The phase equation follows by differentiating the inverse transpose: $\dot{\zeta}_j = -D_{<j}^T \zeta_j$.

On $[-\pi, \pi]$ define the released profile by the following formula and then extend it periodically to $\mathbb{R}$:

$$F(s) = \sin s + \chi(s)(s - \sin s), \quad P' = F, \quad \int_{-\pi}^{\pi} P(s) ds = 0.$$

Here χ is even, supported in $(-\pi, \pi)$, and equals one on $[-s_0, s_0]$, with $0 < s_0 < 1$. Oddness and periodicity give a unique even periodic primitive P with the specified mean. In particular $F(s) = s$ on $|s| \leq s_0$; its primitive there is $P(0) + s^2/2$, with its original constant $P(0)$ retained. Let $f \in C_c^\infty(B_1)$ be radial, $0 \leq f \leq 1$, and $f = 1$ on $B_{1/2}$. Retain the source radii

$$\ell_1 = \frac{s_0}{C_\ell \mu}, \quad \ell_2 = \frac{\lambda_1^{-3}}{\mu}, \quad C_\ell \geq 4.$$

Define the material coordinates and envelopes

$$y_j = M_j^{-1}x, \quad g_j(x, t) = f(y_j/\ell_j), \quad s_j = K_j \zeta_j \cdot x.$$

With $N = 1 + (3 - h_0)/a$, the matrix expression (113) and $\|I + qE_{12}\| \leq 1 + |q|$ show

$$\|M_1\| = \|M_1^{-1}\| = 1, \quad \|M_2\|, \|M_2^{-1}\| \leq N. \quad (114)$$

The second inequality also holds before steering, where $M_2 = I$. The entry construction, (83), selects λ_1 with the explicit strict inequalities

$$N\lambda_1^{-3} < \frac{s_0}{2C_\ell}, \quad N\lambda_1^{-2} < s_0. \quad (115)$$

Consequently every point of the second support satisfies $|x| \leq N\ell_2 < \ell_1/2$ and $|K_1\zeta_1 \cdot x| \leq K_1N\ell_2 < s_0$. It therefore lies in the exact affine cell of the first wave for the whole construction. The first support lies in B_{ℓ_1} ; since $\mu\ell_1 = s_0/C_\ell < s_0$, the base is exactly affine there for every angle. Strictness leaves an open neighborhood of each support on which the corresponding affine identities hold.

6.2 Original datum, fields, and complete residual forces

Retain the original radial cutoff $\chi_0 \in C_c^\infty(B_{R_0})$, equal to one on B_2 , and the exact datum

$$\theta_{\text{in}}(x) = -\frac{A_0}{\mu} \sin(\mu x_2) \chi_0(x), \quad u_{\text{in}} = 0.$$

Choose the radial $H_0 \in C_c^\infty(B_{R_H})$ used in the base construction, where $R_H > R_0 + 1$ and $H_0 = |x|^2/2$ on B_{R_0+1} . Put

$$\begin{aligned} \theta_0 &= -\frac{A_0}{\mu} F(\mu e_0 \cdot x) \chi_0(x), & \theta_{\text{ref}} &= -\frac{A_0}{\mu} F(\mu x_2) \chi_0(x), \\ \theta_b &= \theta_0 + (1 - \eta(\nu t))(\theta_{\text{in}} - \theta_{\text{ref}}), & \psi_b &= \dot{\alpha} H_0, & u_b &= J\nabla \psi_b. \end{aligned} \quad (116)$$

The angle is zero before $1/\nu$; the initial interpolation is therefore exact and the initial velocity is zero. Define

$$\begin{aligned} T_j &= w_j \Theta_j, & Q_j &= \frac{w_j \Omega_j}{K_j^2 r_j^2}, & V_j &= T_j F(s_j) g_j, & \Psi_j &= Q_j P(s_j) g_j, \\ v_j &= J\nabla \Psi_j, & \theta &= \theta_b + V_1 + V_2, & u &= u_b + v_1 + v_2. \end{aligned} \quad (117)$$

The flat activation functions make the zero extensions of the increments smooth, with all spatial and temporal derivatives, at $t = t_{0j}$. The velocity is divergence free because mixed partial derivatives commute.

Hereafter fix a layer and temporarily suppress its index. Write $k = K\zeta$, $\rho = |k|^2 = K^2 r^2$, $D = D_{<j}$, $G = G_{<j}$, and $b = k_1$. All functions of one variable in the formulas below are evaluated at $s = k \cdot x$. Define

$$Q_* = \frac{\dot{w}\Omega + wb\Theta}{\rho} - Q \frac{\dot{\rho}}{\rho}, \quad \dot{\rho} = -2k \cdot D^T k. \quad (118)$$

Thus $\dot{Q} = Q_* - d\rho Q$, directly from (112), without a constant-phase-length assumption. The complete added scalar and vector forces are

$$\begin{aligned} E_\theta &= \dot{w}\Theta Fg + QP(J\nabla g \cdot G) + QT(F^2 - PF')g(Jk \cdot \nabla g) \\ &\quad - dT\{\rho(F + F'')g + 2F'k \cdot \nabla g + F\Delta g\}, \end{aligned} \quad (119)$$

$$E_u = 2Dv + Q_* J\{kFg + P\nabla g\} + (\nabla v)v - TFge_2 - d(\rho v + \Delta v). \quad (120)$$

These are full vector forces at the declared pressure $p = 0$. For completely componentwise evaluation one uses

$$v = QJ\{kFg + P\nabla g\},$$

$$\nabla v = QJ\{k \otimes kF'g + F(k \otimes \nabla g + \nabla g \otimes k) + P\nabla^2 g\}, \quad (121)$$

$$\Delta V = T\{\rho F''g + 2F'k \cdot \nabla g + F\Delta g\}, \quad (122)$$

$$\Delta v = QJ\{\rho kF''g + \rho F'\nabla g + 2F'k(k \cdot \nabla g) + 2F(\nabla^2 g)k + Fk\Delta g + P\nabla\Delta g\}. \quad (123)$$

Thus every phase, envelope, and mixed Laplacian term is displayed. The base forces are

$$\begin{aligned} f_{\theta,b} &= \nu\eta'(\nu t)(\theta_{\text{ref}} - \theta_{\text{in}}) - d\Delta\theta_b, \\ f_{u,b} &= \ddot{\alpha}J\nabla H_0 + \dot{\alpha}^2(\nabla(J\nabla H_0))J\nabla H_0 - \theta_b e_2 - d\dot{\alpha}J\nabla\Delta H_0. \end{aligned} \quad (124)$$

The requested total forces are exactly

$$f_\theta = f_{\theta,b} + E_{\theta,1} + E_{\theta,2}, \quad f_u = f_{u,b} + E_{u,1} + E_{u,2}. \quad (125)$$

Proof of the full spatial equations. For the material derivative $\mathcal{D} = \partial_t + Dx \cdot \nabla$, differentiating $M^{-1}x$ gives $\mathcal{D}(M^{-1}x) = 0$. The phase equation gives $\mathcal{D}(k \cdot x) = 0$ as well. Consequently $\mathcal{D}g = 0$, $\mathcal{D}V = \dot{T}Fg$, and $\mathcal{D}\Psi = \dot{Q}Pg$. On this support the actual older temperature and velocity are $G \cdot x$ and Dx by (115); in particular their gradient and Jacobian are exactly G, D , even while these change. The temperature advection increment is therefore $\dot{T}Fg + v \cdot G + v \cdot \nabla V$. Using $Jk \cdot k = 0$ and $J\nabla g \cdot \nabla g = 0$ gives

$$v \cdot \nabla V = QT(F^2 - PF')g(Jk \cdot \nabla g), \quad v \cdot G = QFg(Jk \cdot G) + QP(J\nabla g \cdot G).$$

Since $\dot{\Theta} = -(Jk \cdot G)\Omega/\rho - d\rho\Theta$, the first term in $v \cdot G$ cancels the coupled part of $\dot{T}Fg$ exactly. Subtraction of $d\Delta V$, computed by the product rule, proves (119).

For every trace-free two by two matrix, direct multiplication gives $DJ + JD^T = 0$. Differentiation of a gradient under $\mathcal{D}$ therefore gives

$$\mathcal{D}(J\nabla\Psi) = DJ\nabla\Psi + J\nabla(\mathcal{D}\Psi).$$

The other linear advection increment is $(v \cdot \nabla)(Dx) = Dv$. Adding it to this formula, adding the self-advection $(\nabla v)v$, and subtracting $Ve_2 + d\Delta v$ proves (120), after using $\dot{Q} = Q_* - d\rho Q$. Differentiating $\Delta\Psi = Q(\rho F'g + 2Fk \cdot \nabla g + P\Delta g)$ and applying J proves all six groups in (123).

The base phase $\mu e_0 \cdot x$ is transported by $\dot{\alpha}Jx$, and its radial cutoff is also transported. Before $1/\nu$ its velocity is zero. This proves the scalar base formula. Differentiating $\dot{\alpha}J\nabla H_0$ proves the vector base formula directly on all of $\mathbb{R}^2$. Finally, adding increments in the order 1, 2 accounts exactly for both directions of each interaction. In the second increment, $\mathcal{D}$ contains $u_b + v_1$, and the terms $v_2 \cdot G_{<2}$ and $D_{<2}v_2$ contain the reverse interactions with that full older state. Outside an increment's support all its derivatives vanish. Thus the residual identities telescope globally to (125), with no omitted interaction and no inverse curl or pressure reconstruction. $\square$

The material-coordinate form makes the original metric equally explicit. Set $C = M^{-1}M^{-T}$ and $\tilde{g}(y) = f(y/\ell)$. The pullback of the spatial derivative and Laplacian on a function of (s, y) are

$$\mathcal{P} = K\zeta\partial_s + M^{-T}\nabla_y, \quad \mathcal{L} = K^2r^2\partial_s^2 + 2K(Cp \cdot \nabla_y)\partial_s + \sum_{l,m} C_{lm}\partial_{y_l}\partial_{y_m}. \quad (126)$$

Indeed $s = Kp \cdot y$, $\nabla_x = M^{-T}\nabla_y$ on a material function, and $|M^{-T}p|^2 = p^T Cp = r^2$; multiplying the two derivative operators proves (126). For example $k \cdot \nabla_x g = K(Cp) \cdot \nabla_y \tilde{g}$. This proves an exact coordinate correspondence for every displayed mixed derivative and keeps the source radii in their original units.

6.3 What the affine cells and sine regions actually satisfy

On $g = 1$ and $|s| < s_0$ the wave is exactly

$$V = Tk \cdot x, \quad v = QJk \otimes kx = \frac{w\Omega}{r^2} J\zeta \otimes \zeta x.$$

Both Laplacians vanish, but their damping contributions to the forces (119)–(120) are

$$-d\rho V, \quad -d\rho v. \quad (127)$$

They do not vanish unless the corresponding wave vanishes. For the first wave the scalar core force is $-dK_1^2(w_1K_1\Theta_1\zeta_1) \cdot x$ after activation, and the curl of its damping vector force is $-dK_1^2w_1\Omega_1$. Thus the evolving affine parent itself has the force required by its changed amplitude equations. On an uncut region where $F(s) = \sin s$, instead, $F'' = -F$ and the two damping contributions are zero: $\rho V + \Delta V = 0$ and $\rho v + \Delta v = 0$. The non-damping vector terms in (120) are still present; this cancellation is not a claim of unforced momentum at $p = 0$. In cutoff regions every term in (122)–(123) must also be kept. These restrictions prove exactly how the damped coefficient dynamics map into the released spatial profile. They differ from retaining an inviscid state and adding only $-d\Delta$ of that state: the present amplitudes, parent, phases and controls have all changed according to (112).

6.4 Explicit bounds at every spatial derivative order

For a scalar, vector, or tensor field U , let $|U|_m = \sup_x \|D_x^m U(x)\|$, using Euclidean multilinear operator norms and $m \geq 0$. For a one-variable profile W write $[W]_m = \sup_s |W^{(m)}(s)|$, and put $F_m^f = \sup_y \|D^m f(y)\|$. Set $N_1 = 1$, $N_2 = N$, $r_{1,-} = r_{1,+} = 1$, $r_{2,-} = h_0$, $r_{2,+} = \sqrt{10}$. Define finite explicit numbers

$$\begin{aligned} \kappa_j^* &= K_j r_{j,+}, & e_{j,m} &= (N_j/\ell_j)^m F_m^f, \\ \mathcal{B}_{j,m}(W) &= \sum_{q=0}^m \binom{m}{q} (\kappa_j^*)^q [W]_q e_{j,m-q}, & \mathcal{C}_{j,m}(P) &= \sum_{q=0}^m \binom{m}{q} (\kappa_j^*)^q [P]_q e_{j,m-q+1}. \end{aligned} \quad (128)$$

Here κ_j^* is merely a frequency bound, not a diffusivity. Repeated differentiation and the chain rule give $|g_j|_m \leq e_{j,m}$, $|W(s_j)g_j|_m \leq \mathcal{B}_{j,m}(W)$, and $|P(s_j)\nabla g_j|_m \leq \mathcal{C}_{j,m}(P)$. To prove the middle inequality, assign each of the m derivatives either to the profile or to the envelope. There are $\binom{m}{q}$ assignments with q profile derivatives, each bounded by the corresponding summand. The other inequalities follow by the same argument. This proof applies to all directions in the defined operator norm, not only coordinate derivatives.

The actual threshold amplitudes are $P_j^* = (\nu^2/\mu)\lambda_j^{-7/8}$. The finite coefficient calculation, (92), supplies the following global bounds through the hold:

$$\begin{aligned} U_{\Theta,1} &= \max\{e^{2+\Lambda_1^{-1}} P_1^*, 2\sigma^2/K_1\}, & U_{\Omega,1} &= \max\{(K_1/\nu)e^{2+\Lambda_1^{-1}} P_1^*, 9\sigma\}, \\ U_{\Theta,2} &= e^{18} P_2^*, & U_{\Omega,2} &= 3K_2 U_{\Theta,2}/\sigma. \end{aligned} \quad (129)$$

In particular these preserve the actual source frequencies and the changed inherited scale. Put

$$A_1 = \sup_{[0,T_f]} |\dot{\alpha}|, \quad A_2 = \sup_{[0,T_f]} |\ddot{\alpha}|, \quad D_1^* = A_1, \quad D_2^* = A_1 + U_{\Omega,1}, \quad G_1^* = A_0, \quad G_2^* = A_0 + K_1 U_{\Theta,1}.$$

Explicit profile bounds for A_1, A_2 are given below. Set $W_{1,1} = \Gamma_1[\eta']_0$ and $W_{2,1} = \Gamma[\eta']_0$. Then

$$\begin{aligned} Q_j^* &= \frac{U_{\Omega,j}}{K_j^2 r_{j,-}^2}, \\ Q_{*,j}^* &= \frac{W_{j,1} U_{\Omega,j} + \kappa_j^* U_{\Theta,j} + 2D_j^* U_{\Omega,j}}{K_j^2 r_{j,-}^2}. \end{aligned} \quad (130)$$

The bounds $|Q_j| \leq Q_j^*$ and $|Q_{*,j}| \leq Q_{*,j}^*$ follow from (118) and $|\dot{\rho}/\rho| \leq 2\|D\|$. Define

$$\begin{aligned}\mathcal{D}_{\theta,j,m} &= dU_{\Theta,j}\{(\kappa_j^*)^2 \mathcal{B}_{j,m}(F) + 2\mathcal{B}_{j,m+2}(F)\}, \\ \mathcal{D}_{u,j,m} &= dQ_j^*\{(\kappa_j^*)^2 \mathcal{B}_{j,m+1}(P) + 2\mathcal{B}_{j,m+3}(P)\}.\end{aligned}\tag{131}$$

These bound the complete damping-plus-diffusion terms in (119)–(120). In particular they include the core damping and every envelope derivative. The two in front of $\mathcal{B}_{m+2}$ or $\mathcal{B}_{m+3}$ is the sum of the two coordinate Laplacian terms; J has operator norm one. The whole added forces satisfy the explicit inequalities

$$\begin{aligned}|E_{\theta,j}|_m &\leq W_{j,1}U_{\Theta,j}\mathcal{B}_{j,m}(F) + Q_j^*G_j^*\mathcal{C}_{j,m}(P) + \mathcal{D}_{\theta,j,m} \\ &\quad + Q_j^*U_{\Theta,j}\sum_{q=0}^m\binom{m}{q}\mathcal{B}_{j,q+1}(P)\mathcal{B}_{j,m-q+1}(F),\end{aligned}\tag{132}$$

$$\begin{aligned}|E_{u,j}|_m &\leq (2D_j^*Q_j^* + Q_{*,j}^*)\mathcal{B}_{j,m+1}(P) + U_{\Theta,j}\mathcal{B}_{j,m}(F) + \mathcal{D}_{u,j,m} \\ &\quad + (Q_j^*)^2\sum_{q=0}^m\binom{m}{q}\mathcal{B}_{j,q+1}(P)\mathcal{B}_{j,m-q+2}(P).\end{aligned}\tag{133}$$

For the scalar nonlinear term it is enough to use its exact alternative expression $v_j \cdot \nabla V_j$; differentiation of this product proves the displayed sum. For the vector nonlinear term differentiate $(\nabla v_j)v_j$ and interchange the summation index to obtain the final sum. These arguments prove all four bounds directly from the exact formulas.

The base has equally explicit bounds. Put $c_m = |\chi_0|_m$, $h_m = |H_0|_m$, and

$$b_m(W) = \sum_{q=0}^m\binom{m}{q}\mu^q[W]_q c_{m-q}, \quad B_m^0 = \frac{A_0}{\mu} \max\{b_m(F), b_m(\sin)\}.$$

The interpolation is a convex combination when $\alpha = 0$; afterwards $|\mu e_0| = \mu$. Hence $|\theta_b|_m \leq B_m^0$. The product rule and (124) give

$$\begin{aligned}|f_{\theta,b}|_m &\leq \frac{\nu A_0}{\mu}[\eta']_0\{b_m(F) + b_m(\sin)\} + 2dB_{m+2}^0, \\ |f_{u,b}|_m &\leq A_2h_{m+1} + B_m^0 + 2dA_1h_{m+3} + A_1^2\sum_{q=0}^m\binom{m}{q}h_{q+2}h_{m-q+1}.\end{aligned}\tag{134}$$

All total force norms are bounded by adding their base and two increment bounds. Integrating any bound over its interval multiplies its constant by that interval's physical length. For example

$$\int_0^{T_f} |\text{wave damping-plus-diffusion force}_j|_m dt \leq (T_f - t_{0j})\mathcal{D}_{\bullet,j,m}.\tag{135}$$

Here $T_f = t_1 + (1 + L_2^{-1} + 1/32)/\Gamma$ is the actual end of the finite hold and the dot denotes the corresponding scalar or vector bound. Every frequency, envelope radius and diffusion coefficient is explicit. No estimate in this section asserts a bound uniform over a sequence $K_j \rightarrow \infty$.

6.5 Control derivatives and every mixed derivative order

The constants A_1, A_2 in the spatial force costs can be bounded without differentiating an unknown shooting root. The root is a fixed number in $[0, 3]$, and differentiation below is in time. For a compression L put

$$Z_0(L) = \max\{1, 3L[\eta_2]_0\}, \quad Z_b(L) = \max\{[\eta^{(b)}]_0, 3L^{b+1}[\eta_2^{(b)}]_0\} \quad (b \geq 1).$$

These bound the derivatives of the source ramp and pulse z ; the hold has zero derivatives. During the first control, $\alpha = s_* - \arcsin(\sin s_* z)$ and $|\sin s_* z| \leq 1/4$. Twice differentiating this displayed formula gives

$$A_{1,\text{first}} = \frac{4}{\sqrt{15}} \Gamma_1 \sin s_* Z_1(\Lambda_1),$$

$$A_{2,\text{first}} = \Gamma_1^2 \left\{ \frac{4}{\sqrt{15}} \sin s_* Z_2(\Lambda_1) + \frac{64}{15\sqrt{15}} \sin^3 s_* Z_0(\Lambda_1) Z_1(\Lambda_1)^2 \right\}.$$

During second steering and holding write $z_b = Z_b(L_2)$ and $y_- = \sqrt{59}/8$. The exact control is

$$\alpha_\tau = \frac{W\zeta_{1,1} - az'}{Y}, \quad Y' = \frac{hW - a^2 z z'}{Y}, \quad \zeta_{1,\tau} = \alpha_\tau J \zeta_1.$$

Define the explicit constants

$$U_\alpha = (9 + az_1)/y_-, \quad U_Y = (27 + az_1/4)/y_-,$$

$$U_{W'} = \frac{128}{63}(\sqrt{10} + 3z_0) + \frac{9}{16},$$

$$U_{\alpha'} = (U_{W'} + 9U_\alpha + az_2)/y_- + (9 + az_1)U_Y/y_-^2.$$

The bounds $Y \geq y_-$, $|az| \leq 1/4$, $h < 3$, $W < 9$, $B \leq 2$, $h^2 + a^2 \geq 63/64$, and $\delta_1 \leq 1/16$ give $|\alpha_\tau| \leq U_\alpha$, $|Y'| \leq U_Y$ and $|W'| \leq U_{W'}$. Differentiating the quotient for α_τ gives $|\alpha_{\tau\tau}| \leq U_{\alpha'}$. Thus one can use

$$A_1 \leq \max\{A_{1,\text{first}}, \Gamma U_\alpha\}, \quad A_2 \leq \max\{A_{2,\text{first}}, \Gamma^2 U_{\alpha'}\}.$$

On intervening growth and transition intervals the angle is constant.

For completeness the following exact finite recursion evaluates any mixed derivative, rather than treating spatial estimates as time estimates. For a fixed layer put $A = M^{-1}/\ell$ and introduce independent variables t, x, s, y . Define operators on smooth expressions in these variables by

$$\mathfrak{T} = \partial_t + (\dot{k} \cdot x) \partial_s + (\dot{A}x) \cdot \nabla_y, \quad \mathfrak{X}_i = \partial_{x_i} + k_i \partial_s + \sum_l A_{li} \partial_{y_l}. \quad (136)$$

Here ∂_t differentiates every time-dependent coefficient and ∂_{x_i} differentiates the explicitly occurring x . If $\iota^* R = R(t, x, k(t) \cdot x, A(t)x)$, the chain rule proves

$$\partial_t^b \partial_x^\gamma (\iota^* R) = \iota^* (\mathfrak{T}^b \mathfrak{X}_1^{\gamma_1} \mathfrak{X}_2^{\gamma_2} R). \quad (137)$$

Induction proves this identity for all nonnegative b, γ_i . For example, for $R = Q(t)P(s)f(y)$ it gives every streamfunction jet; applying one more $\mathfrak{X}$ and J gives every velocity jet. The commutators vanish: in $[\mathfrak{T}, \mathfrak{X}_i]$ the derivative $\dot{k}_i$ is canceled by differentiating $\dot{k} \cdot x$, and $\dot{A}_{li}$ by differentiating $(\dot{A}x)_l$. The $\mathfrak{X}_i$ commute by their displayed coefficients. This checks the order consistency of the recursion.

Apply (137) to each summand of (119)–(124), using a separate pair (s, y) for each layer when necessary. After expansion the answer is a finite sum of terms of the form

$$c(t)x^\beta \prod_{l=1}^q W_l^{(n_l)}(s_j) \prod_{l=1}^r \partial^{\gamma_l} f(y_j),$$

or the analogous base terms with χ_0, H_0 . An explicit bound for each term is obtained by replacing it by

$$\sup_{[0, T_f]} |c(t)| (N_j \ell_j)^{|\beta|} \prod_{l=1}^q [W_l]_{n_l} \prod_{l=1}^r F_{|\gamma_l|}^f,$$

and summing; for a base term use $R_H^{|\beta|}$. Each coefficient jet in c is computed by differentiating (112), the phase equations, (113), and the explicit controls. This is a finite recursion using only addition,

multiplication, the stated profile derivatives and the nonzero denominators a , K_j , Γ , r_j^2 , Y . Their positive lower bounds have already been proved. Induction on the requested order, applying the product and quotient rules to those equations, therefore supplies finite bounds for every coefficient jet from the preceding-order bounds. This specifies every mixed force and diffusion norm by an exact finite procedure, including the time variation of the material cutoff. It does not replace the cutoff by a stationary radial function for layer two.

6.6 Smooth finite slab, compact forces, and exact returned state

All fields and forces are smooth and supported in B_{R_H} on $0 \leq t \leq T_f$. Each velocity is the rotated gradient of a compact streamfunction, so its energy is finite. It also has the usual exact Biot–Savart relation. Indeed for $N_0(x) = (2\pi)^{-1} \log |x|$ and compact smooth $\psi = \psi_b + \Psi_1 + \Psi_2$, distributional integration by parts gives $N_0 * \Delta\psi = (\Delta N_0) * \psi = \psi$; applying $J\nabla$ gives $u = J\nabla N_0 * \omega$ with $\omega = \Delta\psi$. All integrations are legitimate because ψ is compact and N_0 is locally integrable. Odd F , even P , and even cutoffs give odd θ, u, f_θ, f_u and even ω . The pressure $p = 0$ has $p(0, t) = 0$.

At the selected second return and throughout its physical hold of duration $1/(32\Gamma)$, the coefficient calculation proves $\zeta_{2,1} = \Omega_2 = 0$, $w_2 = 1$, and

$$\Theta_2(t) = -P_2(t_b) \exp \left[-dK_2^2 \int_{t_b}^t |\zeta_2(r)|^2 dr \right].$$

Thus $v_2 = 0$ everywhere on this hold, while its temperature and the first parent continue changing. In the inner second affine cell the total fields have exactly

$$\theta = (-A_0 e_0 + K_1 \Theta_1 \zeta_1 + K_2 \Theta_2 \zeta_2) \cdot x, \quad u = (\dot{\alpha} J + \Omega_1 J \zeta_1 \otimes \zeta_1) x.$$

The second holding shear is zero, but the first holding shear is positive at return and evolves by the displayed system. In particular these exact full spatial fields have the origin vorticity calculated in (110):

$$\text{curl } u(0, t) = 2\dot{\alpha} + \Omega_1 = \Omega_1 \left(1 - \frac{2a^2}{h^2 + a^2} \right) \geq \frac{31}{32} \Omega_1 > 0 \quad (t_b \leq t \leq T_f). \quad (138)$$

Indeed on holding $\alpha = s_* + \arctan(a/h)$ and $\dot{h} = a\Omega_1$, so $\dot{\alpha} = -a^2\Omega_1/(h^2 + a^2)$. Moreover $h \geq h_0$, $h_0^2 + a^2 = 1$ and $a \leq 1/8$ prove the numerical lower bound. Positivity of Ω_1 at return persists because the holding equation has positive forcing B/r_2 . These are the actual final states realized by the full spatial equations.

There is also an honest compact extension of the forces in time. For $t < 0$ extend the initial state by its constant values, with $\alpha = 0$ and both increments zero. Flatness of η makes the formulas smooth at zero. To the right, continue $z = 0$ and the exact holding ODE for a collar of physical length $\epsilon_R = 1/(128\Gamma)$. Its extension is quantitative: the total scaled duration is at most $17/16$ since $L_2 \geq 64$; the same integrating-factor estimates give

$$0 \leq W \leq 26/3, \quad h \leq 1 + 16/15 + (26/3)(7/128) = 813/320 < 3, \quad B > 191/512 > 1/4.$$

For the last bound use $B_{\text{in}} \geq 1/2$, $\delta_1 \leq 1/16$, $C_1 \leq 1/32$ and $B = e^{-\delta_1 \tau} B_{\text{in}} - C_1 \int_0^\tau e^{-\delta_1(\tau-r)} W(r) dr$. These strict bounds close the positivity bootstrap and keep all denominators away from zero; the smooth finite-dimensional vector field therefore continues through this collar. The support bounds continue as well. Extend the newest temperature by its exact hold exponential, and the angle and matrices by their exact inverse. In particular no nonstationary endpoint is frozen.

Choose $\epsilon_L = 1/\nu$ and multiply the extended residual forces, not the fields on their required slab, by

$$\beta(t) = \eta(1 + t/\epsilon_L) \eta(1 + (T_f - t)/\epsilon_R).$$

This function is one on $[0, T_f]$, zero outside $(-\epsilon_L, T_f + \epsilon_R)$, and flat at the two external endpoints. The resulting forces belong to $C_c^\infty(\mathbb{R}^2 \times \mathbb{R})$ and still give exactly the solution already proved on

$[0, T_f]$. On the collar the same mixed-derivative recursion applies, and the product rule includes the explicit additional factors $\epsilon_L^{-b}[\eta^{(b)}]_0$ and $\epsilon_R^{-b}[\eta^{(b)}]_0$. No equation for the continued fields outside the required slab is claimed after this force cutoff. This completes the compact finite spatial realization and its full derivative and support costs.

7 Exact third growth on the moving second holding state

The third growth begins at the actual second return. Its inherited first vorticity is positive, its second vorticity is zero, and its second temperature and phase length continue changing. We derive the complete third phase and linear pair on the holding interval already proved in Theorem 2. The original source growth control is $\dot{\alpha} = \mathcal{F}_2$; the different control $\mathcal{F}_3$ is retained below as a computed function. This distinction matters at the next transition. All formulas below use the physical frequency $K_j = \mu\hat{\lambda}_j$, the original $\nu = \nu_{AB} = \sqrt{A_0}$, and the actual changed scales.

7.1 Actual birth data and their branches

Set $t_0 = t_2^b$, $\sigma_1 = \sigma$ and $s_2 = L_2\nu/\sigma_1$, with $a = \sin s_2$ as in (97). Write $h_b = h(t_0)$, $\chi_b = \sqrt{h_b^2 + a^2}$, $P_2 = -\Theta_2$ and $A_1 = -K_1\Theta_1$. On the proved interval

$$I = \left[t_0, t_0 + \frac{1}{32\Gamma_2} \right], \quad \Gamma_2 = \sigma_1 a, \quad (139)$$

the actual physical holding equations and phases are

$$\begin{aligned} \chi &= \sqrt{h^2 + a^2}, \quad \zeta_1 = \frac{(-a, h)}{\chi}, \quad \zeta_2 = (0, \chi), \quad \theta = \arctan(a/h), \quad \alpha = s_* + \theta, \\ \dot{h} &= a\Omega_1, \quad \dot{\Omega}_1 = \frac{aA_1}{\chi} - dK_1^2\Omega_1, \quad \dot{A}_1 = -A_0 \sin s_* \Omega_1 - dK_1^2 A_1, \\ \Omega_2 &= 0, \quad \dot{P}_2 = -dK_2^2 \chi^2 P_2, \quad \dot{\alpha} = -\frac{a^2 \Omega_1}{\chi^2}. \end{aligned} \quad (140)$$

Here A_1, P_2, Ω_1 are strictly positive throughout I ; $h_b \leq h < 3$, $h_b \geq \cos s_2 \geq 15/16$ and $1 \leq \chi < \sqrt{10}$ are proved bounds, not frozen values. The notation θ in this subsection denotes an angle only; the temperature field is denoted ϑ below where needed.

Retain the original recursion

$$\begin{aligned} Q_3 &= Q_* + 3, \quad \hat{\lambda}_3 = \hat{\lambda}_2^{Q_3}, \quad k_3 = \max\{k \in \mathbb{N}_0 : 120k \leq Q_3, K(k) \leq \log \hat{\lambda}_3\}, \\ L_3 &= (k_3 + 41/8) \log \hat{\lambda}_3, \quad \sigma_2^2 = |G_{<3}(t_0)|, \quad s_3 = \frac{L_3 \sigma_1}{\sigma_2}, \quad \Gamma_3 = \sigma_2 \sin s_3, \quad p_3 = e(s_3). \end{aligned} \quad (141)$$

The positive square root defining σ_2 is the source scale evaluated on the actual state. It is not the magnitude of only the second temperature contribution. The set defining k_3 is nonempty because $k = 0$ belongs to it by (72), and finite because $120k \leq Q_3$.

For completeness, even the earlier frequency choice fixes the small positive branch needed for the exact map. Indeed $\alpha \leq 1/4$ on I , so the vertical component of $G_{<3} = -A_0 R_\alpha e_2 - A_1 \zeta_1 - K_2 P_2 \zeta_2$ at birth has three negative summands. Therefore

$$\sigma_2^2 \geq K_2 P_2(t_0) \chi_b \geq A_0 \hat{\lambda}_2^{1/8}, \quad \sigma_2 \geq \nu \hat{\lambda}_2^{1/16}.$$

The threshold and selected gain in Theorem 1 justify the second inequality. Put $y = \log \hat{\lambda}_1$. Equations (79) and (72) give $\sigma_1 \leq C_\sigma \nu e^{y/16}$ and $y \geq Q_2 \geq 202$. Since $\Lambda_1 \geq 6$, one has $C_\sigma = \sqrt{1 + e^{2+\Lambda_1^{-1}}} < 6$. One may verify the numerical bound by $e < 3$, which follows from $n! \geq 2^{n-1}$ for

$n \geq 2$ and strict inequality for $n \geq 3$. Also $Q_3 = Q_2 + 1 \leq 2Q_2$ and $k_3 + 41/8 \leq Q_3/120 + 6 \leq Q_3$; hence $L_3 \leq 8Q_2^3 y$. Consequently

$$0 < s_3 \leq 48Q_2^3 y e^{-(Q_2-1)y/16} \leq 48y^4 e^{-12y} \leq \frac{5760}{12^5 y} < \frac{1}{8}. \quad (142)$$

The penultimate inequality is the fifth positive term in the exponential series. Thus $0 < \theta_b := \arctan(a/h_b) \leq s_2 < 1/8$ and $0 < s_3 + \theta_b < 1/4$. These are statements about the original source angle, including its branch. Stronger angle and physical diffusion bounds needed to reach the third threshold are supplied by the accompanying actual third-growth theorem; none is presumed in the identities below.

Define the oriented constants and the changing inner product by

$$b = \sin(s_3 + \theta_b), \quad g_b = \cos(s_3 + \theta_b), \quad g = g_b + \frac{b}{a}(h - h_b), \quad c = bh_b - ag_b = \chi_b \sin s_3 > 0. \quad (143)$$

The constant b in (143) is dimensionless and must not be confused with the physical buoyancy coefficient $\mathfrak{b}_3$ below. One has $b > 0$, $g_b > 15/16$, $g \geq g_b$, and the exact identity $bh - ag = c$ on I .

7.2 The complete material map, phases and original feedback

For $v \in \mathbb{R}^2$, the matrix convention is $Jv = (-v_2, v_1)$ and $R_\beta = \exp(\beta J)$. The third deformation and its inverse are

$$\begin{aligned} M_3(t) &= M_2(t)M_2(t_0)^{-1} = R_\theta \begin{pmatrix} 1 & -(h - h_b)/a \\ 0 & 1 \end{pmatrix} R_{-\theta_b}, \\ M_3(t)^{-1} &= R_{\theta_b} \begin{pmatrix} 1 & (h - h_b)/a \\ 0 & 1 \end{pmatrix} R_{-\theta}. \end{aligned} \quad (144)$$

These formulas describe linear maps $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ in the original physical coordinates and have determinant one. In particular $M_3(t_0) = I$ retains the source third-birth datum; it does not reset M_1 or M_2 . Their phase pullback gives

$$\begin{aligned} \zeta_3 &= M_3^{-T} p_3 = g\zeta_1 - bJ\zeta_1 = \frac{(c, hg + ab)}{\chi}, \quad R_3 := |\zeta_3|^2 = g^2 + b^2, \\ \det(\zeta_1, \zeta_3) &= -b, \quad \det(\zeta_2, \zeta_3) = -c, \quad \dot{g} = b\Omega_1, \quad \dot{R}_3 = 2bg\Omega_1, \quad \dot{\chi} = \frac{ah\Omega_1}{\chi}. \end{aligned} \quad (145)$$

Thus $R_3(t_0) = 1$ and $R_3 \geq 1$. Its first component is positive and decreases on growth; it is not held constant by the preceding feedback. The lower component is positive and satisfies the explicit bound

$$\zeta_{3,2} = \frac{hg + ab}{\chi} > \frac{225}{256\sqrt{10}} > \frac{1}{4}. \quad (146)$$

For example the last strict comparison follows on squaring from $225^2 > 64^2 10$.

Proof of the geometric map. On holding $\Omega_2 = 0$ and the older cutoffs equal one, so $D_{<3} = D_{<2} = \dot{\alpha}J + \Omega_1 J\zeta_1 \otimes \zeta_1$. The previously proved formula for M_2 is $R_\theta \begin{pmatrix} 1 & -(h - \cos s_2)/a \\ 0 & 1 \end{pmatrix}$. Multiplication by its birth inverse gives (144). Differentiating $M_2(t)M_2(t_0)^{-1}$ gives $\dot{M}_3 = D_{<3}M_3$, including the original initial value. Multiplying the proposed inverse by M_3 on either side gives I .

Since $R_{-\theta_b}e(s_3) = (b, g_b)$, inverse transposition of the shear sends this vector to (b, g) . Subsequent rotation gives $R_\theta(b, g) = (-ag + bh, hg + ab)/\chi$, proving the pullback in (145). Because $\zeta_1 = R_\theta e_2$ and $J\zeta_1 = R_\theta(-e_1)$, this is also $g\zeta_1 - bJ\zeta_1$. Its norm, inner product and signed determinants follow by the displayed orthonormal basis and the two laboratory components. Differentiating $bh - ag = c$ and using $\dot{h} = a\Omega_1$ gives $\dot{g} = b\Omega_1$. Direct differentiation gives the remaining rates. All maps are invertible in their specified domains because $a, h, \chi > 0$; their branch choices were proved above. The lower bound (146) follows from $h, g > 15/16$ and $\chi < \sqrt{10}$. $\square$

Write the original feedback functions at the actual state as

$$\mathcal{F}_2 = -\frac{a^2\Omega_1}{\chi^2}, \quad \mathcal{F}_3 = -\frac{ab\Omega_1}{hg + ab}, \quad \mathcal{F}_3 - \mathcal{F}_2 = -\frac{ach\Omega_1}{\chi^2(hg + ab)}. \quad (147)$$

To verify them from the original sum, use

$$\mathcal{F}_j = -\frac{\sum_{i < j} \Omega_i e_{i,1} (J e_i \cdot \zeta_j)}{\zeta_{j,2}}.$$

Here $e_1 = \zeta_1$, $J\zeta_1 \cdot \zeta_2 = -a$, $J\zeta_1 \cdot \zeta_3 = -b$ and $\Omega_2 = 0$. The two substitutions give the first two formulas in (147). Taking their difference and using $bh - ag = c$ gives the third. In particular the physical controlled-component identity is exactly

$$\dot{\zeta}_{3,1} = (\mathcal{F}_3 - \mathcal{F}_2)\zeta_{3,2} = -\frac{ach\Omega_1}{\chi^3} = \frac{d}{dt} \left(\frac{c}{\chi} \right).$$

No term has been discarded by replacing the growth feedback with the third feedback. Conversely, substituting (147) in the full phase equation reconstructs this same growth map.

7.3 Both inherited temperatures and the exact numerator identity

Put $S_* = A_0 \sin s_*$ and $C_* = A_0 \cos s_*$. These symbols retain the original constants; they are not new adjustable scales. The complete parent gradient can be written in two identical forms:

$$\begin{aligned} G_{<3} &= -A_0 R_\alpha e_2 - A_1 \zeta_1 - K_2 P_2 \zeta_2 \\ &= -(A_1 + C_* + K_2 P_2 h) \zeta_1 - (S_* - a K_2 P_2) J \zeta_1. \end{aligned} \quad (148)$$

The second form follows from $R_\alpha e_2 = \cos s_* \zeta_1 + \sin s_* J \zeta_1$ and $\zeta_2 = h \zeta_1 - a J \zeta_1$. It also supplies the exact source scale formula

$$\sigma_2^4 = (A_1(t_0) + C_* + K_2 P_2(t_0) h_b)^2 + (S_* - a K_2 P_2(t_0))^2. \quad (149)$$

There is no sign restriction on the second bracket, and none is needed in this identity.

Define

$$N_3 = b(A_1 + C_*) + g S_* + c K_2 P_2, \quad E_3 = b K_1^2 A_1 + c K_2^3 \chi^2 P_2. \quad (150)$$

Every summand of N_3 and E_3 is positive on I . The exact scalar-temperature and vorticity couplings are

$$\mathfrak{a}_3 = -\frac{J \zeta_3 \cdot G_{<3}}{K_3 R_3} = \frac{N_3}{K_3 R_3} > 0, \quad \mathfrak{b}_3 = K_3 \zeta_{3,1} = \frac{K_3 c}{\chi} > 0. \quad (151)$$

Their exact rates, including all changing-parent contributions, are

$$\begin{aligned} \dot{N}_3 &= -d E_3, \quad N_3(t) = N_3(t_0) - d \int_{t_0}^t \{b K_1^2 A_1(r) + c K_2^3 \chi(r)^2 P_2(r)\} dr, \\ \frac{d}{dt} \log \mathfrak{a}_3 &= -\frac{d E_3}{N_3} - \frac{2bg\Omega_1}{R_3}, \quad \frac{d}{dt} \log \mathfrak{b}_3 = -\frac{ah\Omega_1}{\chi^2}. \end{aligned} \quad (152)$$

In particular the numerator would be exactly constant at $d = 0$ on this holding geometry. With positive physical diffusivity its replacement is the integral in (152), with both older temperature losses retained. Its positivity follows from the original sum in (150), not from assuming that this integral is small.

Proof. Using (148) and $J \zeta_3 = g J \zeta_1 + b \zeta_1$ gives

$$-J \zeta_3 \cdot G_{<3} = b(A_1 + C_* + K_2 P_2 h) + g(S_* - a K_2 P_2) = N_3.$$

This proves the full coupling and its signs. Next differentiate the three summands of N_3 , using (140) and (145). The two terms $-b S_* \Omega_1$ and $\dot{g} S_* = b S_* \Omega_1$ cancel exactly; the surviving terms are $-db K_1^2 A_1$ and $-dc K_2^3 \chi^2 P_2$. Thus $\dot{N}_3 = -d E_3$. Integrating proves its displayed integral identity. Finally apply the logarithmic derivative to (151) and insert $\dot{R}_3$, $\dot{\chi}$ and $\dot{N}_3$. All logarithms are defined because their arguments have already been proved positive. $\square$

7.4 Exact third amplitudes, inverse maps and fundamental functions

The original signed seed and activation are

$$\begin{aligned} S_3 &= \frac{A_0}{\mu} \hat{\lambda}_3^{-k_3-6}, \quad u_i = \sqrt{\mathfrak{b}_3(t_0)/\mathfrak{a}_3(t_0)}, \\ \Theta_3(t_0) &= -S_3, \quad \Omega_3(t_0) = -S_3 u_i, \quad w_3(t) = \eta(\Gamma_3(t - t_0)). \end{aligned} \quad (153)$$

The homogeneous pair is active in the coefficient construction at t_0 , while its physical increment is multiplied by w_3 , exactly as in source (3.11). The full physical equal-diffusion equations are

$$\dot{\Theta}_3 = \mathfrak{a}_3 \Omega_3 - dK_3^2 R_3 \Theta_3, \quad \dot{\Omega}_3 = \mathfrak{b}_3 \Theta_3 - dK_3^2 R_3 \Omega_3. \quad (154)$$

Here are convergent fundamental functions for the changing coefficients, so the solution does not rely on a constant-coefficient surrogate. Let

$$\begin{aligned} \mathcal{A}(t) &= \begin{pmatrix} 0 & \mathfrak{a}_3(t) \\ \mathfrak{b}_3(t) & 0 \end{pmatrix}, \quad \mathcal{H}_0(t) = I, \quad \mathcal{H}_n(t) = \int_{t_0}^t \mathcal{A}(r) \mathcal{H}_{n-1}(r) dr, \\ \mathcal{H}(t) &= \sum_{n=0}^{\infty} \mathcal{H}_n(t), \quad \Lambda_3(t) = dK_3^2 \int_{t_0}^t R_3(r) dr, \quad \mathcal{U}(t) = e^{-\Lambda_3(t)} \mathcal{H}(t). \end{aligned} \quad (155)$$

Each term is an explicit ordered integral of the actual coefficients. The solution and inverse-data map are

$$\begin{aligned} \begin{pmatrix} \Theta_3(t) \\ \Omega_3(t) \end{pmatrix} &= -S_3 \mathcal{U}(t) \begin{pmatrix} 1 \\ u_i \end{pmatrix}, \quad \begin{pmatrix} \Theta_3(t_0) \\ \Omega_3(t_0) \end{pmatrix} = e^{\Lambda_3(t)} \mathcal{H}(t)^{-1} \begin{pmatrix} \Theta_3(t) \\ \Omega_3(t) \end{pmatrix}, \\ \det \mathcal{U} &= e^{-2\Lambda_3} > 0. \end{aligned} \quad (156)$$

Existence, inversion and signs. All parent coefficients are smooth on the compact interval I , and their denominators have the positive bounds already proved. For any matrix operator norm put $M = \max_I \|\mathcal{A}\|$ and $\ell = |I|$. Repeated integration proves $\|\mathcal{H}_n(t)\| \leq M^n (t - t_0)^n / n!$ by induction. The scalar exponential series therefore proves absolute uniform convergence. Also $\mathcal{H}'_n = \mathcal{A} \mathcal{H}_{n-1}$ for $n \geq 1$; the same bound with index $n - 1$ proves uniform convergence of the derivative series. Termwise differentiation gives $\mathcal{H}' = \mathcal{A} \mathcal{H}$ and $\mathcal{H}(t_0) = I$. If two fundamental solutions had the same data, their difference V would obey $V(t) = \int_{t_0}^t \mathcal{A}(r) V(r) dr$. Iteration bounds its norm by $\sup_I \|V\| (M\ell)^n / n!$ for every n , which tends to zero; this proves uniqueness without assuming invertibility first.

For $\mathcal{H} = \begin{pmatrix} f_1 & f_2 \\ g_1 & g_2 \end{pmatrix}$, differentiation of $f_1 g_2 - f_2 g_1$ with $\dot{f}_j = \mathfrak{a}_3 g_j$, $\dot{g}_j = \mathfrak{b}_3 f_j$ gives zero. Its initial value is one. Thus its inverse is exactly $\begin{pmatrix} g_2 & -f_2 \\ -g_1 & f_1 \end{pmatrix}$. Differentiating $\mathcal{U}$ now proves the full damped matrix equation; multiplying by e^{Λ_3} proves its inverse and the determinant in (156). Every entry in $\mathcal{A}$ is nonnegative, and its two off-diagonal entries are positive. Consequently every ordered-integral term has nonnegative entries, the diagonal of $\mathcal{H}$ is at least one, and its off-diagonal entries are positive for $t > t_0$ because already the first term has positive such entries. Since $S_3, u_i > 0$, both amplitudes in (156) are strictly negative throughout I . Hence the temperature sign and all following quotients are established on the complete interval. $\square$

An equivalent scalar fundamental equation retains the geometry in a particularly direct form. Put $Y = e^{\Lambda_3} \Omega_3$. Then

$$\frac{d}{dt}(\chi \dot{Y}) - \frac{cN_3}{R_3} Y = 0, \quad \Omega_3 = e^{-\Lambda_3} Y, \quad \Theta_3 = e^{-\Lambda_3} \frac{\chi \dot{Y}}{K_3 c}. \quad (157)$$

Indeed the first transformed equation is $\dot{X} = \mathfrak{a}_3 Y$, where $X = e^{\Lambda_3} \Theta_3$, and the second is $\dot{Y} = (K_3 c / \chi) X$. The latter gives $\chi \dot{Y} = K_3 c X$; differentiating it and substituting the former gives

(157). Conversely that scalar equation and its displayed inverse give both transformed first-order equations, and hence (154). In physical damped variables the same operator is exactly

$$(\partial_t + \lambda_3)\{\chi(\partial_t + \lambda_3)\Omega_3\} - \frac{cN_3}{R_3}\Omega_3 = 0, \quad \lambda_3 = dK_3^2 R_3. \quad (158)$$

Its complete expanded left side is

$$\chi\ddot{\Omega}_3 + (\dot{\chi} + 2\chi\lambda_3)\dot{\Omega}_3 + (\chi\dot{\lambda}_3 + \dot{\chi}\lambda_3 + \chi\lambda_3^2 - cN_3/R_3)\Omega_3.$$

In particular $\dot{\lambda}_3 = 2dK_3^2 bg\Omega_1$ is present in the physical second-order equation. The integrating factor cancels the newest diagonal damping on the same evolved parent; it does not replace N_3, h, P_2 by their inviscid counterparts.

7.5 Exact quotient, gain and diffusion accounting

Set $\rho = \Gamma_3(t - t_0)$, $v_3 = \sigma_2\Omega_3/(K_3\Theta_3)$ and $P_3 = -\Theta_3$. This change has inverse $t = t_0 + \rho/\Gamma_3$, $\Theta_3 = -P_3$, $\Omega_3 = -K_3P_3v_3/\sigma_2$. Define

$$z_3 = \frac{\chi b}{\chi}, \quad k_3^{\text{dyn}} = \frac{N_3}{\sigma_2^2 \sin s_3 R_3}, \quad \delta_3 = \frac{dK_3^2}{\Gamma_3}. \quad (159)$$

The symbol k_3^{dyn} is a time-dependent coupling and is distinct from the original integer k_3 in (141). Direct quotient and product differentiation prove the exact mutually inverse equations

$$\frac{dv_3}{d\rho} = z_3 - k_3^{\text{dyn}}v_3^2, \quad \frac{d}{d\rho} \log P_3 = k_3^{\text{dyn}}v_3 - \delta_3 R_3, \quad v_3(0) = \frac{\sigma_2 u_i}{K_3}. \quad (160)$$

Thus z_3 is the changing preceding phase-length ratio, not a prescribed constant control. The direct diffusion term cancels from the quotient only because both physical diffusivities equal d . For unequal physical diffusivities κ and ε in the newest pair on this same parent, the corresponding exact term is $(\kappa - \varepsilon)K_3^2 R_3 v_3/\Gamma_3$, and the logarithmic loss is $\kappa K_3^2 R_3/\Gamma_3$. These are algebraic maps for that pair; they make no assertion that the older equal-diffusion trajectory solves the unequal-diffusion parent equations.

The instantaneous eigenline formulation used by the source is also exact. With

$$\gamma_3 = \sqrt{\mathbf{a}_3 \mathbf{b}_3} = \sqrt{\frac{cN_3}{\chi R_3}}, \quad u_* = \sqrt{\mathbf{b}_3/\mathbf{a}_3} = K_3 \sqrt{\frac{cR_3}{\chi N_3}}, \quad Z = \frac{\Omega_3}{u_* \Theta_3},$$

one has, in physical time,

$$\begin{aligned} \dot{Z} &= \gamma_3(1 - Z^2) - Z \frac{d}{dt} \log u_*, & \frac{d}{dt} \log u_* &= \frac{bg\Omega_1}{R_3} - \frac{ah\Omega_1}{2\chi^2} + \frac{dE_3}{2N_3}, \\ \frac{d}{dt} \log P_3 &= \gamma_3 Z - dK_3^2 R_3. \end{aligned} \quad (161)$$

These follow by dividing (154), substituting (152), and differentiating u_* ; the seed has exactly $Z(t_0) = 1$.

For every $t \in I$ the full new-layer diffusion exponent is

$$\begin{aligned} \Lambda_3(t) &= dK_3^2 \int_{t_0}^t \left\{ \left[g_b + \frac{b}{a}(h(r) - h_b) \right]^2 + b^2 \right\} dr, \\ dK_3^2(t - t_0) &\leq \Lambda_3(t) \leq dK_3^2 R_{3,\max}(t - t_0), \end{aligned} \quad (162)$$

where the finite explicit bound is $R_{3,\max} = [g_b + (b/a)(3 - h_b)]^2 + b^2$. The inequalities follow from $h_b \leq h < 3$ and $b, g_b > 0$. The exact gain is the integral

$$\log \frac{P_3(t)}{S_3} = \int_{t_0}^t \gamma_3(r) Z(r) dr - \Lambda_3(t).$$

The two inherited diagonal damping exponents on this same interval are

$$dK_1^2(t - t_0), \quad dK_2^2 \int_{t_0}^t \chi(r)^2 dr.$$

The first temperature also retains the exact response

$$A_1(t) = e^{-dK_1^2(t-t_0)} A_1(t_0) - S_* \int_{t_0}^t e^{-dK_1^2(t-r)} \Omega_1(r) dr.$$

These expressions retain all three physical frequencies and the phase-length factors. The formula for N_3 records their combined effect on the third gain coefficient. No attainment of the third threshold is inferred from positivity of the cooperative transformed pair alone; actual attainment uses the quantitative estimates proved in the accompanying continuation.

7.6 The exact affine parent equations during third growth

The moving affine parent also has an exact spatial interpretation throughout I . Put

$$D = D_{<3}, \quad B_1 = -A_1 \zeta_1, \quad B_2 = -K_2 P_2 \zeta_2, \quad \vartheta(x, t) = G_{<3}(t) \cdot x, \quad u(x, t) = D(t)x.$$

Then

$$\begin{aligned} \dot{G}_{<3} + D^T G_{<3} &= -dK_1^2 B_1 - dK_2^2 \chi^2 B_2, \quad C := D_{21} - D_{12} = 2\dot{\alpha} + \Omega_1, \\ \dot{C} - G_{<3,1} &= 2\ddot{\alpha} - A_0 \sin \alpha - dK_1^2 \Omega_1. \end{aligned} \tag{163}$$

Thus the full equations with equal physical diffusion are satisfied by the explicitly specified scalar force, vector force and pressure

$$\begin{aligned} f_\vartheta &= (-dK_1^2 B_1 - dK_2^2 \chi^2 B_2) \cdot x, \\ f_u &= \frac{1}{2}(2\ddot{\alpha} - A_0 \sin \alpha - dK_1^2 \Omega_1) Jx, \\ p &= -\frac{1}{2} x^T \text{sym}(\dot{D} + D^2 - e_2 \otimes G_{<3})x. \end{aligned} \tag{164}$$

To verify the new temperature contribution, differentiate B_2 using $\dot{P}_2 = -dK_2^2 \chi^2 P_2$ and $\dot{\zeta}_2 = -D^T \zeta_2$; this gives $\dot{B}_2 + D^T B_2 = -dK_2^2 \chi^2 B_2$. The base-plus-first contribution is the exact identity proved in Section 5, so their sum proves the first equation in (163). The curl identity is unchanged because $\Omega_2 = 0$ and $B_{2,1} = 0$, as the displayed vertical phase shows. The trace of D is zero, so D^2 is scalar. Substitution of the quadratic pressure cancels the symmetric part of $\dot{D} + D^2 - e_2 \otimes G_{<3}$; its skew part is $(\dot{C} - G_{<3,1})J/2$. This proves the entire vector equation. Both affine Laplacians vanish by second differentiation. These forces are the affine-core parent forces. The compact released profiles add their exact profile, envelope and interaction terms; the present identities do not discard those spatial residuals or claim an unforced third-layer superposition.

8 The actual third growth target on the moving second hold

We continue from the actual second returned state, with its first vorticity strictly positive. The third wave sees both older temperature gradients and the changing first shear. This section proves attainment of its original growth threshold, with the original third insertion rule and retained positive physical diffusion. The second parent is held by its original feedback during this interval. The third transition changes that control and requires its own calculation; no third return is inferred from the growth result.

8.1 A further explicit choice in the same source family

Retain every original datum, profile, frequency and constant from Section 4. In particular $\nu = \sqrt{A_0}$, $\mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil$ and $K_j = \mu \hat{\lambda}_j$ are still physical quantities. Write $\sigma_1 = \sigma$ for the actual second-stage scale in (78). Extend the exact source recursion to

$$\begin{aligned} Q_3 &= Q_* + 3, & \hat{\lambda}_3 &= \hat{\lambda}_2^{Q_3}, & k_3 &= \max\{k \in \mathbb{N}_0 : 120k \leq Q_3, K(k) \leq \log \hat{\lambda}_3\}, \\ L_3 &= (k_3 + 41/8) \log \hat{\lambda}_3, & S_3 &= (A_0/\mu) \hat{\lambda}_3^{-k_3-6}, & P_3^* &= (A_0/\mu) \hat{\lambda}_3^{-7/8} = S_3 e^{L_3}. \end{aligned} \quad (165)$$

The finite maximum exists for the same reason as the earlier two. Define fixed numbers

$$\begin{aligned} c_2 &= (41/8)Q_2, & c_3 &= (41/8)Q_2Q_3, \\ C_3 &= (\lfloor Q_3/120 \rfloor + 41/8)Q_2Q_3, \\ C_2 &= e^9 \sqrt{5\sqrt{10}/4}, & C_M &= \frac{256C_2}{15\sqrt{e}}. \end{aligned}$$

Then $L_2 \geq c_2 \log \hat{\lambda}_1$, $L_3 \geq c_3 \log \hat{\lambda}_1$, and $L_3 \leq C_3 \log \hat{\lambda}_1$ for all these exact choices. Using the already defined Y_0, C_σ, C_N , select

$$Y_3 = 1 + \max \left\{ \begin{array}{l} Y_0, \quad 1024/c_2, \quad 16384/c_3, \quad \frac{8}{Q_2-1} \log(12C_\sigma^2), \\ \frac{16}{Q_2-2} \log(2C_\sigma^2 C_3/c_2), \quad \frac{\log(290C_N)}{3Q_2-49/16}, \quad \frac{\log(145\sqrt{10}/s_0)}{2Q_2}, \\ \frac{\log(4C_N C_M)}{47Q_2/16-3}, \quad \frac{\log(\sqrt{11}C_M/s_0)}{(31Q_2+1)/16} \end{array} \right\}. \quad (166)$$

Choose $\hat{\lambda}_1 \geq e^{Y_3}$. All denominators are positive because $Q_2 \geq 202$. These additional lower bounds retain the same source frequency family and integer selection; they do not replace any earlier scale by one. Set

$$\sigma_- = \sqrt{e} \nu \hat{\lambda}_1^{1/16}, \quad d_{3,\max} = \min \left\{ d_{\max}, \frac{\sigma_- \log 2}{160K_2^2}, \frac{L_3 \sigma_-}{128K_3^2} \right\} > 0. \quad (167)$$

For the remainder choose any $0 < d \leq d_{3,\max}$ and construct the actual first and second stages already proved. Every number on the right of (167) is specified before selecting the second root or solving a new growth equation. The lower bound $\sigma_1 \geq \sigma_-$ follows from (79).

8.2 The actual third insertion scale

Let $t_b = t_2^b$ be the selected second return, and use a subscript b for evaluation there. Set $a = \sin s_2$, $h_b = h(t_b)$, $\chi_b = \sqrt{h_b^2 + a^2}$, and $P_{2,b} = -\Theta_2(t_b) > 0$. Define the original second gradient amplitude and next scale at this actual endpoint by

$$A_{2,b} = K_2 \chi_b P_{2,b}, \quad \sigma_2^2 = |G_{<3}(t_b)|, \quad s_3 = \frac{L_3 \sigma_1}{\sigma_2}, \quad \Gamma_3 = \sigma_2 \sin s_3. \quad (168)$$

In particular σ_2 is evaluated after the selected second control, rather than at an inviscid endpoint. The complete older gradient at this time is

$$\begin{aligned} G_{<3}(t_b) &= -A_0 R_{\alpha_b} e_2 - A_{1,b} \zeta_{1,b} - A_{2,b} e_2, \\ A_{1,b} &= -K_1 \Theta_1(t_b), \quad \zeta_{1,b} = \frac{(-a, h_b)}{\chi_b}, \quad \alpha_b = s_* + \arctan(a/h_b). \end{aligned}$$

Theorem 2 gives $\chi_b \geq 1$, $P_{2,b} \geq P_2^*$, $A_{1,b} \leq 2\sigma_1^2$, and $A_0 \leq \sigma_1^2$. The new choice $L_2 > 1024$ and (80) imply

$$0 < a < \frac{1}{12HL_2} \leq \frac{1}{12288}. \quad (169)$$

Because $h_b \geq \cos s_2 \geq 15/16$, $0 < \alpha_b \leq 1/8 + (16/15)a < 1/4$. Both older terms in the displayed $G_{<3}$ have negative vertical component. Its norm is therefore at least $A_{2,b}$, while the triangle inequality bounds it above by $A_{2,b} + 3\sigma_1^2$. Moreover (166) and the original threshold give

$$A_{2,b} \geq A_0 \hat{\lambda}_2^{1/8}, \quad \frac{A_{2,b}}{\sigma_1^2} \geq \frac{\hat{\lambda}_1^{(Q_2-1)/8}}{C_\sigma^2} > 12.$$

Consequently

$$A_{2,b} \leq \sigma_2^2 \leq \frac{5}{4} A_{2,b}, \quad \sigma_2 > \sigma_1, \quad \sigma_2 \geq \nu \hat{\lambda}_2^{1/16}, \quad \frac{4}{5} \leq \frac{A_{2,b}}{\sigma_2^2} \leq 1. \quad (170)$$

The actual third angle obeys an inequality stronger than a branch choice. Since $a \geq L_2 \nu / (2\sigma_1)$, the scale bounds imply

$$\frac{s_3}{a} \leq \frac{2L_3 \sigma_1^2}{L_2 \nu \sigma_2} \leq 2C_\sigma^2 \frac{C_3}{c_2} \hat{\lambda}_1^{-(Q_2-2)/16} < 1.$$

Thus $0 < s_3 < a < 1/8$. The elementary inequality $\sin x \geq x - x^3/6 \geq (15/16)x$ for $0 \leq x \leq 1/8$ follows by integrating $\cos x \geq 1 - x^2/2$. It proves

$$\frac{15}{16} L_3 \sigma_1 \leq \Gamma_3 \leq L_3 \sigma_1. \quad (171)$$

No small-angle identity has replaced the actual sine in the controls.

8.3 Exact moving geometry and the changing coupling numerator

Use physical elapsed time $r = t - t_b$ and the original third growth control $\dot{\alpha} = F_2^{\text{ctl}}$. The second amplitude and its horizontal phase component remain zero in vorticity and phase, respectively, while both older temperatures and the first shear evolve:

$$\begin{aligned} \zeta_1 &= (-a, h)/\chi, & \zeta_2 &= (0, \chi), & \chi &= \sqrt{h^2 + a^2}, & \dot{h} &= a\Omega_1, \\ \dot{A}_1 &= -A_0 \sin s_* \Omega_1 - dK_1^2 A_1, & \dot{P}_2 &= -dK_2^2 \chi^2 P_2, & \dot{\alpha} &= -a^2 \Omega_1 / \chi^2. \end{aligned} \quad (172)$$

These are the previously proved holding equations in physical time. Let

$$\beta_b = \arctan(a/h_b), \quad b = \sin(s_3 + \beta_b), \quad g_b = \cos(s_3 + \beta_b), \quad c = bh_b - ag_b = \chi_b \sin s_3 > 0. \quad (173)$$

Here b is an oriented determinant magnitude, not an amplitude coefficient; the latter will carry its layer index below. Since $s_3 \leq a$ and $\beta_b \leq (16/15)a$, $0 < s_3 + \beta_b < 3a < 1/8$, so $g_b > 0$ and $0 < b < 3a$.

The third material matrix and phase have the exact expressions

$$\begin{aligned} M_3(t) &= M_2(t) M_2(t_b)^{-1}, & M_3(t_b) &= I, \\ g(t) &= g_b + (b/a)(h(t) - h_b), & \zeta_3(t) &= g\zeta_1 - bJ\zeta_1 = \frac{(c, hg + ab)}{\chi}, & R_3 &= |\zeta_3|^2 = g^2 + b^2. \end{aligned} \quad (174)$$

Here is a direct verification preserving the original datum $p_3 = e(s_3)$. Since $\Omega_2 = 0$, the actual matrices $D_{<3}$ and $D_{<2}$ are equal, so differentiation proves the matrix equation for M_3 with its stated initial value. In the basis $\zeta_1, J\zeta_1$, the third phase equation is $\dot{g} = b\Omega_1$, $\dot{b} = 0$; these follow by differentiating its scalar products and using $\dot{\zeta}_1 = \dot{\alpha}J\zeta_1$, $\dot{\zeta}_3 = \dot{\alpha}J\zeta_3 - \Omega_1\zeta_1(J\zeta_1 \cdot \zeta_3)$. Integration and $\dot{h} = a\Omega_1$ give (174). At birth, resolving $e(s_3)$ in this oriented basis gives precisely g_b, b in (173). The identity for c follows from $h_b = \chi_b \cos \beta_b$, $a = \chi_b \sin \beta_b$ and the sine subtraction formula. This verifies both components of the phase formula and its initial datum. Uniqueness of the linear phase equation identifies it with $M_3^{-T} p_3$.

The full inherited gradient and third amplitude coefficients are

$$\begin{aligned} G_{<3} &= -(A_1 + A_0 \cos s_* + K_2 P_2 h) \zeta_1 + (-A_0 \sin s_* + a K_2 P_2) J \zeta_1, \\ N_3 &= b(A_1 + A_0 \cos s_*) + g A_0 \sin s_* + c K_2 P_2, \\ a_3 &= \frac{N_3}{K_3 R_3}, \quad b_3 = K_3 \zeta_{3,1} = \frac{K_3 c}{\chi}. \end{aligned} \quad (175)$$

To verify the first line, use $R_\alpha e_2 = \cos s_* \zeta_1 + \sin s_* J \zeta_1$ and $\zeta_2 = h \zeta_1 - a J \zeta_1$ in the original sum $G_{<3} = -A_0 R_\alpha e_2 - A_1 \zeta_1 - K_2 P_2 \zeta_2$. The scalar product of $J \zeta_3 = g J \zeta_1 + b \zeta_1$ with that sum is $-N_3$, proving the coefficient including the original gravity direction. Differentiating its full numerator gives

$$\dot{N}_3 = -d\{bK_1^2 A_1 + cK_2^3 \chi^2 P_2\}. \quad (176)$$

Indeed the terms $-bA_0 \sin s_* \Omega_1$ from $\dot{A}_1$ and $b\Omega_1 A_0 \sin s_*$ from $\dot{g}$ cancel exactly; the second temperature contributes the other displayed term. Thus the inviscid numerator would be exactly constant despite the changing shear. At positive d both parent losses are retained.

8.4 A common proved horizon before taking an amplitude quotient

Set $R_H = 16/\sigma_1$. Its duration in the second-stage time is $\Gamma_2 R_H = 16a < 1/32$, by (169). Thus the entire closed older system exists on $0 \leq r \leq R_H$ by its already proved second hold, uniformly for the current d . There A_1, P_2 are positive, $\Omega_1 < 9\sigma_1$, $\chi \geq \chi_b \geq 1$, and

$$\begin{aligned} 0 \leq h - h_b &\leq 144a, \quad 0 \leq g - g_b \leq 144b, \\ 1 \leq \sqrt{R_3} &\leq 1 + 432a \leq 265/256 < \sqrt{2}, \quad 1 \leq \frac{\chi}{\chi_b} \leq 259/256 < 2. \end{aligned} \quad (177)$$

The first inequality integrates $\dot{h} = a\Omega_1$. The second uses the exact relation (174). For the third apply the Euclidean triangle inequality to $(g, b) = (g_b, b) + (g - g_b, 0)$, whose first summand has norm one. The fourth follows similarly from $\sqrt{h^2 + a^2} \leq \chi_b + (h - h_b)$ and $\chi_b \geq 1$. The rational constants use $432/12288 = 9/256$ and $144/12288 = 3/256$; $265^2 < 2 \cdot 256^2$ proves the strict square-root bound without a numerical approximation.

All four terms in N_3 are positive on this horizon. Since $K_1 < K_2$ and $\chi^2 < 10$, (176) gives $-10dK_2^2 N_3 \leq \dot{N}_3 \leq 0$. Multiplication by $e^{10dK_2^2 r}$ and integration, followed by (167), proves

$$\frac{1}{2} N_i \leq N_3(r) \leq N_i, \quad \frac{1}{4} a_i \leq a_3(r) \leq a_i, \quad \frac{1}{2} b_i \leq b_3(r) \leq b_i, \quad (178)$$

where $N_i = N_3(t_b)$, $a_i = N_i/K_3$, and $b_i = K_3 \sin s_3$ are the actual birth values. Indeed $10dK_2^2 R_H = 160dK_2^2/\sigma_1 \leq \log 2$; the remaining bounds use (177) in the exact coefficient formulas. No value of a later temperature is assumed in these estimates.

For the actual initial rate $\gamma_i = \sqrt{a_i b_i}$ put $n_i = \gamma_i^2/\Gamma_3^2 = N_i/(\sigma_2^2 \sin s_3)$. The positive newest-parent contribution to its numerator is $cK_2 P_{2,b} = A_{2,b} \sin s_3$. Its remaining two terms satisfy

$$b(A_{1,b} + A_0 \cos s_*) + g_b A_0 \sin s_* \leq 4\sigma_1^2,$$

using $b, g_b \leq 1$, $A_{1,b} \leq 2\sigma_1^2$ and $A_0 \leq \sigma_1^2$. Equations (170) and (171) therefore give

$$\frac{4}{5} \leq n_i \leq 1 + \frac{64\sigma_1}{15L_3\sigma_2} \leq \frac{16}{15}, \quad \gamma_i \geq \frac{7}{8} \Gamma_3 \geq \frac{105}{128} L_3 \sigma_1. \quad (179)$$

For the middle upper bound use $L_3 \geq 64$ and $\sigma_2 > \sigma_1$. The lower rate follows from $4/5 > (7/8)^2$.

The original third seed is exactly $\Theta_3(t_b) = -S_3$, $\Omega_3(t_b) = -S_3 \sqrt{b_i/a_i}$. Solve its complete linear system with the changing coefficients:

$$\dot{\Theta}_3 = a_3 \Omega_3 - dK_3^2 R_3 \Theta_3, \quad \dot{\Omega}_3 = b_3 \Theta_3 - dK_3^2 R_3 \Omega_3. \quad (180)$$

Its coefficients and derivatives are finite on the common horizon, so the iterated integral series provides existence and uniqueness there. Define

$$E(r) = \exp\left(dK_3^2 \int_0^r R_3(t_b + u) du\right), \quad X = -E\Theta_3, \quad Y = -E\Omega_3.$$

Differentiation gives $\dot{X} = a_3 Y$, $\dot{Y} = b_3 X$. Both initial values are positive, and before a hypothetical first zero both derivatives are positive. This excludes the zero and proves $\Theta_3, \Omega_3 < 0$ throughout the horizon. The quotient may now be formed. With $u = \Omega_3/\Theta_3$ and $u_i = \sqrt{b_i/a_i}$, it satisfies $\dot{u} = b_3 - a_3 u^2$. The bounds in (178) make $u_i/\sqrt{2}$ a constant subsolution and $2u_i$ a constant supersolution. The integrating-factor comparison for their differences proves

$$\frac{u_i}{\sqrt{2}} \leq u \leq 2u_i, \quad \sqrt{15/32} \leq v_3 := \frac{\sigma_2 \Omega_3}{K_3 \Theta_3} \leq \sqrt{5} < 3. \quad (181)$$

The exact identity $\sigma_2 u_i / K_3 = 1/\sqrt{n_i}$ and (179) give the latter interval.

8.5 The original third threshold and its full diffusion cost

The last bound in (167) and (179) imply $dK_3^2 \leq L_3 \sigma_1 / 64 \leq \gamma_i / (16\sqrt{2})$. For the last comparison it suffices to use $\gamma_i \geq L_3 \sigma_1 / 2$ and $\sqrt{2} \leq 2$. Thus the actual $P_3 = -\Theta_3$ has the positive physical rate

$$\frac{\gamma_i}{8\sqrt{2}} \leq (\log P_3)' = a_3 u - dK_3^2 R_3 \leq 2\gamma_i < 3\Gamma_3. \quad (182)$$

The lower bound uses $a_3 u \geq a_i u_i / (4\sqrt{2})$, $R_3 < 2$, and the displayed diffusion bound. The upper bound uses $a_3 u \leq 2a_i u_i$ and $n_i \leq 16/15 < 9/4$.

Theorem 3. *For every fixed original datum and profiles, choose the source frequencies by (166) and any $0 < d \leq d_{3,\max}$ in (167). After the actual two returns already constructed, the original third growth control reaches its original target P_3^* exactly once, at $t_3^a = t_b + r_a$, with*

$$\frac{L_3}{2\gamma_i} \leq r_a \leq \frac{8\sqrt{2}L_3}{\gamma_i} \leq \frac{1024\sqrt{2}}{105\sigma_1} < \frac{14}{\sigma_1} < \frac{16}{\sigma_1}. \quad (183)$$

The crossing is transverse, its activation is already complete, and its initial and final physical amplitude ratios obey (181). Both inherited temperatures and the first shear evolve by (172) throughout. The exact logarithmic diffusion cost of the third amplitude is

$$dK_3^2 \int_0^{r_a} R_3(t_b + r) dr, \quad dK_3^2 r_a \leq dK_3^2 \int_0^{r_a} R_3(t_b + r) dr \leq 2dK_3^2 r_a. \quad (184)$$

Its total actual logarithmic gain is exactly L_3 .

Proof. The complete older and new systems were constructed on a fixed horizon before the threshold was chosen. The upper candidate time in (183) is within that horizon: squaring positive numbers gives $(1024\sqrt{2})^2 < 1470^2$. Integrating the lower rate in (182) to that candidate time gives at least L_3 . Continuity and strict increase therefore prove a unique crossing. Integrating the upper rate gives its lower time bound. Because $2\gamma_i < 3\Gamma_3$ and $L_3 \geq 64$, $\Gamma_3 r_a > L_3/3 > 1$, so the original activation $\eta(\Gamma_3 r)$ is one before this crossing. The rate lower bound proves transversality. The positive phase-length bounds in (177) give (184) by integration, with all physical factors retained. The target identity in (165) gives the exact gain L_3 . $\square$

For the two older amplitudes during this third growth, the exact diffusion exponents are $dK_1^2 r_a$ and $dK_2^2 \int_0^{r_a} \chi(t_b + r)^2 dr$. The second one is at most $10dK_2^2 r_a \leq \log 2$ on our common horizon. The additional first-temperature coupling $-A_0 \sin s_* \Omega_1$ is retained in (172); it is not a damping exponent. The numerator's exact total change is the integral of (176), including both parent contributions.

Exact third support nesting. The source third support also fits the changing exact affine cells. Indeed the relative matrix (174) is a product of two rotations and the shear with off-diagonal entry $-(h - h_b)/a$; hence $\|M_3\|, \|M_3^{-1}\| \leq 145$ on this horizon. The older bound is $\|M_2^{-1}\| \leq N \leq C_N \widehat{\lambda}_1^{1/16}$. With the exact source radius $\ell_3 = \widehat{\lambda}_2^{-3}/\mu$, the two growth-support conditions in (166) give

$$2 \cdot 145 N \widehat{\lambda}_2^{-3} < \widehat{\lambda}_1^{-3}, \quad 145 \sqrt{10} \widehat{\lambda}_2^{-2} < s_0.$$

Thus $|M_2^{-1}x| < \ell_2/2$ and $|K_2 \zeta_2 \cdot x| < s_0$ on the entire third support. The earlier proved nesting places it in the first and base affine cells as well. These explicit inequalities justify the complete compact third spatial increment and its full forces below.

This proves a further original growth event with retained diffusion, not a third return or an infinite iteration. The newly permitted range includes a factor K_3^{-2} and retains that dependence. The following transition must evolve both older vorticities and its prescribed blend of F_2^{ctl} and F_3^{ctl} ; the exact third-growth numerator identity identifies the parent losses that must remain in that calculation.

9 The actual third transition and its revived second parent

The released transition is the full physical interval $[t_3^a, t_3^a + \Gamma_3^{-1}]$, with $\dot{\alpha} = (1 - \eta_b)\mathcal{F}_2 + \eta_b\mathcal{F}_3$ and $\eta_b = \eta(\Gamma_3(t - t_3^a))$; see source (3.4)–(3.6), Section 3.7.5, pp. 30–31, and (5.22), p. 46. The following calculation retains that control, both evolving parents, their physical diffusion, and the original laboratory gravity. It does not use the exceptional fixed geometry of transition two. All older activation weights are one on this interval.

9.1 Exact determinant coordinates, coefficients, and inverse

Keep $A_0, \mu, \nu = \sqrt{A_0}$, $K_j = \mu \widehat{\lambda}_j$, s_* , and the actual inherited scales σ_1, σ_2 . Physical viscosity and thermal diffusivity equal $d > 0$. Write

$$a = \sin s_2, \quad c = A_0 \cos s_*, \quad C = A_0 \sin s_*, \quad s = s_3 = L_3 \sigma_1 / \sigma_2, \quad \Gamma = \Gamma_3 = \sigma_2 \sin s. \quad (185)$$

Thus a, s and d are distinct parameters. Let $t_b = t_2^b$, let $r_b = |\zeta_2(t_b)|$, and put $b = r_b \sin s > 0$. The original third datum is $\zeta_3(t_b) = e(s)$ and $M_3(t_b) = I$, while $\zeta_2(t_b) = r_b e_2$. For $e(\phi) = (\sin \phi, \cos \phi)$ and $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, the two adjacent oriented determinants are exactly

$$\det(\zeta_1, \zeta_2) = -a, \quad \det(\zeta_2, \zeta_3) = -b. \quad (186)$$

Both identities persist through growth, transition and later controls whenever their equations exist. Indeed $|\zeta_1| = 1$ and ζ_1 also solves the transport equation of ζ_2 : the added first shear has transpose annihilating ζ_1 . Likewise the added second shear has transpose annihilating ζ_2 , so ζ_2, ζ_3 solve the same third transport equation. The determinant of two solutions of $\dot{z} = -D^T z$ has derivative $-\text{tr}(D)$ times itself, by differentiating its two columns. Every parent matrix has zero trace, proving both assertions.

Use the following coordinates, with their original dimensional factors:

$$\begin{aligned} \zeta_2 &= h\zeta_1 - aJ\zeta_1, & \zeta_3 &= x\zeta_1 - yJ\zeta_1, & D &= h^2 + a^2, \\ g &= \zeta_2 \cdot \zeta_3 = hx + ay, & hy - ax &= b, & x &= \frac{hg - ab}{D}, & y &= \frac{ag + hb}{D}, \\ E_1 &= -K_1\Theta_1 = A_1, & E_2 &= -K_2\Theta_2 = A_2/\sqrt{D}, & R &= |\zeta_3|^2 = \frac{g^2 + b^2}{D}. \end{aligned} \quad (187)$$

Here D and R are scalars, not velocity-gradient matrices. In particular the actual second amplitude is $A_2 = \sqrt{D}E_2$; replacing it by E_2 would lose the evolving phase length. Put $\theta = \phi_2$, so the full phase and angular inverse is

$$\zeta_1 = \frac{1}{\sqrt{D}}(h \sin \theta - a \cos \theta, \quad h \cos \theta + a \sin \theta), \quad (188)$$

$$\begin{aligned}
\zeta_2 &= \sqrt{D}(\sin \theta, \cos \theta), \\
\zeta_3 &= \frac{1}{\sqrt{D}}(g \sin \theta + b \cos \theta, g \cos \theta - b \sin \theta), \\
\phi_1 &= \theta - \arctan(a/h), \quad \alpha = s_* - \phi_1.
\end{aligned}$$

The domain used below has $h, g > 0$ and $g \cos \theta - b \sin \theta > 0$; hence these expressions select the positive second-component branches. The original deformation matrices are

$$M_1 = R_\alpha, \quad M_2 = R_{\alpha-s_*} \begin{pmatrix} 1 & -(h - \cos s_2)/a \\ 0 & 1 \end{pmatrix}, \quad M_3 = \mathcal{L}(t)^{-T} \mathcal{L}(t_b)^T, \quad \mathcal{L} = (\zeta_2 \ \zeta_3). \quad (189)$$

The first two preserve their original activation data, and $\det \mathcal{L} = -b \neq 0$. Differentiating the last formula using $\dot{\mathcal{L}} = -D_{<3}^T \mathcal{L}$ gives $\dot{M}_3 = D_{<3} M_3$, $M_3(t_b) = I$, and $M_3^{-T} e(s) = \zeta_3$. All three have determinant one. Thus the inverse gives the complete original matrix evolution, rather than merely the newest phase direction.

Define the exact temperature numerators

$$N_2 = a(E_1 + c) + Ch, \quad N_3 = (E_1 + c)y + Cx + bE_2. \quad (190)$$

The full older physical equations are

$$\begin{aligned}
\dot{h} &= a\Omega_1, & \dot{g} &= \Omega_1 \frac{2ahg + (h^2 - a^2)b}{D} + b\Omega_2, \\
\dot{E}_1 &= -C\Omega_1 - dK_1^2 E_1, & \dot{\Omega}_1 &= -E_1 \zeta_{1,1} - dK_1^2 \Omega_1, \\
\dot{E}_2 &= -\frac{N_2}{D} \Omega_2 - dK_2^2 D E_2, & \dot{\Omega}_2 &= -E_2 \sqrt{D} \sin \theta - dK_2^2 D \Omega_2, \\
\dot{\theta} &= -\dot{\alpha} - \frac{a^2 \Omega_1}{D}.
\end{aligned} \quad (191)$$

In terms of the actual amplitude A_2 the fifth equation is exactly

$$\dot{A}_2 = \left(\frac{ah\Omega_1}{D} - dK_2^2 D \right) A_2 - \frac{N_2}{\sqrt{D}} \Omega_2.$$

The exact source feedbacks and their factored difference are

$$\begin{aligned}
\mathcal{F}_2 &= \frac{a\Omega_1(h \tan \theta - a)}{D}, \\
\mathcal{F}_3 &= \frac{\Omega_1 y(h \sin \theta - a \cos \theta) + b\Omega_2 \sin \theta}{g \cos \theta - b \sin \theta}, \\
\mathcal{F}_3 - \mathcal{F}_2 &= \frac{b\Omega_1(h \sin \theta - a \cos \theta)(h \cos \theta + a \sin \theta)}{D \cos \theta(g \cos \theta - b \sin \theta)} + \frac{b\Omega_2 \sin \theta}{g \cos \theta - b \sin \theta}.
\end{aligned} \quad (192)$$

With η_b as specified, the final equation in (191) becomes

$$\dot{\theta} = -\frac{ah\Omega_1}{D} \tan \theta - \eta_b(\mathcal{F}_3 - \mathcal{F}_2). \quad (193)$$

At the actual transition entry $\theta = \Omega_2 = 0$, $\mathcal{F}_3 - \mathcal{F}_2 = -abh\Omega_1/(Dg) < 0$. The switching therefore drives a positive second angle and a negative second vorticity, as proved on the full interval below.

Verification of the closed equations and exact invariants. The rotating base and the sum of inherited gradients satisfy

$$\begin{aligned}
R_\alpha e_2 &= \cos s_* \zeta_1 + \sin s_* J \zeta_1, \\
G_{<3} &= -(E_1 + c + hE_2) \zeta_1 - (C - aE_2) J \zeta_1.
\end{aligned}$$

Taking $-J\zeta_2 \cdot G_{<2}$ and $-J\zeta_3 \cdot G_{<3}$ gives respectively N_2, N_3 ; the shear-amplitude coefficients are therefore $a_2 = N_2/(K_2D)$ and $a_3 = N_3/(K_3R)$, with their original signs. Since $\dot{\zeta}_1 = \dot{\alpha}J\zeta_1$, the phase equations in this moving orthonormal basis give, component by component,

$$\dot{h} = a\Omega_1, \quad \dot{x} = \Omega_1 y + \frac{bh}{D}\Omega_2, \quad \dot{y} = \frac{ba}{D}\Omega_2.$$

The derivative of $hy - ax$ is zero, and differentiating $g = hx + ay$ proves its displayed row. The damped amplitude rows then give every remaining row of (191). Substituting these phases in the source feedback formula proves (192); the polar phase equation gives (193). Conversely these calculations reverse: differentiating (188), with (191) and (193), gives the three physical phase equations and the source angular control. Together with (189), the temperature inverse and the vorticity rows, this proves equivalence on the stated domain.

There is an exact cancellation despite the revival of Ω_2 :

$$\dot{N}_2 = -adK_1^2 E_1, \quad \dot{N}_3 = -d(K_1^2 E_1 y + bK_2^2 D E_2). \quad (194)$$

Indeed

$$\begin{aligned} \dot{N}_3 = & (-C\Omega_1 - dK_1^2 E_1)y + (E_1 + c)\frac{ba}{D}\Omega_2 + C\left(\Omega_1 y + \frac{bh}{D}\Omega_2\right) \\ & + b\left(-\frac{a(E_1 + c) + Ch}{D}\Omega_2 - dK_2^2 D E_2\right), \end{aligned}$$

whose two shear groups cancel exactly. Differentiating N_2 gives its first formula. Thus both numerators are conserved when $d = 0$ even though the individual phases, amplitudes and controls evolve. For positive d the two losses in $\dot{N}_3$ are retained explicitly. $\square$

9.2 The general aggregate identity, including activation

The cancellation has an exact form for any finite active family. Write $B_j = w_j K_j \Theta_j \zeta_j$, $r_j = |\zeta_j|$, $E_j = -K_j \Theta_j$, and

$$G_{<q} = -A_0 R_\alpha e_2 + \sum_{j<q} B_j, \quad D_{<q} = \dot{\alpha}J + \sum_{j<q} w_j \Omega_j \frac{J\zeta_j \otimes \zeta_j}{r_j^2}.$$

Use the phase equations and the physical temperature rows with thermal diffusivity d , without added amplitude forcing. Direct pairwise cancellation gives

$$\dot{G}_{<q} + D_{<q}^T G_{<q} = \sum_{j<q} (\dot{w}_j K_j \Theta_j - d w_j K_j^3 r_j^2 \Theta_j) \zeta_j. \quad (195)$$

For precision, the j th shear applied to all lower-index gradient terms is $w_j \Omega_j \zeta_j (J\zeta_j \cdot G_{<j})/r_j^2$. It cancels $w_j K_j \dot{\Theta}_j \zeta_j$ apart from diffusion. The terms with lower shears cancel the phase derivative in B_j , its self-shear vanishes because $J\zeta_j \cdot \zeta_j = 0$, and the rotational base and phase terms cancel exactly. The only terms left are precisely those displayed in (195), including $\dot{w}_j$.

Put $N_q = -J\zeta_q \cdot G_{<q}$ and $\Delta_{jq} = \det(\zeta_j, \zeta_q)$. For any trace-free 2×2 matrix D , direct multiplication gives $JD^T + DJ = 0$. Differentiating the bilinear pairing using $\dot{\zeta}_q = -D_{<q}^T \zeta_q$ thus gives

$$\dot{N}_q = \sum_{j<q} (\dot{w}_j - d w_j K_j^2 r_j^2) E_j (-\Delta_{jq}). \quad (196)$$

Equivalently its diffusion part is $\sum_{j<q} dK_j^2 r_j^2 J\zeta_q \cdot B_j$. The sign is fixed by $\det(\zeta_q, \zeta_j) = -\Delta_{jq}$, not by a convention about unoriented angle sizes. No nonadjacent determinant has been assumed constant. For $q = 3$, $-\Delta_{13} = y$, $-\Delta_{23} = b$, and $w_1 = w_2 = 1$; (196) gives exactly (194). During activation the term $\dot{w}_j E_j (-\Delta_{jq})$ remains, so inviscid constancy is asserted only after those weights become constant.

9.3 Full transition from the actual third-growth endpoint

Choose the original source data and positive physical diffusivity by (166)–(167), and construct the actual third threshold in Theorem 3. Here are the actual endpoint bounds used in the finite proof. They are quantities at the already constructed threshold $t_a = t_3^a$, not assumptions about the subsequent unknown trajectory. The preceding third-growth calculation starts at the proved second return and gives

$$\begin{aligned}
1 \leq r_b \leq \sqrt{10}, \quad h(t_a) \in [\sqrt{1-a^2}, 3), \quad 1 \leq D(t_a) < 10, \quad \theta(t_a) = \Omega_2(t_a) = 0, \\
E_1(t_a)/\sigma_1^2 \in [1/4, 2], \quad 0 < \Omega_1(t_a)/\sigma_1 < 9, \quad A_2(t_a)/\sigma_2^2 \in [2/5, 2], \\
1 \leq R(t_a) \leq 2, \quad \frac{1}{2} \sin s \leq \zeta_{3,1}(t_a) \leq \sin s, \quad \zeta_{3,2}(t_a) \geq \cos s > 15/16, \\
\frac{2}{5} \leq \frac{N_3(t_a)}{\sigma_2^2 \sin s} \leq \frac{16}{15}, \quad \frac{1}{2} \leq \frac{\sigma_2 \Omega_3(t_a)}{K_3 \Theta_3(t_a)} \leq \sqrt{5}, \\
\Theta_3(t_a) = -\frac{A_0}{\mu} \hat{\lambda}_3^{-7/8}, \quad 0 < c/\sigma_1^2 \leq 1/4, \quad 0 < C/\sigma_1^2 \leq a/32.
\end{aligned} \tag{197}$$

In particular $E_2(t_a)/\sigma_2^2 \in [1/8, 2]$: the lower bound follows from $(2/5)/\sqrt{10} > 1/8$. Also $g(t_a) = \sqrt{D(t_a)}\zeta_{3,2}(t_a)$ lies in $[15/16, \sqrt{20}]$. These are the exact converted endpoint bounds needed below. Retain the explicit source restrictions

$$0 < a \leq 1/512, \quad 0 < s \leq a, \quad L_3 \geq 16384, \quad \delta_2 = \frac{dK_2^2}{\Gamma} \leq \frac{1}{128}, \quad \delta_3 = \frac{dK_3^2}{\Gamma} \leq \frac{1}{108}. \tag{198}$$

Every bound in (197) and (198) follows from that explicit construction, as we now verify. The second hold gives $E_1/\sigma_1^2 > 191/512 > 1/4$ and all its other displayed first-parent bounds. Its second-temperature diffusion exponent on the third growth is at most $\log 2$, while its length increases. Consequently

$$\frac{A_2(t_a)}{\sigma_2^2} = \frac{A_{2,b}}{\sigma_2^2} \frac{\sqrt{D(t_a)}}{r_b} \exp\left(-dK_2^2 \int_{t_b}^{t_a} D dt\right) \in [2/5, 2].$$

Here the lower bound uses $A_{2,b}/\sigma_2^2 \geq 4/5$, $\sqrt{D}/r_b \geq 1$ and $e^{-\log 2} = 1/2$; the upper bound uses $A_{2,b}/\sigma_2^2 \leq 1$ and $\sqrt{D}/r_b < 2$. Equations (177), (178) and (179) give the displayed third length, first component and numerator bounds. The positive second component has square $R - \zeta_{3,1}^2 \geq 1 - \sin^2 s = \cos^2 s$, so it is at least $\cos s$. Equation (181) supplies the actual signed pair and its quotient. The two base bounds are exactly (82) after the stated change of notation. Finally (166) gives $L_3 > 16384$ and $a < 1/12288 < 1/512$, and its proved angle satisfies $s \leq a$. The two positive entries in (167), with (171), give

$$\delta_2 \leq \frac{\log 2}{150L_3} < \frac{1}{128}, \quad \delta_3 \leq \frac{1}{120} < \frac{1}{108}.$$

Thus no assumption on an unconstructed transition is present.

Theorem 4. *For every original source datum, with frequencies and positive physical diffusivity chosen by (166)–(167), the actual third threshold constructed above is followed by the entire original third transition, which exists through $t_1 = t_a + \Gamma^{-1}$. Put $\tau = \Gamma(t - t_a)$, $u = \theta/s$, and $V = \Omega_2/\sigma_2$. Throughout $0 \leq \tau \leq 1$,*

$$0 \leq u \leq \frac{640a}{L_3}, \quad -\frac{7040a}{L_3} \leq V \leq 0. \tag{199}$$

Both inequalities have strict signs after the switching profile has become positive on an interval. The full older geometry obeys

$$\begin{aligned}
\sqrt{1-a^2} \leq h < 13/4, \quad 1 \leq D < 11, \quad 3/4 < g < 5, \quad x, y > 0, \\
1/8 < E_1/\sigma_1^2 \leq 2, \quad 0 < \Omega_1/\sigma_1 < 10, \quad 1/16 < E_2/\sigma_2^2 < 3, \\
\cos \theta > 15/16, \quad g \cos \theta - b \sin \theta > 2/3.
\end{aligned} \tag{200}$$

All source feedback denominators are therefore positive, and (189) gives finite original matrices and inverses. The third numerator and phase length satisfy

$$e^{-11\delta_2\tau} N_3(t_a) \leq N_3(t) \leq N_3(t_a), \quad \left| \log \frac{|\zeta_3(t)|}{|\zeta_3(t_a)|} \right| \leq \frac{46000a}{L_3} < \frac{1}{128}. \quad (201)$$

The actual newest amplitude remains negative and its quotient $v = \sigma_2\Omega_3/(K_3\Theta_3)$ satisfies $1/2 \leq v \leq 3$. Its full transition logarithmic gain is bounded by

$$\frac{1}{18} \leq \log \frac{-\Theta_3(t_1)}{-\Theta_3(t_a)} \leq 4. \quad (202)$$

The exact exponent, including physical diffusion, is given in the proof. This theorem completes the transition; it does not assert a third shooting root or an infinite fixed-viscosity cascade.

Proof. Set $L = L_3$, $\epsilon = \sigma_1/\Gamma = s/(L \sin s) \leq 2/L$, $q = \sigma_1/\sigma_2 = s/L$, $E = E_1/\sigma_1^2$, $F = E_2/\sigma_2^2$, $\omega = \Omega_1/\sigma_1$, $\bar{c} = c/\sigma_1^2$, $\bar{C} = C/\sigma_1^2$ and $\delta_1 = dK_1^2/\Gamma \leq \delta_2$. Primes now mean $d/d\tau$ exclusively. The exact dimensionless older equations, retaining each physical ratio, are

$$\begin{aligned} h' &= a\epsilon\omega, & g' &= \epsilon\omega \frac{2ahg + (h^2 - a^2)b}{D} + \frac{b}{\sin s} V, \\ E' &= -\epsilon\bar{C}\omega - \delta_1 E, & \omega' &= -\epsilon E \frac{h \sin(su) - a \cos(su)}{\sqrt{D}} - \delta_1 \omega, \\ F' &= -q\epsilon \frac{a(E + \bar{c}) + \bar{C}h}{D} V - \delta_2 DF, & V' &= -F\sqrt{D} \frac{\sin(su)}{\sin s} - \delta_2 DV. \end{aligned} \quad (203)$$

The exact u' is (193) divided by Γs , with (192); no approximation of either feedback occurs.

Initially impose the temporary closed bounds $0 \leq u \leq 1/1024$, $-1/32 \leq V \leq 0$, $\sqrt{1 - a^2} \leq h \leq 13/4$, $3/4 \leq g \leq 5$, $1/8 \leq E \leq 3$, $0 \leq \omega \leq 10$, and $1/16 \leq F \leq 3$. The sign boundaries are invariant: at $u = 0$, its vector field equals $\eta b \epsilon \omega a h / (s D g) \geq 0$; at $V = 0$ its vector field is nonpositive. Also $h \sin(su) < a \cos(su)$, since $s \leq a$ and $u \leq 1/1024$. The ω equation and its positive datum imply $\omega > 0$. These conclusions can equally be obtained from their integrating factors and a first-exit argument, so tangency of a sign boundary causes no loss of the sign assertion.

The temporary bounds give $D \geq 1$, because $h' \geq 0$ and $D(t_a) \geq 1$, and $D < 11$. They give $\cos(su) > 15/16$ and

$$g \cos(su) - b \sin(su) > \frac{3}{4} \frac{15}{16} - \frac{4a^2}{1024} > \frac{2}{3},$$

using $b \leq \sqrt{10} \sin s < 4s \leq 4a$. Thus every equation is smooth on a fixed neighborhood of this box. Because E decreases and $\omega' \leq \epsilon E a \leq 6a/L$,

$$E \leq 2, \quad 0 < \omega \leq 9 + 6a/L < 10, \quad h \leq 3 + 20a/L < 13/4.$$

Integrating the E row, using the temporary upper bounds, gives

$$E \geq \frac{1}{4} - \frac{20a}{32L} - \frac{3}{128} > \frac{1}{8}.$$

The numerator in the F equation obeys $0 < a(E + \bar{c}) + \bar{C}h < 4a$. Since $V \leq 0$, $F' \geq -11\delta_2 F$, hence $F \geq \frac{1}{8} e^{-11/128} > 1/16$. Its positive derivative contribution is at most $q\epsilon(4a)/32 \leq as/(4L^2)$, so $F < 3$. The older contribution to g' is positive, and

$$g \geq \frac{15}{16} - \frac{\sqrt{10}}{32} > \frac{3}{4}, \quad g \leq \sqrt{20} + \frac{300a}{L} < 5.$$

For the upper estimate use $[2ahg + (h^2 - a^2)b]/D \leq 2ag/h + b < 15a$ and $\epsilon\omega \leq 20/L$. Also $hg - ab > 0$ and $ag + hb > 0$, proving $x, y > 0$.

It remains to strictly improve the two revival bounds. On the box, the first term in (193) is nonpositive. In the factored feedback difference its first numerator has negative first factor; the other factor is positive. Its absolute forcing in u' is at most

$$\frac{2}{L} 4^{10} a \frac{7}{2} \frac{16}{15} \frac{3}{2} = \frac{448a}{L} < \frac{512a}{L}.$$

Here $|h \sin \theta - a \cos \theta| \leq a$, $h \cos \theta + a \sin \theta < 7/2$, $b/s < 4$, and $D \geq 1$. The second-parent feedback term is at most $6|V|u$ because $b/\sin s < 4$, $\sin(su)/s \leq u$, and the second denominator exceeds $2/3$. Therefore

$$u' \leq 512a/L + (3/16)u, \quad u(\tau) \leq (512a/L)e^{3/16} < 640a/L < 1/1024.$$

The elementary estimate $e^{3/16} < 5/4$ follows, for example, from $e^x \leq (1-x)^{-1}$ for $0 \leq x < 1$. For $Z = -V \geq 0$, the integrating factor in its equation gives

$$Z(\tau) \leq \int_0^\tau F \sqrt{D} \frac{\sin(su)}{\sin s} dr < 11 \int_0^\tau u dr \leq 7040a/L < 1/32,$$

since $3\sqrt{11}(16/15) < 11$ and $s/\sin s < 16/15$. Every boundary has now been strictly improved, or has its invariant sign. The original positive data and integrating factors keep the positive quantities away from zero. The vector field remains in a compact subset of its smooth domain; its integral equation and the local existence theorem therefore extend it through $\tau = 1$. This proves the full prescribed interval, not only a small initial piece. Once η is positive on an interval, the strictly positive forcing at $u = 0$ gives $u > 0$ thereafter, and the integral for Z then gives $V < 0$. No instantaneous derivative at a flat endpoint is used to assert this revival.

For the numerator, $E_1, E_2, x, y, c, C > 0$ and $D \geq 1$ show

$$0 \leq K_1^2 E_1 y + b K_2^2 D E_2 \leq K_2^2 D N_3.$$

Equation (194) and its integrating factor give the first part of (201). For the length, the exact polar transport equation gives

$$|(\log |\zeta_3|)'| \leq \epsilon \omega |\sin(\phi_3 - \phi_1)| + \frac{|V|}{\sin s} |\sin(\phi_3 - \phi_2)|.$$

The determinant coordinates give

$$|\sin(\phi_3 - \phi_1)| = \frac{ag + hb}{\sqrt{D}\sqrt{g^2 + b^2}} \leq a + b/g < 7a, \quad |\sin(\phi_3 - \phi_2)| = \frac{b}{\sqrt{g^2 + b^2}} < 6s.$$

The cosine factors in the exact polar equation have absolute value at most one; these inequalities thus retain its full contribution. Consequently

$$|(\log |\zeta_3|)'| \leq 140a/L + (32/5)|V| \leq 45196a/L < 46000a/L < 1/128.$$

The last numerical inequality uses $a \leq 1/512$, $L \geq 16384$. This proves the second part of (201), and in particular $R(t) < 2e^{1/64} < 3$.

For the newest linear pair define the exact scaled coefficients

$$\mathcal{K} = \frac{N_3}{\sigma_2^2 \sin s R}, \quad \mathcal{B} = \frac{\zeta_{3,1}}{\sin s}.$$

At entry $1/5 \leq \mathcal{K}(t_a) \leq 16/15$. The proved variations give

$$\frac{1}{6} < \frac{1}{5} e^{-13/128} \leq \mathcal{K} \leq \frac{16}{15} e^{1/64} < \frac{9}{8}.$$

For the lower strict inequality use $e^{-13/128} \geq 115/128 > 5/6$; for the upper use $e^{1/64} \leq 64/63$. Furthermore $(\log \sqrt{D})' = ah\epsilon\omega/D \leq 22a/L < 1/64$. Because the second phase is vertical at entry,

$$\frac{\zeta_{3,1}(t)}{\zeta_{3,1}(t_a)} = \frac{\sqrt{D(t_a)}}{\sqrt{D}} \left(\cos \theta + \frac{g}{b} \sin \theta \right).$$

Its lower bound is $(63/64)^2$, and its upper bound is $1 + 6u \leq 1 + 3840a/L < 9/8$, using $b \geq \sin s$. Thus

$$1/6 < \mathcal{K} < 9/8, \quad 15/32 < \mathcal{B} < 9/8. \quad (204)$$

Construct the Riccati equation $v' = \mathcal{B} - \mathcal{K}v^2$ with its actual entry value. Its field points strictly inward at $v = 1/2$ and $v = 3$, since $15/32 - (9/8)/4 = 3/16 > 0$ and $9/8 - 9/6 = -3/8 < 0$. It exists on the entire interval with $1/2 \leq v \leq 3$. Then define

$$P_3(t) = P_3(t_a) \exp\left(\int_0^\tau (\mathcal{K}v - \delta_3 R) dr\right), \quad \Theta_3 = -P_3, \quad \Omega_3 = -\frac{K_3}{\sigma_2} P_3 v. \quad (205)$$

Differentiating these expressions gives exactly $\dot{\Theta}_3 = N_3 \Omega_3 / (K_3 R) - dK_3^2 R \Theta_3$ and $\dot{\Omega}_3 = K_3 \zeta_{3,1} \Theta_3 - dK_3^2 R \Omega_3$. The inverse is thus the unique physical linear pair, by its integral equation; it has not assumed that its temperature avoids zero. Finally $\mathcal{K}v \geq 1/12$, $\delta_3 R < 3/108$, and $\mathcal{K}v < 27/8 < 4$ give $1/18 \leq \mathcal{K}v - \delta_3 R \leq 4$. Integrating over the full unit transition proves (202). $\square$

The exact first, second and third diffusion exponents during this transition are respectively δ_1 , $\delta_2 \int_0^1 D d\tau$ and $\delta_3 \int_0^1 R d\tau$. They equal $dK_j^2 \int_{t_a}^{t_1} |\zeta_j(t)|^2 dt$ in physical coordinates. The second phase and amplitude continue evolving inside these integrals. The end control is exactly $\dot{\alpha} = \mathcal{F}_3$, so the source next steering starts smoothly, with its actual values $Z_3(t_1) = \zeta_{3,1}(t_1)$ and $|\zeta_3(t_1)|$. The released periodic profile is not a Laplacian eigenfunction; realizing these damped trajectories with its compact spatial fields still requires the explicitly calculated profile and localization forces. None of the determinant or transition results discards those forces or asserts an unforced spatial superposition.

10 Compact spatial equations through the third threshold

The third growth starts at the actual second return $t_b = t_2^b$. The preceding coefficient construction proves its actual crossing time $t_{a3} := t_3^a > t_b$ in Theorem 3 of the original threshold. This section realizes that construction in the full equations, with the same physical diffusivity $d > 0$ in both equations and the declared pressure $p = 0$:

$$\begin{aligned} \theta_t + u \cdot \nabla \theta &= d\Delta \theta + f_\theta, \\ u_t + (u \cdot \nabla)u &= d\Delta u + \theta e_2 + f_u, \quad \operatorname{div} u = 0. \end{aligned}$$

Every field is defined on $\mathbb{R}^2$ in the original coordinate x . Retain

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad A_0 \geq 1, \quad \nu = \sqrt{A_0}, \quad \mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil, \quad K_j = \mu \hat{\lambda}_j \quad (1 \leq j \leq 3).$$

Here ν is the source time scale, not d . The hats distinguish the source frequencies from their physical multiples K_j . The released recursive frequency, source growth logarithm, inherited scales, angle, and activation are

$$\begin{aligned} \hat{\lambda}_3 &= \hat{\lambda}_2^{Q_3}, \quad L_3 = (k_3 + 5 + 1/8) \log \hat{\lambda}_3, \quad s_3 = L_3 \sigma_1 / \sigma_2, \quad \Gamma_3 = \sigma_2 \sin s_3, \\ S_3 &= (\nu^2 / \mu) \hat{\lambda}_3^{-k_3-6}, \quad P_3^* = S_3 e^{L_3} = (\nu^2 / \mu) \hat{\lambda}_3^{-7/8}, \quad w_3(t) = \eta(\Gamma_3(t - t_b)). \end{aligned} \quad (206)$$

The source step η is smooth, nondecreasing, zero on $(-\infty, 0]$, and one on $[1, \infty)$. The actual coefficient proof, rather than an inviscid comparison, supplies the following bounds through t_{a3} and existence through the larger horizon $t_b + 16/\sigma_1$:

$$\begin{aligned} h_0 \leq h < 3, \quad 0 < \Omega_1 < 9\sigma_1, \quad 1/4 < B \leq 2, \quad a = \sin s_2 \in (0, 1/8], \quad h_0 = \cos s_2 \geq \sqrt{63}/8, \\ \chi = (h^2 + a^2)^{1/2}, \quad r_1 = 1, \quad 1 \leq \chi^2 < 10, \quad 1 \leq r_3^2 \leq 2, \quad 0 < -\Theta_3 \leq P_3^*, \quad |\Omega_3| \leq 3K_3P_3^*/\sigma_2. \end{aligned} \quad (207)$$

The angle branch is the actual small source branch $0 < s_2, s_3 < 1/8$. The old amplitudes remain the actual ones:

$$\Theta_1 = -\sigma_1^2 B/K_1, \quad \Omega_2 = 0, \quad \Theta_2(t) = -P_{2b} \exp \left[-dK_2^2 \int_{t_b}^t \chi(r)^2 dr \right], \quad 0 < P_{2b} \leq e^{18} P_2^*.$$

These equations do not hold either old temperature fixed.

10.1 The exact third material flow and support inclusions

Put $\vartheta = \arctan(a/h)$ and write $h_b = h(t_b)$, $\vartheta_b = \vartheta(t_b)$. The actual held second feedback is

$$\begin{aligned} \alpha &= s_* + \vartheta, \quad \dot{h} = a\Omega_1, \quad \dot{\alpha} = -a^2\Omega_1/\chi^2, \\ \zeta_1 &= (-a, h)/\chi, \quad \zeta_2 = (0, \chi), \\ G_{<3} &= -A_0 R_\alpha e_2 + K_1 \Theta_1 \zeta_1 + K_2 \Theta_2 \zeta_2, \\ D_{<3} &= \dot{\alpha} J + \Omega_1 J \zeta_1 \otimes \zeta_1. \end{aligned} \quad (208)$$

The second velocity increment is zero, but its scalar increment contributes the entire last term in $G_{<3}$. The closed older amplitude equations on this interval are

$$\begin{aligned} \dot{\Theta}_1 &= \frac{A_0 \sin s_*}{K_1} \Omega_1 - dK_1^2 \Theta_1, \quad \dot{\Omega}_1 = -\frac{aK_1}{\chi} \Theta_1 - dK_1^2 \Omega_1, \\ \dot{\Theta}_2 &= -dK_2^2 \chi^2 \Theta_2, \quad \dot{\Omega}_2 = 0. \end{aligned} \quad (209)$$

Indeed $-K_1 \Theta_1 = A_1$ in (172) gives the first equation, and $\zeta_{1,1} = -a/\chi$ in (112) gives the second. The held identities $\Omega_2 = \zeta_{2,1} = 0$ give the remaining two equations. For $S(q) = \begin{pmatrix} 1 & -q \\ 0 & 1 \end{pmatrix}$, the matrix in the exact geometry calculation is

$$M_3(t) = R_\vartheta S((h - h_b)/a) R_{-\vartheta_b} = M_2(t) M_2(t_b)^{-1}, \quad \dot{M}_3 = D_{<3} M_3, \quad M_3(t_b) = I. \quad (210)$$

Its determinant is one. Its inverse is $R_{\vartheta_b} S(-(h - h_b)/a) R_{-\vartheta}$. Indeed the rotation matrices have inverse rotations and $S(q)^{-1} = S(-q)$, while differentiating S gives $\dot{S} = \Omega_1 (J e_2 \otimes e_2) S$. Since $\zeta_1 = R_\vartheta e_2$, multiplication proves the ODE in (210) directly, including its signs. Integrating $\dot{h} = a\Omega_1$ over $16/\sigma_1$ gives $0 \leq (h - h_b)/a \leq 144$. The elementary inequality $\|S(q)\| \leq 1 + |q|$, obtained from $S(q) = I - q e_1 \otimes e_2$, proves

$$\|M_3\|, \|M_3^{-1}\| \leq N_3 := 145. \quad (211)$$

Use the original unit insertion vector $p_3 = e(s_3)$ and put $\zeta_3 = M_3^{-T} p_3$. Differentiating this exact inverse proves $\dot{\zeta}_3 = -D_{<3}^T \zeta_3$. Thus the moving phase and moving material envelope below are transported by the same actual older affine velocity, without changing the insertion coordinates.

The released functions F, P, f are retained without alteration: $F(s) = \sin s + \chi_F(s)(s - \sin s)$ on $[-\pi, \pi]$ and periodically extended, with χ_F smooth and even, $F(s) = s$ on $|s| \leq s_0$, $0 < s_0 < 1$; $P' = F$, P is even and periodic, and $\int_{-\pi}^{\pi} P = 0$. The radial $f \in C_c^\infty(B_1)$ satisfies $0 \leq f \leq 1$ and $f = 1$ on $B_{1/2}$. In particular the constant $P(0)$ on the affine cell is preserved. Keep the source radii

$$\ell_1 = \frac{s_0}{C_\ell \mu}, \quad \ell_2 = \frac{\hat{\lambda}_1^{-3}}{\mu}, \quad \ell_3 = \frac{\hat{\lambda}_2^{-3}}{\mu}, \quad C_\ell \geq 4. \quad (212)$$

Let $N = 1 + (3 - h_0)/a$, so $\|M_2^{\pm 1}\| \leq N$. The actual source frequency selection in (166) proves the strict inequalities established in the [exact third support nesting calculation](#):

$$\begin{aligned} N\widehat{\lambda}_1^{-3} &< s_0/(2C_\ell), & N\widehat{\lambda}_1^{-2} &< s_0, \\ 2N_3N\widehat{\lambda}_2^{-3} &< \widehat{\lambda}_1^{-3}, & N_3\sqrt{10}\widehat{\lambda}_2^{-2} &< s_0. \end{aligned} \quad (213)$$

Here they have an exact spatial meaning. Put

$$y_3 = M_3^{-1}x, \quad g_3 = f(y_3/\ell_3), \quad k_3^{\text{ph}} = K_3\zeta_3, \quad s_3^{\text{ph}} = k_3^{\text{ph}} \cdot x.$$

The superscript distinguishes a phase vector and a phase variable from the source seed exponent k_3 and insertion angle s_3 . On the support of any third increment or derivative, $|x| \leq N_3\ell_3$. Consequently

$$|M_2^{-1}x| \leq NN_3\ell_3 < \ell_2/2, \quad |K_2\zeta_2 \cdot x| \leq K_2\sqrt{10}N_3\ell_3 < s_0.$$

Thus the second envelope is one and its profile is exactly affine. Moreover $x = M_2y_2$ with $|y_2| < \ell_2/2$, whence $|x| < N\ell_2/2$. The first two inequalities in (213) imply $|x| < \ell_1/2$ and $|K_1\zeta_1 \cdot x| < s_0$. This proves inclusion in the first affine cell as well. The base is affine there because $\mu\ell_1 = s_0/C_\ell < s_0$. All inequalities are strict, so the identities hold on an open neighborhood of the supports; differentiating a force introduces no missing boundary term. The actual older fields are therefore exactly $G_{<3} \cdot x$ and $D_{<3}x$ at every point needed below.

10.2 Third fields and the entire scalar and vector forces

Write $k = k_3^{\text{ph}}$, $\rho = |k|^2 = K_3^2 r_3^2$, $g = g_3$, $s = s_3^{\text{ph}}$, $D = D_{<3}$, and $G = G_{<3}$ in this subsection only. The actual third pair satisfies

$$\dot{\Theta}_3 = -\frac{Jk \cdot G}{\rho}\Omega_3 - d\rho\Theta_3, \quad \dot{\Omega}_3 = k_1\Theta_3 - d\rho\Omega_3. \quad (214)$$

Define, with all one-variable profiles evaluated at s ,

$$T = w_3\Theta_3, \quad Q = \frac{w_3\Omega_3}{\rho}, \quad V_3 = TFg, \quad \Psi_3 = QPg, \quad v_3 = J\nabla\Psi_3 = QJ(kFg + P\nabla g). \quad (215)$$

For $t \leq t_b$ these three increments are defined to be zero. The flat activation and the smooth finite ODE solution make this zero extension smooth in space and time. On $[0, t_{a3}]$ set

$$\theta = \theta_b + V_1 + V_2 + V_3, \quad u = u_b + v_1 + v_2 + v_3, \quad p = 0.$$

The base and first two increments are precisely those in (116)–(117), continued by their actual holding equations through the third growth. The initial data remain exactly $\theta(x, 0) = -(A_0/\mu)\sin(\mu x_2)\chi_0(x)$ and $u(x, 0) = 0$. Define

$$Q_* = \frac{\dot{w}_3\Omega_3 + w_3k_1\Theta_3}{\rho} - Q\frac{\dot{\rho}}{\rho}, \quad \dot{\rho} = -2k \cdot D^T k. \quad (216)$$

Then $\dot{Q} = Q_* - d\rho Q$, including the varying phase length. The complete third scalar and full vector force increments are

$$\begin{aligned} E_{\theta,3} &= \dot{w}_3\Theta_3Fg + QP(J\nabla g \cdot G) + QT(F^2 - PF')g(Jk \cdot \nabla g) \\ &\quad - dT\{\rho(F + F'')g + 2F'k \cdot \nabla g + F\Delta g\}, \end{aligned} \quad (217)$$

$$E_{u,3} = 2Dv_3 + Q_*J(kFg + P\nabla g) + (\nabla v_3)v_3 - TFge_2 - d(\rho v_3 + \Delta v_3). \quad (218)$$

In components, every derivative required in these formulas is

$$\nabla v_3 = QJ\{k \otimes kF'g + F(k \otimes \nabla g + \nabla g \otimes k) + P\nabla^2 g\}, \quad (219)$$

$$\Delta V_3 = T\{\rho F''g + 2F'k \cdot \nabla g + F\Delta g\}, \quad (220)$$

$$\begin{aligned} \Delta v_3 = QJ\{\rho k F''g + \rho F' \nabla g + 2F'k(k \cdot \nabla g) + 2F(\nabla^2 g)k \\ + Fk\Delta g + P\nabla\Delta g\}. \end{aligned} \quad (221)$$

The total forces are

$$f_\theta = f_{\theta,b} + \sum_{j=1}^3 E_{\theta,j}, \quad f_u = f_{u,b} + \sum_{j=1}^3 E_{u,j}, \quad (222)$$

with the exact base and old increments of (119)–(124) evaluated at the actual continued coefficients. There is no change of pressure hidden in the vector formula.

Proof of both equations and every interaction direction. For $\mathcal{D} = \partial_t + Dx \cdot \nabla$, (210) and its inverse give $\mathcal{D}(M_3^{-1}x) = 0$, and the phase equation gives $\mathcal{D}(k \cdot x) = 0$. Hence $\mathcal{D}g = 0$, $\mathcal{D}V_3 = \dot{T}Fg$, and $\mathcal{D}\Psi_3 = \dot{Q}Pg$. Subtracting the old scalar equation from the new one gives $\dot{T}Fg + v_3 \cdot G + v_3 \cdot \nabla V_3 - d\Delta V_3$. The two exact contractions are

$$v_3 \cdot G = QFg(Jk \cdot G) + QP(J\nabla g \cdot G), \quad v_3 \cdot \nabla V_3 = QT(F^2 - PF')g(Jk \cdot \nabla g).$$

Substitution of (214) cancels the first contraction against the coupling in $\dot{T}$ and proves (217).

Since $\text{tr } D = 0$, direct multiplication gives $DJ + JD^T = 0$. Therefore $\mathcal{D}(J\nabla\Psi_3) = Dv_3 + J\nabla(\mathcal{D}\Psi_3)$. The reverse interaction is $(v_3 \cdot \nabla)(Dx) = Dv_3$. Together with self-advection, buoyancy, and diffusion, this proves (218) after substituting (216). The ordinary product rule proves (219)–(221); in particular the last display contains all phase, envelope, and mixed terms.

The operator $\mathcal{D}$ uses the full old velocity $u_b + v_1 + v_2$, which equals Dx on the third support. It accounts for $(u_b + v_1 + v_2) \cdot \nabla V_3$ and $(u_b + v_1 + v_2) \cdot \nabla v_3$. The contractions with G and D account for $v_3 \cdot \nabla(\theta_b + V_1 + V_2)$ and $v_3 \cdot \nabla(u_b + v_1 + v_2)$. Although $v_2 = 0$, $v_3 \cdot \nabla V_2$ generally is not zero; it is exactly the $K_2\Theta_2\zeta_2$ part of $v_3 \cdot G$. The earlier increments already account for interactions among the base and first two layers. Outside a third support all its derivatives vanish. Adding these identities therefore proves (222) globally, with every ordered pair counted once. Finally each velocity is a rotated gradient, so mixed partial symmetry proves $\text{div } u = 0$. $\square$

For the exact coordinate correspondence put $C_3 = M_3^{-1}M_3^{-T}$ and $\tilde{g}(y) = f(y/\ell_3)$. The spatial gradient and Laplacian pulled back to (s, y) are

$$\begin{aligned} \mathcal{P}_3 &= K_3\zeta_3\partial_s + M_3^{-T}\nabla_y, \\ \mathcal{L}_3 &= K_3^2r_3^2\partial_s^2 + 2K_3(C_3p_3 \cdot \nabla_y)\partial_s + \sum_{l,m} (C_3)_{lm}\partial_{y_l}\partial_{y_m}. \end{aligned} \quad (223)$$

The chain rule gives the first identity, and multiplying its commuting spatial derivatives gives the second. In particular $k \cdot \nabla_x g = K_3(C_3p_3) \cdot \nabla_y \tilde{g}$ and $\Delta_x g = \text{tr}(C_3\nabla_y^2 \tilde{g})$. This proves the original-coordinate meaning of all mixed terms, without imposing a scalar metric or a fixed cutoff.

10.3 All spatial derivative norms and diffusion costs

For any scalar, vector, or tensor U , set $|U|_m = \sup_x \|D_x^m U(x)\|$ in the Euclidean multilinear operator norm, $m \geq 0$. For one-variable W set $[W]_m = \sup_s |W^{(m)}(s)|$, and set $f_m = \sup_y \|D^m f(y)\|$. Define only explicit, finite constants

$$\begin{aligned} \kappa_3^* &= \sqrt{2}K_3, & e_{3,m} &= (N_3/\ell_3)^m f_m, \\ \mathcal{B}_{3,m}(W) &= \sum_{q=0}^m \binom{m}{q} (\kappa_3^*)^q [W]_q e_{3,m-q}, & \mathcal{C}_{3,m}(P) &= \sum_{q=0}^m \binom{m}{q} (\kappa_3^*)^q [P]_q e_{3,m-q+1}. \end{aligned} \quad (224)$$

The symbol κ_3^* is a frequency bound, not a diffusivity. Repeated product and chain rules prove

$$|g_3|_m \leq e_{3,m}, \quad |W(s_3^{\text{ph}})g_3|_m \leq \mathcal{B}_{3,m}(W), \quad |P(s_3^{\text{ph}})\nabla g_3|_m \leq \mathcal{C}_{3,m}(P).$$

Indeed choose the q of m directional derivatives assigned to the profile, producing $\binom{m}{q}$ terms, each bounded by the corresponding summand. The derivative of a linear phase is k , and that of $M_3^{-1}x/\ell_3$ is M_3^{-1}/ℓ_3 ; their higher spatial derivatives vanish. This proves the stated operator norm bounds for arbitrary directions.

The actual coefficient bounds give

$$\begin{aligned} U_\Theta &= P_3^*, & U_\Omega &= 3K_3P_3^*/\sigma_2, & W_1 &= \Gamma_3[\eta']_0, \\ D^* &= 9\sigma_1(1+a^2), & G^* &= A_0 + 2\sigma_1^2 + \sqrt{10}K_2e^{18}P_2^*, \\ Q^* &= \frac{U_\Omega}{K_3^2}, & Q_*^* &= \frac{W_1U_\Omega + \kappa_3^*U_\Theta + 2D^*U_\Omega}{K_3^2}. \end{aligned} \quad (225)$$

In fact $\|D\| \leq |\dot{\alpha}| + \Omega_1$ and $|\dot{\alpha}| = a^2\Omega_1/\chi^2 \leq 9a^2\sigma_1$, which prove the bound for D . The three summands in G give its bound. The denominator $\rho \geq K_3^2$ and $|\dot{\rho}/\rho| \leq 2D^*$ prove the bounds for Q, Q_* . In particular activation contributes its exact physical factor $\Gamma_3[\eta']_0$ to both forces.

The full damping-plus-diffusion costs are bounded by

$$\begin{aligned} \mathcal{D}_{\theta,3,m} &= dU_\Theta\{(\kappa_3^*)^2\mathcal{B}_{3,m}(F) + 2\mathcal{B}_{3,m+2}(F)\}, \\ \mathcal{D}_{u,3,m} &= dQ^*\{(\kappa_3^*)^2\mathcal{B}_{3,m+1}(P) + 2\mathcal{B}_{3,m+3}(P)\}. \end{aligned} \quad (226)$$

These include $-d\rho V_3$ and $-d\rho v_3$ from modal damping as well as the Laplacians of the changed fields. The factor two bounds the two coordinate traces defining Δ in dimension two. Keeping this factor explicitly avoids identifying a Laplacian with a single second derivative. The complete third force estimates are

$$\begin{aligned} |E_{\theta,3}|_m &\leq W_1U_\Theta\mathcal{B}_{3,m}(F) + Q^*G^*\mathcal{C}_{3,m}(P) + \mathcal{D}_{\theta,3,m} \\ &\quad + Q^*U_\Theta \sum_{q=0}^m \binom{m}{q} \mathcal{B}_{3,q+1}(P)\mathcal{B}_{3,m-q+1}(F), \end{aligned} \quad (227)$$

$$\begin{aligned} |E_{u,3}|_m &\leq (2D^*Q^* + Q_*^*)\mathcal{B}_{3,m+1}(P) + U_\Theta\mathcal{B}_{3,m}(F) + \mathcal{D}_{u,3,m} \\ &\quad + (Q^*)^2 \sum_{q=0}^m \binom{m}{q} \mathcal{B}_{3,q+1}(P)\mathcal{B}_{3,m-q+2}(P). \end{aligned} \quad (228)$$

The nonlinear scalar term is bounded as $v_3 \cdot \nabla V_3$, and the nonlinear vector term as $(\nabla v_3)v_3$. The product rule with the displayed allocation of derivatives proves the two sums. All other terms follow individually from (217)–(221) and (224); hence the estimates include every activation, parent-interaction, self-interaction, buoyancy and diffusion term in the full vector equation. For time integrals one has, in particular,

$$\int_{t_b}^{t_{a3}} |\text{third damping-plus-diffusion force}|_m dt \leq \frac{16}{\sigma_1} \mathcal{D}_{\bullet,3,m}, \quad (229)$$

with the matching scalar or vector bound. The identical physical length factor integrates either full-force estimate.

All frequency and localization dependences are retained:

$$\frac{N_3}{\ell_3} = 145\mu\hat{\lambda}_2^3, \quad \kappa_3^* = \sqrt{2}\mu\hat{\lambda}_3, \quad dU_\Theta K_3^2 = d\nu^2\mu\hat{\lambda}_3^{9/8}, \quad dQ^* K_3^3 = \frac{3d\nu^2\mu}{\sigma_2}\hat{\lambda}_3^{9/8}. \quad (230)$$

The last two equalities identify coefficients occurring in the costs, not lower bounds for the entire forces. No cancellation of unrelated vector terms has been assumed. The finite positive- d range from the actual coefficient theorem is not replaced here by a bound uniform over an infinite frequency sequence.

For completeness, the bounds for the old and base forces remain fully quantitative on the slab ending at the third threshold. The formulas (128)–(133) hold with the original $N_1 = 1, N_2 = N$,

radii, and global amplitude bounds (129); on the added interval the sharper $|\Theta_1| \leq 2\sigma_1^2/K_1$, $|\Omega_1| \leq 9\sigma_1$, $|\Theta_2| \leq e^{18}P_2^*$, $\Omega_2 = 0$ prove their continued validity. If $A_1^{\text{old}}, A_2^{\text{old}}$ are the explicit earlier control derivative bounds, one may use the following bounds on this slab:

$$\begin{aligned} A_1^{\text{ext}} &= \max\{A_1^{\text{old}}, 9a^2\sigma_1\}, \\ A_2^{\text{ext}} &= \max\{A_2^{\text{old}}, a^2(4\sigma_1^2 + 9dK_1^2\sigma_1) + 486a^3\sigma_1^2\}. \end{aligned} \quad (231)$$

To verify the second number, the holding equations give $\dot{\Omega}_1 = \sigma_1^2 aB/\chi - dK_1^2\Omega_1$, so $|\dot{\Omega}_1| \leq 4\sigma_1^2 + 9dK_1^2\sigma_1$. Differentiating $\dot{\alpha} = -a^2\Omega_1/\chi^2$ and using $\dot{\chi}^2 = 2ha\Omega_1$, $h < 3$, and $\chi^2 \geq 1$ gives the stated additional term $486a^3\sigma_1^2$. Substitution into the explicit base norms (134) and old force norms completes the estimates for both total forces in (222), at every spatial order. Their integrals use this slab's actual interval length t_{a3} .

10.4 Mixed derivatives, affine core, and the threshold state

No stationary-envelope convention is used to estimate time derivatives. Set $A(t) = M_3(t)^{-1}/\ell_3$, introduce independent variables (t, x, s, y) , and define

$$\mathfrak{T} = \partial_t + (\dot{k} \cdot x)\partial_s + (\dot{A}x) \cdot \nabla_y, \quad \mathfrak{X}_i = \partial_{x_i} + k_i\partial_s + \sum_l A_{li}\partial_{y_l}. \quad (232)$$

For $\iota^*R = R(t, x, k(t) \cdot x, A(t)x)$, the chain rule gives

$$\partial_t^b \partial_x^\gamma \iota^*R = \iota^*(\mathfrak{T}^b \mathfrak{X}_1^{\gamma_1} \mathfrak{X}_2^{\gamma_2} R). \quad (233)$$

Induction on $b + |\gamma|$ proves this equality. The commutators vanish: $\dot{k}_i$ cancels the x_i derivative of $\dot{k} \cdot x$, and $\dot{A}_{li}$ cancels the corresponding derivative of $\dot{A}x$. This also proves consistency of the order of mixed derivatives. Apply these operators to the exact third fields and forces, starting for example with $Q(t)P(s)f(y)$ for the streamfunction. Every resulting term is a finite product of coefficient jets, explicit powers x^β , profile derivatives, and envelope derivatives. Its absolute value is bounded by replacing those factors by their coefficient suprema on $[t_b, t_{a3}]$, $(N_3\ell_3)^{|\beta|}$, $[P]_q$, $[F]_q$, and f_m , then summing all terms produced by (233). The coefficient jets themselves are evaluated recursively by (208), (209), (210), (214), and the explicit activation. Product and quotient rules, in increasing time order, use only the positive denominators a, K_3, χ^2, ρ and the fixed source scales. Their lower bounds in (207) therefore yield a finite explicit bound at every prescribed mixed order. The same recursion from (136)–(137) handles the base and old layers with the extended control bounds. This procedure includes every time derivative of the diffusion terms and of the changing material metric.

On a third affine cell, where $g_3 = 1$ and $|s_3^{\text{ph}}| < s_0$, the retained primitive is exactly $P(s) = P(0) + s^2/2$. Thus $\Psi_3 = Q(P(0) + s^2/2)$ there; taking its gradient, with the original constant still present in the full cutoff formula, gives

$$V_3 = TK_3\zeta_3 \cdot x, \quad v_3 = \frac{w_3\Omega_3}{r_3^2}J\zeta_3 \otimes \zeta_3 x, \quad \Delta V_3 = \Delta v_3 = 0. \quad (234)$$

The damping parts of the forces are still $-dK_3^2r_3^2V_3$ and $-dK_3^2r_3^2v_3$ there. On an uncut sine-profile region instead, $F'' = -F$ makes these damping-plus-diffusion contributions zero exactly. The remaining vector terms in (218) are retained at $p = 0$. This proves precisely how the actual damped pair maps into the released profile; it is not an unforced eigenfunction assertion.

At the actual third threshold,

$$\Theta_3(t_{a3}) = -P_3^*, \quad \Omega_3(t_{a3}) < 0,$$

with the exact phase $M_3(t_{a3})^{-T}e(s_3)$, the exact moving cutoff, and $w_3(t_{a3}) = 1$. The last equality follows from

$$\Gamma_3(t_{a3} - t_b) > L_3/3 > 1,$$

proved in Theorem 3. In the common affine cell the complete fields are

$$\theta = (G_{<3} + w_3 K_3 \Theta_3 \zeta_3) \cdot x, \quad u = \left(D_{<3} + \frac{w_3 \Omega_3}{r_3^2} J \zeta_3 \otimes \zeta_3 \right) x. \quad (235)$$

These are actual states of the full PDE just proved. The third vorticity has not returned to zero at this threshold; no third steering or return statement follows from the growth calculation.

10.5 Smooth finite-slab realization and honest force continuation

Let R_{base} denote the original spatial support radius called R_H in (116), as distinct from the temporal horizon $16/\sigma_1$ of the third-growth construction. The fields and forces are smooth and spatially supported in the same $B_{R_{\text{base}}}$ as the base. All third derivatives are supported within $|x| \leq N_3 \ell_3$; the proved inclusions put this ball inside the old base support. The velocity has finite energy since it is smooth and compact. For $N_0(x) = (2\pi)^{-1} \log|x|$, the compact smooth streamfunction $\psi = \psi_b + \Psi_1 + \Psi_2 + \Psi_3$ satisfies $N_0 * \Delta\psi = (\Delta N_0) * \psi = \psi$ by distributional integration by parts. Applying $J\nabla$ gives exactly $u = J\nabla N_0 * \omega$, $\omega = \Delta\psi$. Odd F , even P , and even material envelopes give odd θ, u, f_θ, f_u and even ω .

There is a smooth right collar at the threshold. The actual coefficient theorem continues the same F_2 growth equations through $t_b + 16/\sigma_1$, with t_{a3} strictly before that horizon. By (183) its remaining interval exceeds $2/\sigma_1$. Choose the explicit positive collar length $\epsilon_R = 1/\sigma_1$. Continue the exact coefficient solution and fields to $t_{a3} + \epsilon_R$; do not freeze their nonstationary endpoint. The upper rate in (182) holds on that entire horizon, as does the quotient bound (181). Consequently on the collar one has the explicit bounds

$$-\Theta_3 \leq e^{3\Gamma_3/\sigma_1} P_3^*, \quad |\Omega_3| \leq \frac{3K_3}{\sigma_2} e^{3\Gamma_3/\sigma_1} P_3^*. \quad (236)$$

Indeed integrate $(\log(-\Theta_3))' < 3\Gamma_3$ from the threshold to each collar time and then use the quotient bound. All spatial collar costs therefore follow from (225)–(228) by replacing U_Θ, U_Ω by the right sides of (236); the same coefficient-jet recursion proves the mixed costs. This does not impose the pre-threshold bound $-\Theta_3 \leq P_3^*$ after the crossing. The strict support inequalities continue on the proved horizon. Extend the original initial state constantly to the left of zero, with zero angle and inactive increments as before, and put $\epsilon_L = 1/\nu$. Multiplying the extended total residual forces by

$$\beta(t) = \eta(1 + t/\epsilon_L)\eta(1 + (t_{a3} - t)/\epsilon_R) \quad (237)$$

leaves them exactly unchanged on $[0, t_{a3}]$ and makes them smooth and compact in time as well as space. Flatness at the two outer endpoints proves smoothness of their zero extensions. Every additional mixed derivative follows from the ordinary product rule, with the explicit factors $\epsilon_L^{-b}[\eta^{(b)}]_0$ and $\epsilon_R^{-b}[\eta^{(b)}]_0$. The spatial fields solve the prescribed equations on the required slab $[0, t_{a3}]$; no equation outside that slab is asserted after the force cutoff. This proves the complete compact realization through the actual third threshold and retains the precise finite scope of the result.

11 Full compact spatial realization through the third transition

We continue the actual third-growth fields through the entire third transition just proved. The second vorticity now revives with negative sign. Both older velocities and temperatures enter the spatial equations. The construction below uses the actual coefficient solution and proves its complete spatial realization, including a smooth finite extension of its forces. Its endpoint is the third transition endpoint; no third return is asserted here.

11.1 The original objects and a uniform material bound

The physical equations, coordinates, and orientation remain

$$\begin{aligned} \theta_t + u \cdot \nabla \theta &= d\Delta \theta + f_\theta, & x &\in \mathbb{R}^2, \\ u_t + (u \cdot \nabla)u &= d\Delta u + \theta e_2 + f_u, & \operatorname{div} u &= 0, \quad p = 0, \\ J &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, & e(s) &= (\sin s, \cos s). \end{aligned} \quad (238)$$

Here $d > 0$ is the common physical viscosity and thermal diffusivity; $A_0 \geq 1$, $\nu = \sqrt{A_0}$, and $\mu = \lambda_0 = \lceil \nu/2 \rceil$ retain their original values. In particular ν is a time scale. Retain $K_j = \mu \hat{\lambda}_j$, the exact source integer recursion, the source step η , and the actual scales σ_1, σ_2 . Put

$$t_b = t_2^b, \quad t_a = t_3^a, \quad s = s_3 = L_3 \sigma_1 / \sigma_2, \quad \Gamma = \Gamma_3 = \sigma_2 \sin s, \quad t_1 = t_a + \Gamma^{-1}, \quad I = [t_b, t_1]. \quad (239)$$

These are the actual second return, third threshold, and third transition endpoint already constructed. Their proved bounds include $0 < t_a - t_b < 14/\sigma_1$, $L_3 \geq 16384$, $0 < s \leq a := \sin s_2 \leq 1/512$, and $\Gamma \geq (15/16)L_3 \sigma_1$. Thus

$$|I| = (t_a - t_b) + \Gamma^{-1} < 15/\sigma_1. \quad (240)$$

All earlier activation weights are one on I ; the third is $w_3(t) = \eta(\Gamma(t - t_b))$ and is one before t_a . The exact source control on I is $\dot{\alpha} = (1 - \lambda)\mathcal{F}_2 + \lambda\mathcal{F}_3$, where $\lambda(t) = \eta(\Gamma(t - t_a))$ is zero during growth. The two switches w_3, λ have different original start times.

Use the determinant coordinates of the third-transition theorem:

$$\begin{aligned} h_0 &= \cos s_2, \quad D = h^2 + a^2, \quad r_b = |\zeta_2(t_b)|, \quad b = r_b \sin s, \quad R = |\zeta_3|^2 = (g^2 + b^2)/D, \\ \zeta_1 &= \frac{(h \sin \vartheta - a \cos \vartheta, h \cos \vartheta + a \sin \vartheta)}{\sqrt{D}}, \\ \zeta_2 &= \sqrt{D}(\sin \vartheta, \cos \vartheta), \\ \zeta_3 &= \frac{(g \sin \vartheta + b \cos \vartheta, g \cos \vartheta - b \sin \vartheta)}{\sqrt{D}}, \quad \vartheta = su. \end{aligned} \quad (241)$$

The symbol D in this display is a scalar, not a matrix. The actual growth and transition estimates give, on all of I ,

$$\begin{aligned} h_0 &\leq h < 13/4, \quad 1 \leq D < 11, \quad 3/4 < g < 5, \quad e^{-1/64} \leq R < 3, \\ 0 &\leq u \leq 640a/L_3 < 1/1024, \quad \cos \vartheta > 15/16, \quad g \cos \vartheta - b \sin \vartheta > 2/3, \\ 0 &< \Omega_1 < 10\sigma_1, \quad -7040a\sigma_2/L_3 \leq \Omega_2 \leq 0, \quad 0 < -\Theta_1 \leq 2\sigma_1^2/K_1, \\ 0 &< -\Theta_2 \leq 3\sigma_2^2/K_2, \quad 0 < -\Theta_3 \leq e^4 P_3^*, \quad |\Omega_3| \leq 3e^4 K_3 P_3^*/\sigma_2, \\ P_3^* &= (A_0/\mu)\hat{\lambda}_3^{-7/8}, \quad 1 \leq r_b \leq \sqrt{10}. \end{aligned} \quad (242)$$

For the bound on R , growth has $1 \leq R \leq 2$, and the transition changes $\log |\zeta_3|$ by less than $1/128$. The amplitude bounds follow from the actual growth target, the transition gain at most four, and the proved quotient at most three. They are consequences of the constructed trajectory, not additional conditions imposed on an unknown flow.

For completeness the exact material equations for all three layers are

$$\begin{aligned} G_{<j} &= -A_0 R_\alpha e_2 + \sum_{i < j} w_i K_i \Theta_i \zeta_i, & D_{<j} &= \dot{\alpha} J + \sum_{i < j} \frac{w_i \Omega_i}{r_i^2} J \zeta_i \otimes \zeta_i, \\ \dot{M}_j &= D_{<j} M_j, & M_j(t_{0j}) &= I, \quad \zeta_j = M_j^{-T} p_j, \quad r_j = |\zeta_j|, \\ \dot{\Theta}_j &= -\frac{J \zeta_j \cdot G_{<j}}{K_j r_j^2} \Omega_j - d K_j^2 r_j^2 \Theta_j, & \dot{\Omega}_j &= K_j \zeta_{j,1} \Theta_j - d K_j^2 r_j^2 \Omega_j. \end{aligned} \quad (243)$$

Here $p_1 = e(s_*)$, $p_2 = e(s_2)$, $p_3 = e(s)$, and $t_{03} = t_b$; the first two original insertion times are unchanged. The scalar r_j^2 is strictly positive. Each matrix $D_{<j}$ has trace zero, since $\zeta_i \cdot J\zeta_i = 0$. The exact matrices are

$$\begin{aligned} M_1 &= R_\alpha, & M_2 &= R_{\alpha-s_*} \begin{pmatrix} 1 & -(h-h_0)/a \\ 0 & 1 \end{pmatrix}, \\ \mathcal{L}(t) &= (\zeta_2(t) \ \zeta_3(t)), & M_3 &= \mathcal{L}(t)^{-T} \mathcal{L}(t_b)^T, & M_3^{-1} &= \mathcal{L}(t_b)^{-T} \mathcal{L}(t)^T. \end{aligned} \quad (244)$$

Indeed ζ_2 is annihilated by the transpose of its own added shear. Thus both columns of $\mathcal{L}$ solve $\dot{z} = -D_{<3}^T z$. Their determinant is $\det \mathcal{L} = -b$, including at birth. Differentiating the last two expressions proves $\dot{M}_3 = D_{<3} M_3$, $M_3(t_b) = I$, and $M_3^{-T} p_3 = \zeta_3$. The same calculation, or the displayed triangular form, proves the first two material equations and their original data. All matrices have determinant one, and hence define global linear diffeomorphisms of the original $\mathbb{R}^2$.

Let $\|\cdot\|$ be Euclidean operator norm and $\|\cdot\|_F$ Frobenius norm. For a two by two matrix its adjugate has the same Frobenius norm: it permutes its four entries and changes two signs. Consequently

$$\|\mathcal{L}^{-1}\| \leq \|\mathcal{L}\|_F/b, \quad \|\mathcal{L}\|_F^2 = D + R < 14, \quad \|\mathcal{L}(t_b)\|_F^2 = r_b^2 + 1 \leq 11.$$

Applying these inequalities to both products in (244) gives the uniform bounds

$$\begin{aligned} N_1 &= 1, & N &= 1 + (3 - h_0)/a, & N_2 &= \tilde{N} = 1 + (13/4 - h_0)/a \leq 2N, \\ N_3 &= \mathcal{M} = 16/\sin s, & \|M_j\|, \|M_j^{-1}\| &\leq N_j \quad (1 \leq j \leq 3). \end{aligned} \quad (245)$$

For $j = 3$ the product bound is $\sqrt{154}/b < 16/\sin s$. For $j = 2$, write its triangular factor as $I - qe_1 \otimes e_2$ and its inverse as $I + qe_1 \otimes e_2$, so both norms are at most $1 + |q|$. Finally $2N - \tilde{N} = 1 + (11/4 - h_0)/a > 0$. The third estimate is valid when Ω_2 is nonzero; it does not use the shorter growth-only shear formula.

11.2 The exact source supports remain nested

The birth bound $P_{2,b} \leq e^{18}(A_0/\mu)\hat{\lambda}_2^{-7/8}$ gives $A_{2,b} = K_2 r_b P_{2,b} \leq \sqrt{10}e^{18}\nu^2\hat{\lambda}_2^{1/8}$. The already proved scale relation $\sigma_2^2 \leq 5A_{2,b}/4$ therefore gives

$$\begin{aligned} C_2 &= e^9 \sqrt{5\sqrt{10}/4}, & \sigma_2 &\leq C_2 \nu \hat{\lambda}_2^{1/16}, \\ \sigma_1 &\geq \sqrt{e} \nu \hat{\lambda}_1^{1/16}, & C_M &= \frac{256C_2}{15\sqrt{e}}, & \mathcal{M} &\leq C_M \hat{\lambda}_1^{(Q_2-1)/16}. \end{aligned} \quad (246)$$

For the last estimate use $\sin s \geq 15L_3\sigma_1/(16\sigma_2)$, $L_3 \geq 1$, and the unchanged recursion $\hat{\lambda}_2 = \hat{\lambda}_1^{Q_2}$. With $N \leq C_N \hat{\lambda}_1^{1/16}$, the strengthened actual source selection includes the strict inequalities

$$\hat{\lambda}_1^{47Q_2/16-3} > 4C_N C_M, \quad \hat{\lambda}_1^{(31Q_2+1)/16} > \sqrt{11}C_M/s_0. \quad (247)$$

Their exponents are positive because $Q_2 \geq 202$. Multiplication by the corresponding negative frequency powers gives

$$\begin{aligned} 2N_2 N_3 \hat{\lambda}_2^{-3} &< \hat{\lambda}_1^{-3}, & \sqrt{11}N_3 \hat{\lambda}_2^{-2} &< s_0, \\ N_2 \hat{\lambda}_1^{-3} &< s_0/(2C_\ell), & N_2 \hat{\lambda}_1^{-2} &< s_0. \end{aligned} \quad (248)$$

For the first, the exponent of $\hat{\lambda}_1$ in $4C_N C_M \hat{\lambda}_1^{Q_2/16-3Q_2+3}$ is $-(47Q_2/16 - 3)$. For the second its exponent is $(Q_2 - 1)/16 - 2Q_2 = -(31Q_2 + 1)/16$. The last two follow from the earlier actual first-source margins $N\hat{\lambda}_1^{-3} < s_0/(4C_\ell)$ and $N\hat{\lambda}_1^{-2} < s_0/2$, together with $N_2 \leq 2N$. Thus the doubling needed in the third transition is absorbed by proved strict margins.

Retain the original profile, primitive, and spatial cutoffs. Namely $F(s) = \sin s + \chi_F(s)(s - \sin s)$ on $[-\pi, \pi]$ is extended periodically; χ_F is even, smooth, supported away from the endpoints, and

equals one on $[-s_0, s_0]$, $0 < s_0 < 1$. The even periodic primitive is uniquely specified by $P' = F$ and $\int_{-\pi}^{\pi} P = 0$. On the affine cell it is $P(s) = P(0) + s^2/2$, with its original constant $P(0)$ retained. The radial $f \in C_c^\infty(B_1)$ equals one on $B_{1/2}$. Keep the exact source radii and material envelopes

$$\begin{aligned} \ell_1 &= s_0/(C_\ell \mu), \quad \ell_2 = \widehat{\lambda}_1^{-3}/\mu, \quad \ell_3 = \widehat{\lambda}_2^{-3}/\mu, \quad C_\ell \geq 4, \\ g_j(x, t) &= f(M_j^{-1}x/\ell_j), \quad k_j = K_j \zeta_j, \quad \xi_j = k_j \cdot x, \quad \rho_j = |k_j|^2. \end{aligned} \quad (249)$$

The vectors k_j here are physical phases, not the source integer seed exponents. Every third field and derivative is supported in $|x| \leq N_3 \ell_3$. At such points

$$|M_2^{-1}x| \leq N_2 N_3 \ell_3 < \ell_2/2, \quad |K_2 \zeta_2 \cdot x| \leq K_2 \sqrt{11} N_3 \ell_3 < s_0.$$

Thus the full second field is affine on an open neighborhood of the third support. Every point of the second support satisfies $|x| \leq N_2 \ell_2 < \ell_1/2$ and $|K_1 \zeta_1 \cdot x| < s_0$ by the last two inequalities in (248); M_1 is a rotation. The first support lies in B_{ℓ_1} , where the original radial base cutoff is one and $|\mu R_\alpha e_2 \cdot x| < s_0$. This proves every inclusion in the original laboratory coordinates. Strictness ensures an open affine neighborhood for every support and every derivative support, not only an identity at its center.

11.3 All fields, original initial data, and full forces

Retain the original radial $\chi_0 \in C_c^\infty(B_{R_0})$, equal to one on B_2 , and the radial $H_0 \in C_c^\infty(B_{R_H})$, $R_H > R_0 + 1$, with $H_0 = |x|^2/2$ on B_{R_0+1} . Define

$$\begin{aligned} \theta_{\text{in}} &= -\frac{A_0}{\mu} \sin(\mu x_2) \chi_0, & \theta_{\text{ref}} &= -\frac{A_0}{\mu} F(\mu x_2) \chi_0, \\ \theta_b &= -\frac{A_0}{\mu} F(\mu R_\alpha e_2 \cdot x) \chi_0 + (1 - \eta(\nu t))(\theta_{\text{in}} - \theta_{\text{ref}}), & \psi_b &= \dot{\alpha} H_0, \quad u_b = J \nabla \psi_b. \end{aligned} \quad (250)$$

The angle is zero before $1/\nu$. Thus this is exactly the original sine initial datum and zero initial velocity. The interpolation has finished before the first insertion and hence before I . For $j = 1, 2, 3$ define the active increments

$$\begin{aligned} T_j &= w_j \Theta_j, \quad Q_j = w_j \Omega_j / \rho_j, \quad V_j = T_j F(\xi_j) g_j, \quad \Psi_j = Q_j P(\xi_j) g_j, \quad v_j = J \nabla \Psi_j, \\ \theta &= \theta_b + \sum_{j=1}^3 V_j, \quad \psi = \psi_b + \sum_{j=1}^3 \Psi_j, \quad u = J \nabla \psi. \end{aligned} \quad (251)$$

For $t \leq t_{0j}$ the j th increment is zero. Flatness of the original η makes this zero extension smooth. Before t_b these formulas give the earlier constructed spatial solution exactly. On I they use the actual revived Ω_2 and its corresponding streamfunction; none of the older fields is frozen.

For each layer suppress its index temporarily and set $k = k_j$, $\rho = \rho_j$, $\xi = k \cdot x$, $g = g_j$, $D_* = D_{<j}$, $G_* = G_{<j}$. All profiles in the next formulas are evaluated at ξ . Set

$$Q_r = \frac{\dot{w} \Omega + w k_1 \Theta}{\rho} - Q \frac{\dot{\rho}}{\rho}, \quad \dot{\rho} = -2k \cdot D_*^T k, \quad \dot{Q} = Q_r - d\rho Q. \quad (252)$$

The exact force increments, with no pressure reconstruction, are

$$\begin{aligned} E_{\theta,j} &= \dot{w} \Theta F g + Q P(J \nabla g \cdot G_*) + Q T(F^2 - P F') g (J k \cdot \nabla g) \\ &\quad - dT \{ \rho(F + F'') g + 2F' k \cdot \nabla g + F \Delta g \}, \end{aligned} \quad (253)$$

$$E_{u,j} = 2D_* v + Q_r J(k F g + P \nabla g) + (\nabla v) v - T F g e_2 - d(\rho v + \Delta v). \quad (254)$$

The complete spatial differentiations in these expressions are

$$v = Q J(k F g + P \nabla g),$$

$$\begin{aligned}
\nabla v &= QJ\{k \otimes kF'g + F(k \otimes \nabla g + \nabla g \otimes k) + P\nabla^2 g\}, \\
\Delta V &= T\{\rho F''g + 2F'k \cdot \nabla g + F\Delta g\}, \\
\Delta v &= QJ\{\rho k F''g + \rho F'\nabla g + 2F'k(k \cdot \nabla g) + 2F(\nabla^2 g)k + Fk\Delta g + P\nabla\Delta g\}.
\end{aligned} \tag{255}$$

Keep the full base residuals

$$\begin{aligned}
f_{\theta,b} &= \nu\eta'(\nu t)(\theta_{\text{ref}} - \theta_{\text{in}}) - d\Delta\theta_b, \\
f_{u,b} &= \ddot{\alpha}J\nabla H_0 + \dot{\alpha}^2(\nabla(J\nabla H_0))J\nabla H_0 - \theta_b e_2 - d\dot{\alpha}J\nabla\Delta H_0, \\
f_\theta &= f_{\theta,b} + \sum_{j=1}^3 E_{\theta,j}, \quad f_u = f_{u,b} + \sum_{j=1}^3 E_{u,j}.
\end{aligned} \tag{256}$$

Verification of every equation and interaction. On a j th support the proved nesting makes the full older fields exactly $G_* \cdot x$ and D_*x . For $\mathcal{D}_j = \partial_t + D_*x \cdot \nabla$, differentiating M_j^{-1} gives $\mathcal{D}_j(M_j^{-1}x) = 0$. Likewise $\dot{k} = -D_*^T k$ implies $\mathcal{D}_j\xi = 0$. Therefore $\mathcal{D}_jg = 0$, $\mathcal{D}_jV = \dot{T}Fg$, and $\mathcal{D}_j\Psi = \dot{Q}Pg$. Direct contractions give

$$\begin{aligned}
v \cdot G_* &= QFg(Jk \cdot G_*) + QP(J\nabla g \cdot G_*), \\
v \cdot \nabla V &= QT(F^2 - PF')g(Jk \cdot \nabla g).
\end{aligned}$$

The second identity uses $Jk \cdot k = 0$, $J\nabla g \cdot \nabla g = 0$, and $J\nabla g \cdot k = -Jk \cdot \nabla g$. The coupling in $\dot{\Theta}$ in (243) cancels exactly the first term of the first identity. Adding $\dot{w}\Theta Fg$ and subtracting the complete $d\Delta V$ proves (253).

For an arbitrary trace-free two by two D_* , direct entrywise multiplication gives $D_*J + JD_*^T = 0$. Hence

$$\mathcal{D}_j(J\nabla\Psi) = D_*v + J\nabla(\mathcal{D}_j\Psi).$$

The reverse velocity interaction is $(v \cdot \nabla)(D_*x) = D_*v$. Adding it, the exact self-advection, and the negative buoyancy and diffusion terms gives (254), including its factor two. The product rule applied to $\Psi = QPg$ proves all entries of (255); differentiating $\Delta\Psi = Q(\rho F'g + 2Fk \cdot \nabla g + P\Delta g)$ in particular produces all six vector Laplacian groups.

These are full ordered interactions. In the third increment, $\mathcal{D}_3$ contains both $v_1 \cdot \nabla V_3$ and $v_2 \cdot \nabla V_3$, and both corresponding advections of v_3 . The term $v_3 \cdot G_{<3}$ contains both $v_3 \cdot \nabla V_1$ and $v_3 \cdot \nabla V_2$; the term $D_{<3}v_3$ contains both reverse velocity interactions. The same statements for $j = 2$ include all pairs of the first two layers, and $j = 1$ includes the base. Thus adding in order 1, 2, 3 counts every ordered pair exactly once. Outside a layer's support its derivatives vanish; the identities consequently telescope on all of $\mathbb{R}^2$, not merely on an affine neighborhood.

The base radial cutoff is transported by $\dot{\alpha}Jx$, and so is the base phase $\mu R_{\alpha}e_2 \cdot x$. On its scalar support $u_b = \dot{\alpha}Jx$; before $1/\nu$ the angle and velocity are zero. This proves the scalar base formula, including the initial interpolation. Differentiating $u_b = \dot{\alpha}J\nabla H_0$ proves its vector formula globally. The sum therefore solves (238) on $[0, t_1]$ with the original data. Since $u = J\nabla\psi$, equality of mixed spatial partial derivatives gives divergence zero. $\square$

The exact spatial coordinate correspondence is also retained. For each j put $C_j = M_j^{-1}M_j^{-T}$ and $\tilde{g}_j(y) = f(y/\ell_j)$. On a function of independent variables (ξ, y) , pulled back by $(\xi, y) = (K_j\zeta_j \cdot x, M_j^{-1}x)$, the chain rule gives

$$\begin{aligned}
\nabla_x &= K_j\zeta_j\partial_\xi + M_j^{-T}\nabla_y, \\
\Delta_x &= K_j^2r_j^2\partial_\xi^2 + 2K_j(C_jp_j \cdot \nabla_y)\partial_\xi + \sum_{l,m}(C_j)_{lm}\partial_{y_l}\partial_{y_m}.
\end{aligned} \tag{257}$$

Multiplying the commuting constant spatial derivatives proves the second equality from the first; in particular it retains all mixed terms and the actual time-dependent metric.

On an affine cell $g_j = 1$, $|\xi_j| < s_0$, the exact fields are

$$V_j = T_j K_j \zeta_j \cdot x, \quad v_j = \frac{w_j \Omega_j}{r_j^2} J \zeta_j \otimes \zeta_j x, \quad \Delta V_j = \Delta v_j = 0. \quad (258)$$

This follows by differentiating the original $Q_j(P(0) + \xi_j^2/2)$. The constant remains in the full cutoff field; it was not discarded from its definition. The damping forces in the core remain exactly $-d\rho_j V_j$ and $-d\rho_j v_j$. On an uncut sine-profile region instead, $F'' = -F$ gives $\rho_j V_j + \Delta V_j = 0$ and $\rho_j v_j + \Delta v_j = 0$. All other vector residual terms in (254) remain. Thus the core and sine calculations give the exact map from the damped amplitude trajectories to the released profile, including the forces that the profile requires.

11.4 Quantitative spatial costs for every layer and both forces

For any scalar, vector or tensor field U , write $|U|_m = \sup_x \|D_x^m U\|$ in the Euclidean multilinear norm. Let $[W]_q = \sup_\xi |W^{(q)}(\xi)|$ for a periodic profile and $f_m = \sup_y \|D^m f(y)\|$. Define fixed bounds on I by

$$\begin{aligned} r_{-,j}^2 &= (1, 1, 63/64)_j, & r_{+,j}^2 &= (1, 11, 3)_j, & \kappa_j &= K_j r_{+,j}, \\ U_{\Theta,j} &= (2\sigma_1^2/K_1, 3\sigma_2^2/K_2, e^4 P_3^*)_j, \\ U_{\Omega,j} &= (10\sigma_1, 7040a\sigma_2/L_3, 3e^4 K_3 P_3^*/\sigma_2)_j, \\ W_{j,1} &= (0, 0, \Gamma[\eta']_0)_j, & \vartheta_* &= 640as/L_3, & A_* &= 270a^2\sigma_1 + 6aU_{\Omega,2}\vartheta_*. \end{aligned} \quad (259)$$

The lower third length follows from $e^{-1/64} \geq 1 - 1/64 = 63/64$. To verify the control bound $|\dot{\alpha}| \leq A_*$, write

$$y = \frac{ag + hb}{D}, \quad \mathcal{F}_2 = \frac{a\Omega_1(h \tan \vartheta - a)}{D}, \quad \mathcal{F}_3 = \frac{\Omega_1 y (h \sin \vartheta - a \cos \vartheta) + b\Omega_2 \sin \vartheta}{g \cos \vartheta - b \sin \vartheta}.$$

On the proved box $0 \leq h \tan \vartheta < a$, $y < 18a$, $b < 4a$, and the denominator of $\mathcal{F}_3$ exceeds $2/3$. Thus $|\mathcal{F}_2| \leq 10a^2\sigma_1$ and $|\mathcal{F}_3| \leq 270a^2\sigma_1 + 6aU_{\Omega,2}\vartheta_*$. Their convex combination satisfies the asserted bound, including the preceding growth interval.

For each layer set the explicit constants

$$\begin{aligned} D_j^* &= A_* + \sum_{i < j} U_{\Omega,i}, & G_j^* &= A_0 + \sum_{i < j} K_i U_{\Theta,i} r_{+,i}, \\ Q_j^* &= \frac{U_{\Omega,j}}{K_j^2 r_{-,j}^2}, & Q_{r,j}^* &= \frac{W_{j,1} U_{\Omega,j} + \kappa_j U_{\Theta,j} + 2D_j^* U_{\Omega,j}}{K_j^2 r_{-,j}^2}. \end{aligned} \quad (260)$$

The rank-one operator $J\zeta_i \otimes \zeta_i/r_i^2$ has norm one, proving $\|D_{<j}\| \leq D_j^*$. The gradient sum in (243) proves $|G_{<j}| \leq G_j^*$. The identity $|\dot{\rho}_j/\rho_j| \leq 2D_j^*$ proves $|Q_j| \leq Q_j^*$ and $|Q_{r,j}| \leq Q_{r,j}^*$. In particular the second-parent velocity contributes the complete $U_{\Omega,2}$ to D_3^* , with no sign-based removal.

Define finite sums

$$\begin{aligned} e_{j,m} &= (N_j/\ell_j)^m f_m, \\ B_{j,m}(W) &= \sum_{q=0}^m \binom{m}{q} \kappa_j^q [W]_q e_{j,m-q}, \\ C_{j,m}(P) &= \sum_{q=0}^m \binom{m}{q} \kappa_j^q [P]_q e_{j,m-q+1}. \end{aligned} \quad (261)$$

The q derivatives allocated to the linear phase have size at most κ_j^q ; the other $m - q$ derivatives of the linear material coordinate have size at most $(N_j/\ell_j)^{m-q}$. The product rule with all $\binom{m}{q}$

allocations therefore proves $|g_j|_m \leq e_{j,m}$, $|W(\xi_j)g_j|_m \leq B_{j,m}(W)$, and $|P(\xi_j)\nabla g_j|_m \leq C_{j,m}(P)$. These bounds hold for every choice of unit derivative directions.

The full damping-plus-diffusion costs are

$$\begin{aligned}\mathfrak{D}_{\theta,j,m} &= dU_{\Theta,j}\{\kappa_j^2 B_{j,m}(F) + 2B_{j,m+2}(F)\}, \\ \mathfrak{D}_{u,j,m} &= dQ_j^*\{\kappa_j^2 B_{j,m+1}(P) + 2B_{j,m+3}(P)\}.\end{aligned}\tag{262}$$

They bound, respectively, $-d(\rho_j V_j + \Delta V_j)$ and $-d(\rho_j v_j + \Delta v_j)$. The explicit factor two bounds the sum of the two coordinate traces defining the Laplacian; it is not an identification with a single second derivative. The complete force estimates are

$$\begin{aligned}|E_{\theta,j}|_m &\leq W_{j,1}U_{\Theta,j}B_{j,m}(F) + Q_j^*G_j^*C_{j,m}(P) + \mathfrak{D}_{\theta,j,m} \\ &\quad + Q_j^*U_{\Theta,j}\sum_{q=0}^m\binom{m}{q}B_{j,q+1}(P)B_{j,m-q+1}(F),\end{aligned}\tag{263}$$

$$\begin{aligned}|E_{u,j}|_m &\leq (2D_j^*Q_j^* + Q_{r,j}^*)B_{j,m+1}(P) + U_{\Theta,j}B_{j,m}(F) + \mathfrak{D}_{u,j,m} \\ &\quad + (Q_j^*)^2\sum_{q=0}^m\binom{m}{q}B_{j,q+1}(P)B_{j,m-q+2}(P).\end{aligned}\tag{264}$$

For the scalar nonlinear term apply the product rule to $v_j \cdot \nabla V_j$; for the vector term apply it to $(\nabla v_j)v_j$. The displayed derivative allocations give exactly the two sums. All other terms follow individually from (253)–(255). This proves the estimates with activation, all parent interactions, self-advection, buoyancy, moving phase length, and diffusion included.

For the base on I , let $c_m = \sup_x \|D^m \chi_0(x)\|$, $H_m = |J\nabla H_0|_m$, and

$$B_{0,m}(F) = \sum_{q=0}^m \binom{m}{q} \mu^q [F]_q c_{m-q}.$$

Write A_{**} for the explicit bound on $|\ddot{\alpha}|$ constructed in (271) below. Then

$$\begin{aligned}|f_{\theta,b}|_m &\leq 2d(A_0/\mu)B_{0,m+2}(F), \\ |f_{u,b}|_m &\leq A_{**}H_m + A_*^2\sum_{q=0}^m\binom{m}{q}H_{q+1}H_{m-q} + (A_0/\mu)B_{0,m}(F) + 2dA_*H_{m+2}.\end{aligned}\tag{265}$$

These follow directly from (256), since the interpolation is complete on I . Adding these bounds to (263) and (264) gives explicit bounds for both total forces at every spatial order. The previously proved formulas cover $[0, t_b]$ with the same initial interpolation; the maximum of its bounds and these new ones covers $[0, t_1]$. Every bound on I integrates with the explicit length $(t_a - t_b) + \Gamma^{-1} < 15/\sigma_1$ from (240).

In particular all localization and frequency factors remain visible:

$$\begin{aligned}N_1/\ell_1 &= C_\ell \mu/s_0, \quad N_2/\ell_2 = \tilde{N} \mu \hat{\lambda}_1^3, \quad N_3/\ell_3 = \frac{16\mu \hat{\lambda}_2^3}{\sin s}, \\ dU_{\Theta,3}K_3^2 &= de^4 A_0 \mu \hat{\lambda}_3^{9/8}, \quad dQ_3^*K_3^3 = \frac{64e^4 dA_0 \mu \hat{\lambda}_3^{9/8}}{21\sigma_2}.\end{aligned}\tag{266}$$

These are exact factors in upper costs, not lower bounds for the total force. The exact modal diffusion exponents on I are

$$dK_1^2|I|, \quad dK_2^2 \int_I D \, dt, \quad dK_3^2 \int_I R \, dt.\tag{267}$$

The second lies between $dK_2^2|I|$ and $11dK_2^2|I|$. The third splits exactly into the proved growth cost and $\delta_3 \int_0^1 R \, d\tau$, whose transition part lies between $(63/64)\delta_3$ and $3\delta_3$, with $\delta_3 = dK_3^2/\Gamma$. Older coupling terms remain in (243); they have not been replaced by damping.

11.5 An explicit recursion for every temporal and mixed cost

We give the coefficient recursion as well as the spatial chain rule, so no stationary-envelope convention enters a time derivative. Use $\tau = \Gamma(t - t_a)$, $L = L_3$, $q = \sigma_1/\sigma_2 = s/L$, $\epsilon = \sigma_1/\Gamma$, $\delta_j = dK_j^2/\Gamma$, and the dimensionless state

$$\begin{aligned} z &= (h, g, u, E, F, \omega, V, X, Y), \\ E &= -K_1\Theta_1/\sigma_1^2, \quad F = -K_2\Theta_2/\sigma_2^2, \quad \omega = \Omega_1/\sigma_1, \quad V = \Omega_2/\sigma_2, \\ X &= -\Theta_3/P_3^*, \quad Y = -\sigma_2\Omega_3/(K_3P_3^*). \end{aligned} \quad (268)$$

The two last coordinates are signed linear amplitudes; forming them does not divide by a temperature. Put $\bar{c} = A_0 \cos s_*/\sigma_1^2$ and $\bar{C} = A_0 \sin s_*/\sigma_1^2$, and define

$$\begin{aligned} x_c &= (hg - ab)/D, \quad y_c = (ag + hb)/D, \\ N_3 &= \sigma_1^2(E + \bar{c})y_c + \sigma_1^2\bar{C}x_c + b\sigma_2^2F, \\ \mathcal{K} &= \frac{N_3}{\sigma_2^2 \sin s R}, \quad \mathcal{B} = \frac{g \sin(su) + b \cos(su)}{\sqrt{D} \sin s}. \end{aligned}$$

Let $\Phi(z, \lambda) = (1-\lambda)\mathcal{F}_2 + \lambda\mathcal{F}_3$, with the exact feedbacks displayed above and substitutions $\Omega_1 = \sigma_1\omega$, $\Omega_2 = \sigma_2V$, $\vartheta = su$. The complete explicit dimensionless vector field $\mathcal{V}(z, \lambda)$ is

$$\begin{aligned} h' &= a\epsilon\omega, & g' &= \epsilon\omega \frac{2ahg + (h^2 - a^2)b}{D} + \frac{b}{\sin s}V, \\ u' &= \frac{-\Phi - a^2\sigma_1\omega/D}{\Gamma s}, & E' &= -\epsilon\bar{C}\omega - \delta_1E, \\ F' &= -q\epsilon \frac{a(E + \bar{c}) + \bar{C}h}{D}V - \delta_2DF, & \omega' &= -\epsilon E \frac{h \sin(su) - a \cos(su)}{\sqrt{D}} - \delta_1\omega, \\ V' &= -F\sqrt{D} \frac{\sin(su)}{\sin s} - \delta_2DV, & X' &= \mathcal{K}Y - \delta_3RX, \\ Y' &= \mathcal{B}X - \delta_3RY. \end{aligned} \quad (269)$$

Every prime in this display is $d/d\tau$. Substitution of (268) into the physical equations proves each row, including both factors $q\epsilon$ in the second-temperature row. Conversely those substitutions reverse, giving the physical pair, phases, and feedback. This explicitly proves the coordinate map and its inverse wherever the displayed denominators are positive.

For bounds let $\mathcal{Z}$ be the closed rectangular box given by

$$\begin{aligned} h &\in [h_0, 13/4], \quad g \in [3/4, 5], \quad u \in [0, 1/1024], \\ E &\in [1/8, 2], \quad F \in [1/16, 3], \quad \omega \in [0, 10], \\ V &\in [-1/32, 0], \quad X \in [0, e^4], \quad Y \in [0, 3e^4]. \end{aligned}$$

The proved actual solution on I belongs to this box. All feedback denominators are bounded away from zero on the whole box by the same elementary inequalities used above. In particular $D \geq 1$ and $g \cos(su) - b \sin(su) > 2/3$ there. Introduce independent symbols $\lambda^{[0]}, \lambda^{[1]}, \dots$ for the jets of the control switch and define on each finite jet expression

$$\mathcal{C} = \sum_{l=1}^9 \mathcal{V}_l(z, \lambda^{[0]}) \partial_{z_l} + \sum_{r \geq 0} \lambda^{[r+1]} \partial_{\lambda^{[r]}}. \quad (270)$$

The second sum is finite on every expression to which it is applied. The chain rule proves that the physical r th time derivative of any state function $H(z, \lambda)$ is $\Gamma^r \mathcal{C}^r H$ evaluated at $\lambda^{[j]} = \eta^{(j)}(\Gamma(t - t_a))$. Before the transition these are zero for every $j \geq 0$; the same identity therefore holds on all of I . Consequently the explicit control constants are

$$A_{r+1} = \Gamma^r \max \left\{ |\mathcal{C}^r \Phi| : z \in \mathcal{Z}, \quad 0 \leq \lambda^{[0]} \leq 1, \quad |\lambda^{[j]}| \leq [\eta^{(j)}]_0 \quad (1 \leq j \leq r) \right\}, \quad A_{**} = A_2. \quad (271)$$

This is a maximum of a fully specified continuous elementary function over an explicit compact rectangle; it is finite and depends only on the retained parameters and profile derivative bounds. It is not a supremum over an unspecified later solution. The equality of derivatives just proved yields $|\alpha^{(r+1)}| \leq A_{r+1}$ at every order. The sharper A_* in (259) bounds its first derivative.

For a complete algebraic mixed-jet evaluation, set $A_j = M_j^{-1}/\ell_j$ and introduce independent (t, x, ξ, y) . Define

$$\mathfrak{T}_j = \partial_t + (\dot{k}_j \cdot x) \partial_\xi + (\dot{A}_j x) \cdot \nabla_y, \quad \mathfrak{X}_{j,i} = \partial_{x_i} + k_{j,i} \partial_\xi + \sum_l (A_j)_{li} \partial_{y_l}. \quad (272)$$

For $\iota_j^* H = H(t, x, k_j(t) \cdot x, A_j(t)x)$, the ordinary chain rule, iterated one derivative at a time, gives

$$\partial_t^n \partial_x^\gamma \iota_j^* H = \iota_j^* (\mathfrak{T}_j^n \mathfrak{X}_{j,1}^{\gamma_1} \mathfrak{X}_{j,2}^{\gamma_2} H). \quad (273)$$

The commutators vanish: the time derivative of $k_{j,i}$ cancels the x_i derivative of $\dot{k}_j \cdot x$, and the two corresponding terms for A_j cancel as well. Thus the identity also verifies the consistency of all mixed derivative orders. Every coefficient jet in this recursion is evaluated by (269)–(271) and

$$\dot{M}_j = D_{<j} M_j, \quad \dot{A}_j = -A_j D_{<j}, \quad \dot{k}_j = -D_{<j}^T k_j,$$

together with product differentiation of (243) and the exact activation jets $\Gamma^r \eta^{(r)}$ for w_3 . For example $\partial_t^{n+1} A_j = -\sum_{q=0}^n \binom{n}{q} (\partial_t^q A_j) (\partial_t^{n-q} D_{<j})$; the other two linear equations have the identical product recursion with their displayed order and signs. Their zeroth bounds are (245), so induction gives explicit finite matrix and phase jet bounds at every prescribed order.

To obtain an actual numerical expression for a mixed norm, expand (273) on every displayed field or residual, keeping its finitely many summands. Replace each coefficient jet by the preceding recursive bound, each x^β by $(N_j \ell_j)^{|\beta|}$ on its layer support, each periodic derivative by $[F]_r$ or $[P]_r$, and each envelope derivative by f_m . Sum the absolute products with their explicit binomial coefficients. The triangle and product inequalities prove the resulting bound. For the base use its support radius R_H and its displayed functions instead. This procedure computes all mixed costs, including every time derivative of $d\rho_j$ and the changing Laplacian metric; its recursion has finite depth $n + |\gamma|$.

11.6 The transition endpoint and a proved smooth force collar

At t_1 all weights are one, $\Theta_3(t_1) < 0$ and $\Omega_3(t_1) < 0$, with the exact inverse amplitudes of the transition theorem. In the common affine cell the complete state is

$$\theta = \left(-A_0 R_\alpha e_2 + \sum_{j=1}^3 K_j \Theta_j \zeta_j \right) \cdot x, \quad u = \left(\dot{\alpha} J + \sum_{j=1}^3 \frac{\Omega_j}{r_j^2} J \zeta_j \otimes \zeta_j \right) x. \quad (274)$$

The second vorticity has the revived negative sign proved above; the first remains positive. Formula (274) and the full cutoff formulas give actual states of (238), with their exact material matrices and laboratory gravity.

To continue smoothly to the right, use the explicit vector field (269) with the constant control value $\lambda = 1$. Let $z_1 = z(t_1)$ and take the closed cube $\mathcal{C}_1 = \{z : \|z - z_1\|_\infty \leq 1/100\}$. Its interior is in the smooth domain of this vector field. Indeed $h > h_0 - 1/100 > 3/4$, $g > 3/4 - 1/100 > 7/10$, and $|u| < 1/1024 + 1/100 < 1/50$ there. Since $s \leq a \leq 1/512$ and $b < 4s$, these imply $\cos(su) > 15/16$ and

$$g \cos(su) - b \sin(su) > (7/10)(15/16) - 4s^2/50 > 3/5.$$

Thus D , R , $\sin s$ and every feedback denominator remain strictly positive on a neighborhood of the cube. No amplitude quotient occurs in its vector field. Define the concrete finite maxima

$$B = \max_{\mathcal{C}_1} \|\mathcal{V}(z, 1)\|_\infty, \quad L_{\text{lip}} = \max_{\mathcal{C}_1} \|D_z \mathcal{V}(z, 1)\|_{\infty \rightarrow \infty}, \quad \Delta\tau = \min \left\{ \frac{1}{400(1+B)}, \frac{1}{2(1+L_{\text{lip}})} \right\} > 0. \quad (275)$$

On continuous paths in this cube beginning at z_1 , the Picard map $z(\tau) \mapsto z_1 + \int_1^\tau \mathcal{V}(z(r), 1) dr$ moves points by at most $B\Delta\tau < 1/400 < 1/100$. The mean-value formula on the convex cube makes its Lipschitz constant at most $L_{\text{lip}}\Delta\tau < 1/2$ in the uniform norm. Successive iterates therefore converge uniformly: their successive differences are bounded by a geometric series with ratio below $1/2$. The limit satisfies the integral equation, and the same contraction inequality proves uniqueness. Differentiation of this equation first gives C^1 , and repeated differentiation of the smooth vector field gives C^∞ . This constructs an actual right collar of physical length

$$\epsilon_R = \Delta\tau/\Gamma > 0. \quad (276)$$

At t_1 the original switching profile is flat and equal to one, so the control and every derivative match this continuation. The matrices and phase fields continue by their exact linear ODEs. Their determinant stays one by the trace identity, or by the explicit formulas with $\det \mathcal{L} = -b$. The angle continues by $\dot{\alpha} = \mathcal{F}_3$. Hence the fields have a smooth right extension with the actual endpoint jets, not a frozen state.

On this collar define the forces by the raw full residuals

$$f_\theta^{\text{raw}} = \theta_t + u \cdot \nabla \theta - d\Delta\theta, \quad f_u^{\text{raw}} = u_t + (u \cdot \nabla)u - d\Delta u - \theta e_2. \quad (277)$$

They agree with (256) on the entire required slab by the proof above. Thus no additional nesting assertion outside that slab is needed for the collar. They are smooth and compact in space, since the fields are smooth compact fields and their matrices vary continuously on a compact time interval. All collar mixed costs are explicitly computed by the same derivation and pullback recursions, replacing $\mathcal{Z}$ by $\mathcal{C}_1$ and λ by one in their maxima, and using the raw expressions (277). The compact matrix bounds follow directly from (244) on that cube; for example $h < 13/4 + 1/100$, $g < 5 + 1/100$, and its positive denominators bound $\mathcal{L}$ and $\mathcal{L}^{-1}$ there. These explicit finite maxima provide every derivative order without assuming the original pre-endpoint amplitude bounds on the collar.

Extend the original sine state constantly to negative times, with zero velocity and inactive increments, and put $\epsilon_L = 1/\nu$. Use the raw residuals of these extensions and set

$$\beta(t) = \eta(1 + t/\epsilon_L)\eta(1 + (t_1 - t)/\epsilon_R), \quad \tilde{f}_\bullet = \beta f_\bullet^{\text{raw}}. \quad (278)$$

They equal the proved forces on $[0, t_1]$. Flatness of η makes their zero extension outside $[-\epsilon_L, t_1 + \epsilon_R]$ smooth. Their support is compact in space and time. More explicitly, with

$$B_{\beta,r} = \sum_{q=0}^r \binom{r}{q} \epsilon_L^{-q} \epsilon_R^{-(r-q)} [\eta^{(q)}]_0 [\eta^{(r-q)}]_0,$$

the product rule gives for every multi-index γ

$$\|\partial_t^n \partial_x^\gamma \tilde{f}_\bullet\|_\infty \leq \sum_{q=0}^n \binom{n}{q} B_{\beta,q} \|\partial_t^{n-q} \partial_x^\gamma f_\bullet^{\text{raw}}\|_\infty. \quad (279)$$

The right-hand mixed norms have the explicit finite recursion just proved. The PDE is asserted on $[0, t_1]$, where the cutoff is one.

Finally, the streamfunction is compact and smooth; hence the velocity has finite energy. If $N_0(x) = (2\pi)^{-1} \log |x|$, then $N_0 * \Delta\psi = (\Delta N_0) * \psi = \psi$ by distributional integration by parts, and therefore $u = J\nabla N_0 * \omega$, $\omega = \Delta\psi$, exactly. Odd F , even P , and even material envelopes make θ, u, f_θ, f_u odd and ω even. Every field and force on the required slab has the original spatial support inside B_{R_H} by the proved nesting. This completes the full compact spatial extension through the actual third transition with every retained physical diffusion cost and both older interaction directions.

12 The exact third steering map with two evolving parents

The actual third transition supplies a nonvertical second phase and a nonzero second vorticity. This section derives the original next steering system from that state. Its prescribed horizontal component determines all three phases explicitly. The resulting six older equations retain both parent temperatures, both parent vorticities, and their different physical diffusion rates. We prove an explicit positive interval of the actual steering evolution and the exact shooting and holding equations. A third return root is not asserted here; its endpoint estimates are a subsequent calculation.

12.1 Actual source data and prescribed horizontal component

Use the parameters and actual endpoint t_1 of Theorem 4. In particular retain

$$a = \sin s_2, \quad b = r_b \sin s_3, \quad c = A_0 \cos s_*, \quad C = A_0 \sin s_*,$$

$$s = s_3 = \frac{L_3 \sigma_1}{\sigma_2}, \quad L = L_3, \quad \Gamma = \Gamma_3 = \sigma_2 \sin s.$$

The constants a, b, c, C are positive. The quantity $\nu = \sqrt{A_0}$ remains the source time scale, whereas $d > 0$ is physical viscosity and thermal diffusivity. The original physical frequencies remain $K_j = \mu \hat{\lambda}_j$ with $\mu = \lambda_0 = \lceil \sqrt{A_0}/2 \rceil$. Put

$$Z_1 = \zeta_{3,1}(t_1) > 0, \quad \chi_1 = |\zeta_3(t_1)| > 0, \quad P_{3,1} = -\Theta_3(t_1) > 0, \quad v_1 = \frac{\sigma_2 \Omega_3(t_1)}{K_3 \chi_1 \Theta_3(t_1)} > 0. \quad (280)$$

All these numbers are the actual diffusive transition values; neither χ_1 nor $Z_1/\sin s$ is replaced by one. Write $\tau = \Gamma(t - t_1)$, $T = 1 + L^{-1}$, and retain exactly the source profiles η, η_2 and $H = \|\eta_2\|_\infty$. Thus η is a smooth nondecreasing step from zero to one on $[0, 1]$, and $\eta_2 \in C_c^\infty((0, 1))$ is nonnegative with integral one. For the source trial parameter $m \in [0, 3]$, set

$$Z_m(\tau) = \begin{cases} Z_1(1 - \eta(\tau)), & 0 \leq \tau \leq 1, \\ -mL \sin s \chi_1 \eta_2(L(\tau - 1)), & 1 \leq \tau \leq T, \\ 0, & \tau \geq T. \end{cases} \quad (281)$$

The source pulse parameter in (3.12) is $h_3 = mL \sin s$; its full range $[0, 3L \sin s]$ is preserved. The multiplier χ_1 in the negative pulse is required by that equation. Flatness of the step and interior support of the pulse make (281) smooth at the two joins, with every derivative zero there apart from its displayed constant value at $\tau = 0$. The prescribed family is independent of m before $\tau = 1$. For every integer $r \geq 1$ its physical derivatives on the two pieces are exactly

$$\partial_t^r Z_m = \begin{cases} -Z_1 \Gamma^r \eta^{(r)}(\tau), & 0 \leq \tau \leq 1, \\ -m \chi_1 \sin s L (\Gamma L)^r \eta_2^{(r)}(L(\tau - 1)), & 1 \leq \tau \leq T. \end{cases} \quad (282)$$

In particular the pulse derivative in the angular control contains ΓL^2 , with its full original coefficient.

12.2 Complete physical coordinate equations and inverse

Let $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Use the exact determinant coordinates of (187):

$$D = h^2 + a^2, \quad x = \frac{hg - ab}{D}, \quad y = \frac{ag + hb}{D}, \quad R = x^2 + y^2 = \frac{g^2 + b^2}{D},$$

$$E_1 = -K_1 \Theta_1, \quad E_2 = -K_2 \Theta_2, \quad N_2 = a(E_1 + c) + Ch, \quad N_3 = (E_1 + c)y + Cx + bE_2. \quad (283)$$

Here $D = |\zeta_2|^2$ and $R = |\zeta_3|^2$ are scalars. For a prescribed $Z = Z_m(\Gamma(t - t_1))$ put

$$\begin{aligned} Y &= \sqrt{R - Z^2}, & \zeta_3 &= (Z, Y), \\ \zeta_2 &= \left(\frac{gZ - bY}{R}, \frac{gY + bZ}{R} \right), & \zeta_1 &= \left(\frac{xZ - yY}{R}, \frac{xY + yZ}{R} \right). \end{aligned} \quad (284)$$

The square root specifies the original positive third second component. On the component domain used below, $h, g, x, y > 0$ and all three second components are positive. The exact angular inverse is

$$\begin{aligned} \phi_3 &= \arcsin(Z/\sqrt{R}), & \phi_2 &= \phi_3 - \arctan(b/g), \\ \phi_1 &= \phi_2 - \arctan(a/h), & \alpha &= s_* - \phi_1. \end{aligned} \quad (285)$$

Both inverse trigonometric functions take their principal real branches. These formulas retain the original angular lift at t_1 ; continuity in this positive-component domain precludes a change by 2π . Indeed the actual transition has $\phi_3 = \theta + \arctan(b/g) \in (-\pi/2, \pi/2)$, as follows from (199), $g > 3/4$ and $b < 4a$. The principal inverse therefore agrees with the actual transition lift, rather than selecting a new representative. An alternative unwrapped initial lift would retain its fixed initial 2π multiple throughout the same formulas.

The complete closed older system in physical time is

$$\begin{aligned} \dot{h} &= a\Omega_1, \\ \dot{g} &= \Omega_1 \frac{2ahg + (h^2 - a^2)b}{D} + b\Omega_2, \\ \dot{E}_1 &= -C\Omega_1 - dK_1^2 E_1, \\ \dot{\Omega}_1 &= -E_1 \frac{xZ - yY}{R} - dK_1^2 \Omega_1, \\ \dot{E}_2 &= -\frac{N_2}{D} \Omega_2 - dK_2^2 D E_2, \\ \dot{\Omega}_2 &= -E_2 \frac{gZ - bY}{R} - dK_2^2 D \Omega_2. \end{aligned} \quad (286)$$

Its six initial values are inherited from t_1 without resetting any older amplitude. Neither Ω_3 nor Θ_3 enters this older system, because the third shear does not enter its own parent. The exact angular control reconstructed from it is

$$\begin{aligned} \mathcal{F}_3 &= \frac{\Omega_1 y \zeta_{1,1} + (b\Omega_2/D) \zeta_{2,1}}{Y}, & \dot{\alpha} &= \mathcal{F}_3 - \frac{\dot{Z}}{Y}, \\ \dot{Z} &= \Gamma \partial_\tau Z_m. \end{aligned} \quad (287)$$

Thus the prescribed component and the source control (3.12) are exactly equivalent; no independent angular forcing was substituted.

The complete original deformation matrices are

$$\begin{aligned} M_1 &= R_\alpha, & M_2 &= R_{\alpha-s_*} \begin{pmatrix} 1 & -(h - \cos s_2)/a \\ 0 & 1 \end{pmatrix}, \\ \mathcal{L} &= (\zeta_2 \ \zeta_3), & M_3 &= \mathcal{L}(t)^{-T} \mathcal{L}(t_b)^T. \end{aligned} \quad (288)$$

Here $t_b = t_2^b$ is the original third insertion time, not the third steering-entry time. All three matrices retain their original activation data, have determinant one, and obey their respective physical transport equations.

Exact forward and inverse calculation. The elementary identities

$$hx + ay = g, \quad hy - ax = b, \quad x^2 + y^2 = R$$

follow by expanding (283). The inverse (284) gives

$$|\zeta_1|^2 = 1, \quad |\zeta_2|^2 = D, \quad |\zeta_3|^2 = R, \quad \zeta_2 = h\zeta_1 - aJ\zeta_1, \quad \zeta_3 = x\zeta_1 - yJ\zeta_1.$$

For example $\zeta_1 = (xI + yJ)\zeta_3/R$, whose squared length is $(x^2 + y^2)|\zeta_3|^2/R^2 = 1$. Also $\zeta_2 = (gI + bJ)\zeta_3/R$; multiplying $(hI - aJ)(xI + yJ) = (gI + bJ)$ proves the first relative formula. Taking their oriented determinants gives

$$\det(\zeta_1, \zeta_2) = -a, \quad \det(\zeta_2, \zeta_3) = -b.$$

The dot products are respectively h and g , so the construction is inverse to the determinant coordinates, with the original signs. The polar expressions for these same identities give (285), since multiplication by J rotates the vector $(\sin \phi, \cos \phi)$ in the direction of decreasing ϕ .

Differentiating x, y with the first two equations of (286) gives the exact formulas

$$\dot{x} = \Omega_1 y + \frac{bh}{D}\Omega_2, \quad \dot{y} = \frac{ba}{D}\Omega_2, \quad \dot{R} = 2\Omega_1 xy + \frac{2bg}{D}\Omega_2. \quad (289)$$

In particular the second parent's vorticity changes y ; it has not disappeared in this representation. Differentiating (285) gives

$$\dot{\alpha} = -\frac{\dot{Z}}{Y} + \frac{Z\dot{R}}{2RY} - \frac{b\dot{g}}{g^2 + b^2} - \frac{a\dot{h}}{D}.$$

For an explicit check write $A = ax + hy$, so $\dot{g} = \Omega_1 A + b\Omega_2$. Then

$$bA + a^2 R = (hy - ax)(ax + hy) + a^2(x^2 + y^2) = Dy^2.$$

The last two terms in the angular derivative are consequently $-\Omega_1 y^2/R - b^2\Omega_2/(DR)$. Combining these with $Z\dot{R}/(2RY)$ gives exactly $(\Omega_1 y(xZ - yY) + (b\Omega_2/D)(gZ - bY))/(RY)$, which equals $(\Omega_1 y\zeta_{1,1} + (b\Omega_2/D)\zeta_{2,1})/Y$. This proves (287).

For a direct check of the phase dynamics define

$$D_{<2} = \dot{\alpha}J + \Omega_1 J\zeta_1 \otimes \zeta_1, \quad D_{<3} = D_{<2} + \frac{\Omega_2}{D}J\zeta_2 \otimes \zeta_2.$$

Since $\phi_1 + \alpha = s_*$, one has $\dot{\zeta}_1 = \dot{\alpha}J\zeta_1$. Differentiating the two relative formulas above using (289) gives, without dropping either shear,

$$\dot{\zeta}_2 = \dot{\alpha}J\zeta_2 + a\Omega_1\zeta_1 = -D_{<2}^T\zeta_2, \quad \dot{\zeta}_3 = \dot{\alpha}J\zeta_3 + \Omega_1 y\zeta_1 + \frac{b\Omega_2}{D}\zeta_2 = -D_{<3}^T\zeta_3.$$

Their first third component is $\dot{Z}$ by (287); their second is the derivative of the selected square root. The older physical gradient is

$$G_{<3} = -(E_1 + c + hE_2)\zeta_1 - (C - aE_2)J\zeta_1.$$

Hence $-J\zeta_2 \cdot G_{<2} = N_2$ and $-J\zeta_3 \cdot G_{<3} = N_3$, proving the coefficients $a_2 = N_2/(K_2D)$ and $a_3 = N_3/(K_3R)$. The damped physical amplitude rows and $\dot{\Omega}_j = K_j\zeta_{j,1}\Theta_j - dK_j^2|\zeta_j|^2\Omega_j$ are now exactly the last four equations of (286).

The added second shear has transpose annihilating ζ_2 . Thus the columns of $\mathcal{L}$ both solve the third transport equation, and $\det \mathcal{L} = -b$. Differentiation of $\mathcal{L}^{-T}\mathcal{L}(t_b)^T$ proves $\dot{M}_3 = D_{<3}M_3$ and its original value I at t_b . Differentiating the two earlier matrices proves their equations as in (189); the constants a and $\cos s_2$ are their original data. Conversely, the physical phase equations give the first two rows of (286) by differentiating the dot products; the amplitude rows give the other four. These calculations prove the full forward and inverse equivalence. $\square$

The exact loss equations continue through the prescribed steering:

$$\dot{N}_2 = -adK_1^2 E_1, \quad \dot{N}_3 = -d(K_1^2 E_1 y + bK_2^2 DE_2). \quad (290)$$

Indeed differentiating N_2 cancels $-aC\Omega_1$ with $Ca\Omega_1$. Differentiating N_3 gives the shear terms

$$-C\Omega_1 y + C\Omega_1 y + \frac{b\Omega_2}{D} \{a(E_1 + c) + Ch - N_2\} = 0;$$

the remaining terms are precisely (290). The value of $\dot{Z}$, and therefore the possibly large derivative of the compressed pulse, does not enter either cancellation. An additional exact algebraic relation useful for continuing the older temperature calculation is

$$E_1 = \frac{N_2 - Ch}{a} - c, \quad E_2 = \frac{N_3}{b} - \frac{N_2 y}{ab} + \frac{C}{a}. \quad (291)$$

To check the second relation insert the first in N_3 and use $ax - hy = -b$. Thus one may equally use N_2, N_3 instead of E_1, E_2 in the older system, with the displayed inverse and the explicit diffusion losses, without assuming either numerator constant at positive d .

12.3 An explicit actual interval of smooth steering

Theorem 5. *For every original datum and positive physical diffusivity constructed by (166)–(167), the actual third transition is followed by a unique smooth solution of (286) and the source steering control on*

$$0 \leq \tau \leq \tau_0 := \frac{\sin s}{8192}, \quad t_1 \leq t \leq t_1 + \frac{1}{8192\sigma_2}.$$

All $m \in [0, 3]$ give the same solution there. Every older temperature is negative, all source phase second components are positive, and the original matrices and their inverses are smooth. On this interval the third temperature and vorticity remain strictly negative and the exact source quotient below is positive.

Proof. This interval is proved from the actual entry margins, without assuming bounds on a future trial. Use the dimensionless state

$$U = (h, g, E, F, \omega, V) = \left(h, g, \frac{E_1}{\sigma_1^2}, \frac{E_2}{\sigma_2^2}, \frac{\Omega_1}{\sigma_1}, \frac{\Omega_2}{\sigma_2} \right).$$

Define $\epsilon = \sigma_1/\Gamma \leq 2/L$, $q = \sigma_1/\sigma_2 = s/L$, $\bar{c} = c/\sigma_1^2 \leq 1/4$, $\bar{C} = C/\sigma_1^2 \leq a/32$, and $\delta_j = dK_j^2/\Gamma$. Primes in this proof denote τ derivatives. The exact scaled older equations are

$$\begin{aligned} h' &= a\epsilon\omega, \\ g' &= \epsilon\omega \frac{2ahg + (h^2 - a^2)b}{D} + \frac{b}{\sin s} V, \\ E' &= -\epsilon\bar{C}\omega - \delta_1 E, \quad \omega' = -\epsilon E\zeta_{1,1} - \delta_1 \omega, \\ F' &= -q\epsilon \frac{a(E + \bar{c}) + \bar{C}h}{D} V - \delta_2 DF, \quad V' = -\frac{F\zeta_{2,1}}{\sin s} - \delta_2 DV. \end{aligned} \quad (292)$$

Keep the closed cube $|U - U(t_1)|_\infty \leq 1/128$. The actual bounds (200) and (199) imply throughout this cube

$$\begin{aligned} 63/64 &< h < 4, \quad 11/16 < g < 6, \quad 15/128 < E < 3, \quad 7/128 < F < 4, \\ |\omega| &< 11, \quad |V| < 1/16, \quad 15/16 < D < 17. \end{aligned}$$

Here the lower h bound uses $a \leq 1/512$ and $h(t_1) \geq \sqrt{1 - a^2}$. The bound on V uses $7040a/L < 1/32$. For $0 \leq \tau \leq 1$, (281) and (204) give

$$0 \leq Z \leq Z_1 < \frac{9}{8} \sin s < \frac{1}{128}, \quad b < 4 \sin s \leq 4a.$$

Consequently

$$R > \frac{(11/16)^2}{17} > \frac{1}{64}, \quad Y^2 > \frac{1}{64} - \frac{1}{128^2} > \frac{1}{81}, \quad x > \frac{1}{32}, \quad y > 0.$$

For x use $hg - ab > (63/64)(11/16) - 4a^2 > 17/32$ and $D < 17$. Thus $Y > 1/9$, $gY + bZ > 0$, and $xY + yZ > 0$ on this entire known cube and time interval. The original positive-component domain has a fixed positive margin; R and D never vanish. All right-hand sides are smooth on a neighborhood of this cube.

Their sizes can be bounded before solving. The first four rows of (292) have absolute value less than one: use $\epsilon \leq 2/L$, $L \geq 16384$, $\delta_1 \leq 1/128$, $|\zeta_{1,1}| \leq 1$, and

$$\left| \frac{2ahg + (h^2 - a^2)b}{D} \right| \leq \frac{2ag}{h} + b < 17a.$$

The fifth row also has absolute value less than one, since $a(E + \bar{c}) + \bar{C}h < 4a$, $q = s/L \leq a/L$, and $\delta_2 DF < 68/128$. Finally $|\zeta_{2,1}| \leq \sqrt{D} < 5$ gives

$$|V'| < \frac{20}{\sin s} + \frac{17}{2048} < \frac{21}{\sin s}.$$

Thus the norm of the complete vector field is strictly less than $M = 32/\sin s$ on this cube. Local existence follows from its smooth integral equation. On any initial solution inside the cube, $|U(\tau) - U(0)|_\infty < M\tau$. At $\tau = \tau_0$ this is at most $1/256$, strictly smaller than its radius $1/128$. A first exit is impossible, and the same smooth compact-domain bound extends the solution through τ_0 . The integral equation and its bounded first derivatives prove uniqueness. Since $\tau_0 < 1$, every trial has the same prescribed Z there. The reconstructed original matrices and control are smooth because the denominators just bounded are positive. They join the preceding transition smoothly: there $\dot{\zeta}_{3,1} = (1 - \eta_b)(\mathcal{F}_3 - \mathcal{F}_2)\zeta_{3,2}$, whose derivatives of every order vanish at t_1 . The new prescribed Z has the same constant value and the same vanishing derivatives there by (282). Both controls consequently have the common endpoint jet of $\mathcal{F}_3$; differentiating their smooth physical equations successively matches every state derivative.

On this actual interval $E_1, E_2, x, y > 0$, so $N_3 > 0$. The newest linear pair has nonnegative off-diagonal coefficients $A = N_3/(K_3R) > 0$ and $B = K_3Z \geq 0$. The three coefficients $A, B, \lambda = dK_3^2R$ are bounded on the proved compact older trajectory. In the displayed original pair coordinates, $C_* = \max(\lambda + A + B)$ bounds the induced matrix 1-norm. Its integral equation gives $|\Theta_3(t)| + |\Omega_3(t)| \leq (|\Theta_3(t_1)| + |\Omega_3(t_1)|)e^{C_*(t-t_1)}$ by Gronwall's inequality. This finite bound extends the unique linear pair through the same closed interval. Removing the common factor $\exp(-dK_3^2 \int R dt)$ gives a pair whose negative temperature and negative vorticity coordinates have nonnegative derivatives whenever both are positive. Their strictly positive initial values and their integral equations therefore keep both positive through this interval; a first zero contradicts that integral representation. This proves the final sign assertion. $\square$

12.4 The source quotient, pulse equation, and exact return submanifold

The newest physical pair associated to any constructed steering geometry is

$$\frac{d}{dt} \begin{pmatrix} \Theta_3 \\ \Omega_3 \end{pmatrix} = \begin{pmatrix} -dK_3^2R & N_3/(K_3R) \\ K_3Z & -dK_3^2R \end{pmatrix} \begin{pmatrix} \Theta_3 \\ \Omega_3 \end{pmatrix}. \quad (293)$$

For a fundamental matrix equal to I at t_1 , direct differentiation of its two-column determinant gives

$$\det \mathcal{U}(t) = \exp \left(-2dK_3^2 \int_{t_1}^t R(r) dr \right) > 0. \quad (294)$$

Indeed differentiating a determinant column by column multiplies it by the trace of the displayed coefficient matrix; solving that scalar equation gives (294). A nonzero pair therefore never becomes

the zero vector. This does not by itself prevent an individual temperature zero, which is why a finite quotient bound must still be proved when extending through the full negative pulse.

In the chart $\Theta_3 < 0$ define the exact source variables

$$\begin{aligned} v &= \frac{\sigma_2 \Omega_3}{K_3 \chi_1 \Theta_3}, & k &= \frac{\chi_1 N_3}{\sigma_2^2 \sin s R}, & f_m &= \frac{Z_m}{\chi_1 \sin s}, & \delta_3 &= \frac{dK_3^2}{\Gamma}, \\ v' &= f_m - kv^2, & P_3 &= P_{3,1} \exp \left(\int_0^\tau (kv - \delta_3 R) dr \right), \\ \Theta_3 &= -P_3, & \Omega_3 &= -\frac{K_3 \chi_1}{\sigma_2} P_3 v. \end{aligned} \quad (295)$$

Differentiating the quotient cancels the two equal newest damping terms and gives its displayed Riccati row. Differentiating the exponential and product gives exactly (293), including both newest diffusion terms. Thus the map is an exact inverse on every finite quotient interval; on the actual interval of Theorem 5 all its divisions and signs have already been proved.

Put $z_1 = Z_1/(\chi_1 \sin s)$. The forcing is exactly

$$f_m(\tau) = \begin{cases} z_1(1 - \eta(\tau)), & 0 \leq \tau \leq 1, \\ -mL\eta_2(L(\tau - 1)), & 1 \leq \tau \leq T, \\ 0, & \tau \geq T. \end{cases} \quad (296)$$

In particular the source normalization removes χ_1 from the negative pulse exactly, while retaining it in k , in z_1 , and in the amplitude inverse. The actual transition estimates imply

$$\frac{63}{64} < \chi_1 < \frac{3}{2}, \quad \frac{1}{3} < v_1 < \frac{64}{21}, \quad \frac{5}{16} < z_1 < \frac{8}{7}.$$

The length bounds follow from (201), $1 \leq R(t_a) \leq 2$, and $e^{-1/128} > 63/64$, $2e^{1/64} < 9/4$. The bounds for v_1 use $1/2 \leq \sigma_2 \Omega_3/(K_3 \Theta_3) \leq 3$; the bounds for z_1 use (204). They are entry data, not estimates for the unconstructed full pulse.

The exact pulse integral equation, written on its quotient chart, is

$$\begin{aligned} V_m(y) &= v(1 + y/L; m), & H_2(y) &= \int_0^y \eta_2(r) dr, \\ V_m(y) &= v(1; m) - mH_2(y) - \frac{1}{L} \int_0^y k(1 + r/L; m)V_m(r)^2 dr, & 0 \leq y \leq 1. \end{aligned} \quad (297)$$

It follows by substituting $\tau = 1 + y/L$ in the Riccati equation and integrating, retaining the trial dependence of every older coefficient. The required shooting endpoint is precisely $v(T; m) = 0$, because the temperature inverse is strictly negative on a finite quotient interval. Formula (297) specifies that endpoint without a frozen coefficient or a second-stage substitution. Establishing its opposite endpoint signs and exclusion of temperature zeros requires full-interval geometry and quotient estimates. Those estimates have not been supplied by the local theorem.

The exact older forcing signs identify the additional analytic work. For $Z \geq 0$ in the positive x, y, g, Y domain,

$$(gZ - bY)(gZ + bY) = R(DZ^2 - b^2), \quad (xZ - yY)(xZ + yY) = R(Z^2 - y^2).$$

Thus the second parent's horizontal phase changes sign precisely at $Z = b/\sqrt{D}$, and the first parent's at $Z = y$. For the negative pulse $Z \leq 0$, both horizontal components are strictly negative throughout this domain. Their vorticity equations then have strictly positive temperature forcing while still retaining their old vorticity data and their damping. In particular the small negative transition value of Ω_2/σ_2 is not a bound to carry unchanged through steering. Its full coupled growth must be estimated from (286) before an endpoint sign is claimed.

There is an exact invariant return submanifold for the holding equations: prescribe $Z = 0$ and set $\Omega_3 = 0$. The vorticity row in (293) then vanishes identically, while

$$\begin{aligned}\zeta_3 &= (0, \sqrt{R}), & \zeta_2 &= \frac{(-b, g)}{\sqrt{R}}, & \zeta_1 &= \frac{(-y, x)}{\sqrt{R}}, \\ \dot{\Omega}_1 &= \frac{E_1 y}{\sqrt{R}} - dK_1^2 \Omega_1, & \dot{\Omega}_2 &= \frac{E_2 b}{\sqrt{R}} - dK_2^2 D \Omega_2, \\ \dot{\Theta}_3 &= -dK_3^2 R \Theta_3, & \dot{\alpha} &= -\frac{\Omega_1 y^2}{R} - \frac{\Omega_2 b^2}{DR}.\end{aligned}\tag{298}$$

The h, g, E_1, E_2 rows remain those of (286), and the two numerator loss rows remain (290). This restriction of the full vector field proves invariance of the return submanifold, and exhibits the continued motion of both parents. It does not assert that a source trial has reached it. The total local velocity curl on this submanifold is exactly

$$2\dot{\alpha} + \Omega_1 + \Omega_2 = \Omega_1 \left(1 - \frac{2y^2}{R}\right) + \Omega_2 \left(1 - \frac{2b^2}{DR}\right).\tag{299}$$

This follows by taking the curl of the rotational matrix and of each rank-one shear; their respective curls are $2\dot{\alpha}$, Ω_1 , and Ω_2 . No sign for the full third-return curl follows without proving that endpoint's parent amplitudes.

Finally the exact temperature gain and diffusion exponents on any constructed finite quotient interval $[0, \tau_f]$ are

$$\begin{aligned}\log \frac{P_3(\tau_f)}{P_{3,1}} &= \int_0^{\tau_f} kv \, d\tau - \delta_3 \int_0^{\tau_f} R \, d\tau, \\ \delta_1 \tau_f, & \quad \delta_2 \int_0^{\tau_f} D \, d\tau, & \quad \delta_3 \int_0^{\tau_f} R \, d\tau.\end{aligned}\tag{300}$$

Each is exactly $dK_j^2 \int |\zeta_j(t)|^2 dt$ in original physical time. The two explicit losses in (290) also remain part of the older response; common damping of the newest quotient has not removed them. Compact original-profile PDE realization additionally uses the scalar and full vector localization forces already calculated from the exact affine background. Nothing in this steering coordinate map discards those spatial forces.

13 Persistence and completion of the third shooting return

The preceding section constructed the exact positive-diffusion third steering equations and proved an initial interval. This section completes the finite third shooting return. The argument uses the strict margins of the original zero-diffusion stage and uniform continuation of the entire finite trial family. It retains both parent responses, all three diffusion losses, the actual third entry data, and the source pulse normalization. No coefficient is frozen at its entry value.

Fix once and for all the source profiles and the finite frequency data selected in (166). Extend the already constructed finite systems to the closed parameter value $d = 0$ by setting every displayed diffusion term to zero. This extension is an exact member of the same equations. The number $d_{3,\max} > 0$ remains the value in (167).

13.1 The compact entry graph

For $0 \leq d \leq d_{3,\max}$, let $\mathfrak{R}_d \subset [0, 3]^2$ be the set of pairs (m_1, m_2) whose first and second steering endpoint vorticities vanish. The two coordinates refer to the two already constructed pulse families; no uniqueness is imposed. Put

$$\mathfrak{R} = \{(d, m_1, m_2) : 0 \leq d \leq d_{3,\max}, (m_1, m_2) \in \mathfrak{R}_d\}.$$

Lemma 1. *The set $\mathfrak{R}$ is compact and every fibre $\mathfrak{R}_d$ is nonempty. The map which starts from a point of $\mathfrak{R}$, runs the subsequent holds, the third growth to its actual threshold, and the full third transition, has a continuous image $\mathcal{E}$. Hence the set $\mathcal{E}$ of all possible third-steering entry states is compact. If $\mathcal{E}_d$ denotes its fibre at diffusion d , then*

$$\lim_{r \downarrow 0} \sup_{0 \leq d \leq r} \sup_{e \in \mathcal{E}_d} \text{dist}(e, \mathcal{E}_0) = 0. \quad (301)$$

The actual least-root construction at each d is one point of this graph.

Proof. The first and second return theorems construct their complete trial families on fixed compact time intervals, with uniform compact bounds for every $0 \leq d \leq d_{3,\max}$. Their integral equations depend continuously on the initial data, d , and the pulse parameters. The first endpoint-vorticity map is continuous on its full parameter box. The second endpoint-vorticity map is continuous on the resulting closed first-zero graph, which is precisely the relative domain on which the second stage starts from a returned first state. The nested simultaneous zero set is therefore closed in the compact box $[0, d_{3,\max}] \times [0, 3]^2$, so it is compact. Nonemptiness of each fibre is the first- and second-return existence result, including its $d = 0$ specialization.

After a return, the holding equations are ordinary differential equations on the proved compact domain. The third-growth stopping time is the unique transverse crossing proved in Theorem 3; the rate lower bound there keeps the crossing transverse at $d = 0$ as well. Continuous dependence of an ODE flow and the elementary transverse-hitting time lemma therefore make the holds, the third threshold state, and the fixed-duration transition continuous on $\mathfrak{R}$. Their image $\mathcal{E}$ is compact.

To prove (301), suppose it failed. Then there would be $d_n \downarrow 0$ and $e_n \in \mathcal{E}_{d_n}$ whose distance from $\mathcal{E}_0$ is bounded below by a positive number. Compactness of $\mathfrak{R}$ supplies a convergent subsequence of the associated root triples, with limit $(0, m_{1,0}, m_{2,0}) \in \mathfrak{R}$. Continuity of the entry map sends that subsequence to a point of $\mathcal{E}_0$, contradicting the assumed distance. This proof does not require continuity of the least root itself. In particular a jump among zeroes causes no gap. $\square$

At $d = 0$, every point of $\mathcal{E}_0$ is an admissible source entry. Indeed all damping terms vanish, the coordinate maps in the preceding sections are exact inverses, and every earlier endpoint zero obeys the same source endpoint estimates whether or not it is the least zero. Thus the source's later-stage comparison applies to every element of $\mathcal{E}_0$, not only to one chosen root history.

13.2 A nonsingular pair coordinate retaining diffusion

For a third-steering entry $e \in \mathcal{E}_d$, keep its actual numbers $P_{3,1}, \sigma_2, K_3, \chi_1, s$ and define

$$p = -\frac{\Theta_3}{P_{3,1}}, \quad o = -\frac{\sigma_2 \Omega_3}{K_3 P_{3,1}}, \quad \mathcal{K} = \frac{N_3}{\sigma_2^2 \sin s R}. \quad (302)$$

Then $p(0) = 1$ and $o(0) = \chi_1 v_1$. Direct substitution in (293), with $\tau = \Gamma(t - t_1)$, gives the exact system

$$p' = \mathcal{K}o - \delta_3 R p, \quad o' = \frac{Z_m}{\sin s} p - \delta_3 R o, \quad \text{quad} \delta_3 = \frac{dK_3^2}{\Gamma}. \quad (303)$$

No division by Θ_3 occurs in this system. On the open set $p > 0$,

$$v = \frac{o}{\chi_1 p}, \quad k = \chi_1 \mathcal{K}, \quad (304)$$

and differentiating gives, without discarding the common diffusion,

$$v' = \frac{Z_m}{\chi_1 \sin s} - k v^2, \quad (\log p)' = k v - \delta_3 R. \quad (305)$$

These are exactly (295). The first identity records the cancellation of equal newest-layer damping in the quotient; the second retains its complete contribution to the temperature gain.

Combine (292) with (303). Algebraic quantities $D, x, y, R, Y, \zeta_1, \zeta_2, \zeta_3, N_2, N_3$ retain the definitions in (283)–(284). The resulting dimensionless state can be written

$$X = (h, g, E, F, \omega, V, p, o).$$

Together with the constant entry parameters, it has a smooth vector field on the open component domain where the three phase second components, D, R, Y, N_2, N_3 , and the control denominators are positive. The pulse Z_m is smooth at both joins. The vector field is jointly continuous in d, e, m, τ and locally Lipschitz in X , uniformly on compact subsets of that domain.

13.3 Strict zero-diffusion margins

Let $T = 1 + L^{-1}$ and let $F_0(e, m)$ be the source quotient $v(T)$ at $d = 0$. Write $W_0(e, m) = \Omega_2(T)/\sigma_2$ and $G_0(e, m) = \log p(T)$. At a zero of F_0 , set

$$C_0(e, m) = \frac{2\dot{\alpha}(T) + \Omega_1(T) + \Omega_2(T)}{\sigma_2}.$$

The third term $\Omega_3(T)$ is zero there. The constant $c_W > 0$ is the fixed source constant in (3.14) and (3.19).

Lemma 2. *For every $e \in \mathcal{E}_0$, the complete source trial family exists on $[0, T]$ in one common compact subset of the smooth component domain, and $p > 0$. Its endpoint margins are*

$$F_0(e, 0) \geq \frac{1}{8}, \quad F_0(e, 3) \leq -1. \quad (306)$$

The zero set

$$\mathcal{Z}_0 = \{(e, m) \in \mathcal{E}_0 \times [0, 3] : F_0(e, m) = 0\}$$

is compact, and every $(e, m) \in \mathcal{Z}_0$ satisfies

$$W_0(e, m) \geq \frac{1}{12}, \quad G_0(e, m) \geq \frac{1}{6}, \quad C_0(e, m) \geq c_W. \quad (307)$$

Proof. At $d = 0$ the complete finite equations reduce exactly to the source controlled system. The frequency choices remain in the same source family, and the schedule choice places the model error δ_{mdl} inside Lemma 3.9 of the source. In that lemma the comparison constant C_{cmp} and the chosen tolerance obey $C_{\text{cmp}}\delta_{\text{mdl}} \leq 1/4$.

On the first steering part, source (3.40) at $\tau = 1$ gives

$$v(1) \geq \frac{1 - C_{\text{cmp}}\delta_{\text{mdl}}}{2} \geq \frac{3}{8}, \quad v(1) < \bar{v} = 2.$$

Source (3.44) then gives

$$v(T; 0) \geq v(1) - C_{\text{cmp}}\delta_{\text{mdl}} \geq \frac{1}{8}, \quad v(T; 3) \leq v(1) - 3 < -1,$$

which proves (306); weakening the last strict inequality to the displayed non-strict bound is harmless.

For every endpoint zero, source (3.45) gives

$$W_0(T) \geq \frac{1}{3} - C_{\text{cmp}}\delta_{\text{mdl}} \geq \frac{1}{12}.$$

Source (3.46), including $I_0 \geq 2/9$, gives

$$G_0 = \int_0^T kv \, d\tau \geq (1 - C_{\text{cmp}}\delta_{\text{mdl}})I_0 \geq \frac{3}{4} \frac{2}{9} = \frac{1}{6}.$$

At a return the proof of source (3.19) uses this positive preceding-layer vorticity and the complete older feedback sum to give $2\dot{\alpha} + \Omega_1 + \Omega_2 \geq c_W \sigma_2$. That calculation uses only the endpoint bounds shared by every zero, so it does not depend on which zero is selected.

Source Section 3.7.6 first constructs the geometry for every trial in a common compact subset of its smooth domain. It then solves the newest linear pair and excludes a temperature zero by its bounded quotient. Consequently $p > 0$ for every trial. Compactness of $\mathcal{E}_0 \times [0, 3] \times [0, T]$ and continuous dependence make the union of these trajectories compact and give a positive minimum for p . Continuity of F_0 makes $\mathcal{Z}_0$ closed in a compact set, proving its compactness. $\square$

The strict inequalities on $\mathcal{Z}_0$ also control nearby almost-zero endpoints. More precisely, compactness supplies an open neighborhood $\mathcal{U}$ of $\mathcal{Z}_0$ in $\mathcal{E}_0 \times [0, 3]$ on which

$$W_0 > \frac{1}{24}, \quad G_0 > \frac{1}{12}, \quad C_0 > \frac{c_W}{2}. \quad (308)$$

Because F_0 has no zero in the compact complement of $\mathcal{U}$, the number

$$\eta_* = \min_{(e,m) \in (\mathcal{E}_0 \times [0,3]) \setminus \mathcal{U}} |F_0(e, m)| \quad (309)$$

is positive.

13.4 Uniform continuation and the selected return

Theorem 6. *There is a number*

$$0 < d_{\text{ret}} \leq d_{3,\text{max}},$$

determined by the fixed source profiles, the finite frequency data, and the already constructed first two return families before a third root is selected, with the following property. For every $0 < d \leq d_{\text{ret}}$, the entire actual third trial family $m \in [0, 3]$ exists through $\tau = T$. All original matrices and controls are smooth, every source phase second component is positive, and $\Theta_3 < 0$ on every trial. Its endpoint quotient obeys

$$v(T; 0) \geq \frac{1}{16}, \quad v(T; 3) \leq -\frac{15}{16}. \quad (310)$$

Hence its endpoint-zero set is a nonempty compact subset of $(0, 3)$ and has a least member $m_{3,}(d)$.*

Every endpoint zero, including this least one, satisfies

$$\frac{\Omega_2(T)}{\sigma_2} \geq \frac{1}{48}, \quad \log \frac{-\Theta_3(T)}{-\Theta_3(0)} \geq \frac{1}{24}, \quad \frac{2\dot{\alpha}(T) + \Omega_1(T) + \Omega_2(T)}{\sigma_2} \geq \frac{c_W}{4}. \quad (311)$$

At the selected endpoint,

$$Z_{m_{3,*}}(T) = 0, \quad \Omega_3(T) = 0, \quad \Theta_3(T) < 0. \quad (312)$$

Thus the actual third return, its positive preceding-layer vorticity, its positive temperature gain, and its positive origin curl all hold at one fixed positive physical viscosity and thermal diffusivity.

Proof. Let $\mathcal{T}_0$ be the union of the inviscid trajectories in Lemma 2. It is compact and lies in the open smooth component domain. Choose a compact neighborhood $\mathcal{N}$ of $\mathcal{T}_0$ still contained in that domain. Let $p_* > 0$ be the minimum of p on $\mathcal{T}_0$. The vector field and its state derivative are bounded on $\mathcal{N}$, uniformly in $e \in \mathcal{E}_0$, $m \in [0, 3]$, and $\tau \in [0, T]$.

We first prove uniform existence and convergence. If no positive diffusion interval had this property, there would be $d_n \downarrow 0$, entries $e_n \in \mathcal{E}_{d_n}$, and trials $m_n \in [0, 3]$ whose maximal solutions either left $\mathcal{N}$ or failed to reach T . By Lemma 1, after taking a subsequence, $e_n \rightarrow e_0 \in \mathcal{E}_0$; also $m_n \rightarrow m_0 \in [0, 3]$. The integral equations and the uniform state-Lipschitz bound give the usual Gronwall estimate on every common initial interval. Since the vector fields and entries converge, the d_n solutions converge uniformly to the complete inviscid solution from (e_0, m_0) . That solution remains in the interior of $\mathcal{N}$ and has $p \geq p_*$. For large n the d_n solution therefore cannot be the

first one to reach the boundary of $\mathcal{N}$ or $p = p_*/2$; the local continuation theorem then extends it past the alleged endpoint. This contradiction proves that, for all sufficiently small positive d , every trial exists through T , remains in $\mathcal{N}$, and has $p \geq p_*/2$. The same contradiction argument proves uniform convergence of the complete trial family to $\mathcal{T}_0$ as $d \downarrow 0$.

The endpoint observables $F = v = o/(\chi_1 p)$, $W = V(T)$, $G = \log p(T)$, and $\mathcal{C}$ are continuous on this compact family. Uniform convergence therefore permits one common positive radius in d for which

$$\begin{aligned} |F_d(e, m) - F_0(e_0, m)| &< \min\{1/16, \eta_*/2\}, \\ |W_d(e, m) - W_0(e_0, m)| &< 1/48, \\ |G_d(e, m) - G_0(e_0, m)| &< 1/24, \\ |\mathcal{C}_d(e, m) - \mathcal{C}_0(e_0, m)| &< c_W/4, \end{aligned}$$

whenever $e \in \mathcal{E}_d$ and a nearest inviscid entry e_0 is chosen as in (301). Shrink that radius further, if needed, to retain the already proved positive phase-component and matrix domains. Let $\mathcal{A}$ be the set of radii $r \in (0, d_{3,\max}]$ for which these four uniform estimates, all smooth-component inequalities, and $p \geq p_*/2$ hold simultaneously for every $0 \leq d \leq r$. Define

$$d_{\text{ret}} = \frac{1}{2} \sup \mathcal{A}. \quad (313)$$

The preceding argument proves that $\mathcal{A}$ contains a positive interval, so $0 < d_{\text{ret}} \leq d_{3,\max}/2$. Its definition uses only fixed data and complete trial families, before choosing the third root.

Apply the first observable estimate at $m = 0, 3$ and use (306). This gives

$$F_d(e, 0) \geq \frac{1}{8} - \frac{1}{16} = \frac{1}{16}, \quad F_d(e, 3) \leq -1 + \frac{1}{16} = -\frac{15}{16},$$

proving (310). The endpoint map is continuous in m by its integral equation, so the intermediate value theorem gives a zero. Its zero set is closed, excludes both endpoints, and is contained in the compact interval $[0, 3]$; hence it is compact and has a least element.

If $F_d(e, m) = 0$, the first observable estimate gives $|F_0(e_0, m)| < \eta_*$. Definition (309) forces $(e_0, m) \in \mathcal{U}$. Combining (308) with the other three observable estimates proves (311). Finally $p(T) > 0$ means $\Theta_3(T) < 0$. Since $v(T) = 0$ and every factor in (304) is finite and nonzero, $o(T) = 0$ and therefore $\Omega_3(T) = 0$. The pulse is supported in the interior of its interval, so $Z_m(T) = 0$. This proves (312) and all assertions. $\square$

13.5 Exact holding continuation

On the selected endpoint, prescribe $Z = 0$. The restriction (298) then gives

$$\Omega_3(t) = 0, \quad \Theta_3(t) = \Theta_3(t_3^b) \exp\left(-dK_3^2 \int_{t_3^b}^t R(r) dr\right) < 0 \quad (314)$$

for as long as the older system exists. This proves invariance of the return submanifold with the full temperature diffusion retained. The parents continue according to (298); neither is frozen. Since Z is flat at the pulse endpoint and $\Omega_3 = 0$, repeated differentiation of the vorticity equation shows that all right-end derivatives of Ω_3 vanish there. The join is smooth, while Θ_3 follows the nonstationary decay in (314).

The proof is finite-stage. It neither supplies a diffusion interval uniform in the layer number nor establishes the terminal smooth-force extension. Those remain the next separate calculations.

The exact algebra in (303)–(305) and all rational margins in (306)–(311) are checked by `third_return/checks/third_return_persistence_check.py`. The compact-family mechanism and the endpoint implications are displayed in Figure 1.

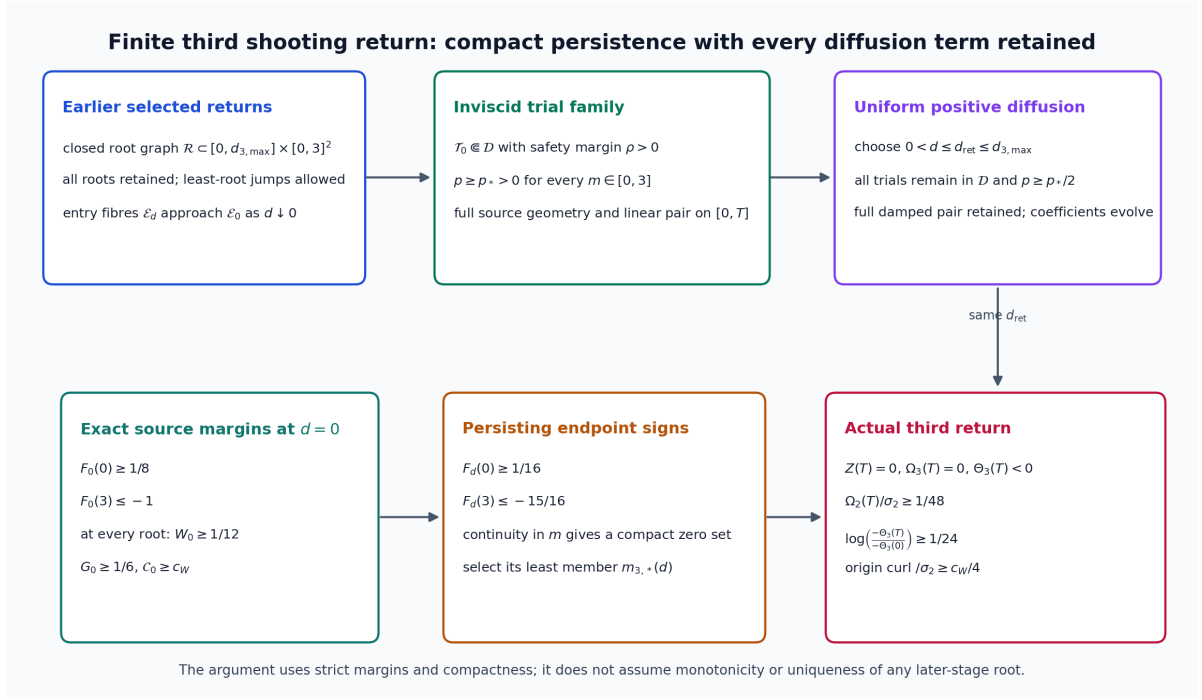


Figure 1: The exact finite persistence argument. The inviscid entry and trial set has a compact trajectory image strictly inside the smooth component domain. The closed earlier-root graph and uniform ODE continuation keep the positive-diffusion family in that interior. The source endpoint margins persist, forcing a third endpoint zero and the displayed return bounds. No monotonicity of the shooting map is used.

14 Compact spatial realization of the actual local third steering

The interval in Theorem 5 has a complete compact realization with the original periodic profile and physical diffusion. We prove the substituted material, support, amplitude, and force bounds here. Both older vorticities remain variables in this realization. We also construct an explicit right continuation of the same steering, which supplies a smooth compact time extension of its forces. The required endpoint in this section is the local steering endpoint, without an assertion about the full negative pulse.

14.1 A proved interval, amplitude bounds, and material matrices

Retain all the data of Theorem 5, and put

$$\begin{aligned} \tau &= \Gamma(t - t_1), \quad \tau_0 = \sin s / 8192, \quad t_* = t_1 + \frac{1}{8192\sigma_2}, \\ \epsilon_R &= \frac{1}{16384\sigma_2}, \quad I_* = [t_1, t_*], \quad I_+ = [t_1, t_* + \epsilon_R]. \end{aligned} \tag{315}$$

Thus $0 \leq \tau \leq 3\tau_0/2$ on I_+ , and the original source steering remains its first ramp $Z = Z_1(1 - \eta(\tau))$ throughout. In particular every original trial $m \in [0, 3]$ gives the same solution on this interval.

The existence argument in Theorem 5 extends to this entire I_+ . Indeed on the same closed cube of radius $1/128$ about its actual six-dimensional entry state, the norm of the scaled vector field is strictly less than $32/\sin s$. On $0 \leq \tau \leq 3\tau_0/2$ its integral can therefore move the state by at most

$$\frac{32}{\sin s} \frac{3 \sin s}{2 \cdot 8192} = \frac{3}{512} < \frac{1}{128}.$$

A first exit from that cube is impossible. Its fixed positive denominator margins and bounded smooth derivatives extend the unique solution to this closed interval, by the same integral-equation

argument. This proves a continuation of the actual source control, including all its endpoint derivatives.

Write the dimensionless state as in (292), with $E = E_1/\sigma_1^2$, $F_2^{\text{amp}} = E_2/\sigma_2^2$, $\omega = \Omega_1/\sigma_1$ and $V = \Omega_2/\sigma_2$. To avoid confusing the second amplitude with the original periodic profile F , denote F_2^{amp} by F_a below. On all of I_+ the already verified cube gives

$$\begin{aligned} 63/64 < h < 4, \quad 11/16 < g < 6, \quad 15/128 < E < 3, \quad 7/128 < F_a < 4, \\ |\omega| < 11, \quad |V| < 1/16, \quad 15/16 < D = h^2 + a^2 < 17, \\ 0 \leq Z \leq Z_1 < (9/8) \sin s < 1/128, \\ b = r_b \sin s < 4 \sin s, \quad 1 \leq r_b \leq \sqrt{10}. \end{aligned} \quad (316)$$

Use the exact x, y, R, Y and phases in (283)–(284). Then

$$\frac{1}{64} < R = \frac{g^2 + b^2}{D} < 39, \quad Y > 1/9, \quad 0 < x < 7, \quad 0 < y < 7. \quad (317)$$

For the upper bound on R , use $(36 + 16/512^2)/(15/16) < 39$. Its lower bound and that of Y are the inequalities proved on the same cube in Theorem 5. Positivity of x, y is also proved there; $x^2 + y^2 = R < 39 < 49$ gives their upper bounds. All three second components stay positive by the exact inverse, without imposing a sign on either older vorticity.

We prove an explicit bound for the newest amplitudes. Set

$$X = -\Theta_3/P_3^*, \quad W = -\sigma_2\Omega_3/(K_3P_3^*), \quad \mathcal{K} = \frac{N_3}{\sigma_2^2 \sin s R}, \quad \mathcal{B} = \frac{Z}{\sin s}.$$

The exact pair (293) becomes

$$X' = \mathcal{K}W - \delta_3 R X, \quad W' = \mathcal{B}X - \delta_3 R W, \quad \delta_3 = dK_3^2/\Gamma. \quad (318)$$

Its entry values satisfy $0 < X(0) \leq e^4$ and $0 < W(0) \leq 3e^4$ by the actual transition theorem. The older geometry is already constructed on the compact interval I_+ , so the coefficient matrix of (318) is smooth and bounded there. Its iterated integral series converges uniformly: if its matrix norm is bounded by B , the term with n integrals is bounded by $(B\tau)^n/n!$ times the initial norm. The series solves the integral equation, and the same iterated estimate on the difference of two solutions proves uniqueness. It therefore constructs the newest pair through all of I_+ independently of the following sign argument. On the cube $N_3 > 0$, since E_1, E_2, c, C, x, y, b are positive. Removing the common damping and integrating the resulting positive off-diagonal linear system proves $X, W > 0$ throughout I_+ . More explicitly, a first zero of either positive coordinate would contradict its positive initial value plus the integral of a nonnegative multiple of the other coordinate.

There is a useful bound on $\mathcal{K}$ that retains the source scale ratio. With $q = \sigma_1/\sigma_2 = s/L$, $\bar{c} = c/\sigma_1^2 \leq 1/4$ and $\bar{C} = C/\sigma_1^2 \leq a/32$, the exact numerator is

$$\frac{N_3}{\sigma_2^2 \sin s} = \frac{q^2}{\sin s} \{ (E + \bar{c})y + \bar{C}x \} + r_b F_a.$$

The expression in braces is less than $7(13/4 + 1/16384) < 25$. Furthermore

$$\frac{q^2}{\sin s} = \frac{s^2}{L^2 \sin s} \leq \frac{16s}{15L^2} \leq \frac{16}{15 \cdot 512 \cdot 16384^2}, \quad \frac{25 \cdot 16}{15 \cdot 512 \cdot 16384^2} < 1.$$

Here $\sin s \geq 15s/16$ is the established source angle estimate. Thus $0 < N_3/(\sigma_2^2 \sin s) < 17$ and

$$0 < \mathcal{K} < 1088, \quad 0 \leq \mathcal{B} < 9/8. \quad (319)$$

Since $X, W > 0$, addition of (318) gives $(X + W)' \leq 1089(X + W)$. Multiplication by $e^{-1089\tau}$ and integration show

$$X + W \leq 4e^4 e^{1089\tau}, \quad 1089\tau \leq \frac{3267 \sin s}{16384} < \frac{1}{4}.$$

The power series gives $e^z \leq \sum_{n \geq 0} z^n = (1 - z)^{-1} \leq 4/3$ for $0 \leq z \leq 1/4$. Consequently on the required interval and its proved right collar,

$$0 < -\Theta_3 < 6e^4 P_3^*, \quad |\Omega_3| < 6e^4 K_3 P_3^* / \sigma_2. \quad (320)$$

This uses the signed linear pair before any quotient estimate.

The exact matrices remain (288); all retain their original birth times. With $h_0 = \cos s_2$ and $N = 1 + (3 - h_0)/a$, define

$$\mathbf{N}_1 = 1, \quad \mathbf{N}_2 = 2N, \quad \mathbf{N}_3 = 32/\sin s. \quad \|M_j\|, \|M_j^{-1}\| \leq \mathbf{N}_j. \quad (321)$$

For the second matrix, the triangular factor and its inverse have norm at most $1 + |h - h_0|/a$. Since $h > 0$, $h < 4$, and $h_0 < 1$, one has $|h - h_0| < 4 - h_0$. The difference $2N - [1 + (4 - h_0)/a] = 1 + (2 - h_0)/a$ is positive. For the third, use $\mathcal{L} = (\zeta_2 \ \zeta_3)$. Its determinant is $-b$ and $\|\mathcal{L}\|_F^2 = D + R < 56$, whereas $\|\mathcal{L}(t_b)\|_F^2 = r_b^2 + 1 \leq 11$. The adjugate identity for two by two matrices therefore bounds both products defining M_3 and its inverse by $\sqrt{616}/b < 32/\sin s$. This calculation allows both older shears to evolve. The matrices have determinant one and define the same global linear diffeomorphisms of $\mathbb{R}^2$.

14.2 Support nesting with the unchanged source choice

Let C_M, C_N and Y_3 retain their exact values in (166) and (246). Put

$$p = 47Q_2/16 - 3, \quad r = (31Q_2 + 1)/16.$$

The literal outer $+1$ in (166) gives the strict stronger margins

$$\widehat{\lambda}_1^p \geq e^p (4C_N C_M), \quad \widehat{\lambda}_1^r \geq e^r \sqrt{11} C_M / s_0. \quad (322)$$

Indeed $\log \widehat{\lambda}_1 \geq Y_3$ is at least one plus either logarithmic summand divided by its positive exponent. Also $N \leq C_N \widehat{\lambda}_1^{1/16}$ and $\mathbf{N}_3 \leq 2C_M \widehat{\lambda}_1^{(Q_2-1)/16}$, by (246). Multiplying these exact powers proves

$$\begin{aligned} \frac{2\mathbf{N}_2 \mathbf{N}_3 \widehat{\lambda}_2^{-3}}{\widehat{\lambda}_1^{-3}} &\leq 2e^{-p} < 1, \\ \frac{\sqrt{17} \mathbf{N}_3 \widehat{\lambda}_2^{-2}}{s_0} &\leq 2\sqrt{17/11} e^{-r} < 1. \end{aligned}$$

For the last strict inequalities, $Q_2 \geq 202$ implies $p > 1$ and $r > 2$. The elementary inequality $e^z \geq 1 + z$ therefore gives $e^p > 2$ and $e^r > 3$, and $2\sqrt{17/11} < 3$ follows from $68 < 99$. The earlier source margins give

$$\begin{aligned} 2\mathbf{N}_2 \mathbf{N}_3 \widehat{\lambda}_2^{-3} &< \widehat{\lambda}_1^{-3}, \quad \sqrt{17} \mathbf{N}_3 \widehat{\lambda}_2^{-2} < s_0, \\ \mathbf{N}_2 \widehat{\lambda}_1^{-3} &< s_0 / (2C_\ell), \quad \mathbf{N}_2 \widehat{\lambda}_1^{-2} < s_0. \end{aligned} \quad (323)$$

For the last two use $\mathbf{N}_2 = 2N$ and the proved first-source margins $N \widehat{\lambda}_1^{-3} < s_0 / (4C_\ell)$ and $N \widehat{\lambda}_1^{-2} < s_0 / 2$.

Keep exactly the radii, periodic F , its even zero-mean primitive P , and material cutoffs in (249). For every $t \in I_+$ a third support point satisfies $|x| \leq \mathbf{N}_3 \ell_3$. Thus

$$|M_2^{-1}x| \leq \mathbf{N}_2 \mathbf{N}_3 \ell_3 < \ell_2/2, \quad |K_2 \zeta_2 \cdot x| < K_2 \sqrt{17} \mathbf{N}_3 \ell_3 < s_0.$$

The entire second field is affine in a neighborhood of that point. For every second support point the last two inequalities in (323) give $|x| < \ell_1/2$ and $|K_1 \zeta_1 \cdot x| < s_0$. The first material matrix is a rotation, so the first envelope is one there and its profile is linear. Finally the original first support remains inside the original affine cell of the radial base. The same inclusions hold for every derivative support because their containing closed supports are unchanged. All the inequalities are strict, so these are open affine neighborhoods in the original physical coordinates.

14.3 Full fields and exact scalar and vector equations

Define the base and all three increments by (250) and (251), using the exact solution on I_+ . All activation weights are one on this interval. The fields agree at t_1 , with every derivative, with the already constructed fields through the third transition: the exact control and phase jets match by the proof of Theorem 5, and repeated differentiation of the amplitude and material equations matches their jets. They therefore give one smooth field on $[0, t_* + \epsilon_R]$, with the original sine initial temperature and zero initial velocity.

Use the exact scalar and full vector increments (252)–(255), and the base and total forces (256). These formulas remain the full residuals on I_+ , as can be checked directly here. On a j th support the proved older field is precisely $G_{<j} \cdot x$ and $D_{<j}x$ with the unchanged sums in (243). The identities $\partial_t k_j = -D_{<j}^T k_j$ and $\partial_t M_j^{-1} = -M_j^{-1} D_{<j}$ give

$$(\partial_t + D_{<j}x \cdot \nabla)\xi_j = 0, \quad (\partial_t + D_{<j}x \cdot \nabla)g_j = 0.$$

In the scalar equation, the Fg_j part of $v_j \cdot G_{<j}$ cancels the exact $\dot{\Theta}_j$ coupling, leaving the cutoff term, self-advection, modal damping, and full Laplacian exactly as in (253). In the vector equation, trace zero gives $D_{<j}J + JD_{<j}^T = 0$, whence the material derivative of $J\nabla\Psi_j$ contributes $D_{<j}v_j$. The reverse interaction $(v_j \cdot \nabla)(D_{<j}x)$ is a second $D_{<j}v_j$. Adding its other terms gives (254), including both parent velocities, the full buoyancy, and the moving-length term in Q_r . Adding the three identities counts every ordered older/newer interaction once; outside their supports every derivative vanishes. The identities therefore hold globally on $\mathbb{R}^2$.

Together with the base residual this proves exactly (238) on $[0, t_* + \epsilon_R]$, with $p = 0$. The streamfunction identity gives $\operatorname{div} u = 0$. Every spatial support remains in B_{RH} by the proved nesting. The phase-coordinate Laplacian is still precisely (257), with its evolving positive matrix $M_j^{-1}M_j^{-T}$ and all mixed derivatives. On a linear profile cell the viscous force still includes $-d\rho_j V_j$ and $-d\rho_j v_j$; the corresponding Laplacians vanish there by (258). The full fields retain the constant $P(0)$ in their cutoff region. Thus the original profile has not been replaced by a modal eigenfunction.

14.4 Actual all-order spatial and diffusion costs

The following explicit data hold on the whole proved I_+ :

$$\begin{aligned} r_{-,j}^2 &= (1, 15/16, 1/64)_j, & r_{+,j}^2 &= (1, 17, 39)_j, & \kappa_j &= K_j r_{+,j}, \\ U_{\Theta,j} &= (3\sigma_1^2/K_1, 4\sigma_2^2/K_2, 6e^4 P_3^*)_j, \\ U_{\Omega,j} &= (11\sigma_1, \sigma_2/16, 6e^4 K_3 P_3^*/\sigma_2)_j, & W_{j,1} &= 0, \\ A_* &= 693\sigma_1 + 12\Gamma + (81/8)\Gamma \sin s [\eta']_0. \end{aligned} \tag{324}$$

The older data are the actual cube estimates, rather than their smaller preceding-transition bounds. To verify $|\dot{\alpha}| \leq A_*$, use (287). Here $Y > 1/9$, $y < 7$, $|\zeta_{1,1}| \leq 1$, $|\zeta_{2,1}| < 5$, $b/D < (64/15) \sin s$, $|\Omega_1| < 11\sigma_1$, and $|\Omega_2| < \sigma_2/16$. Therefore

$$|\mathcal{F}_3| < 9 \left(77\sigma_1 + \frac{64 \sin s}{15} \frac{\sigma_2}{16} 5 \right) = 693\sigma_1 + 12\Gamma.$$

The other control term has size $|\dot{Z}|/Y \leq (81/8)\Gamma \sin s [\eta']_0$, which proves the stated bound using the exact source derivative.

The following finite sums specify every spatial cost, with $f_m, [F]_m, [P]_m, H_m, c_m$ defined as in Section 11:

$$\begin{aligned} D_j &= A_* + \sum_{i < j} U_{\Omega,i}, & G_j &= A_0 + \sum_{i < j} K_i U_{\Theta,i} r_{+,i}, \\ Q_j &= \frac{U_{\Omega,j}}{K_j^2 r_{-,j}^2}, & Q_{r,j} &= \frac{\kappa_j U_{\Theta,j} + 2D_j U_{\Omega,j}}{K_j^2 r_{-,j}^2}, \end{aligned} \tag{325}$$

$$\begin{aligned}
\mathbf{e}_{j,m} &= (\mathbf{N}_j/\ell_j)^m f_m, & \mathbf{B}_{j,m}(W) &= \sum_{q=0}^m \binom{m}{q} \kappa_j^q [W]_q \mathbf{e}_{j,m-q}, \\
\mathbf{C}_{j,m}(P) &= \sum_{q=0}^m \binom{m}{q} \kappa_j^q [P]_q \mathbf{e}_{j,m-q+1}.
\end{aligned}$$

In the occurrences $\mathbf{B}_{j,m}(W)$ the argument is a periodic test profile, not the signed amplitude W of (318). The operator norm of $J\zeta_i \otimes \zeta_i/r_i^2$ equals one, so $\|D_{<j}\| \leq D_j$ with the entire second vorticity included. The gradient sum similarly gives $|G_{<j}| \leq G_j$. Differentiating ρ_j gives $|\dot{\rho}_j/\rho_j| \leq 2D_j$. These prove the bounds $|Q_j| \leq Q_j$ and $|Q_{r,j}| \leq Q_{r,j}$. The product rule with its $\binom{m}{q}$ allocations and the linear material bound (321) proves $|W(\xi_j)g_j|_m \leq \mathbf{B}_{j,m}(W)$ and $|P(\xi_j)\nabla g_j|_m \leq \mathbf{C}_{j,m}(P)$.

In particular the complete damping and diffusion terms obey

$$\begin{aligned}
D_{\theta,j,m} &= d\mathbf{U}_{\Theta,j} \{ \kappa_j^2 \mathbf{B}_{j,m}(F) + 2\mathbf{B}_{j,m+2}(F) \}, \\
D_{u,j,m} &= d\mathbf{Q}_j \{ \kappa_j^2 \mathbf{B}_{j,m+1}(P) + 2\mathbf{B}_{j,m+3}(P) \}.
\end{aligned} \tag{326}$$

The two coordinate traces of the Laplacian account for the factor two in each expression. Direct termwise application to the full residuals proves the bounds

$$\begin{aligned}
\mathbf{E}_{\theta,j,m} &= \mathbf{Q}_j \mathbf{G}_j \mathbf{C}_{j,m}(P) + D_{\theta,j,m} \\
&\quad + \mathbf{Q}_j \mathbf{U}_{\Theta,j} \sum_{q=0}^m \binom{m}{q} \mathbf{B}_{j,q+1}(P) \mathbf{B}_{j,m-q+1}(F), \\
\mathbf{E}_{u,j,m} &= (2D_j \mathbf{Q}_j + \mathbf{Q}_{r,j}) \mathbf{B}_{j,m+1}(P) + \mathbf{U}_{\Theta,j} \mathbf{B}_{j,m}(F) + D_{u,j,m} \\
&\quad + \mathbf{Q}_j^2 \sum_{q=0}^m \binom{m}{q} \mathbf{B}_{j,q+1}(P) \mathbf{B}_{j,m-q+2}(P), \\
|E_{\theta,j}|_m &\leq \mathbf{E}_{\theta,j,m}, \quad |E_{u,j}|_m \leq \mathbf{E}_{u,j,m}.
\end{aligned} \tag{327}$$

Indeed the scalar product sum bounds $v_j \cdot \nabla V_j$; the vector sum bounds $(\nabla v_j)v_j$, allocating all m derivatives in each case. The remaining terms are exactly the linear terms of (253) and (254). There is no activation derivative on I_+ because all weights are one.

Let $B_{0,m}(F) = \sum_{q=0}^m \binom{m}{q} \mu^q [F]_q c_{m-q}$. The control constant $\mathbf{A}_2$ bounding $|\ddot{\alpha}|$ is explicitly defined below. The base and sums now give global norms on the original $\mathbb{R}^2$:

$$\begin{aligned}
|\theta|_m &\leq (A_0/\mu) B_{0,m}(F) + \sum_{j=1}^3 \mathbf{U}_{\Theta,j} \mathbf{B}_{j,m}(F), \\
|u|_m &\leq \mathbf{A}_* H_m + \sum_{j=1}^3 \mathbf{Q}_j \mathbf{B}_{j,m+1}(P), \\
|f_\theta|_m &\leq 2d(A_0/\mu) B_{0,m+2}(F) + \sum_{j=1}^3 \mathbf{E}_{\theta,j,m}, \\
|f_u|_m &\leq \mathbf{A}_2 H_m + \mathbf{A}_*^2 \sum_{q=0}^m \binom{m}{q} H_{q+1} H_{m-q} + (A_0/\mu) B_{0,m}(F) + 2d\mathbf{A}_* H_{m+2} + \sum_{j=1}^3 \mathbf{E}_{u,j,m}.
\end{aligned} \tag{328}$$

These follow from the displayed fields and base residual; the initial interpolation is already complete on I_+ . They bound the entire fields and forces, not only their common affine cell. For each fixed bound in this display, integration over the required interval multiplies it by $1/(8192\sigma_2)$; over I_+ the factor is $3/(16384\sigma_2)$. The earlier proved norms together with these, taking the maximum on their two adjacent intervals, give the corresponding bounds through t_* .

The exact retained localization and newest diffusion factors include

$$\begin{aligned} N_1/\ell_1 &= C_\ell \mu/s_0, & N_2/\ell_2 &= 2N\mu\hat{\lambda}_1^3, & N_3/\ell_3 &= 32\mu\hat{\lambda}_2^3/\sin s, \\ dU_{\Theta,3}K_3^2 &= 6e^4 dA_0\mu\hat{\lambda}_3^{9/8}, & dQ_3K_3^3 &= 384e^4 dA_0\mu\hat{\lambda}_3^{9/8}/\sigma_2. \end{aligned} \quad (329)$$

They are explicit factors in upper force costs. The exact three modal exponents over I_* are

$$dK_1^2|I_*|, \quad dK_2^2 \int_{I_*} D dt, \quad dK_3^2 \int_{I_*} R dt. \quad (330)$$

The second is between $(15/16)dK_2^2|I_*|$ and $17dK_2^2|I_*|$, and the third between $dK_3^2|I_*|/64$ and $39dK_3^2|I_*|$. The exact newest gain is (300) with $\tau_f = \tau_0$; all divisions there are valid by positivity of the pair just proved. Both older numerator-loss terms in (290) remain present, separately from these modal exponents.

14.5 Exact source jets, mixed costs, and compact time forces

We give an explicit finite recursion for all control derivatives. Let $z = (h, g, E, F_a, \omega, V, X, W)$ and introduce the switch jets $\lambda^{[j]}$. Define $Z = Z_1(1 - \lambda^{[0]})$, and use the exact inverse (284) with this Z . The eight-dimensional vector field $\mathcal{V}^{\text{loc}}$ uses the coordinate right-hand sides in (292) with $F = F_a$, ordered as $(h, g, E, F_a, \omega, V)$, followed by the two right-hand sides in (318). In particular it depends on $\lambda^{[0]}$, not its derivatives. Define the exact physical angular function

$$\Phi(z, \lambda^{[0]}, \lambda^{[1]}) = \frac{\sigma_1 \omega y \zeta_{1,1} + (b\sigma_2 V/D) \zeta_{2,1} + \Gamma Z_1 \lambda^{[1]}}{Y}. \quad (331)$$

Substituting $\lambda^{[j]} = \eta^{(j)}(\tau)$ gives exactly $\dot{\alpha}$ in (287), because $\dot{Z} = -\Gamma Z_1 \eta'(\tau)$. Thus every source-ramp derivative retains its original factor $\Gamma^r Z_1$. No constant switch is used on the collar.

On the explicit compact rectangle

$$\begin{aligned} h &\in [63/64, 4], & g &\in [11/16, 6], & E &\in [15/128, 3], & F_a &\in [7/128, 4], \\ \omega &\in [-11, 11], & V &\in [-1/16, 1/16], & X, W &\in [0, 6e^4], & \lambda^{[0]} &\in [0, 1], \end{aligned} \quad (332)$$

the estimates for D, R, Y used above are uniform and positive. The same elementary inequalities apply at its boundary, since their final margins are strict even with all displayed weak bounds. The vector field and Φ are consequently smooth on a neighborhood of this rectangle for any finite control-jet bounds. Define the derivation

$$\mathcal{C}_{\text{loc}} = \sum_{l=1}^8 \mathcal{V}_l^{\text{loc}} \partial_{z_l} + \sum_{j \geq 0} \lambda^{[j+1]} \partial_{\lambda^{[j]}}. \quad (333)$$

Each application to a finite expression uses only finitely many terms. The chain rule gives, by induction on n ,

$$\begin{aligned} \alpha^{(n+1)} &= \Gamma^n \mathcal{C}_{\text{loc}}^n \Phi|_{\lambda^{[j]} = \eta^{(j)}(\tau)}, \\ A_{n+1} &= \Gamma^n \max \left\{ |\mathcal{C}_{\text{loc}}^n \Phi| : z, \lambda^{[0]} \text{ in (332)}, |\lambda^{[j]}| \leq [\eta^{(j)}]_0, 1 \leq j \leq n+1 \right\}, \quad n \geq 0. \end{aligned} \quad (334)$$

These are finite maxima of specified elementary expressions on specified compact rectangles. They contain no supremum over a future trajectory. The preceding identity proves $|\alpha^{(n+1)}| \leq A_{n+1}$; the sharper A_* in (324) may be used at first order. In particular the base constant in (328) is precisely A_2 .

For clarity, the induction behind the displayed identity differentiates a function of z along its eight exact state rows and differentiates each $\eta^{(j)}$ to $\eta^{(j+1)}$. Physical time adds a factor Γ at each step. Conversely substituting the state scalings into these rows recovers the six older physical

equations and newest physical pair; their inverse scalings are exact and all scale denominators are positive.

All mixed space and time norms are then evaluated by the complete pullback operators (272) and their identity (273), replacing the transition derivation by (333) and its rectangle by (332). Explicitly start from the zeroth constants (321) and (324) and differentiate

$$\dot{M}_j = D_{<j} M_j, \quad \dot{A}_j = -A_j D_{<j}, \quad \dot{k}_j = -D_{<j}^T k_j, \quad A_j = M_j^{-1} / \ell_j.$$

For example $\partial_t^{n+1} A_j = -\sum_{q=0}^n \binom{n}{q} (\partial_t^q A_j) (\partial_t^{n-q} D_{<j})$. The product rule gives the other two recursions with the displayed matrix order and signs. Coefficient jets of the parent sums are given by (333); all activation jets on I_+ vanish. On every layer support bound each factor x^β in the pullback expression by $(N_j \ell_j)^{|\beta|}$, each periodic derivative by its original $[F]_r$ or $[P]_r$, and each envelope derivative by f_r . Sum the resulting finite absolute products with the binomial coefficients produced by these recursions. The chain and product rules prove the resulting bound for every prescribed mixed order. On the base use R_H , its fixed profile derivatives, and (334) instead. The recursion has finite depth equal to the requested derivative order and retains every time derivative of the actual Laplacian metric and modal diffusion.

Finally extend the original sine datum constantly to negative time with zero velocity as in Section 11, and take $\epsilon_L = 1/\nu$. Let $f_\bullet^{\text{raw}}$ be the full residuals (277) of these fields on $[-\epsilon_L, t_* + \epsilon_R]$. On the nonnegative interval they equal the proved telescoping force formulas. Define

$$\beta_*(t) = \eta(1 + t/\epsilon_L) \eta(1 + (t_* - t)/\epsilon_R), \quad \tilde{f}_\bullet = \beta_* f_\bullet^{\text{raw}}. \quad (335)$$

Here the right collar is the actual source steering continuation already constructed, with physical length ϵ_R in (315). The cutoff equals one on $[0, t_*]$; flatness of the original step makes extension by zero smooth at both outer times. The forces are therefore smooth and compact in space and time, and solve the exact PDE on the entire required $[0, t_*]$. Their complete cutoff costs are

$$\begin{aligned} B_{\beta,r} &= \sum_{q=0}^r \binom{r}{q} \epsilon_L^{-q} \epsilon_R^{-(r-q)} [\eta^{(q)}]_0 [\eta^{(r-q)}]_0, \\ \|\partial_t^n \partial_x^\gamma \tilde{f}_\bullet\|_\infty &\leq \sum_{q=0}^n \binom{n}{q} B_{\beta,q} \|\partial_t^{n-q} \partial_x^\gamma f_\bullet^{\text{raw}}\|_\infty. \end{aligned} \quad (336)$$

The raw mixed norms on I_+ have the explicit recursion just proved, those on the preceding slab have the proved earlier recursions, and those on the constant left extension are obtained by differentiating its fixed sine datum and residual. Taking their maximum gives every norm in (336).

Oddness of the original scalar profile, evenness of its primitive and cutoffs, and the linear material maps give the same odd temperature, velocity and forces and even vorticity as before. The compact smooth streamfunction has finite-energy velocity and still satisfies the exact Biot–Savart identity proved in Section 11. Thus the local source steering, its moving endpoint, and its proved right collar have a complete compact original-profile PDE realization with all physical diffusion, localization, and control costs specified.

15 Exact coupled feedback and the scope of common damping

Retain the source layer frequencies λ_j , and write $K_j = \mu \lambda_j$ for physical frequencies. Let Θ_j, Ω_j denote physical amplitudes. For each activated layer j , let $\zeta_j \neq 0$, $r_j = |\zeta_j|$, $e_j = \zeta_j / r_j$, $S_j = J e_j \otimes e_j$, and $A_j = -K_j r_j \Theta_j$. The exact physical counterpart of source (3.3)–(3.5) is

$$\begin{aligned} G_{<q} &= -A_0 e_0 + \sum_{j<q} w_j K_j \Theta_j \zeta_j, & D_{<q} &= \dot{A} J + \sum_{j<q} w_j \Omega_j S_j, \\ \dot{M}_q &= D_{<q} M_q, \quad M_q(t_q^{\text{in}}) = I, & \zeta_q &= M_q^{-T} p_q, \end{aligned}$$

$$\dot{\zeta}_q = -D_{<q}^T \zeta_q, \quad \dot{\Theta}_q = -\frac{J\zeta_q \cdot G_{<q}}{K_q r_q^2} \Omega_q, \quad \dot{\Omega}_q = K_q \zeta_{q,1} \Theta_q. \quad (337)$$

Each sum begins at $j = 1$, and contains finitely many terms at every finite preterminal time. The rotating base coefficient is the original A_0 , not a reset unit value. The physical growth-scale definition is $\sigma_q^2 = |G_{<q+1}(t_q^b)|$, so $\sigma_q = \nu \tilde{\sigma}_q$.

For $\zeta_{j,2} \neq 0$ define

$$\mathcal{F}_j = -\frac{\sum_{i<j} w_i \Omega_i e_{i,1} (J e_i \cdot \zeta_j)}{\zeta_{j,2}}, \quad \mathcal{F}_0 = 0. \quad (338)$$

Transposing S_i gives $S_i^T \zeta_j = e_i (J e_i \cdot \zeta_j)$, and the first component of $-\dot{\alpha} J^T \zeta_j$ is $-\dot{\alpha} \zeta_{j,2}$. Hence

$$\dot{\zeta}_{j,1} = (\mathcal{F}_j - \dot{\alpha}) \zeta_{j,2}. \quad (339)$$

It follows by direct differentiation that $\dot{\alpha} = \mathcal{F}_j - \dot{Z}/\zeta_{j,2}$ gives $\dot{\zeta}_{j,1} = \dot{Z}$; equality of initial values gives $\zeta_{j,1} = Z$ on its entire interval of definition. This proves the control map exactly, including its denominator and the laboratory component that drives buoyancy.

There is no omitted parent evolution in (337). For example, differentiation of the definitions gives

$$\dot{G}_{<q} = -A_0 \dot{\alpha} J e_0 + \sum_{j<q} K_j \{ (\dot{w}_j \Theta_j + w_j \dot{\Theta}_j) \zeta_j - w_j \Theta_j D_{<j}^T \zeta_j \}, \quad (340)$$

$$\dot{D}_{<q} = \ddot{\alpha} J + \sum_{j<q} \{ (\dot{w}_j \Omega_j + w_j \dot{\Omega}_j) S_j + w_j \Omega_j \dot{S}_j \}, \quad (341)$$

$$\dot{S}_j = -\frac{J D_{<j}^T \zeta_j \otimes \zeta_j + J \zeta_j \otimes D_{<j}^T \zeta_j}{r_j^2} + \frac{2 \zeta_j \cdot D_{<j}^T \zeta_j}{r_j^2} S_j. \quad (342)$$

To verify the last line, use $S_j = J \zeta_j \otimes \zeta_j / r_j^2$, $\dot{\zeta}_j = -D_{<j}^T \zeta_j$, and $(r_j^2)' = -2 \zeta_j \cdot D_{<j}^T \zeta_j$. These three identities give exactly the three displayed terms by the quotient rule. Equations (340)–(342), together with the amplitude rows, are exact equations for both inherited backgrounds, including activation and the common angular control.

15.1 Damped amplitude equations as an exact controlled realization

For a sine profile on an affine background, retaining physical diffusion gives the amplitude rows

$$\begin{aligned} \dot{\Theta}_j &= -\kappa K_j^2 r_j^2 \Theta_j - \frac{J \zeta_j \cdot G_{<j}}{K_j r_j^2} \Omega_j + c_{\theta,j}, \\ \dot{\Omega}_j &= -\varepsilon K_j^2 r_j^2 \Omega_j + K_j \zeta_{j,1} \Theta_j + c_{\omega,j}. \end{aligned} \quad (343)$$

For an actual finite family these equations are coupled to the phase rows, D, G , and the control; the symbols $c_{\theta,j}$ and $c_{\omega,j}$ here explicitly record added amplitude forcing. Inserting them in (340)–(341) produces the exact diffusion contributions

$$\begin{aligned} (\dot{G}_{<q})_{\text{amplitude diffusion}} &= -\kappa \sum_{j<q} w_j K_j^3 r_j^2 \Theta_j \zeta_j, \\ (\dot{D}_{<q})_{\text{amplitude diffusion}} &= -\varepsilon \sum_{j<q} w_j K_j^2 r_j^2 \Omega_j S_j, \end{aligned} \quad (344)$$

and additional terms $\sum w_j K_j c_{\theta,j} \zeta_j$ and $\sum w_j c_{\omega,j} S_j$, respectively. The phase geometry and feedback then use those changing backgrounds, rather than their inviscid values.

For the released general F , (343) has a different precise meaning: after substitution into the exact uncut scalar and vorticity residuals, the wave forces are

$$R_{\theta,j} = c_{\theta,j} F - \kappa K_j^2 r_j^2 \Theta_j (F + F''),$$

$$R_{\omega,j} = c_{\omega,j}F' - \varepsilon K_j^2 r_j^2 \Omega_j (F' + F'''). \quad (345)$$

These formulas follow because $\Delta[\Theta_j F(K_j \zeta_j \cdot x)] = K_j^2 r_j^2 \Theta_j F''$ and $\Delta[\Omega_j F'(K_j \zeta_j \cdot x)] = K_j^2 r_j^2 \Omega_j F'''$. On an affine core they are $c_{\theta,j}s - \kappa K_j^2 r_j^2 \Theta_j s$ and $c_{\omega,j} - \varepsilon K_j^2 r_j^2 \Omega_j$. Thus damping these amplitudes for this profile requires force *inside* the affine core. Taking $c_{\theta,j} = \kappa K_j^2 r_j^2 \Theta_j$ and $c_{\omega,j} = \varepsilon K_j^2 r_j^2 \Omega_j$ restores exactly the original coupled amplitude rows and leaves instead $-\kappa K_j^2 r_j^2 \Theta_j F''$ and $-\varepsilon K_j^2 r_j^2 \Omega_j F'''$. These vanish in the core and are precisely the compensation used in (11)–(12), with all cutoff terms additionally retained there.

15.2 The exact defect of multiplying all old amplitudes by damping

To quantify a proposed map, let a finite inviscid coupled trajectory be decorated with E , and suppose for this calculation that $\kappa = \varepsilon = \delta > 0$. Keep its phases and angle and put

$$d_j(t) = \exp \left[-\delta K_j^2 \int_{t_{\text{in}}^j}^t r_j(s)^2 ds \right], \quad \Theta_j^d = d_j \Theta_j^E, \quad \Omega_j^d = d_j \Omega_j^E.$$

The backgrounds computed from these candidate fields differ by

$$\Delta G_{<q} = \sum_{j<q} w_j K_j (d_j - 1) \Theta_j^E \zeta_j, \quad \Delta D_{<q} = \sum_{j<q} w_j (d_j - 1) \Omega_j^E S_j. \quad (346)$$

Differentiating the candidate fields and using the inviscid equations gives the exact defects

$$\dot{\zeta}_q + (D_{<q}^d)^T \zeta_q = \Delta D_{<q}^T \zeta_q, \quad (347)$$

$$\dot{\Theta}_q^d + \delta K_q^2 r_q^2 \Theta_q^d + \frac{J \zeta_q \cdot G_{<q}^d}{K_q r_q^2} \Omega_q^d = \frac{J \zeta_q \cdot \Delta G_{<q}}{K_q r_q^2} \Omega_q^d, \quad (348)$$

$$\dot{\Omega}_q^d + \delta K_q^2 r_q^2 \Omega_q^d - K_q \zeta_{q,1} \Theta_q^d = 0. \quad (349)$$

For instance the derivative of d_q cancels the diagonal diffusion term, leaving the difference of the two temperature couplings; this is precisely (348), with its positive sign. Also the feedback computed at these same phases has the exact difference

$$\mathcal{F}_j^d - \mathcal{F}_j^E = - \frac{\sum_{i<j} w_i (d_i - 1) \Omega_i^E e_{i,1} (J e_i \cdot \zeta_j)}{\zeta_{j,2}}. \quad (350)$$

Changing $\dot{\alpha}$ by this expression cancels the first component of (347) for that controlled index, as follows from (339); it does not in general cancel its second component or the equations for other phases. Equations (347)–(350) are an exact relation between the two candidate systems, rather than an assertion that the systems are unrelated.

The defects are nonzero even in a particularly transparent actual holding geometry. After the first stage, $\zeta_1 = e_2$, $\Omega_1 = 0$, and $\Theta_1 < 0$. At the start of the next growth interval $\zeta_2 = e(s_2)$, $s_2 > 0$. For the damping candidate with $d_1 < 1$,

$$J \zeta_2 \cdot \Delta G_{<2} = (d_1 - 1) K_1 \Theta_1 \sin s_2 \neq 0.$$

All factors on the right are nonzero with the indicated signs. Thus even when the old shear vanishes exactly on a hold, its decaying temperature changes the next layer's amplification coefficient. The common scalar damping identity for a fixed prescribed D, G has not proved a conjugacy of this coupled system.

For the compact finite stage of Section 1, the stronger exact forced relation is available: retain the original coupled fields and add the explicitly calculated Laplacian forces. It preserves both backgrounds and all controls, and quantifies the entire cost. The next section records precisely why this particular relation has no smooth-force extension along the unchanged infinite source trajectory.

16 The exact cost along the released infinite trajectory

The finite construction above proves no infinite viscous singularity. There is a precise calculation for the particular proposed extension that retains the released infinite fields and changes only the forces by their Laplacians. We record both that calculation and its scope.

The source trajectory in this section means the specific construction of Alpöge–Buckmaster, including their compact corrections. Its existence and infinite inviscid conclusions are the source's Theorems 3.2, 7.3, 8.2, and 1.1, not new conclusions of this note. We use the following exact properties established in those proofs: at the growth crossing t_q^a , activation is complete and

$$\Theta_q(t_q^a) = -\frac{\nu^2}{\mu} \lambda_q^{-7/8}, \quad r_q(t_q^a) \geq r_* > \frac{1}{2}, \quad \left| \frac{\Omega_q(t_q^a)}{\Theta_q(t_q^a)} \right| \geq \frac{K_q}{2\sigma_{q-1}}. \quad (351)$$

The first is the definition of the stopping time, source (3.8) and item (i) after (3.11), p. 10. Here $r_* := 1 - C_r \epsilon_{\text{sch}} > 1/2$ is the improved phase-length bound, with the fixed constant C_r from the source estimate; it is not the separate bootstrap constant called ζ_- in source (3.14). The second is source (3.18), p. 13 and the tolerance choice on p. 12. The last is the tracking estimate in the complete growth proof, pp. 29–30, between (3.63) and (3.64), with the low-order tolerance decreased within its permitted fixed range. For clarity the extraction of a numerical factor does not presume equality with the reference growing eigenline. With $u = \Omega/\Theta$, $u_* = (b/a)^{1/2}$, and $z = u/u_*$, direct differentiation gives

$$z' = \sqrt{ab}(1 - z^2) - z(\log u_*)'.$$

The source seed is chosen on the growing eigenline, so $z(t_q^{\text{in}}) = 1$. For $q = 1$ the eigenline and the ratio are exact throughout growth, as proved in the finite-stage section; the following tracking computation applies to later stages. If the source error bound is at most $\delta \leq 1/4$, its Riccati barrier gives $z \geq 1 - \delta$: at $z = 1 - \delta$, $z' \geq \sqrt{ab}\{1 - (1 - \delta)^2 - \delta(1 - \delta)\} = \delta\sqrt{ab} > 0$. At $z = 1 + \delta$ the corresponding upper derivative is $-\delta\sqrt{ab} < 0$. A first exit is therefore impossible. To extract the coefficient ratio, use the exact source (3.55)–(3.56): during growth the preceding direction is vertical, $a_q = \mathcal{G}_q/(K_q r_q)$ and $b_q = K_q d_q$, where $d_q = r_q \sin(\phi_q - \phi_{q-1})$ and $\mathcal{G}_q = \sum_{i < q} w_i A_i \sin(\phi_q - \phi_i)$ includes the base term with A_0 and $w_0 = 1$. Hence $u_* = K_q \sqrt{d_q r_q / \mathcal{G}_q}$. The three source estimates are $r_q \geq 1 - \delta$, $d_q / \sin s_q \geq 1 - \delta$, and $\mathcal{G}_q / (\sigma_{q-1}^2 \sin s_q) \leq 1 + \delta$. Together they give $u_* \geq (K_q / \sigma_{q-1})(1 - \delta) / \sqrt{1 + \delta}$. Their product is at least $K_q / (2\sigma_{q-1})$, because $(3/4)^2 / \sqrt{5/4} > 1/2$. This proves the displayed extraction from the source's explicit tracking estimates without assigning a sign to an uncontrolled error.

The source also gives

$$\sigma_{q-1} \leq \nu \sqrt{C_\sigma} \lambda_{q-1}^{1/16}, \quad \lambda_q = \lambda_{q-1}^{Q_q}, \quad Q_q = Q_* + q, \quad Q_* \geq 200, \quad (352)$$

by (3.7), (3.18). The material radius is $\ell_q^{\text{phys}} = \lambda_{q-1}^{-3} / \mu$ for $q \geq 2$. Each older layer and the base are affine on the full new support (Lemma 4.3, pp. 37–38), while all corrections to the newest layer vanish on its entire material plateau (Proposition 6.1, pp. 54–55). These spatial statements are essential to the following pointwise computation: a bound on a central affine ball alone would not provide the points used here.

Choose one fixed $s_\dagger \in (-\pi, \pi)$ with $F''(s_\dagger) \neq 0$. Such a point exists for the released profile. Otherwise F'' would vanish on the full period and by periodicity everywhere, so F would be an affine periodic function, hence constant; oddness would make it zero, contradicting $F'(0) = 1$. Fix this point once for every layer. At the growth crossing put

$$a_q = \frac{s_\dagger}{K_q} p_q, \quad x_q = M_q(t_q^a) a_q. \quad (353)$$

The activation vectors have $|p_q| = 1$ and $\zeta_q = M_q^{-T} p_q$, so $K_q \zeta_q \cdot x_q = s_\dagger$ exactly. The point belongs to the plateau as soon as

$$\frac{|s_\dagger|}{K_q} < \frac{\ell_q^{\text{phys}}}{2}, \quad \text{equivalently} \quad 2|s_\dagger| < \lambda_{q-1}^{Q_q-3}.$$

This holds for all sufficiently large q by (352). The entire neighborhood of this point in the plateau has constant envelope, zero correction fields, and affine older fields. No later layer is yet activated at t_q^a . Consequently at this point the Laplacians of the *full released fields*, not merely those of an isolated formal wave, are exactly

$$\begin{aligned}\Delta\theta_E(x_q, t_q^a) &= \Theta_q K_q^2 r_q^2 F''(s_\dagger), \\ \Delta u_E(x_q, t_q^a) &= \Omega_q K_q J \zeta_q F''(s_\dagger).\end{aligned}\tag{354}$$

For the second line use $u_q = \Omega_q J \zeta_q F(s)/(K_q r_q^2)$ on the plateau; applying the Laplacian contributes $K_q^2 r_q^2$ and cancels the displayed denominator. This verifies both the vector direction and every frequency and phase-length factor.

Inserting (351) gives

$$|\kappa \Delta\theta_E(x_q, t_q^a)| \geq \kappa \nu^2 \mu r_*^2 |F''(s_\dagger)| \lambda_q^{9/8}, \tag{355}$$

$$|\varepsilon \Delta u_E(x_q, t_q^a)| \geq \frac{\varepsilon \nu \mu r_*}{2\sqrt{C_\sigma}} |F''(s_\dagger)| \lambda_q^{9/8} \lambda_{q-1}^{-1/16}. \tag{356}$$

For the second line first use $|\Omega_q| \geq |\Theta_q| K_q / (2\sigma_{q-1})$ and $|J \zeta_q| = r_q$, and then (352). For fixed positive κ the first right-hand side tends to infinity. For fixed positive ε the second does also, since

$$\lambda_q^{9/8} \lambda_{q-1}^{-1/16} = \lambda_{q-1}^{9Q_q/8-1/16} \longrightarrow \infty.$$

In particular even the case $\kappa = 0$, $\varepsilon > 0$ has an explicit diverging vector diffusion cost along these unchanged fields.

The original forces extend smoothly through the source terminal time and have fixed compact support, and hence are bounded; this is Proposition 9.4(b),(c), pp. 72–73, proved using Lemma 9.3, pp. 71–72. For the specific exact extension

$$f_\theta^{\kappa, \varepsilon} = f_\theta^E - \kappa \Delta\theta_E, \quad f_u^{\kappa, \varepsilon} = f_u^E - \varepsilon \Delta u_E, \quad p^{\kappa, \varepsilon} = p_E,$$

the reverse triangle inequality and (355)–(356) therefore prove unbounded forces at the corresponding positive coefficient, before the terminal time. They cannot have a continuous, and thus cannot have a smooth, terminal extension. This proves failure of the specified unchanged-pressure map. There is a stronger, pressure-independent conclusion for these same fields. Choose $s_\dagger$ with $F'''(s_\dagger) \neq 0$. It exists: otherwise F'' would be constant, and periodicity would successively force $F''' = F'' = 0$, contradicting $F'(0) = 1$. Use the plateau point in (353) with this phase. The vorticity there equals a spatially constant older vorticity plus $\Omega_q F'(s)$, so

$$\Delta\omega_E = \Omega_q K_q^2 r_q^2 F'''(s_\dagger), \quad |\varepsilon \Delta\omega_E| \geq \frac{\varepsilon \nu \mu^2 r_*^2}{2\sqrt{C_\sigma}} |F'''(s_\dagger)| \lambda_q^{17/8} \lambda_{q-1}^{-1/16} \longrightarrow \infty.$$

For any smooth pressure p_{new} and any vector force producing the same u_E, θ_E in the viscous equation, subtraction of the inviscid momentum equation and application of curl give the exact identity

$$\text{curl } f_{u, \text{new}} = \text{curl } f_u^E - \varepsilon \Delta\omega_E,$$

because mixed derivatives of $p_{\text{new}} - p_E$ commute. The source curl force is bounded. The displayed divergence therefore excludes a terminal extension with bounded first spatial derivatives for *every* pressure and force that retain these exact state fields. In fact the failure is local at the origin and terminal time. Source Theorem 3.2(iv) gives $\|M_q\| \leq \sqrt{C_\sigma} \lambda_{q-1}^{1/16}$, so the test points satisfy

$$|x_q| \leq \frac{|s_\dagger| \sqrt{C_\sigma}}{\mu} \lambda_{q-1}^{1/16 - Q_q} \longrightarrow 0, \quad t_q^a \longrightarrow T_*.$$

A continuous derivative on a neighborhood of $(0, T_*)$ is bounded on a smaller compact neighborhood, contradicting the computed curl divergence there. If a vector field has bounded first spatial

derivatives then its curl is bounded by their sum, which proves this implication explicitly. This still does not exclude a modified coupled trajectory, a different state construction, or viscous blowup; none of those claims follows from evaluating the unchanged source state.

The completed result of this note is the actual finite forced stage proved above and below, including its original datum, control, background feedback, return, hold, support and every diffusion term. The released infinite inviscid sequence is reconstructed as source mathematics. A new infinite viscous sequence with forces smooth through its terminal time is not established by these calculations.

17 Fixed-diffusion similarity and the force-flat cascade space

The finite third-return theorem supplies one positive interval of equal thermal diffusivity and viscosity. This section proves that the interval does not shrink merely because a finite packet is placed at a smaller physical scale. The exact Boussinesq similarity transfers any one value in that interval to any prescribed positive physical diffusion.

The same map also identifies the next infinite-stage defect. A raw sequence of rescaled copies has force jets which diverge at exact powers of the spatial radius. This obstruction defines the weighted force-flat space that a corrected sequence must occupy. The definition retains the scalar force, the vector force, every space-time derivative, the physical diffusion, and the full finite return. It is not an assertion that the required corrected packet sequence has already been constructed.

17.1 Exact two-parameter similarity with both forces retained

Let $(\tilde{\theta}, \tilde{u}, \tilde{p}, \tilde{f}_\theta, \tilde{f}_u)$ be smooth on $\mathbb{R}_y^2 \times [0, \tilde{T}]$ and solve

$$\partial_\tau \tilde{\theta} + \tilde{u} \cdot \nabla_y \tilde{\theta} = \delta \Delta_y \tilde{\theta} + \tilde{f}_\theta, \quad (357)$$

$$\partial_\tau \tilde{u} + (\tilde{u} \cdot \nabla_y) \tilde{u} + \nabla_y \tilde{p} = \delta \Delta_y \tilde{u} + \tilde{\theta} e_2 + \tilde{f}_u, \quad \operatorname{div}_y \tilde{u} = 0. \quad (358)$$

No coefficient in these equations is normalized away: $\delta \geq 0$ is the common dimensionless diffusion.

Fix $r, c > 0$, a spatial centre x_* , and a starting time t_* . Put

$$y = \frac{x - x_*}{r}, \quad \tau = \frac{t - t_*}{c}, \quad (359)$$

and define

$$\theta(x, t) = \frac{r}{c^2} \tilde{\theta}(y, \tau), \quad u(x, t) = \frac{r}{c} \tilde{u}(y, \tau), \quad p(x, t) = \frac{r^2}{c^2} \tilde{p}(y, \tau), \quad (360)$$

$$f_\theta(x, t) = \frac{r}{c^3} \tilde{f}_\theta(y, \tau), \quad f_u(x, t) = \frac{r}{c^2} \tilde{f}_u(y, \tau), \quad d = \frac{r^2}{c} \delta. \quad (361)$$

Proposition 1. *The tuple in (360)–(361) solves the same Boussinesq system on $\mathbb{R}_x^2 \times [t_*, t_* + c\tilde{T}]$ with equal physical diffusion d . The map is invertible. Spatial supports are translated by x_* and multiplied by r ; time intervals are multiplied by c . The vorticity obeys*

$$\omega(x, t) = \frac{1}{c} \tilde{\omega}(y, \tau). \quad (362)$$

Proof. Every term is kept. The scalar terms are

$$\partial_t \theta = \frac{r}{c^3} \partial_\tau \tilde{\theta}, \quad u \cdot \nabla_x \theta = \frac{r}{c^3} \tilde{u} \cdot \nabla_y \tilde{\theta}, \quad d \Delta_x \theta = \frac{d}{rc^2} \Delta_y \tilde{\theta} = \frac{r}{c^3} \delta \Delta_y \tilde{\theta}.$$

For momentum,

$$\partial_t u = \frac{r}{c^2} \partial_\tau \tilde{u}, \quad (u \cdot \nabla_x) u = \frac{r}{c^2} (\tilde{u} \cdot \nabla_y) \tilde{u}, \quad \nabla_x p = \frac{r}{c^2} \nabla_y \tilde{p},$$

while

$$d\Delta_x u = \frac{d}{rc}\Delta_y \tilde{u} = \frac{r}{c^2}\delta\Delta_y \tilde{u}, \quad \theta e_2 = \frac{r}{c^2}\tilde{\theta}e_2.$$

Thus the scalar equation has common factor r/c^3 and the vector equation has common factor r/c^2 . Also $\operatorname{div}_x u = c^{-1}\operatorname{div}_y \tilde{u}$. All scale factors are positive, so reversing (359) and reciprocating the factors gives the inverse. One spatial derivative of u proves (362). The support and interval assertions follow directly from the affine coordinate map. $\square$

17.2 The finite return at arbitrary physical diffusion and scale

Choose once and for all

$$0 < \delta_* \leq d_{\text{ret}}, \quad (363)$$

where d_{ret} is the positive number constructed in Theorem 6. For any prescribed physical diffusion $d > 0$ and any $r > 0$, set

$$c(r; d, \delta_*) = \frac{\delta_*}{d}r^2. \quad (364)$$

Then $dc/r^2 = \delta_*$ exactly.

The controls and amplitudes carried by the finite packet transform with the fields. If a tilde denotes the packet before similarity, then

$$\alpha(t) = \tilde{\alpha}(\tau), \quad K_j = r^{-1}\tilde{K}_j, \quad \Theta_j(t) = \frac{r}{c^2}\tilde{\Theta}_j(\tau), \quad \Omega_j(t) = \frac{1}{c}\tilde{\Omega}_j(\tau), \quad \sigma_j = \frac{1}{c}\tilde{\sigma}_j. \quad (365)$$

Indeed $K_j\Theta_j$ is a temperature-gradient amplitude and therefore has factor c^{-2} , while Ω_j , $\dot{\alpha}$, and σ_j are vorticity-scale quantities and therefore have factor c^{-1} .

Theorem 7. *For every $d > 0$ and $r > 0$, the similarity image of the complete finite third-return solution at diffusion δ_* is a complete finite third-return solution at physical diffusion d , supported at spatial scale r and running for $c(r; d, \delta_*)\tilde{T}$. It retains*

$$Z(T) = 0, \quad \Omega_3(T) = 0, \quad \Theta_3(T) < 0,$$

and the dimensionless endpoint bounds

$$\frac{\Omega_2(T)}{\sigma_2} \geq \frac{1}{48}, \quad \log \frac{-\Theta_3(T)}{-\Theta_3(0)} \geq \frac{1}{24}, \quad \frac{2\dot{\alpha}(T) + \Omega_1(T) + \Omega_2(T)}{\sigma_2} \geq \frac{c_W}{4}.$$

Thus fixed positive diffusion creates no new finite-stage scale restriction after parabolic similarity.

Proof. Apply Proposition 1 with $c = \delta_*r^2/d$. The normalized solution exists and returns by Theorem 6. Positive scale factors preserve all zeroes and signs. The first four identities in (365) give the zero, sign, and temperature-ratio assertions. Every vorticity and $\dot{\alpha}$ is multiplied by c^{-1} by (362) and (365). The complete temperature-gradient vector is multiplied by c^{-2} ; hence each growth scale σ_j , defined as the positive square root of its magnitude, is multiplied by c^{-1} . The two normalized vorticity bounds are therefore unchanged. Supports and duration follow from the same similarity map. $\square$

This is stronger than a stagewise choice of ever smaller physical diffusion. One fixed $d > 0$ works at every radius. The cost appears in the amplitudes and forces, which the next calculation makes exact.

17.3 Every field and force jet

Let $\gamma \in \mathbb{N}_0^2$ and $b \in \mathbb{N}_0$. Direct differentiation of (360)–(361) gives

$$\partial_x^\gamma \partial_t^b u = r^{1-|\gamma|} c^{-1-b} (\partial_y^\gamma \partial_\tau^b \tilde{u})(y, \tau), \quad (366)$$

$$\partial_x^\gamma \partial_t^b \theta = r^{1-|\gamma|} c^{-2-b} (\partial_y^\gamma \partial_\tau^b \tilde{\theta})(y, \tau), \quad (367)$$

$$\partial_x^\gamma \partial_t^b p = r^{2-|\gamma|} c^{-2-b} (\partial_y^\gamma \partial_\tau^b \tilde{p})(y, \tau), \quad (368)$$

$$\partial_x^\gamma \partial_t^b f_u = r^{1-|\gamma|} c^{-2-b} (\partial_y^\gamma \partial_\tau^b \tilde{f}_u)(y, \tau), \quad (369)$$

$$\partial_x^\gamma \partial_t^b f_\theta = r^{1-|\gamma|} c^{-3-b} (\partial_y^\gamma \partial_\tau^b \tilde{f}_\theta)(y, \tau). \quad (370)$$

At the fixed-diffusion clock (364), these factors are respectively

$$(d/\delta_*)^{1+b} r^{-1-|\gamma|-2b}, \quad (d/\delta_*)^{2+b} r^{-3-|\gamma|-2b}, \quad (371)$$

$$(d/\delta_*)^{2+b} r^{-2-|\gamma|-2b}, \quad (d/\delta_*)^{2+b} r^{-3-|\gamma|-2b},$$

$$(d/\delta_*)^{3+b} r^{-5-|\gamma|-2b}. \quad (372)$$

The first row contains the velocity and scalar factors; the second row contains pressure, vector force, and scalar force in that order. Every power of r, d, δ_* is retained.

17.4 Why raw repeated packets fail

The complete finite packet has a nonzero normalized vector force. During its initial interpolation the base velocity is zero and $\dot{\alpha} = 0$. Equations (5) and (6) give

$$\tilde{f}_u = -\tilde{\theta}_b e_2.$$

The retained initial temperature is nonzero, so there is a normalized space-time point $(y_\dagger, \tau_\dagger)$ and a number $a_\dagger > 0$ with

$$|\tilde{f}_u(y_\dagger, \tau_\dagger)| = a_\dagger. \quad (373)$$

Proposition 2. *Let $r_q \downarrow 0$, let $t_q \rightarrow T_*$, and place complete similarity copies with the fixed clock $c_q = \delta_* r_q^2/d$ at the common spatial centre x_* . At the corresponding points*

$$x_q = x_* + r_q y_\dagger, \quad s_q = t_q + c_q \tau_\dagger,$$

one has

$$|f_{u,q}(x_q, s_q)| = a_\dagger (d/\delta_*)^2 r_q^{-3} \longrightarrow \infty. \quad (374)$$

Consequently $f_{u,q}$ does not tend to zero locally uniformly near (x_*, T_*) . The series $\sum_q f_{u,q}$ therefore cannot converge uniformly on any compact neighborhood of that point, because uniform convergence of a function series requires its terms to tend uniformly to zero. Repeating the finite return without a force-flattening correction therefore fails the packet-series convergence required by the infinite construction.

Proof. The points converge to (x_*, T_*) because $r_q, c_q \rightarrow 0$. Insert (373) into (369) with $b = |\gamma| = 0$, then use (364). This is exactly (374). Fix a compact neighborhood K of (x_*, T_*) . For every sufficiently large q , $(x_q, s_q) \in K$, and therefore

$$\sup_K |f_{u,q}| \geq a_\dagger (d/\delta_*)^2 r_q^{-3} \longrightarrow \infty.$$

If the series of forces converged uniformly on K , the differences of successive partial sums, namely $f_{u,q}$, would converge uniformly to zero. The displayed lower bound rules this out. $\square$

This obstruction is distinct from the unchanged-state Laplacian obstruction in Section 16. The similarity map absorbs the physical diffusion into each finite packet; the new defect is the scale of the packet's remaining force.

17.5 The force-flat space defined by the obstruction

For a sequence of normalized corrected packets, write

$$\varepsilon_{q,b,\gamma}^u = \sup |\partial_y^\gamma \partial_\tau^b \tilde{f}_{u,q}|, \quad \varepsilon_{q,b,\gamma}^\theta = \sup |\partial_y^\gamma \partial_\tau^b \tilde{f}_{\theta,q}|. \quad (375)$$

The supremum is taken on the full normalized support and time interval; no phase, cutoff, pressure, or joining region is omitted.

Definition (the force-flat packet space). At physical diffusion d and normalized diffusion δ_* , the force-flat packet space $\mathfrak{F}(d, \delta_*)$ consists of sequences $(r_q, \tilde{f}_{\theta,q}, \tilde{f}_{u,q})$ with $r_q \downarrow 0$ and normalized forces in $C_c^\infty(\mathbb{R}^2 \times \mathbb{R})$ such that, for every $m \in \mathbb{N}_0$,

$$\sum_q \max_{b+|\gamma| \leq m} (d/\delta_*)^{2+b} r_q^{-3-|\gamma|-2b} \varepsilon_{q,b,\gamma}^u < \infty, \quad (376)$$

$$\sum_q \max_{b+|\gamma| \leq m} (d/\delta_*)^{3+b} r_q^{-5-|\gamma|-2b} \varepsilon_{q,b,\gamma}^\theta < \infty. \quad (377)$$

These are the complete force weights imposed by the PDE similarity, not chosen regularity exponents. Translate and rescale each compactly supported force by (359)–(361), and extend it by zero. Then (376)–(377) give uniform convergence of every derivative on compact sets by the Weierstrass test. The fundamental theorem of calculus in each coordinate identifies the limits of the derivative series with the derivatives of the force limit. Hence the force sum is smooth. Spatial and temporal support, state convergence, nonlinear residual convergence, endpoint matching, and the blowup property remain additional equations of the construction; the force-flat definition does not conceal them.

The weight space has nonzero elements. Choose fixed nonzero functions $\Phi_u \in C_c^\infty(\mathbb{R}^2 \times (0, 1); \mathbb{R}^2)$ and $\Phi_\theta \in C_c^\infty(\mathbb{R}^2 \times (0, 1))$, and set

$$r_q = 2^{-q}, \quad \tilde{f}_{u,q} = 2^{-q^2} \Phi_u, \quad \tilde{f}_{\theta,q} = 2^{-q^2} \Phi_\theta \quad (q \geq 1). \quad (378)$$

For each fixed derivative pair, the corresponding quantities in (375) equal 2^{-q^2} times the finite derivative supremum of Φ_u or Φ_θ . The maximum of these finitely many profile constants for $b + |\gamma| \leq m$ is a finite number C_m independent of q . For $b + |\gamma| \leq m$, the worst vector-force radius power is r_q^{-3-2m} and the worst scalar-force radius power is r_q^{-5-2m} . Thus their variable parts are bounded by

$$2^{-q^2+(3+2m)q}, \quad 2^{-q^2+(5+2m)q}.$$

The successive-term ratios are

$$2^{-2q+2m+2}, \quad 2^{-2q+2m+4}.$$

They are at most $1/2$ for all $q \geq m+2$ and $q \geq m+3$, respectively. Multiplication by C_m and the fixed powers of d/δ_* does not affect convergence. Both series converge for every fixed m . This constructs nonzero elements with a simultaneous superalgebraic budget for every force jet.

17.6 First realization attempt and the exact next calculation

The raw finite packet has $\varepsilon_{q,0,0}^u \geq a_\dagger$, so Proposition 2 proves that it does not belong to $\mathfrak{F}(d, \delta_*)$. The direct unchanged-state force repair also does not belong: Section 16 gives its stronger pointwise scalar, vector, and curl divergences. The current finite source-chart estimates do not repair this. They contain negative powers of the chart parameter and no sequence of normalized residual gains $\varepsilon_{q,b,\gamma}$.

The calculation nevertheless changes the infinite problem in two exact ways. First, fixed positive diffusion is compatible with every finite return scale through (364). This removes the isolated packet's scale restriction; it does not prove a diffusion bound uniform for the coupled infinite

sequence. Second, the target of the correction cycle is now explicit: after all pressure, mean, cutoff, old-wave, quadratic, evaluated-phase, and joining terms have been assembled, its normalized force jets must land in (376)–(377). The concrete budget (378) shows the amount of decay which suffices.

The next construction step is therefore to retain the returned parent, calculate the type-compatible Boussinesq increment residual, and construct an actual correction of that residual. The separate three-dimensional Navier–Stokes moment and stress operations cannot be applied here without first proving their receiving map for these scalar and vector fields. The based residual and state map are derived in Section 18; the first compact local correction is constructed in Section 19. The remaining all-stage calculation must put every actual force jet of order at most q below 2^{-q^2} and prove the complete next-stage state match.

The similarity identities, radius exponents, and ratio tests in this section are checked independently by `infinite_modified/checks/parabolic_similarity_check.py`. Figure 2 displays the exact finite-to-infinite separation.

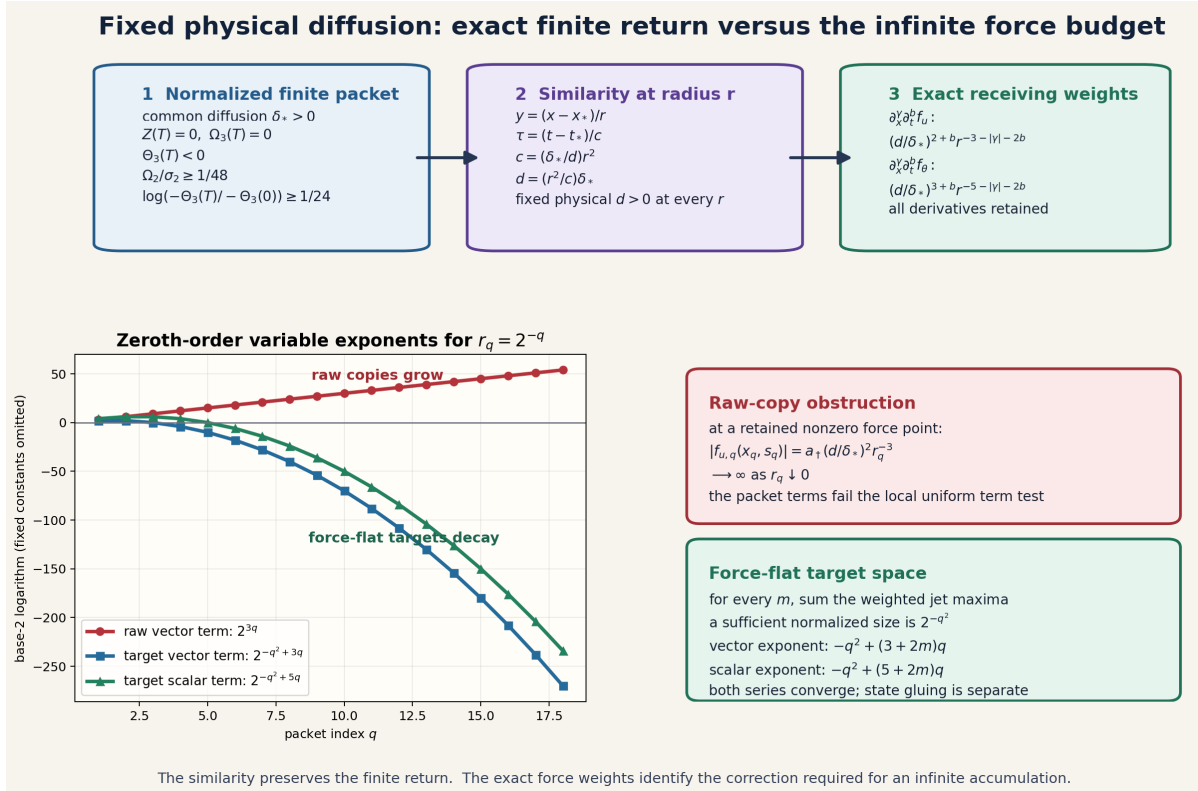


Figure 2: The fixed-diffusion similarity window. Parabolic scaling transfers the complete finite return to every radius, while the exact force-jet powers show that raw copies fail the local uniform term test. The force-flat weights are the receiving space for the still-required correction and gluing operations.

18 The based cascade: exact residual array and state interface

The preceding similarity theorem concerns isolated complete packets. An infinite construction cannot be obtained by putting one complete packet after another. This section proves the state obstruction, replaces packet reset by an increment over the retained parent, and calculates the complete normalized residual of that increment. The calculation keeps the parent velocity, velocity gradient, temperature gradient, pressure, buoyancy, diffusion, and every space-time derivative.

18.1 A complete packet cannot be reset at the next scale

The stationary left extension of the complete finite packet has $u = 0$. Hence its vorticity and its origin curl are zero. At the selected third return, (311) gives

$$2\dot{\alpha}(T) + \Omega_1(T) + \Omega_2(T) \geq \frac{c_W}{4}\sigma_2 > 0. \quad (379)$$

The third vorticity is zero there, so this is the full origin curl of the three-layer returned state.

Proposition 3. *No two similarity images of the complete finite packet can be joined in C^1 by identifying the returned endpoint of the first with the stationary initial endpoint of the second. This holds for every pair of positive radii, every pair of positive clocks, and every choice of spatial centre.*

Proof. Similarity multiplies vorticity by the positive factor c^{-1} , by (362). Thus the returned packet has origin curl at least $c^{-1}c_W\sigma_2/4 > 0$. The stationary initial state of every complete packet has zero velocity everywhere and therefore zero curl at every possible centre. Equality of the first spatial derivatives at a join would imply equality of their curls. The strict inequality rules this out. $\square$

The obstruction concerns reset. The returned state can instead be retained as the parent of a new increment. A based cascade performs exactly that retention.

18.2 Increment over the full retained parent

Let $(\theta^{<q}, u^{<q}, p^{<q})$ solve the equal-diffusion Boussinesq system with physical diffusion $d > 0$ and force $(f_\theta^{<q}, f_u^{<q})$. Let $(\vartheta_q, v_q, \pi_q)$ be a smooth increment with $\text{div } v_q = 0$, and set

$$\theta^{[q]} = \theta^{<q} + \vartheta_q, \quad u^{[q]} = u^{<q} + v_q, \quad p^{[q]} = p^{<q} + \pi_q. \quad (380)$$

Proposition 4. *The state (380) solves the same equation with force $(f_\theta^{<q} + e_{\theta,q}, f_u^{<q} + e_{u,q})$, where*

$$e_{\theta,q} = \partial_t \vartheta_q + u^{<q} \cdot \nabla \vartheta_q + v_q \cdot \nabla \theta^{<q} + v_q \cdot \nabla \vartheta_q - d \Delta \vartheta_q, \quad (381)$$

$$e_{u,q} = \partial_t v_q + (u^{<q} \cdot \nabla) v_q + (v_q \cdot \nabla) u^{<q} + (v_q \cdot \nabla) v_q + \nabla \pi_q - d \Delta v_q - \vartheta_q e_2. \quad (382)$$

Proof. Insert (380) into both Boussinesq equations. Subtract the two equations for the parent. Bilinearity gives the three displayed transport terms in each row, pressure is additive, diffusion is linear, and the buoyancy difference is $\vartheta_q e_2$. Moving that buoyancy difference to the force side gives the last sign in (382). No parent term remains unlisted. $\square$

Inside the finite construction, each newly activated controlled layer is multiplied by a switching factor that is flat and zero at its left endpoint. A future based correction must retain this property for its relative increment. The switching factor supplies smooth birth of that increment; it does not supply the state match calculated below. Keeping every completed increment in the parent avoids the reset in Proposition 3.

18.3 The normalized parent and the exact scaled residual

Fix a packet centre (x_q, t_q) , radius $\rho_q > 0$, and

$$c_q = \frac{\delta_*}{d} \rho_q^2, \quad y = \frac{x - x_q}{\rho_q}, \quad \tau = \frac{t - t_q}{c_q}. \quad (383)$$

Write the increment with the complete similarity weights

$$\vartheta_q = \frac{\rho_q}{c_q^2} \tilde{\vartheta}_q, \quad v_q = \frac{\rho_q}{c_q} \tilde{v}_q, \quad \pi_q = \frac{\rho_q^2}{c_q^2} \tilde{\pi}_q. \quad (384)$$

All tilde fields in this section are evaluated at (y, τ) . Evaluate the parent at $(x_q + \rho_q y, t_q + c_q \tau)$ and define

$$\mathcal{U}_q = \frac{c_q}{\rho_q} u^{<q}, \quad \mathcal{D}_q = c_q \nabla_x u^{<q}, \quad \mathcal{G}_q = c_q^2 \nabla_x \theta^{<q}. \quad (385)$$

Proposition 5. *The physical residual increments have the exact form*

$$e_{\theta,q} = \frac{\rho_q}{c_q^3} \tilde{e}_{\theta,q}, \quad e_{u,q} = \frac{\rho_q}{c_q^2} \tilde{e}_{u,q}, \quad (386)$$

where

$$\tilde{e}_{\theta,q} = \partial_\tau \tilde{\vartheta}_q + (\mathcal{U}_q + \tilde{v}_q) \cdot \nabla_y \tilde{\vartheta}_q + \tilde{v}_q \cdot \mathcal{G}_q - \delta_* \Delta_y \tilde{\vartheta}_q, \quad (387)$$

$$\tilde{e}_{u,q} = \partial_\tau \tilde{v}_q + (\mathcal{U}_q + \tilde{v}_q) \cdot \nabla_y \tilde{v}_q + \mathcal{D}_q \tilde{v}_q + \nabla_y \tilde{\pi}_q - \delta_* \Delta_y \tilde{v}_q - \tilde{\vartheta}_q e_2. \quad (388)$$

Consequently every physical residual jet has exactly the force weights (372).

Proof. The time, self-transport, pressure, diffusion, and buoyancy terms have the common factors already proved in Proposition 1. For the parent scalar terms,

$$u^{<q} \cdot \nabla_x \vartheta_q = \frac{\rho_q}{c_q^3} \mathcal{U}_q \cdot \nabla_y \tilde{\vartheta}_q, \quad v_q \cdot \nabla_x \theta^{<q} = \frac{\rho_q}{c_q^3} \tilde{v}_q \cdot \mathcal{G}_q.$$

For momentum,

$$(u^{<q} \cdot \nabla_x) v_q = \frac{\rho_q}{c_q^2} (\mathcal{U}_q \cdot \nabla_y) \tilde{v}_q, \quad (v_q \cdot \nabla_x) u^{<q} = \frac{\rho_q}{c_q^2} \mathcal{D}_q \tilde{v}_q.$$

Substitution proves (386)–(388). Differentiating (386) gives the last assertion with every space and time factor retained. $\square$

18.4 All-order residual arrays

Let $\alpha = (\gamma_1, \gamma_2, b) \in \mathbb{N}_0^3$, write $\partial^\alpha = \partial_{y_1}^{\gamma_1} \partial_{y_2}^{\gamma_2} \partial_\tau^b$, and define

$$\binom{\alpha}{\beta} = \prod_{j=1}^3 \binom{\alpha_j}{\beta_j}, \quad (A \star B)_\alpha = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} A_\beta B_{\alpha-\beta}.$$

Let S_1, S_2, S_t shift the corresponding derivative index by one. Write $S_i^2 = S_i \circ S_i$ for the twofold spatial shift. On the complete normalized support and time interval, set

$$\begin{aligned} \mathsf{T}_\alpha &= \|\partial^\alpha \tilde{\vartheta}_q\|_\infty, & \mathsf{V}_{i,\alpha} &= \|\partial^\alpha \tilde{v}_{q,i}\|_\infty, & \mathsf{P}_\alpha &= \|\partial^\alpha \tilde{\pi}_q\|_\infty, \\ \mathsf{U}_{i,\alpha} &= \|\partial^\alpha \mathcal{U}_{q,i}\|_\infty, & \mathsf{D}_{ki,\alpha} &= \|\partial^\alpha \mathcal{D}_{q,ki}\|_\infty, & \mathsf{G}_{i,\alpha} &= \|\partial^\alpha \mathcal{G}_{q,i}\|_\infty. \end{aligned}$$

Repeated Leibniz differentiation of (387)–(388) gives the complete finite majorants

$$\mathsf{E}^\theta = S_t \mathsf{T} + \sum_{i=1}^2 (\mathsf{U}_i + \mathsf{V}_i) \star S_i \mathsf{T} + \sum_{i=1}^2 \mathsf{V}_i \star \mathsf{G}_i + \delta_* \sum_{i=1}^2 S_i^2 \mathsf{T}, \quad (389)$$

$$\mathsf{E}_k^u = S_t \mathsf{V}_k + \sum_{i=1}^2 (\mathsf{U}_i + \mathsf{V}_i) \star S_i \mathsf{V}_k + \sum_{i=1}^2 \mathsf{D}_{ki} \star \mathsf{V}_i + S_k \mathsf{P} + \delta_* \sum_{i=1}^2 S_i^2 \mathsf{V}_k + \mathbf{1}_{k=2} \mathsf{T}. \quad (390)$$

Thus

$$\|\partial^\alpha \tilde{e}_{\theta,q}\|_\infty \leq \mathsf{E}_\alpha^\theta, \quad \|\partial^\alpha \tilde{e}_{u,q,k}\|_\infty \leq \mathsf{E}_{k,\alpha}^u. \quad (391)$$

Every parent cross term and every self-interaction appears in these arrays. These are unsigned majorants, not the signed residual itself. They give sufficient estimates but may lose exact pressure, buoyancy, and transport cancellations. The correction cycle must reduce the actual residual jets; a norm of the new field alone is not a force estimate. The exact signed array and the failure of a necessary unsigned target are proved in Section 19.

For reference, the parent jets in (385) are exactly

$$\partial_y^\gamma \partial_\tau^b \mathcal{U}_q = \rho_q^{|\gamma|-1} c_q^{b+1} (\partial_x^\gamma \partial_t^b u^{<q}), \quad (392)$$

$$\partial_y^\gamma \partial_\tau^b \mathcal{D}_q = \rho_q^{|\gamma|} c_q^{b+1} (\partial_x^\gamma \partial_t^b \nabla_x u^{<q}), \quad (393)$$

$$\partial_y^\gamma \partial_\tau^b \mathcal{G}_q = \rho_q^{|\gamma|} c_q^{b+2} (\partial_x^\gamma \partial_t^b \nabla_x \theta^{<q}). \quad (394)$$

With $c_q = (\delta_*/d)\rho_q^2$, their radius powers are respectively

$$\rho_q^{|\gamma|+2b+1}, \quad \rho_q^{|\gamma|+2b+2}, \quad \rho_q^{|\gamma|+2b+4}, \quad (395)$$

times the displayed fixed powers of δ_*/d and the corresponding physical parent derivatives.

18.5 The state-renormalization equation

The positive powers in (395) do not permit one to discard the parent. The return mechanism requires a nonzero normalized temperature gradient and a specified normalized velocity gradient. The normalized entry profile carried by the retained parent is, by definition, the triple $(\mathcal{U}_q, \mathcal{D}_q, \mathcal{G}_q)$ in (385). For a prescribed next-stage profile $(\mathcal{U}_q^{\text{in}}, \mathcal{D}_q^{\text{in}}, \mathcal{G}_q^{\text{in}})$, equality with that profile is equivalent to the three exact inverse equations

$$\begin{aligned} u^{<q}(x_q + \rho_q y, t_q + c_q \tau) &= \frac{\rho_q}{c_q} \mathcal{U}_q^{\text{in}}(y, \tau), \\ \nabla_x u^{<q}(x_q + \rho_q y, t_q + c_q \tau) &= c_q^{-1} \mathcal{D}_q^{\text{in}}(y, \tau), \\ \nabla_x \theta^{<q}(x_q + \rho_q y, t_q + c_q \tau) &= c_q^{-2} \mathcal{G}_q^{\text{in}}(y, \tau) \end{aligned} \quad (396)$$

on the required normalized space-time support. If a proved next-stage entry set contains more than one such profile, admissibility means that the actual triple from (385) belongs to that set; the inverse equations remain exact for the profile it realizes. At the centre, the three physical scale factors are

$$\frac{d}{\delta_*} \rho_q^{-1}, \quad \frac{d}{\delta_*} \rho_q^{-2}, \quad \left(\frac{d}{\delta_*} \right)^2 \rho_q^{-4}. \quad (397)$$

A parent with uniformly bounded first derivatives tends to zero in the last two normalized coordinates and cannot reproduce a fixed nonzero entry tuple. Reproduction of fixed nonzero gradient entries requires the precise ρ_q^{-2} velocity-gradient and ρ_q^{-4} temperature-gradient growth encoded by (396).

Equation (396) is vector- and matrix-valued. Choosing ρ_q from a single norm does not prove it: directions, signs, phase lengths, parent amplitudes, and every derivative required by (389)–(390) must also match. The third-return theorem proves the strict output signs and bounds (311); it does not yet prove that its complete output, after one dilation, lies in a valid fourth-stage entry set.

This answers the next-step question exactly. A successful infinite construction needs a based return-renormalization map satisfying (396), together with corrections that place the actual signed residuals (387)–(388) in the force-flat bounds (376)–(377). For the concrete choice $\rho_q = 2^{-q}$, bounds

$$\max_{|\alpha| \leq q} \|\partial^\alpha \tilde{e}_{\theta,q}\|_\infty \leq 2^{-q^2}, \quad \max_{k \in \{1,2\}, |\alpha| \leq q} \|\partial^\alpha \tilde{e}_{u,q,k}\|_\infty \leq 2^{-q^2} \quad (398)$$

are sufficient by the summability calculation following (378). The stronger bounds with $\mathbf{E}$ in place of the displayed norms are also sufficient, but are not required. Neither bound is asserted for the present finite packet. The complete-copy construction fails Proposition 3; the based residual and state maps above are the exact objects that replace it.

The similarity factors, parent-jet powers, residual terms, and convolution indices are replayed in `infinite_modified/checks/based_cascade_check.py`. Figure 3 shows the state and force interfaces together.

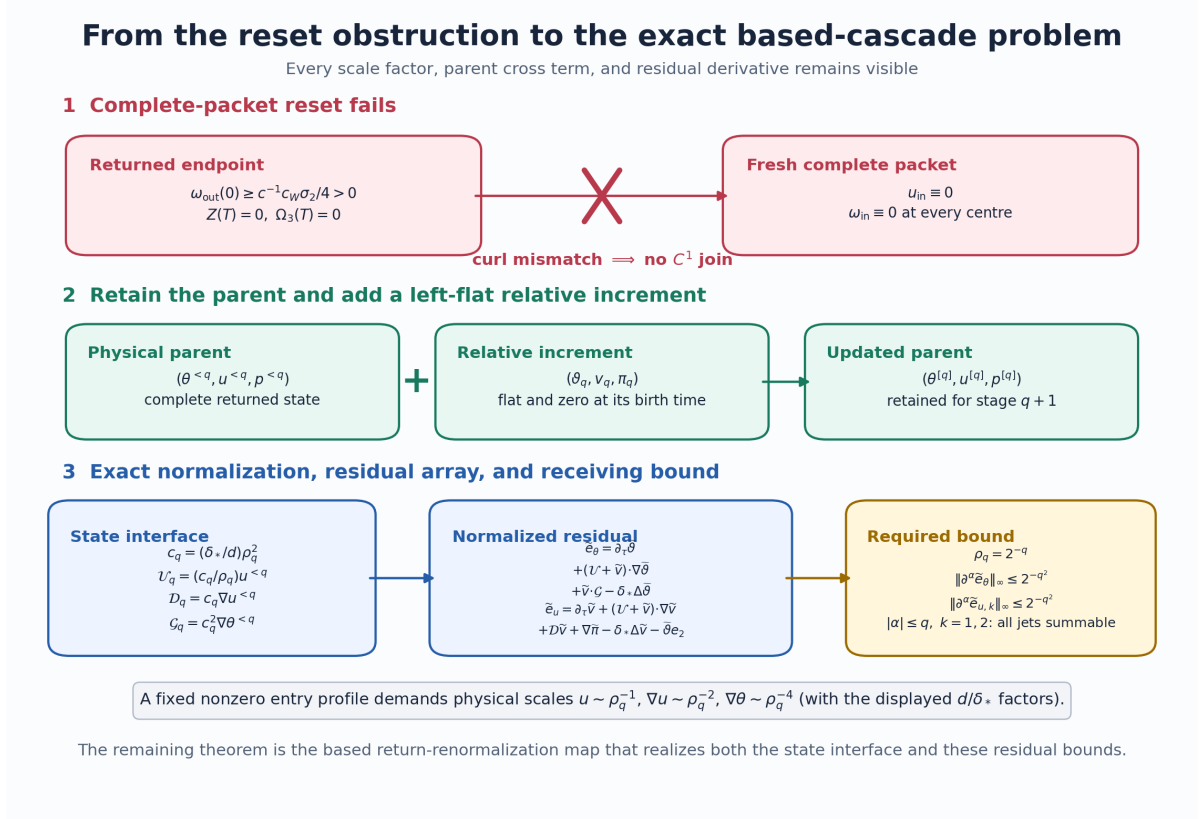


Figure 3: Complete packets fail at the state join because returned curl is positive while every reset packet starts with zero curl. A based cascade retains the returned parent, rescales its complete velocity and temperature jets through (396), and corrects the exact incremental signed residual into the force-flat receiving space.

19 Signed residuals, the exact returned core, and a compact correction

This section corrects the unsigned target in the earlier continuation. It then calculates the complete chart transition and the actual affine state at the selected third return. That state supplies an explicit unforced local continuation. A flat interpolation and a compact streamfunction correction connect it to the retained finite field, with every transition and cutoff contribution displayed. The construction is finite and local; its force cost is not claimed to satisfy an infinite-stage budget.

19.1 The signed force jets and the unsigned-target defect

Use the three-index notation of Section 18. In this subsection only, write $T_\alpha = \partial^\alpha \tilde{\vartheta}$, $V_{i,\alpha} = \partial^\alpha \tilde{v}_i$, $P_\alpha = \partial^\alpha \tilde{\pi}$, $U_{i,\alpha} = \partial^\alpha \mathcal{U}_i$, $D_{ki,\alpha} = \partial^\alpha \mathcal{D}_{ki}$ and $G_{i,\alpha} = \partial^\alpha \mathcal{G}_i$. These are signed functions, with no norm taken. Let e_1, e_2, e_t denote index shifts, distinct from the physical vector $e_2 = (0, 1)$. Repeated differentiation gives the exact signed arrays

$$R_\alpha^\theta = T_{\alpha+e_t} + \sum_{i=1}^2 \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} [(U_{i,\beta} + V_{i,\beta})T_{\alpha-\beta+e_i} + V_{i,\beta}G_{i,\alpha-\beta}] - \delta_* \sum_{i=1}^2 T_{\alpha+2e_i}, \quad (399)$$

$$\begin{aligned} R_{k,\alpha}^u &= V_{k,\alpha+e_t} + \sum_{i=1}^2 \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} [(U_{i,\beta} + V_{i,\beta})V_{k,\alpha-\beta+e_i} + D_{ki,\beta}V_{i,\alpha-\beta}] \\ &\quad + P_{\alpha+e_k} - \delta_* \sum_{i=1}^2 V_{k,\alpha+2e_i} - \mathbf{1}_{k=2} T_\alpha. \end{aligned} \quad (400)$$

They equal $\partial^\alpha \tilde{e}_\theta$ and $\partial^\alpha \tilde{e}_{u,k}$, respectively, by the product rule. Applying the triangle inequality after these equalities gives (389)–(391).

Proposition 6. *The unsigned majorants are not necessary force bounds. On the closed unit ball, the exact fields*

$$\tilde{\vartheta}(y, \tau) = y_2, \quad \tilde{v} = 0, \quad \tilde{\pi} = y_2^2/2, \quad \mathcal{U} = \mathcal{D} = \mathcal{G} = 0$$

have identically zero scalar and vector residuals for every $\delta_* > 0$, whereas $\mathbf{E}_{2,0}^u = 2$. Furthermore, on a duration- L cylinder, bounds $\max_k \mathbf{E}_{k,0}^u \leq \epsilon$ imply $\|\tilde{\vartheta}\|_\infty \leq \epsilon$ and, for an increment starting at $\tilde{v}(\cdot, 0) = 0$, $\|\tilde{v}_k(\cdot, L)\|_\infty \leq L\epsilon$.

Proof. Here $\nabla \tilde{\pi} = y_2 e_2 = \tilde{\vartheta} e_2$, $\Delta \tilde{\vartheta} = 0$, and every velocity term is zero. The signed residuals therefore vanish. The unsigned second momentum row contains $\|\partial_2 \tilde{\pi}\|_\infty + \|\tilde{\vartheta}\|_\infty = 1 + 1$, with all its other terms zero. For the second assertion, all summands in $\mathbf{E}$ are nonnegative. Its second momentum row contains $\mathbf{T}_0$, and each momentum row contains $\|\partial_\tau \tilde{v}_k\|_\infty$. Integration from zero to L proves the asserted endpoint bound. Thus requiring these majorants to tend to zero on uniformly bounded durations excludes increments with a fixed nonzero temperature or velocity amplitude, even when their actual force cancels. The example is local and noncompact; it is not asserted to be a compact packet. $\square$

Consequently (398) now refers to the norms of (399)–(400). The old unsigned target is retained only as a sufficient estimate, with its earlier source version preserved in the main's provenance directory.

19.2 Exact receiving map for a correction

Let $\mathcal{B} = (c_q^2/\rho_q)\theta^{<q}$ and $\mathcal{P} = (c_q^2/\rho_q^2)p^{<q}$ in the same chart. Thus $\nabla \mathcal{B} = \mathcal{G}$ and $\nabla \mathcal{U} = \mathcal{D}$. Set

$$\mathcal{T} = \mathcal{B} + \tilde{\vartheta}, \quad \mathcal{V} = \mathcal{U} + \tilde{v}.$$

For a correction (h, z, κ) with $\operatorname{div} z = 0$, replace $(\tilde{\vartheta}, \tilde{v}, \tilde{\pi})$ by $(\tilde{\vartheta} + h, \tilde{v} + z, \tilde{\pi} + \kappa)$. Subtraction of the complete equations gives

$$\Delta \tilde{e}_\theta = h_\tau + \mathcal{V} \cdot \nabla h + z \cdot \nabla \mathcal{T} + z \cdot \nabla h - \delta_* \Delta h, \quad (401)$$

$$\Delta \tilde{e}_u = z_\tau + (\mathcal{V} \cdot \nabla)z + (z \cdot \nabla)\mathcal{V} + (z \cdot \nabla)z + \nabla \kappa - \delta_* \Delta z - h e_2. \quad (402)$$

Every jet of the updated residual is obtained by differentiating these identities and adding it to the old signed jet. In particular, the correction equation is an equation for the actual signed functions, before any majorant is introduced. These formulas also prove the compatibility constraints on a proposed entry tuple: $\mathcal{D} = \nabla \mathcal{U}$, $\mathcal{G} = \nabla \mathcal{B}$, $\operatorname{tr} \mathcal{D} = 0$, and $\partial_1 \mathcal{G}_2 = \partial_2 \mathcal{G}_1$. They follow by direct differentiation of the defined pullbacks; three independently chosen arrays need not be a realizable tuple.

19.3 The complete chart transition

Keep the physical $d > 0$ and $\delta_* > 0$ fixed. Let two charts have radii $\rho, \rho' > 0$, clocks $c = (\delta_*/d)\rho^2$ and $c' = (\delta_*/d)(\rho')^2$, and centres (x_0, t_0) and (x'_0, t'_0) . Define

$$a = \rho'/\rho > 0, \quad \xi = (x'_0 - x_0)/\rho, \quad \tau_0 = (t'_0 - t_0)/c.$$

Write (Θ, U, P) for the complete scalar, velocity and pressure pullbacks in the first chart, including every retained layer and cutoff. The second chart is exactly

$$\begin{aligned} \Theta'(y, s) &= a^3 \Theta(\xi + ay, \tau_0 + a^2 s), \\ U'(y, s) &= a U(\xi + ay, \tau_0 + a^2 s), \\ P'(y, s) &= a^2 P(\xi + ay, \tau_0 + a^2 s). \end{aligned} \quad (403)$$

This follows from the ratios c'^2/ρ' to c^2/ρ , c'/ρ' to c/ρ , and $c'^2/(\rho')^2$ to c^2/ρ^2 . It proves, with every argument retained,

$$D' = a^2 D(\xi + ay, \tau_0 + a^2 s), \quad G' = a^4 G(\xi + ay, \tau_0 + a^2 s). \quad (404)$$

Space-time jets of Θ, U, P acquire respectively $a^{3+|\gamma|+2b}$, $a^{1+|\gamma|+2b}$ and $a^{2+|\gamma|+2b}$; those of D, G acquire $a^{2+|\gamma|+2b}$ and $a^{4+|\gamma|+2b}$. The complete scalar and vector forces acquire $a^{5+|\gamma|+2b}$ and $a^{3+|\gamma|+2b}$. The chain rule proves each identity; no periodic argument, support boundary, pressure contribution or older layer is replaced. The inverse uses a^{-1} , $-\xi/a$, and $-\tau_0/a^2$.

For a centred affine snapshot $U(y) = Dy$, $\Theta(y) = G \cdot y$, the action becomes $(D, G) \mapsto (a^2 D, a^4 G)$.

Proposition 7. *On the exact state space $\{(D, G) : D \in \mathbb{R}^{2 \times 2}, \operatorname{tr} D = 0, G \in \mathbb{R}^2 \setminus \{0\}\}$, two centred affine snapshots are related by a positive parabolic dilation precisely when*

$$\frac{G'}{|G'|} = \frac{G}{|G|}, \quad \frac{D'}{\sqrt{|G'|}} = \frac{D}{\sqrt{|G|}}. \quad (405)$$

The radius ratio is then uniquely $a = (|G'|/|G|)^{1/4}$. It is a shrinking dilation precisely when $|G'| < |G|$.

Proof. The action in (404) preserves both displayed quantities since $a > 0$. Conversely, their equality and the displayed choice of a give $G' = a^4 G$ and $D' = a^2 D$ entry by entry. This proves the full orbit classification, including the direction and every matrix entry. For $\operatorname{curl} W = D_{21} - D_{12}$ it also gives the invariant $W/\sqrt{|G|}$. Thus selection of a radius from $|G|$ cannot change the remaining state coordinates. $\square$

19.4 The actual third-return affine state

At the selected return time t_R , retain all original constants and the determinant coordinates of (283). In the following display $D_{\text{ph}} = h^2 + a_{\text{ph}}^2$ is the source scalar D ; $a_{\text{ph}} = \sin s_2$ distinguishes it from a radius ratio. The other source names $b, x, y, g, R, c = A_0 \cos s_*$ and $C = A_0 \sin s_*$ are unchanged. On a common affine neighbourhood, the full physical fields are

$$\theta = G_R \cdot x, \quad u = \mathcal{A}_R x, \quad p = 0, \quad (406)$$

where

$$G_R = -A_0 R_\alpha e_2 + \sum_{j=1}^3 K_j \Theta_j \zeta_j, \quad (407)$$

$$\mathcal{A}_R = \dot{\alpha} J + \sum_{j=1}^3 \frac{\Omega_j}{|\zeta_j|^2} J \zeta_j \otimes \zeta_j. \quad (408)$$

These are restrictions of the complete periodic-profile and material-cutoff fields, rather than new definitions of those fields. Indeed $F(s) = s$ on $|s| < s_0$, $P'(s) = F(s)$, and every material cutoff equals one in a neighbourhood of the origin. The sums therefore give exactly their scalar and streamfunction derivatives there.

The return and hold equations give $\zeta_3 = (0, \sqrt{R})$, $\Omega_3 = 0$, $\zeta_1 = (-y, x)/\sqrt{R}$, and $\zeta_2 = (-b, g)/\sqrt{R}$. Since $\dot{\zeta}_{3,1} = 0$ on the hold and $\dot{\zeta}_3 = -D_{<3}^T \zeta_3$, one has $(D_{<3})_{21} = 0$. At the endpoint it follows also by substitution of the displayed angular control in (298). Tracelessness and (311) yield

$$\mathcal{A}_R = \begin{pmatrix} a_R & -W_R \\ 0 & -a_R \end{pmatrix}, \quad a_R = \frac{\Omega_1 xy}{R} + \frac{\Omega_2 bg}{D_{\text{ph}} R}, \quad W_R = 2\dot{\alpha} + \Omega_1 + \Omega_2 \geq c_W \sigma_2 / 4 > 0. \quad (409)$$

No strain sign is imposed. The angular inverse gives $\alpha = s_* + \arctan(y/x)$ at this endpoint. Therefore, retaining the base and all three temperature terms,

$$\begin{aligned} (G_R)_1 &= \frac{(E_1 + c)y + Cx + bE_2}{\sqrt{R}} = \frac{N_3}{\sqrt{R}} > 0, \\ (G_R)_2 &= -\frac{(E_1 + c)x - Cy + gE_2}{\sqrt{R}} + K_3 \Theta_3 \sqrt{R}. \end{aligned} \quad (410)$$

Positivity uses the proved positive component N_3 , not an unproved sign for $(G_R)_2$. The nonzero G_R lies in the state space of Proposition 7. At a physical radius ρ its chart coefficients are exactly $c_\rho \mathcal{A}_R$ and $c_\rho^2 G_R$; a further radius ratio multiplies these by a^2 and a^4 . This preserves all physical diffusion and clock factors in (383).

19.5 An explicit unforced affine continuation

Put $a = a_R$, $b_0 = -W_R$, $g_1 = (G_R)_1 > 0$ and $g_2 = (G_R)_2$, without resetting their values. For $s = t - t_R$ set

$$\begin{aligned} \Phi_k(a, s) &= \int_0^s e^{-kar} dr = \begin{cases} (1 - e^{-kas})/(ka), & a \neq 0, \\ s, & a = 0, \end{cases} \quad k = 1, 2, 3, \\ \Xi(a, s) &= \int_0^s e^{-2ar} \Phi_1(a, r) dr = \begin{cases} (\Phi_2(a, s) - \Phi_3(a, s))/a, & a \neq 0, \\ s^2/2, & a = 0. \end{cases} \end{aligned} \quad (411)$$

Define the complete coefficients

$$\begin{aligned} g_1(s) &= g_1 e^{-as}, & b(s) &= b_0 - g_1 \Phi_1(a, s), \\ g_2(s) &= e^{as} [g_2 - g_1 b_0 \Phi_2(a, s) + g_1^2 \Xi(a, s)], & A(s) &= \begin{pmatrix} a & b(s) \\ 0 & -a \end{pmatrix}, \\ G(s) &= (g_1(s), g_2(s)), & H(s) &= \begin{pmatrix} -a^2 & g_1(s) \\ g_1(s) & g_2(s) - a^2 \end{pmatrix}. \end{aligned} \quad (412)$$

Theorem 8. For every physical $d > 0$, the fields $\theta_A(x, t) = G(s) \cdot x$, $u_A(x, t) = A(s)x$ and $p_A(x, t) = x^T H(s)x/2$ solve the unforced equal-diffusion Boussinesq equations on $\mathbb{R}^2 \times \mathbb{R}$. Their temperature and velocity at t_R agree exactly with (406) on its affine neighbourhood. For $s \geq 0$,

$$(\nabla \theta_A)_1 = g_1 e^{-as} > 0, \quad \text{curl } u_A = W_R + g_1 \Phi_1(a, s) > 0.$$

All coefficients are finite at every finite s .

Proof. The integral formulas, including their $a = 0$ values, are smooth in a, s by their integral definitions. Differentiation gives $g'_1 = -ag_1(s)$, $b' = -g_1(s)$, and $g'_2 = -b(s)g_1(s) + ag_2(s)$. Hence $G' + A^T G = 0$. Direct multiplication retains the full matrix $A^2 = a^2 I$ and gives $A' + A^2 + H - e_2 G^T = 0$. These two identities are precisely the scalar and momentum equations; $\Delta(G \cdot x) = 0$, $\Delta(Ax) = 0$, and $\text{div}(Ax) = \text{tr } A = 0$. The pressure is symmetric and supplies its displayed gradient. At $s = 0$, $\Phi_k = \Xi = 0$, so the temperature and velocity coefficients are the original ones. For $s \geq 0$, $\Phi_1 \geq 0$, which proves both signs. Exponential and finite integrals prove finiteness at every finite s . This solution is noncompact. Its initial pressure need not equal the old zero pressure, and the equations need not match the old time jets; the next construction repairs the join. $\square$

19.6 A flat join with every transition term retained

The selected finite return has a smooth holding continuation on some compact interval $I = [t_R, t_R + \epsilon]$: local existence of its smooth finite vector field and the interior component bounds prove this. Denote its affine coefficients by $A_h, G_h, H_h = 0$. They retain the nonstationary temperature diffusion and all older responses in (298)–(314). Choose $0 < \ell < \epsilon$ and $\beta(t) = \eta((t - t_R)/\ell)$. Set $\Delta A = A - A_h$, $\Delta G = G - G_h$ and $\Delta H = H - H_h$, and define

$$A_\beta = A_h + \beta \Delta A, \quad G_\beta = G_h + \beta \Delta G, \quad H_\beta = H_h + \beta \Delta H. \quad (413)$$

Let the old affine residual coefficients be $r_h^\theta = G'_h + A_h^T G_h$ and $r_h^u = A'_h + A_h^2 + H_h - e_2 G_h^T$. The total interpolated core residual is exactly

$$r_\beta^\theta = (1 - \beta)r_h^\theta + \beta' \Delta G - \beta(1 - \beta)(\Delta A)^T \Delta G, \quad (414)$$

$$r_\beta^u = (1 - \beta)r_h^u + \beta' \Delta A - \beta(1 - \beta)(\Delta A)^2. \quad (415)$$

The scalar force is $r_\beta^\theta \cdot x$ and the vector force is $r_\beta^u x$. To verify these identities, expand $A_\beta^T G_\beta$ and A_β^2 and compare their quadratic terms with $(1 - \beta)$ times the old expression and β times the new zero residual. The difference is $-\beta(1 - \beta)$ times the two displayed products. Derivatives of the interpolation give exactly the β' terms. The pressure and buoyancy are linear and are both retained. For $t \leq t_R$ the interpolated fields equal the old ones with every time jet; for $t \geq t_R + \ell$ they equal the explicit unforced core with every time jet. Flatness of η proves both joins, including the pressure join.

19.7 Compact realization over the actual retained parent

The affine neighbourhood can be chosen uniformly on I . Indeed all phase and material matrix norms have finite maxima there. Writing $L_j = \sup_I \|M_j^{-1}\|$ and $Z_j = \sup_I |\zeta_j|$, one possible positive radius is

$$r_c = \min \left(1, \frac{s_0}{2\mu}, \min_{1 \leq j \leq 3} \frac{\ell_j}{4L_j}, \min_{1 \leq j \leq 3} \frac{s_0}{2K_j Z_j} \right).$$

The unchanged ℓ_j are the original material envelope radii. On B_{r_c} the base and every activated profile are exactly affine, and every material cutoff is one. This proves the needed neighbourhood directly, without extending a local support estimate to the full third trial family.

Take $\chi \in C_c^\infty(B_{r_c})$ equal to one on $B_{r_c/2}$. With $B = -J\beta\Delta A$ symmetric, put

$$h = \chi \beta \Delta G \cdot x, \quad Q = x^T B x / 2, \quad z = J \nabla(\chi Q), \quad \kappa = \chi x^T (\beta \Delta H) x / 2. \quad (416)$$

These are physical fields. Symmetry of B follows from $\text{tr}(\beta\Delta A) = 0$ by multiplying its four entries; $JB = \beta\Delta A$. The correction is compact and divergence free, is flat at birth, and equals the difference between (413) and the held core on $B_{r_c/2}$.

For completeness write $L = \beta\Delta A$, $g = \beta\Delta G$, $K = \beta\Delta H$, so that $B = -JL$ and $Q = x^T Bx/2$. Its complete first and diffusion derivatives are

$$\begin{aligned} z &= \chi Lx + QJ\nabla\chi, \\ \nabla z &= \chi L + (Lx) \otimes \nabla\chi + (J\nabla\chi) \otimes (Bx) + QJ\nabla^2\chi, \end{aligned} \tag{417}$$

$$\Delta z = (\Delta\chi)Lx + 2L\nabla\chi + (\text{tr } B)J\nabla\chi + 2J(\nabla^2\chi)Bx + QJ\nabla\Delta\chi, \tag{418}$$

$$\begin{aligned} \nabla h &= \chi g + (g \cdot x)\nabla\chi, & \Delta h &= (g \cdot x)\Delta\chi + 2g \cdot \nabla\chi, \\ \nabla\kappa &= \chi Kx + \frac{1}{2}(x^T Kx)\nabla\chi, \\ h_t &= \chi g' \cdot x, & z_t &= \chi L'x + \frac{1}{2}(x^T B'x)J\nabla\chi. \end{aligned} \tag{419}$$

Substituting these fields and derivatives into the physical identities (381)–(382) with the full retained parent gives their exact global force increment, including both $d\Delta$ terms. The product rule gives all higher jets; no derivative of χ is discarded.

Proposition 8. *The actual held finite field admits the compact smooth based correction (416). The resulting fields have the same complete initial time jets at t_R as the old finite field. On $B_{r_c/2}$ their total force is exactly (414)–(415) during the transition and is zero on $[t_R + \ell, t_R + \epsilon]$. Outside B_{r_c} both state and force equal the retained parent's.*

Proof. The streamfunction proves $\text{div } z = 0$ by equality of mixed derivatives. Flatness of β proves all initial jets, including those of the compact pressure. Where $\chi = 1$ the state is exactly the interpolated affine state, so the proved residual identities apply. Where $\beta = 1$ that state is the explicit unforced solution. Outside the compact support every correction derivative is zero, and the exact physical increment residual vanishes. This also proves support and smoothness of the full added force. $\square$

The zero assertion concerns the *total* core force. After the transition the force increment there equals the negative old force; it need not be small. The transition terms and cutoff annulus also have no established 2^{-q^2} bound. The operation now constructed is a compact, left-flat state correction with an exact force map. The next calculation is to evaluate that signed force on the full support, including the annulus, and construct a subsequent return whose complete profile satisfies (405) or the broader exact entry equation (396). The explicit constant-strain affine solution has no finite-time singularity; it supplies a correction of the local force, not the missing infinite blowup construction.

The identities are replayed in `infinite_modified/checks/return_correction_check.py`. Figure 4 shows the actual state and the full transition-force decomposition. The original profiles and control sums are those of Alpöge–Buckmaster, source Sections 3.3 and 3.7.10 and equations (4.9)–(4.10), retained with their exact receiving formulas above.

The retained third return and its exact local correction

Physical state, exact chart map, signed transition force and the full cutoff region

1 Exact physical return state

$\theta = G_R \cdot X, \quad u = A_R X, \quad p = 0$
 $(A_R)_{21} = 0, \quad (A_R)_{11} = a_R, \quad (A_R)_{22} = -a_R$
 $(A_R)_{12} = -W_R, \quad W_R \geq c_W \sigma_2/4 > 0$
 $(G_R)_1 = N_3/\sqrt{R} > 0$
 All base and three-layer gradient terms remain present.
 Proof: irc:return-G through irc:exact-return-temperature

2 Exact parabolic chart transition

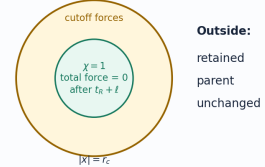
$a = \rho'/\rho > 0, \quad c'/c = a^2$
 $U' = aU(\xi + ay, \tau_0 + a^2s)$
 $\Theta' = a^3\Theta(\xi + ay, \tau_0 + a^2s)$
 $D' = a^2D, \quad G' = a^4G$ at the centred core
 Invariants: $G/|G|, \quad D/\sqrt{|G|}$
 Radius selection preserves every matrix shape and gradient direction.

3 Explicit affine evolution: zero total force

$g_1(s) = g_1 e^{-a_R s}, \quad b(s) = -W_R - g_1 \Phi_1(a_R, s)$
 $g_2(s) = e^{a_R s} [g_2 + g_1 W_R \Phi_2(a_R, s) + g_1^2 \Xi(a_R, s)]$
 $\Phi_k(a, s) = \int_0^s e^{-kar} dr, \quad \Xi(a, s) = \int_0^s e^{-2ar} \Phi_1(a, r) dr$
 $A_{11} = a_R, \quad A_{12} = b, \quad A_{21} = 0, \quad A_{22} = -a_R$
 $G' + A^T G = 0, \quad A' + A^2 + H - e_2 G^T = 0$
 The pressure H cancels the full momentum gradient. This affine solution has no finite-time singularity.

4 Flat interpolation: exact force cost

$r_\beta^\theta = (1 - \beta)r_\beta^\theta + \beta^* \Delta G - \beta(1 - \beta)(\Delta A)^T \Delta G$
 $r_\beta^\mu = (1 - \beta)r_\beta^\mu + \beta^* \Delta A - \beta(1 - \beta)(\Delta A)^2$
 $h = \chi \beta \Delta G \cdot X, \quad z = \nabla(\chi Q), \quad Q = \chi^T(-\beta \Delta A)X/2$
 Initial jets match. Every derivative of the cutoff remains in the full force.



Proofs: irc:full-chart, irc:orbit-classification, irc:affine-continuation, irc:compact-core-repair. No all-stage force budget is asserted.

Figure 4: Exact third-return state and its compact local correction. The physical $G_1 = N_3/\sqrt{R}$ and curl are positive, while every strain and vertical-gradient contribution remains present. Parabolic dilation preserves the displayed full state invariants. The explicit affine evolution cancels its total force. A flat transition and a streamfunction cutoff preserve the old initial jets and localize the correction; their displayed signed terms are the force cost still to be controlled.

A Full source controls and aggregate background identities

Source and scope. The source is Levent Alpöge and Tristan Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, the supplied 76-page released PDF. Page numbers below are the printed PDF page numbers. The companion source receipt identifies the exact local PDF by SHA-256. The equations and proofs here reconstruct the finite-dimensional system in Section 3, especially pp. 8–10 and 26–33, and its exact affine realization in Section 4, pp. 37–39. The source’s local comparison bounds are identified as source results where mentioned; the new claims proved below are the coordinate, operator, feedback, and aggregate-background identities. No infinite retained-viscosity construction is asserted.

Original data and the distinct parameters. Keep the original data of source (1.4), p. 2:

$$A_0 \geq 1, \quad \lambda_0 = \left\lceil \frac{\sqrt{A_0}}{2} \right\rceil, \quad \theta_{\text{in}}(x) = -\frac{A_0}{\lambda_0} \sin(\lambda_0 x_2) \chi_0(x), \quad u_{\text{in}} = 0.$$

Here $\chi_0 \in C_c^\infty(B_{R_0})$ is radial, $R_0 > 2$, $0 \leq \chi_0 \leq 1$, and $\chi_0 = 1$ on B_2 . Write

$$\mu = \lambda_0, \quad \nu_{\text{AB}} = \sqrt{A_0}, \quad X = \mu x, \quad T = \nu_{\text{AB}} t.$$

The symbol ν_{AB} is precisely the time-scaling parameter called ν in source Lemma 1.2, pp. 6–7. It is not physical viscosity. Physical momentum viscosity is $\varepsilon_v \geq 0$ and thermal diffusivity is $\kappa_{\text{th}} \geq 0$. The source uses μ again locally for its shooting amplitude; we display that independent parameter as μ_p , and never identify it with $\mu = \lambda_0$.

Use a hat for a source Section 3 quantity, which is expressed there in (X, T) . The exact dictionary, keeping both presentations, is

$$\begin{aligned} \theta(x, t) &= \frac{\nu_{\text{AB}}^2}{\mu} \hat{\theta}(X, T), & u(x, t) &= \frac{\nu_{\text{AB}}}{\mu} \hat{u}(X, T), & p(x, t) &= \frac{\nu_{\text{AB}}^2}{\mu^2} \hat{p}(X, T), \\ \omega(x, t) &= \nu_{\text{AB}} \hat{\omega}(X, T), & \psi(x, t) &= \frac{\nu_{\text{AB}}}{\mu^2} \hat{\psi}(X, T), \\ f_\theta(x, t) &= \frac{\nu_{\text{AB}}^3}{\mu} \hat{f}_\theta(X, T), & f_u(x, t) &= \frac{\nu_{\text{AB}}^2}{\mu} \hat{f}_u(X, T). \end{aligned} \quad (420)$$

The transformed diffusivities are

$$\widehat{\kappa}_{\text{th}} = \kappa_{\text{th}} \frac{\mu^2}{\nu_{\text{AB}}}, \quad \widehat{\varepsilon}_v = \varepsilon_v \frac{\mu^2}{\nu_{\text{AB}}}. \quad (421)$$

These statements follow directly from $\partial_t = \nu_{\text{AB}} \partial_T$, $\nabla_x = \mu \nabla_X$, and $\Delta_x = \mu^2 \Delta_X$: both scalar transport terms have factor ν_{AB}^3/μ , both momentum transport terms and gravity have factor ν_{AB}^2/μ , and the respective Laplacian terms have factors $\kappa_{\text{th}} \nu_{\text{AB}}^2 \mu$ and $\varepsilon_v \nu_{\text{AB}} \mu$. The divergence scales by ν_{AB} , curl by ν_{AB} , and $J \nabla_x \psi = (\nu_{\text{AB}}/\mu) J \nabla_X \hat{\psi}$. Thus the laboratory gravity vector $e_2 = (0, 1)$ is unchanged. The inverse transformations divide by the displayed nonzero factors and evaluate at $(x, t) = (X/\mu, T/\nu_{\text{AB}})$, so these are bijections of the corresponding smooth fields, with support radii divided by μ and time intervals divided by ν_{AB} .

Layer dictionary. For $j \geq 1$ retain the original source $\hat{\lambda}_j$, and define its actual spatial frequency $\lambda_j = \mu \hat{\lambda}_j$. The remaining dictionary is

$$\begin{aligned} M_j(t) &= \widehat{M}_j(\nu_{\text{AB}} t), & \zeta_j(t) &= \widehat{\zeta}_j(\nu_{\text{AB}} t), & p_j &= \widehat{p}_j, \\ \Theta_j(t) &= \frac{\nu_{\text{AB}}^2}{\mu} \widehat{\Theta}_j(\nu_{\text{AB}} t), & \Omega_j(t) &= \nu_{\text{AB}} \widehat{\Omega}_j(\nu_{\text{AB}} t), & \alpha(t) &= \widehat{\alpha}(\nu_{\text{AB}} t), \\ A_j(t) &= \nu_{\text{AB}}^2 \widehat{A}_j(\nu_{\text{AB}} t), & \sigma_j &= \nu_{\text{AB}} \widehat{\sigma}_j, & \Gamma_j &= \nu_{\text{AB}} \widehat{\Gamma}_j, \\ D_{<j}(t) &= \nu_{\text{AB}} \widehat{D}_{<j}(\nu_{\text{AB}} t), & G_{<j}(t) &= \nu_{\text{AB}}^2 \widehat{G}_{<j}(\nu_{\text{AB}} t), & \ell_j &= \widehat{\ell}_j/\mu. \end{aligned} \quad (422)$$

In particular $\sigma_0 = \nu_{\text{AB}}$, $A_0 = \nu_{\text{AB}}^2$, and the original $\lambda_0 = \mu$ has not been reset. All unadorned derivatives below are in original physical time t .

The exact coupled variables. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad e(\phi) = (\sin \phi, \cos \phi), \quad e_0 = R_\alpha e_2 = e(-\alpha), \quad w_0 = 1.$$

For $j \geq 1$ set $r_j = |\zeta_j|$, $e_j = \zeta_j/r_j = e(\phi_j)$, $P_j = -\Theta_j$, $A_j = \lambda_j r_j P_j$, and $B_j = \lambda_j \Theta_j \zeta_j = -A_j e_j$. The positive-amplitude variables are used only on intervals where their positivity has been established. Put $B_0 = -A_0 e_0$, and

$$\begin{aligned} G_{<j} &= B_0 + \sum_{1 \leq i < j} w_i B_i, & D_{<j} &= \dot{\alpha} J + \sum_{1 \leq i < j} w_i S_i, & S_i &= \Omega_i J e_i \otimes e_i, \\ \dot{M}_j &= D_{<j} M_j, & M_j(t_j^{\text{in}}) &= I, & \zeta_j &= M_j^{-T} p_j, \\ \dot{\Theta}_j &= a_j \Omega_j, & a_j &= -\frac{J \zeta_j \cdot G_{<j}}{\lambda_j r_j^2}, & \dot{\Omega}_j &= b_j \Theta_j, \quad b_j = \lambda_j \zeta_{j,1}. \end{aligned} \quad (423)$$

These are source (3.2)–(3.3), p. 8, in original units. To check their scaling, $a_j = (\nu_{\text{AB}}^2/\mu)\hat{a}_j$ and $b_j = \mu\hat{b}_j$; consequently both amplitude equations have their correct factors ν_{AB}^3/μ and ν_{AB}^2 . Since $\text{tr } J = \text{tr}(J e_i \otimes e_i) = 0$, Jacobi's determinant formula gives $\det M_j = 1$ throughout every finite existence interval.

Derivation of the wave equations. Let F be the exact odd smooth periodic profile of source (4.1), and let $\mathcal{P}$ be its even mean-zero primitive. Put $s = \lambda_j \zeta_j \cdot x$ and

$$\vartheta_j = \Theta_j F(s), \quad \psi_j = \frac{\Omega_j}{\lambda_j^2 r_j^2} \mathcal{P}(s), \quad v_j = \frac{\Omega_j}{\lambda_j r_j^2} J \zeta_j F(s), \quad \varpi_j = \Omega_j F'(s).$$

Differentiating $M_j^{-T} p_j$ gives $\dot{\zeta}_j = -D_{<j}^T \zeta_j$, hence $(\partial_t + D_{<j} x \cdot \nabla)s = 0$. Moreover $J \zeta_j \cdot \zeta_j = 0$, so $v_j \cdot \nabla \vartheta_j = v_j \cdot \nabla \varpi_j = 0$. On an affine earlier state $G_{<j} \cdot x, D_{<j} x$, the added scalar residual is $(\dot{\Theta}_j + \Omega_j J \zeta_j \cdot G_{<j}/(\lambda_j r_j^2))F$, and the added vorticity residual is $(\dot{\Omega}_j - \lambda_j \zeta_{j,1} \Theta_j)F'$. This proves the exact map from (423) to the unactivated local wave equations, including the fixed sign and gravity component.

The control maps and every stage. Source (3.4), p. 9, defines

$$F_j^{\text{ctl}} = -\frac{\sum_{1 \leq i < j} w_i \Omega_i e_{i,1} (J e_i \cdot \zeta_j)}{\zeta_{j,2}}, \quad F_0^{\text{ctl}} = 0. \quad (424)$$

The superscript distinguishes the paper's feedback F_j from its periodic profile F . Transposing $D_{<j}$ in (423) and taking component one gives

$$\dot{\zeta}_{j,1} = (F_j^{\text{ctl}} - \dot{\alpha}) \zeta_{j,2}.$$

Thus $\dot{\alpha} = F_j^{\text{ctl}} - \dot{Z}/\zeta_{j,2}$ implies $\zeta_{j,1}(t) = Z(t)$ whenever equality holds initially and the second component is nonzero. This proves the feedback map; the second component is not replaced by a constant.

Here is the complete source schedule (3.7)–(3.12), pp. 9–10, preserving its units. Fix $\beta = 1/8$, $Q_* \geq 200$, $Q_j = Q_* + j$, and

$$\begin{aligned} \hat{\lambda}_1 &\gg 1, & \hat{\lambda}_j &= \hat{\lambda}_{j-1}^{Q_j} \quad (j \geq 2), \\ \hat{\ell}_1 &= s_0/C_\ell, \quad C_\ell \geq 4, & \hat{\ell}_j &= \hat{\lambda}_{j-1}^{-3} \quad (j \geq 2), \\ \hat{\Pi}_1 &= (\log \hat{\lambda}_1)^{10}, & \hat{\Pi}_j &= \hat{\lambda}_{j-1}^4 (\log \hat{\lambda}_j)^{10} \quad (j \geq 2), \\ k_j &= \max\{k \in \mathbb{N}_0 : 120k \leq Q_j, \quad K(k) \leq \log \hat{\lambda}_j\}, \end{aligned} \quad (425)$$

$$J_j = 2k_j + 8, \quad L_j = (k_j + 5 + \beta) \log \hat{\lambda}_j.$$

The prescribed seed is $\Theta_j^{\text{seed}} = -(\nu_{\text{AB}}^2/\mu)\hat{\lambda}_j^{-k_j-6}$, and its gain target is

$$P_j(t_j^a) = \frac{\nu_{\text{AB}}^2}{\mu} \hat{\lambda}_j^{-7/8}, \quad \frac{P_j(t_j^a)}{|\Theta_j^{\text{seed}}|} = e^{L_j}.$$

In particular the physical frequency recursion is $\lambda_j = \mu(\lambda_{j-1}/\mu)^{Q_j}$, not $\lambda_{j-1}^{Q_j}$. Let η be smooth nondecreasing, zero on $(-\infty, 0]$ and one on $[1, \infty)$; let $\eta_2 \in C_c^\infty((0, 1))$, $\eta_2 \geq 0$, $\int_0^1 \eta_2 = 1$, and $H_2 = \|\eta_2\|_\infty$. The fixed design is

$$c_p = 3, \quad \bar{v} = 2, \quad \Lambda_1 \geq 2c_p H_2, \quad 0 < s_* \leq 1/8, \quad c_p \Lambda_1 H_2 \sin s_* \leq 1/4.$$

Set $t_1^{\text{in}} = 1/\nu_{\text{AB}}$, $s_1 = s_*$, $\Gamma_1 = \nu_{\text{AB}} \sin s_*$, and for $j \geq 2$

$$t_j^{\text{in}} = t_{j-1}^b, \quad s_j = L_j \frac{\sigma_{j-2}}{\sigma_{j-1}}, \quad \Gamma_j = \sigma_{j-1} \sin s_j, \quad \Lambda_j = L_j.$$

At activation $p_j = e(s_j)$, $\Theta_j = \Theta_j^{\text{seed}}$, $\Omega_j = \sqrt{b_j/a_j} \Theta_j$, where source Theorem 3.2 proves $a_j, b_j > 0$ on the constructed growth interval. The activation multiplier is $w_j(t) = \eta(\Gamma_j(t - t_j^{\text{in}}))$.

1. Growth: $\dot{\alpha} = F_{j-1}^{\text{ctl}}$ until the first amplitude threshold t_j^a displayed above.
2. Transition: $t_j^1 = t_j^a + \Gamma_j^{-1}$, $\eta_b = \eta(\Gamma_j(t - t_j^a))$, and $\dot{\alpha} = (1 - \eta_b)F_{j-1}^{\text{ctl}} + \eta_b F_j^{\text{ctl}}$.
3. Steering: write $z_1 = \zeta_{j,1}(t_j^1)$ and $\chi_1 = r_j(t_j^1)$. For $0 \leq h \leq h_j^{\text{max}} = c_p \Lambda_j \sin s_j$ define

$$Z_j(t) = z_1[1 - \eta(\Gamma_j(t - t_j^1))] - h\chi_1\eta_2(\Lambda_j\Gamma_j(t - t_j^1 - \Gamma_j^{-1})),$$

and $\dot{\alpha} = F_j^{\text{ctl}} - \dot{Z}_j/\zeta_{j,2}$. The fixed endpoint is $t_j^b = t_j^1 + \Gamma_j^{-1}(1 + \Lambda_j^{-1})$. Select the least h in the compact zero set $\Omega_j(t_j^b; h) = 0$ proved nonempty in the source. No uniqueness is asserted for $j \geq 2$.

4. Holding: use $\dot{\alpha} = F_j^{\text{ctl}}$ through t_{j+1}^a , concurrently with next-layer growth; evaluate $\sigma_j^2 = |G_{<j+1}(t_j^b)|$ only after selecting the stage. Since $\zeta_{j,1} = \Omega_j = 0$, both $\dot{\Omega}_j = 0$ and $\dot{\Theta}_j = 0$. This proves stationarity of this amplitude pair, while the older state and M_j may continue to evolve.

All these control maps are in physical time; $F_j^{\text{ctl}} = \nu_{\text{AB}} \hat{F}_j^{\text{ctl}}$, and Z_j is dimensionless. The source's smooth time joins follow from flatness of η , interior support of η_2 , and uniqueness for the stationary endpoint pair.

Exact aggregate parent evolution. The following identities explicitly give the background feedback rather than prescribing D, G independently.

Proposition 9. *For every finite activated system (423), put $G = G_{<q}$, $D = D_{<q}$, and $C = D_{21} - D_{12} = 2\dot{\alpha} + \sum_{i<q} w_i \Omega_i$. Then*

$$\dot{G} + D^T G = \sum_{1 \leq i < q} \dot{w}_i B_i, \tag{426}$$

$$\dot{C} - G_1 = 2\ddot{\alpha} - A_0 \sin \alpha + \sum_{1 \leq i < q} \dot{w}_i \Omega_i, \tag{427}$$

$$\begin{aligned} \dot{D} = \ddot{\alpha} J + \sum_{1 \leq i < q} (\dot{w}_i \Omega_i + w_i b_i \Theta_i) J e_i \otimes e_i \\ + \sum_{1 \leq i < q} w_i \Omega_i (\dot{\alpha} + \Sigma_i) (J e_i \otimes J e_i - e_i \otimes e_i), \end{aligned} \tag{428}$$

$$\Sigma_i = \sum_{1 \leq k < i} w_k \Omega_k \sin^2(\phi_i - \phi_k).$$

Proof. Differentiate $B_i = \lambda_i \Theta_i \zeta_i$ and use (423):

$$\dot{B}_i + D_{<i}^T B_i = -\Omega_i e_i (J e_i \cdot G_{<i}) = -S_i^T G_{<i}.$$

Also $\dot{B}_0 + (\dot{\alpha} J)^T B_0 = 0$. Expand $\dot{G} + D^T G$, insert these equalities, and keep all cross terms. Its nonactivation terms are

$$\sum_i w_i S_i^T B_0 - \sum_i w_i S_i^T \left(B_0 + \sum_{k<i} w_k B_k \right) + \sum_i w_i \sum_{k \geq i} w_k S_k^T B_i.$$

The B_0 terms cancel. The terms with $k > i$ in the last sum cancel the preceding double sum after exchanging its indices. Its diagonal terms are $w_i^2 S_i^T B_i = 0$, because $J e_i \cdot e_i = 0$. This proves (426) with every activation derivative retained. Next $\dot{C} = 2\ddot{\alpha} + \sum_i \dot{w}_i \Omega_i + \sum_i w_i \lambda_i \zeta_{i,1} \Theta_i$, while $G_1 = B_{0,1} + \sum_i w_i \lambda_i \Theta_i \zeta_{i,1}$ and $B_{0,1} = A_0 \sin \alpha$. Subtraction proves (427). Finally direct projection of $\dot{\zeta}_i = -D_{<i}^T \zeta_i$ onto e_i and $J e_i$ gives $\dot{e}_i = (\dot{\alpha} + \Sigma_i) J e_i$. Differentiate $S_i = \Omega_i J e_i \otimes e_i$, use $J^2 = -I$, and sum with the activation derivatives. This proves (428). $\square$

For completeness these affine dynamics have an exact momentum realization. On an affine region set $u = Dx$, $\theta = G \cdot x$, $H = D' + D^2 - e_2 \otimes G$, and

$$p = -\frac{1}{2} x^T (\text{sym } H) x, \quad f_\theta = (\dot{G} + D^T G) \cdot x, \quad f_u = \frac{1}{2} (\dot{C} - G_1) J x.$$

Indeed $H - \text{sym } H$ is the antisymmetric matrix with entry difference $H_{21} - H_{12} = \dot{C} - G_1$; here $D^2 = -\det(D)I$ for a trace-free 2×2 matrix, verified by direct matrix multiplication. Thus $u_t + u \cdot \nabla u + \nabla p - \theta e_2 = f_u$. Both Laplacians $\Delta \theta$ and Δu are identically zero, so these same fields and forces solve the retained-viscosity equations on that region for every $\kappa_{\text{th}}, \varepsilon_v \geq 0$.

Relative geometry and its proof. Set $\rho_j = \log r_j$, $d_{ij} = \phi_j - \phi_i$, $\phi_0 = -\alpha$, and

$$\mathcal{G}_j = \sum_{0 \leq i < j} w_i A_i \sin d_{ij}.$$

Projection of the phase equation, using $J e_i \cdot e_j = -\sin d_{ij}$ and $e_i \cdot e_j = \cos d_{ij}$, gives

$$\dot{\rho}_j = \sum_{1 \leq i < j} w_i \Omega_i \sin d_{ij} \cos d_{ij}, \quad \dot{\phi}_j = -\dot{\alpha} - \Sigma_j, \quad \dot{d}_{ij} = \Sigma_i - \Sigma_j, \quad (429)$$

$$a_j = \mathcal{G}_j / (\lambda_j r_j), \quad \dot{\Omega}_j = -A_j \sin \phi_j, \quad (\log P_j)' = -\mathcal{G}_j \Omega_j / A_j,$$

$$F_j^{\text{ctl}} = -\Sigma_j + \dot{\rho}_j \tan \phi_j = \frac{\sum_{1 \leq i < j} w_i \Omega_i \sin \phi_i \sin d_{ij}}{\cos \phi_j}.$$

The last equality follows from $\sin \phi_i = \sin \phi_j \cos d_{ij} - \cos \phi_j \sin d_{ij}$. These are source (3.6), p. 9, and (3.55), p. 26, with A_0 retained.

For adjacent phases $\vartheta_j = \phi_j - \phi_{j-1}$, the matrix difference is $D_{<j} - D_{<j-1} = w_{j-1} \Omega_{j-1} J e_{j-1} \otimes e_{j-1}$. Its transpose annihilates ζ_{j-1} , so both ζ_{j-1} and ζ_j solve the same transport matrix equation. Its trace is zero, hence the derivative of their determinant is zero. At activation $\zeta_{j-1} = r_{j-1} (t_j^{\text{in}}) e_2$ and $\zeta_j = e(s_j)$, and the determinant in that order is $-r_{j-1} (t_j^{\text{in}}) \sin s_j$. Therefore, with the sign of this orientation kept,

$$d_j^{\text{adj}} := r_j \sin \vartheta_j = \sin s_j e^{-\rho_{j-1}(t) + \rho_{j-1}(t_j^{\text{in}})}. \quad (430)$$

This is source (3.56), p. 26; its symbol d_j^{adj} denotes the source's separate $\mathfrak{d}_j$ rather than an angle d_{ij} .

For $q \geq 2$, all older activation weights are one on stage q . Define

$$E_2 = \sum_{i=1}^{q-2} \Omega_i \sin d_{iq} \cos d_{iq}, \quad E_3 = \dot{\rho}_{q-1}, \quad E_1 = \sin \vartheta_q \sum_{i=1}^{q-2} \Omega_i \sin(d_{iq} + d_{i,q-1}),$$

and $R_q = \sum_{i=0}^{q-2} A_i \sin d_{iq}$. Then

$$\dot{\vartheta}_q = -E_1 - \Omega_{q-1} \sin^2 \vartheta_q, \quad \dot{\rho}_q = E_2 + \Omega_{q-1} \sin \vartheta_q \cos \vartheta_q, \quad \mathcal{G}_q = A_{q-1} \sin \vartheta_q + R_q. \quad (431)$$

The first equality follows by subtracting the two angle equations and using $\sin^2 a - \sin^2 b = \sin(a-b)\sin(a+b)$; the other two are the corresponding split sums. At $q = 2$ the shear sums E_1, E_2, E_3 vanish, but $R_2 = A_0 \sin d_{02}$ is retained. The source makes this exceptional base term explicit on p. 28.

Exact later-stage model, including parent response. Use $\sigma = \sigma_{q-1}$, $s = s_q$, $\Gamma = \sigma \sin s$, $L = L_q$, $\tau = \Gamma(t - t_q^1)$, $\chi = r_q$, $\chi_1 = r_q(t_q^1)$, $W = \Omega_{q-1}/\sigma$, and $\mu_p = h/(L \sin s)$. Write $d = d_q^{\text{adj}}$, $d_1^* = d(t_q^1)$, and $e_4 = z_1/d_1^* - 1$. To avoid collision with the error coefficient d_1 , the asterisk here marks the fixed adjacent determinant factor at steering entry. Then source (3.66), p. 31, is exactly

$$\begin{aligned} c_1 &= (d/\sin s) \cos \vartheta_q, & d_1 &= \chi E_2/\Gamma, \\ c_2 &= c_5 = (A_{q-1}/\sigma^2)(d/\sin s), & c_3 &= \sin s/d, \\ d_3 &= 1 - (d_1^*/d)(1 + e_4), & c_4 &= (d_1^*/\sin s)(1 + e_4) = z_1/\sin s, \\ d_4 &= \frac{\chi_1 R_q}{\chi \sigma^2 \sin s}, & d_2 &= \frac{A_{q-1}}{\sigma^2 \sin s} \frac{d(\cos \phi_q - 1) - Z(\cos \vartheta_q - 1)}{\chi}. \end{aligned} \quad (432)$$

Define $\tilde{k} = c_5 \chi_1 / \chi^2 + d_4$ and

$$\begin{aligned} g(\tau) &= \begin{cases} \eta(\tau) + d_3(1 - \eta(\tau)), & 0 \leq \tau \leq 1, \\ 1 + \mu_p c_3 L \chi_1 \eta_2(L(\tau - 1)), & 1 \leq \tau \leq 1 + L^{-1}, \end{cases} \\ f(\tau) &= \begin{cases} (c_4/\chi_1)(1 - \eta(\tau)), & 0 \leq \tau \leq 1, \\ -\mu_p L \eta_2(L(\tau - 1)), & 1 \leq \tau \leq 1 + L^{-1}. \end{cases} \end{aligned}$$

On every interval where $\Theta_q \neq 0$, put $v = \sigma \Omega_q / (\lambda_q \chi_1 \Theta_q)$. The exact system is

$$\chi' = c_1 W + d_1, \quad W' = c_2 g / \chi + d_2, \quad v' = f - \tilde{k} v^2, \quad (\log P_q)' = \tilde{k} v, \quad (433)$$

where these primes mean τ derivatives.

Proof. From (431), $\dot{r}_q = r_q E_2 + \Omega_{q-1} d \cos \vartheta_q$, proving the first equation. Next

$$\dot{\Omega}_{q-1} = A_{q-1} \sin(\vartheta_q - \phi_q) = \frac{A_{q-1}}{\chi} (d \cos \phi_q - Z \cos \vartheta_q).$$

Split its last parentheses as $d - Z + d(\cos \phi_q - 1) - Z(\cos \vartheta_q - 1)$. The profile gives $1 - Z/d = g$ on each of its two intervals, proving the second equation including the full d_2 . Also

$$\frac{a_q \lambda_q \chi_1}{\sigma \Gamma} = \frac{\chi_1}{\sigma^2 \sin s} \left(\frac{A_{q-1} d}{\chi^2} + \frac{R_q}{\chi} \right) = \tilde{k}, \quad \frac{\sigma b_q}{\Gamma \lambda_q \chi_1} = \frac{Z}{\chi_1 \sin s} = f.$$

Differentiate the quotient and $\log P_q$ to obtain the last equations. This proves the exact substitution without freezing $A_{q-1}, \Omega_{q-1}, \rho_{q-1}$, or any older layer. $\square$

Where source selection is proved. The source first proves interval-uniform bounds for (433) in Lemma 3.9, pp. 18–21, then excludes zeros of Θ_q in Proposition 3.10, pp. 21–22. For the actual stage it first constructs the closed older/geometry subsystem on a common compact subset of

$$\det M_j > 1/2, \quad |M_j^{-T} p_j| > 1/4, \quad \zeta_{q-1,2} > z_*/2, \quad \zeta_{q,2} > z_*/2,$$

by the simultaneous bounds in Sections 3.7.2–3.7.6, pp. 26–32. In original units the pair data and coefficient sizes change by the dictionary above, while this geometric domain is unchanged.

The newest pair is absent from that subsystem: F_q^{ctl} uses only older vorticities and phases, and M_q uses only $D_{<q}$. One then solves the newest linear pair over the constructed subsystem. This order matters for the exact map (433). The source's $c_i = 1 + O(\epsilon_{\text{sch}})$, $d_i = O(\epsilon_{\text{sch}})$ estimates are consequences of its coupled induction, not independent prescriptions available after changing those dynamics.

For reference, source (3.67), p. 32, gives on the selected actual stage

$$\frac{\Omega_{q-1}(t_q^b)}{\sigma} \in [1/3 - C\epsilon_{\text{sch}}, 1 + \bar{v} + C\epsilon_{\text{sch}}], \quad \log \frac{P_q(t_q^b)}{(\nu_{\text{AB}}^2/\mu)\hat{\lambda}_q^{-7/8}} \in [1 + I_0 - C\epsilon_{\text{sch}}, 1 + \bar{v} + C\epsilon_{\text{sch}}],$$

where $I_0 = \int_0^1 [(1 + \tau^2/2)^2(1 + \tau)]^{-1} d\tau \in [2/9, \log 2]$. Source (3.58), p. 27, gives the separate first-stage gain interval $[1 + \log 2, 2 + \Lambda_1^{-1}]$ and exact duration $(L_1 + 2 + \Lambda_1^{-1})/(\nu_{\text{AB}} \sin s_*)$. These are source bounds, not claims that the diffusive evolution below satisfies them without further estimates.

An explicit growth-end vorticity lower bound. The source growth proof is Section 3.7.5, pp. 29–30, including (3.63)–(3.64), not the later steering quotient alone. Its identities admit the following explicit extraction from the source's choice of low-order tolerance. Put $u = \Omega_q/\Theta_q$, $u_* = (b_q/a_q)^{1/2}$, $\gamma = (a_q b_q)^{1/2}$, and $z = u/u_*$. The source chooses its fixed error tolerance before its frequency threshold; require the following finite additional strict comparison when making that same choice: all the growth errors used here have absolute value at most $\delta \leq 1/4$. Thus its proved geometric and coefficient estimates give

$$r_q \geq 1 - \delta, \quad \frac{d_q^{\text{adj}}}{\sin s_q} \geq 1 - \delta, \quad \frac{\mathcal{G}_q}{\sigma_{q-1}^2 \sin s_q} \leq 1 + \delta, \quad |(\log u_*)'| \leq \delta\gamma.$$

The original source's error rates in (3.63) are $C\epsilon_{\text{sch}}\Gamma_q/L_q$, while $\gamma = (1 + O(\epsilon_{\text{sch}}))\Gamma_q$, so this comparison is obtained by the same fixed-data choice of ϵ_{sch} ; it imposes no new order-dependent frequency condition.

Here is the complete quotient proof used for the extraction. Direct differentiation before any possible zero of Θ_q gives

$$u' = b_q - a_q u^2, \quad z' = \gamma(1 - z^2) - z(\log u_*)', \quad z(t_q^{\text{in}}) = 1.$$

At $z = 1 + \delta$, the right side is at most $\gamma[-2\delta - \delta^2 + \delta(1 + \delta)] = -\delta\gamma < 0$; at $z = 1 - \delta$ it is at least $\gamma[2\delta - \delta^2 - \delta(1 - \delta)] = \delta\gamma > 0$. The first-exit argument proves $1 - \delta \leq z \leq 1 + \delta$. This bound also excludes a first zero of Θ_q : at such a zero it would force $\Omega_q = 0$, contradicting backward uniqueness of the continuous-coefficient linear system from its nonzero entry datum. Consequently the bound holds through t_q^a . On growth the preceding phase is vertical, so $b_q = \lambda_q d_q^{\text{adj}}$ and $a_q = \mathcal{G}_q/(\lambda_q r_q)$. It follows exactly that

$$u_* = \lambda_q \sqrt{\frac{d_q^{\text{adj}} r_q}{\mathcal{G}_q}}, \quad \frac{|\Omega_q|}{|\Theta_q|} = z u_* \geq \frac{\lambda_q}{\sigma_{q-1}} \frac{(1 - \delta)^2}{\sqrt{1 + \delta}} \geq \frac{\lambda_q}{2\sigma_{q-1}}.$$

For the last inequality the function $(1 - \delta)^2/\sqrt{1 + \delta}$ is decreasing on $[0, 1/4]$ by direct logarithmic differentiation, and its value at $1/4$ is $9/(8\sqrt{5}) > 1/2$ because $81 > 80$. Thus the exact physical growth threshold satisfies

$$|\Omega_q(t_q^a)| \geq \frac{\lambda_q}{2\sigma_{q-1}} \frac{\nu_{\text{AB}}^2}{\mu} \hat{\lambda}_q^{-7/8} = \frac{\nu_{\text{AB}}^2}{2\sigma_{q-1}} \hat{\lambda}_q^{1/8}, \quad r_q(t_q^a) \geq 3/4. \quad (434)$$

For $q = 1$ the exact growing eigenline instead gives equality $|\Omega_1/\Theta_1| = \lambda_1/\nu_{\text{AB}}$ and $r_1 = 1$, which is stronger than the displayed lower bound. No unproved viscous tracking assertion is used in this extraction.

The actual profile and retained viscosity. The source profile, (4.1), p. 37, is

$$F(s) = \sin s + \chi(s)(s - \sin s), \quad |s| \leq \pi,$$

periodically continued, where χ is even, smooth, compactly supported in $(-\pi, \pi)$ and one on $[-s_0, s_0]$. Thus $F = s$ on that interval. Direct differentiation of the local waves gives the retained-diffusion residual increments

$$\begin{aligned} \mathcal{R}_\theta &= (\dot{\Theta}_j - a_j \Omega_j) F(s) - \kappa_{\text{th}} \lambda_j^2 r_j^2 \Theta_j F''(s), \\ \mathcal{R}_\omega &= (\dot{\Omega}_j - b_j \Theta_j) F'(s) - \varepsilon_v \lambda_j^2 r_j^2 \Omega_j F'''(s). \end{aligned} \quad (435)$$

Consequently the actual fixed-profile, finite-stage retained-viscosity realization uses the original coupled system and the explicit additional forces $-\kappa_{\text{th}} \Delta \theta$ and $-\varepsilon_v \Delta u$. The signs in (435) are the signs of these forces when diffusion is placed on the right of the PDE. They are not positive damping terms in the forcing.

This repair preserves the exact actual D, G in Proposition 9: on every older affine region, $F'' = F''' = 0$, all envelope derivatives vanish, and the source's finite corrections vanish on that plateau. Hence the repaired force increments vanish in each affine core, and both Laplacians of the full older affine fields are zero. In physical coordinates the affine core of layer j contains

$$B_{r_j^{\text{aff}}(t)}, \quad r_j^{\text{aff}}(t) = \min \left\{ \frac{\ell_j}{2 \|M_j^{-1}\|}, \frac{s_0}{\lambda_j r_j} \right\}.$$

The exact source nesting criterion is $K_j K_q \ell_q < \ell_j/2$ and $\lambda_j r_j K_q \ell_q < s_0$ for every older layer $j < q$; scaling by μ preserves both inequalities. The full-vector repair and the unchanged support follow because differentiation of a smooth compactly supported field does not enlarge its support. Quantitative force costs are provided separately in the parent calculation; the algebraic identity alone gives no uniform terminal force bound.

The exact feedback terms in a scalar-damped comparison. For transparency, compute also the altered finite-dimensional equations obtained from Fourier eigenprofiles. They are not the released fixed profile. Let $d_j^\theta = \kappa_{\text{th}} \lambda_j^2 r_j^2$ and $d_j^\omega = \varepsilon_v \lambda_j^2 r_j^2$, and replace only the two amplitude equations by

$$\dot{\Theta}_j = a_j \Omega_j - d_j^\theta \Theta_j, \quad \dot{\Omega}_j = b_j \Theta_j - d_j^\omega \Omega_j, \quad (436)$$

while computing all D, G, F_j^{ctl} from the altered amplitudes and phases. For a sine profile these are precisely the homogeneous local wave equations, since $F'' = -F$ and $F''' = -F'$. Repeating the proof of Proposition 9 gives exactly

$$\begin{aligned} \dot{G} + D^T G &= \sum_{i < q} \dot{w}_i B_i - \sum_{i < q} w_i d_i^\theta B_i, \\ \dot{C} - G_1 &= 2\ddot{\alpha} - A_0 \sin \alpha + \sum_{i < q} \dot{w}_i \Omega_i - \sum_{i < q} w_i d_i^\omega \Omega_i. \end{aligned} \quad (437)$$

In (428), replace $b_i \Theta_i$ by $b_i \Theta_i - d_i^\omega \Omega_i$. The geometric equations (429), (430), and (431) retain their formulas but now use altered amplitudes. The exact gradient-amplitude drift becomes

$$\frac{\dot{A}_i}{A_i} = \dot{\rho}_i - \frac{\mathcal{G}_i \Omega_i}{A_i} - d_i^\theta. \quad (438)$$

This follows by differentiating $A_i = \lambda_i r_i P_i$ and (436). In particular

$$\dot{\mathcal{G}}_j = A_0 \cos d_{0j} (\Sigma_0 - \Sigma_j) + \sum_{1 \leq i < j} \dot{w}_i A_i \sin d_{ij}$$

$$+ \sum_{1 \leq i < j} w_i A_i \left[\left(\dot{\rho}_i - \frac{\mathcal{G}_i \Omega_i}{A_i} - d_i^\theta \right) \sin d_{ij} + \cos d_{ij} (\Sigma_i - \Sigma_j) \right], \quad \Sigma_0 = 0.$$

No base term or activation derivative has been discarded.

The feedback derivative is also explicit. With $N_{ij} = w_i \Omega_i \sin \phi_i \sin d_{ij}$,

$$\begin{aligned} \dot{F}_j^{\text{ctl}} = & \frac{1}{\cos \phi_j} \sum_{i < j} [(\dot{w}_i \Omega_i + w_i (b_i \Theta_i - d_i^\omega \Omega_i)) \sin \phi_i \sin d_{ij} \\ & + w_i \Omega_i \cos \phi_i (-\dot{\alpha} - \Sigma_i) \sin d_{ij} + w_i \Omega_i \sin \phi_i \cos d_{ij} (\Sigma_i - \Sigma_j)] + F_j^{\text{ctl}} \tan \phi_j (-\dot{\alpha} - \Sigma_j). \end{aligned}$$

The direct damping term is therefore $-\sum_{i < j} w_i d_i^\omega \Omega_i \sin \phi_i \sin d_{ij} / \cos \phi_j$, in addition to all changes in the evolving state and the other terms displayed above.

For the later-stage variables and coefficients (432), direct differentiation gives the exact altered model

$$\begin{aligned} \chi' &= c_1 W + d_1, & W' &= c_2 g / \chi + d_2 - \frac{d_{q-1}^\omega}{\Gamma} W, \\ v' &= f - \tilde{k} v^2 + \frac{d_q^\theta - d_q^\omega}{\Gamma} v, & (\log P_q)' &= \tilde{k} v - \frac{d_q^\theta}{\Gamma}. \end{aligned} \quad (439)$$

To prove the extra terms, $W' = \dot{\Omega}_{q-1} / (\sigma \Gamma)$ supplies its damping term; differentiating Ω_q / Θ_q supplies $(d_q^\theta - d_q^\omega) \Omega_q / \Theta_q$; differentiating $\log(-\Theta_q)$ supplies $-d_q^\theta$. The previous derivation supplies all remaining terms. These are identities on every interval where their displayed denominators are nonzero, and do not assert a new existence theorem from assumed coefficient bounds.

Even when $\kappa_{\text{th}} = \varepsilon_v$, only the explicit damping term in the quotient equation cancels. The preceding-layer term $-d_{q-1}^\omega W / \Gamma$ remains, A_{q-1} and R_q obey the altered gradient drifts, and hence c_2, c_5, d_4 evolve differently. On holding, $\zeta_{q,1} = \Omega_q = 0$ still solves those two equations, but

$$P_q(t) = P_q(t_q^b) \exp \left(-\kappa_{\text{th}} \lambda_q^2 \int_{t_q^b}^t r_q(s)^2 ds \right)$$

is no longer stationary if $\kappa_{\text{th}} > 0$. This follows by solving $\dot{P}_q = -d_q^\theta P_q$; r_q retains its evolving value inside the integral.

Finite-stage gain and exact diffusion integrals. For the damped comparison, (439) proves the exact identity

$$\log \frac{P_q(t_b)}{P_q(t_a)} = \int_{t_a}^{t_b} \left(-\frac{\mathcal{G}_q \Omega_q}{A_q} \right) dt - \kappa_{\text{th}} \lambda_q^2 \int_{t_a}^{t_b} r_q^2 dt.$$

For a stage partitioned into growth, transition, positive steering, negative steering, and holding, the last integral is the sum over all five actual intervals. Its original-unit coefficient is

$$\kappa_{\text{th}} \lambda_q^2 \int r_q^2 dt = \frac{\kappa_{\text{th}} \mu^2}{\nu_{\text{AB}}} \hat{\lambda}_q^2 \int \hat{r}_q^2 dT.$$

On any finite interval of length H with proved bounds $m \leq r_q \leq M$, its exact value is bracketed by $\kappa_{\text{th}} \lambda_q^2 m^2 H$ and $\kappa_{\text{th}} \lambda_q^2 M^2 H$. The inequality is simply the integrated pointwise inequality $m^2 \leq r_q^2 \leq M^2$. It does not permit retaining the original source gain integral when the coupled parent trajectory has changed. Conversely, in the actual fixed-profile force repair the trajectory and the gain integral remain exactly those of the source, and the diffusion cost is paid by the additional explicit force, not by a damping loss in that trajectory.

Status of the infinite sequence. The source constructs its inviscid infinite sequence after fixing low-order design bounds, then all coefficient/correction constants, then the nondecreasing majorant K of (7.21), p. 63, and finally one initial frequency threshold in (8.1)–(8.6), pp. 66–67. It proves the coupled stages before evaluating each new σ_q , and it proves full-lifetime bounds before applying the finite corrections. The retained-viscosity finite-stage force repair proved here commutes exactly with every finite activated sum. Extending its force series smoothly to the limiting time requires estimates for the additional Laplacian terms; no such terminal estimates have been established in this document. The exact Fourier comparison (439) does not supply them, and its altered parent terms prevent importing the source’s inviscid coefficient bounds without a new calculation. Neither statement is a disproof of another retained-viscosity construction.

B Exact periodic profile evolution and the affine region

This calculation concerns the precise periodic profile in Alpöge–Buckmaster, *Blowup for the Boussinesq equations with smooth forcing*, released 76-page source, Sections 3.1 and 4.1, equations (3.1), (3.2), (4.1)–(4.4). It proves exact identities and a profile evolution for a prescribed affine background, including the exact force that keeps the original profile. It also identifies the change to the affine region when that force is omitted. No conclusion about an infinite viscous controlled sequence follows from these profile identities alone.

B.1 The original profile and all coefficients

Let $0 < s_0 < 1$ and let $\chi \in C_c^\infty((-\pi, \pi))$ be even and equal to one on $[-s_0, s_0]$. The actual source profile is the 2π -periodic extension of

$$F(s) = \sin s + \chi(s)(s - \sin s), \quad -\pi < s < \pi. \quad (440)$$

It is smooth because it equals $\sin s$ near the endpoints of this period; it is odd because χ is even. Hence its period mean is zero. It equals s on $[-s_0, s_0]$, in particular $F'(0) = 1$. Let P be its unique mean-zero periodic primitive. Indeed $\tilde{P}(s) = \int_0^s F(r) dr$ is even and periodic, and $P = \tilde{P} - \langle \tilde{P} \rangle$ has the claimed properties. Here $\langle h \rangle = (2\pi)^{-1} \int_{-\pi}^{\pi} h(s) ds$.

Fix a closed finite time interval $I = [t_0, t_1]$, smooth $D : I \rightarrow \mathbb{R}^{2 \times 2}$ with $\text{tr } D = 0$, smooth $G : I \rightarrow \mathbb{R}^2$, a frequency $\lambda > 0$, and a nonzero initial vector $\zeta(t_0)$. Define $J(x_1, x_2) = (-x_2, x_1)$ and

$$\dot{\zeta} = -D^T \zeta, \quad s(x, t) = \lambda \zeta(t) \cdot x, \quad \kappa(t) = |\zeta(t)|^2, \quad K(t) = \lambda^2 \kappa(t). \quad (441)$$

The notation $\kappa = |\zeta|^2$ is the source's geometric notation. The physical temperature diffusivity is $\kappa_{\text{th}} \geq 0$, and the physical momentum viscosity is $\nu_{\text{phys}} \geq 0$; neither is the paper's time-scaling parameter named ν . These constants are not set to one. This calculation does not rescale A_0, λ_0, μ , the physical coordinates, time, or laboratory gravity $e_2 = (0, 1)$.

The nonvanishing of ζ follows directly from the invertible fundamental matrix of $\dot{M} = DM$, $M(t_0) = \text{Id}$: differentiation gives $\zeta = M^{-T} \zeta(t_0)$, and differentiation of the determinant gives $\det M = \exp \int_{t_0}^t \text{tr } D = 1$. Write

$$a(t) = -\frac{J\zeta \cdot G}{\lambda \kappa}, \quad b(t) = \lambda \zeta_1. \quad (442)$$

In particular the buoyancy coefficient uses the first laboratory component ζ_1 .

For any smooth scalar θ and divergence-free velocity u , with $\omega = \partial_1 u_2 - \partial_2 u_1$, define the full retained-diffusion residuals by

$$\mathcal{R}_\theta(\theta, u) = \partial_t \theta + u \cdot \nabla \theta - \kappa_{\text{th}} \Delta \theta, \quad \mathcal{R}_\omega(\theta, u) = \partial_t \omega + u \cdot \nabla \omega - \partial_1 \theta - \nu_{\text{phys}} \Delta \omega. \quad (443)$$

Thus these residuals are the required scalar force and curl of the required momentum force. On the region under consideration the older fields are $\theta_{\text{old}} = G \cdot x$, $u_{\text{old}} = Dx$; they need not have zero residual. All statements below subtract their actual residuals.

B.2 Exact residual and failure of constant-profile damping

Proposition 1. *For arbitrary smooth real amplitudes Θ, Ω , define the source wave*

$$\vartheta = \Theta F(s), \quad \psi = \frac{\Omega}{\lambda^2 \kappa} P(s), \quad v = \nabla^\perp \psi = \frac{\Omega}{\lambda \kappa} J\zeta F(s), \quad \varpi = \Delta \psi = \Omega F'(s). \quad (444)$$

The scalar and curl residual increments are exactly

$$\mathcal{R}_\theta(\theta_{\text{old}} + \vartheta, u_{\text{old}} + v) - \mathcal{R}_\theta(\theta_{\text{old}}, u_{\text{old}}) = (\dot{\Theta} - a\Omega)F - \kappa_{\text{th}} K \Theta F'', \quad (445)$$

$$\mathcal{R}_\omega(\theta_{\text{old}} + \vartheta, u_{\text{old}} + v) - \mathcal{R}_\omega(\theta_{\text{old}}, u_{\text{old}}) = (\dot{\Omega} - b\Theta)F' - \nu_{\text{phys}} K \Omega F'''. \quad (446)$$

Every profile in these formulas is evaluated at $s(x, t)$.

Proof. Equation (441) gives $(\partial_t + Dx \cdot \nabla)s = \lambda(\dot{\zeta} + D^T \zeta) \cdot x = 0$. The orthogonality $J\zeta \cdot \zeta = 0$ gives both $v \cdot \nabla \vartheta = 0$ and $v \cdot \nabla \varpi = 0$. The older vorticity $D_{21} - D_{12}$ is spatially constant, so $v \cdot \nabla \omega_{\text{old}} = 0$. Moreover $v \cdot G = -a\Omega F$, $\partial_1 \vartheta = b\Theta F'$, $\Delta \vartheta = K\Theta F''$, and $\Delta \varpi = K\Omega F'''$. Expanding (443), canceling the older residual, and retaining all these terms proves (445)–(446). The displayed velocity and vorticity in (444) follow by differentiating $P' = F$ twice; their signs follow from the specified J . $\square$

The profile derivatives, including their transition-region terms, are

$$F' = \cos s + \chi'(s)(s - \sin s) + \chi(s)(1 - \cos s), \quad (447)$$

$$F'' = (\chi - 1) \sin s + 2\chi'(1 - \cos s) + \chi''(s - \sin s), \quad (448)$$

$$F''' = (\chi - 1) \cos s + 3\chi' \sin s + 3\chi''(1 - \cos s) + \chi'''(s - \sin s). \quad (449)$$

These follow by the product rule on $(-\pi, \pi)$ and extend smoothly across the period endpoints.

Proposition 2. *There is no constant $c \in \mathbb{R}$ such that $F'' = cF$ on the circle, and no constant $c \in \mathbb{R}$ such that $F''' = cF'$ on the circle. Consequently, at a time with $\kappa_{\text{th}} > 0$ and $\Theta \neq 0$, no choice of scalar $\dot{\Theta} - a\Omega$ makes (445) vanish for every phase. At a time with $\nu_{\text{phys}} > 0$ and $\Omega \neq 0$, no choice of scalar $\dot{\Omega} - b\Theta$ makes (446) vanish for every phase.*

Proof. On $(-s_0, s_0)$, $F(s) = s$ and $F'' = 0$. An identity $F'' = cF$ therefore implies $cs = 0$ for every such s , so $c = 0$. Then $F'' = 0$ everywhere would make F globally affine. Its periodicity would force its slope to be zero, contradicting $F'(0) = 1$. If $F''' = cF'$, evaluation inside that same interval gives $0 = c$, since $F' = 1$ there. Then $F''' = 0$ everywhere makes F a quadratic polynomial on $\mathbb{R}$. Periodicity forces both its quadratic and linear coefficients to vanish, again contradicting $F'(0) = 1$. Since $K > 0$, each asserted residual cancellation with a nonzero amplitude would imply precisely one of these excluded constant-coefficient identities, by division by $\kappa_{\text{th}}K\Theta$ or $\nu_{\text{phys}}K\Omega$. $\square$

For example a proposed common damping equation $\dot{\Theta} = a\Omega - d\Theta$, $\dot{\Omega} = b\Theta - d\Omega$ gives $-d\Theta F - \kappa_{\text{th}}K\Theta F''$ and $-d\Omega F' - \nu_{\text{phys}}K\Omega F'''$ as its two residuals. On the exact affine core they are $-d\Theta s$ and $-d\Omega$. This proves failure of that fixed-profile amplitude prescription, not failure of a viscous controlled construction.

B.3 An exact evolving-profile map for all diffusivities

We now construct the profile evolution instead of replacing the missing profile dynamics by scalar damping. Let $T(s, t), V(s, t)$ be smooth, mean-zero periodic functions, and let $Q_s = V$, $\langle Q \rangle = 0$. Set

$$\vartheta = T(s, t), \quad \psi = \frac{Q(s, t)}{\lambda^2 \kappa}, \quad v = \frac{J\zeta}{\lambda \kappa} V(s, t), \quad \varpi = V_s(s, t). \quad (450)$$

The time derivative T_t here and below holds phase s fixed.

Proposition 3. *The residual increments for (450) are*

$$T_t - aV - \kappa_{\text{th}}KT_{ss}, \quad \partial_s(V_t - bT - \nu_{\text{phys}}KV_{ss}). \quad (451)$$

Their vanishing is equivalent in the mean-zero class to

$$T_t = aV + \kappa_{\text{th}}KT_{ss}, \quad V_t = bT + \nu_{\text{phys}}KV_{ss}. \quad (452)$$

For the initial data $T(s, t_0) = \Theta_0 F(s)$, $V(s, t_0) = \Omega_0 F(s)$, this system has a unique smooth periodic solution on I given by the following exact Fourier evolution. Write

$$F(s) = \sum_{n \geq 1} f_n \sin(ns), \quad f_n = \frac{1}{\pi} \int_{-\pi}^{\pi} F(s) \sin(ns) ds. \quad (453)$$

For each $n \geq 1$, let $U_n(t, t_0)$ solve

$$\partial_t U_n = \begin{pmatrix} -\kappa_{\text{th}} K n^2 & a \\ b & -\nu_{\text{phys}} K n^2 \end{pmatrix} U_n, \quad U_n(t_0, t_0) = \text{Id}. \quad (454)$$

Then

$$\begin{pmatrix} T(s, t) \\ V(s, t) \end{pmatrix} = \sum_{n \geq 1} f_n U_n(t, t_0) \begin{pmatrix} \Theta_0 \\ \Omega_0 \end{pmatrix} \sin(ns). \quad (455)$$

Proof. The phase and tangency computations in Proposition 1 remain valid, including $v \cdot \nabla T = v \cdot \nabla V_s = 0$. Now $v \cdot G = -aV$ and $\partial_1 T = bT_s$, which proves (451). The expression inside its second derivative has zero period mean. If its phase derivative vanishes, it is constant in s , and its zero mean forces this constant to be zero. This proves the exact equivalence to (452).

For completeness, each matrix solution in (454) can be defined by the convergent iterated-integral series $U_n = \text{Id} + \sum_{j \geq 1} \int_{t_0 < r_j < \dots < r_1 < t} A_n(r_1) \cdots A_n(r_j) dr_j \cdots dr_1$, where A_n is the displayed matrix. For a fixed n , its j th term is bounded by $(\|I\| \sup_I \|A_n\|)^j / j!$, so termwise differentiation gives the ODE. Uniqueness follows by integrating the difference and applying the iterated integral estimate, which tends to zero as $j \rightarrow \infty$. For its column solution $y = (y_1, y_2)$, direct differentiation gives

$$\frac{1}{2} \partial_t |y|^2 = -K n^2 (\kappa_{\text{th}} y_1^2 + \nu_{\text{phys}} y_2^2) + (a + b) y_1 y_2 \leq \frac{1}{2} (|a| + |b|) |y|^2.$$

Multiplication by $\exp(-\int_{t_0}^t (|a| + |b|))$ and differentiation therefore proves the bound, uniform in n ,

$$|y(t)| \leq e^{\frac{1}{2} \int_{t_0}^t (|a| + |b|)} |y(t_0)|.$$

Repeated differentiation of (454) gives, for each fixed j , $\sup_I |\partial_t^j y| \leq C_j (1+n)^{2j} |y(t_0)|$, by induction: the coefficient derivatives of A_n are bounded by constants times $(1+n)^2$, and the differentiated product has finitely many terms.

Integration by parts any prescribed number of times in (453) shows $|f_n| \leq C_N n^{-N}$ for every N , without endpoint terms by periodicity. Thus (455) and every mixed derivative converge uniformly on $\mathbb{T} \times I$, and termwise differentiation proves the equations and the prescribed initial values. The Fourier expansion indeed equals F : its absolutely convergent series has the same Fourier coefficients as F , and a continuous function with all Fourier coefficients zero is zero. To see this last assertion directly, convolve with the Fejér kernel $\mathfrak{F}_N(r) = N^{-1}(\sin(Nr/2)/\sin(r/2))^2$. It has integral 2π , is nonnegative, and its integral outside every fixed neighborhood of 0 tends to zero because its numerator is at most one and the denominator is bounded below there. Uniform continuity therefore gives uniform convergence of its convolution to the continuous function. When all Fourier coefficients are zero each such convolution is zero, proving the assertion. The same Fourier coefficient ODE and its uniqueness show uniqueness among smooth solutions of (452). $\square$

Proposition 4. *The map in Proposition 3 also realizes the full momentum equation. Let $S_s = T$, $\langle S \rangle = 0$, and retain $Q_s = V$, $\langle Q \rangle = 0$. For a solution of (452), the exact pressure increment is*

$$p^{\text{wave}}(x, t) = -\frac{2\zeta \cdot DJ\zeta}{\lambda^2 \kappa^2} Q(s, t) + \frac{\zeta_2}{\lambda \kappa} S(s, t). \quad (456)$$

With this pressure increment, adding (T, v) to any affine older state preserves its actual vector momentum residual, including retained viscosity ν_{phys} and the unchanged laboratory buoyancy θe_2 .

Proof. Put $c = J\zeta/(\lambda\kappa)$, a vector in this proof. The vector momentum residual increment before adding pressure is

$$(\dot{c} + Dc)V + c(V_t - \nu_{\text{phys}} K V_{ss}) - T e_2.$$

Indeed the transported phase has zero material derivative under Dx , $v \cdot \nabla v = 0$, $v \cdot \nabla(Dx) = Dv$, and $\Delta v = KcV_{ss}$. For every trace-free 2×2 matrix, $DJ = -JD^T$; multiplying its four entries verifies

the identity. It gives $\dot{c} = Dc - c\dot{\kappa}/\kappa$ and $\dot{\kappa} = -2\zeta \cdot D\zeta$. The identity $(J\zeta) \cdot D(J\zeta) + \zeta \cdot D\zeta = \kappa \operatorname{tr} D = 0$ shows that $\dot{c} + Dc$ is orthogonal to $J\zeta$. Its component along ζ is therefore

$$\dot{c} + Dc = \frac{2\zeta \cdot DJ\zeta}{\lambda\kappa^2} \zeta.$$

Also $bc - e_2 = -\zeta_2\zeta/\kappa$, by the two explicit components of $J\zeta = (-\zeta_2, \zeta_1)$. Substituting the second profile equation reduces the residual to

$$\zeta \left[\frac{2\zeta \cdot DJ\zeta}{\lambda\kappa^2} V - \frac{\zeta_2}{\kappa} T \right].$$

The spatial gradient of (456) is its negative, because $\nabla s = \lambda\zeta$, $Q_s = V$ and $S_s = T$. This proves the full momentum statement. Mean-zero periodic primitives exist because the profiles have zero mean, by the same integration argument used for P above. $\square$

Proposition 5. *If $\kappa_{\text{th}} = \nu_{\text{phys}} = \delta \geq 0$, put*

$$\tau(t) = \delta\lambda^2 \int_{t_0}^t |\zeta(r)|^2 dr, \quad F_\tau(s) = \mathcal{H}_\tau F(s) = \sum_{n \geq 1} f_n e^{-n^2\tau} \sin(ns). \quad (457)$$

Let $\dot{\Theta} = a\Omega$, $\dot{\Omega} = b\Theta$ with the given initial amplitudes. The exact solution in Proposition 3 is

$$T = \Theta F_\tau, \quad V = \Omega F_\tau, \quad Q = \Omega \mathcal{H}_\tau P, \quad \varpi = \Omega \partial_s F_\tau. \quad (458)$$

In particular this is an exact retained-diffusion map with the same prescribed D, G, ζ , all original frequency factors, and laboratory gravity. Each nonzero temperature mode has exact gain

$$\frac{T_n(t)}{T_n(t_0)} = \frac{\Theta(t)}{\Theta_0} e^{-n^2\tau(t)} \quad (f_n \Theta_0 \neq 0). \quad (459)$$

For vorticity the corresponding ratio is $(\Omega(t)/\Omega_0)e^{-n^2\tau(t)}$ when $nf_n\Omega_0 \neq 0$.

Proof. The matrix in (454) is $B(t) - \delta K(t)n^2 \text{Id}$, where $B = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}$. If U solves $\dot{U} = BU$, differentiation of $e^{-n^2\tau(t)}U(t, t_0)$ gives exactly (454), including its initial identity. Hence uniqueness proves this factorization and (458). Since derivatives commute termwise with the Fourier series, $(\mathcal{H}_\tau P)_s = \mathcal{H}_\tau F$ and its mean remains zero. This verifies the streamfunction as well as the scalar and vorticity profiles. Formula (459) is division of the actual mode coefficients, retaining its stated nonzero denominator. $\square$

The exact temperature phase norm is

$$\|T(\cdot, t)\|_{L^2(-\pi, \pi)}^2 = \pi \Theta(t)^2 \sum_{n \geq 1} f_n^2 e^{-2n^2\tau(t)}. \quad (460)$$

This follows by integrating the uniformly convergent squared series and using the exact orthogonality identity

$$\int_{-\pi}^{\pi} \sin(ns) \sin(ms) ds = \pi \mathbf{1}_{n=m}.$$

Thus $\|F_\tau\|_2 \leq e^{-\tau} \|F\|_2$, and for any index with $f_n \neq 0$ one has $\|F_\tau\|_2 \geq \sqrt{\pi} |f_n| e^{-n^2\tau}$. These are inequalities about phase profiles; an unlocalized affine wave need not be integrable over $\mathbb{R}^2$.

B.4 Positive heat time and the affine core

Proposition 6. *For every $\tau > 0$, $\mathcal{H}_\tau F$ has no affine restriction to any nonempty open interval of $\mathbb{R}$. In particular it cannot retain the source's nontrivial exact linear phase interval. Also F has infinitely many nonzero Fourier modes.*

Proof. For any $R > 0$ and any nonnegative integer m , the series of m th complex derivatives of $\sum_{n \geq 1} f_n e^{-n^2 \tau} \sin(nz)$ is bounded on $|\operatorname{Im} z| \leq R$ by $\sum_{n \geq 1} |f_n| n^m e^{-n^2 \tau + nR} < \infty$. For instance boundedness of f_n and completion of the square make the tail bounded by a polynomial times $e^{-\tau n^2/2}$. The sum therefore defines an entire function $h(z)$ and has its termwise derivatives. If $h(s) = cs + d$ on a nonempty real open interval, the entire function $h(z) - cz - d$ vanishes identically. The identity principle used here follows from Taylor expansion: a nonzero holomorphic function has a first nonzero Taylor coefficient at each zero where its Taylor series is not identically zero, hence that zero is isolated; an accumulating set of zeros forces the Taylor series to vanish locally, and this local vanishing propagates through the connected domain. Periodicity now gives $2\pi c = h(z + 2\pi) - h(z) = 0$. Oddness gives $d = 0$. Thus every Fourier coefficient $f_n e^{-n^2 \tau}$ would vanish, implying $f_n = 0$ for all n . The Fourier uniqueness argument in Proposition 3 would imply $F = 0$, contradicting $F'(0) = 1$. If F had only finitely many nonzero modes, the same entire-function argument, applied to its finite trigonometric sum at heat time zero, would contradict its interval $F(s) = s$. $\square$

There is nevertheless an exact estimate of the defect deep inside the original interval. Define, for $\tau > 0$ and $d > 0$,

$$\mathcal{E}(d, \tau) = \frac{2}{\sqrt{\pi}} \int_{d/(2\sqrt{\tau})}^{\infty} e^{-r^2} dr \leq \frac{2\sqrt{\tau}}{d\sqrt{\pi}} e^{-d^2/(4\tau)}. \quad (461)$$

The inequality follows from $r \geq d/(2\sqrt{\tau})$ and $\int_a^{\infty} r e^{-r^2} dr = e^{-a^2}/2$.

Proposition 7. *For $0 < r_0 < s_0$, $d = s_0 - r_0$, and $|s| \leq r_0$,*

$$|\partial_s F_\tau(s) - 1| \leq \|F' - 1\|_\infty \mathcal{E}(d, \tau), \quad (462)$$

$$|F_\tau(s) - s| \leq |s| \|F' - 1\|_\infty \mathcal{E}(d, \tau), \quad (463)$$

$$|\partial_s^m F_\tau(s)| \leq \|F^{(m)}\|_\infty \mathcal{E}(d, \tau) \quad (m \geq 2). \quad (464)$$

Proof. The periodized Gaussian kernel of $\mathcal{H}_\tau$ is $\sum_{k \in \mathbb{Z}} (4\pi\tau)^{-1/2} e^{-(r+2\pi k)^2/(4\tau)}$. To verify this representation, its n th Fourier multiplier is the Gaussian integral $\int_{\mathbb{R}} (4\pi\tau)^{-1/2} e^{-r^2/(4\tau)} e^{-inr} dr = e^{-n^2 \tau}$. One direct verification of this integral differentiates it with respect to n , integrates the Gaussian derivative by parts, and obtains $I'(n) = -2\tau n I(n)$ with $I(0) = 1$; the last normalization is the usual Gaussian integral, obtained by squaring it and integrating in polar coordinates. Fourier uniqueness then proves the representation. For any periodic bounded function h vanishing on $[-s_0, s_0]$, its periodic lift satisfies $h(s - r) = 0$ whenever $|s| \leq r_0$ and $|r| < d$. The Gaussian representation on the line therefore gives

$$|\mathcal{H}_\tau h(s)| \leq \|h\|_\infty \int_{|r| \geq d} (4\pi\tau)^{-1/2} e^{-r^2/(4\tau)} dr = \|h\|_\infty \mathcal{E}(d, \tau).$$

Apply this to $h = F' - 1$ and $h = F^{(m)}$ for $m \geq 2$, using commutation of derivatives with $\mathcal{H}_\tau$ and $\mathcal{H}_\tau 1 = 1$. Finally $F_\tau(0) = 0$ by oddness; integrating the first bound between 0 and s proves the second bound with the same d . $\square$

The source background in equation (3.3) is obtained because each older layer is exactly affine on the newer support, by equations (4.3)–(4.6). For a heat-evolved older layer with nonzero scalar or velocity amplitude, Proposition 6 removes this exact affine premise. Its gradient at the origin can still be computed, but its values away from the origin are not that gradient times x . The tail estimates give explicit defects rather than an identification of those two fields. Consequently Proposition 5 alone supplies no coupled feedback law for the heat-modified older background.

B.5 Exact preservation by force and the localized diffusion costs

For the original inviscid amplitude equations $\dot{\Theta} = a\Omega$, $\dot{\Omega} = b\Theta$, Proposition 1 shows that the exact additional scalar and curl forces are

$$f_{\theta, \text{diff}} = -\kappa_{\text{th}} K \Theta F'', \quad E_{\text{diff}} = -\nu_{\text{phys}} K \Omega F'''. \quad (465)$$

They vanish on the entire affine interval $|s| < s_0$. This retains exactly the original core, amplitudes, phase and hence its local affine feedback; it pays a nonzero diffusion force in the rest of the profile. In the unlocalized wave this force is periodic in phase and is not a compactly supported force on $\mathbb{R}^2$.

Here is the exact compact counterpart for the leading fields in source equation (4.2). Let M solve $\dot{M} = DM$, $M(t_0) = \text{Id}$, let $p = \zeta(t_0)$, and let $\ell > 0$. Fix the source's radial $f \in C_c^\infty(B_1)$, $f = 1$ on $B_{1/2}$, and put

$$\begin{aligned} a_{\text{mat}} &= M^{-1}x, & g(a_{\text{mat}}) &= f(a_{\text{mat}}/\ell), \\ \theta^L &= w\Theta F(s)g(a_{\text{mat}}), & \psi^L &= \frac{w\Omega}{\lambda^2\kappa} P(s)g(a_{\text{mat}}), & u^L &= \nabla^\perp \psi^L. \end{aligned} \quad (466)$$

The activation $w(t)$ remains the original smooth scalar function; no time differentiation of it is omitted from the existing inviscid residual. The following formulas give exactly the additional diffusion terms, with w retained. At a fixed time put

$$B = M^{-1}, \quad H = BB^T, \quad q = B\zeta, \quad \mathcal{L} = \lambda^2\kappa\partial_s^2 + 2\lambda q \cdot \nabla_a \partial_s + H : \nabla_a^2. \quad (467)$$

Although $s = \lambda p \cdot a_{\text{mat}}$ upon evaluation, s, a are independent variables in this differential operator.

Proposition 8. *For every smooth profile $A(s, a)$, $\Delta_x[A(\lambda\zeta \cdot x, Bx)] = (\mathcal{L}A)(\lambda\zeta \cdot x, Bx)$. Consequently the exact added forces for (466) are*

$$f_{\theta, \text{diff}}^L = -\kappa_{\text{th}} w \Theta [\lambda^2 \kappa F'' g + 2\lambda F' q \cdot \nabla g + F(H : \nabla^2 g)], \quad (468)$$

$$f_{u, \text{diff}}^L = -\nu_{\text{phys}} \Delta u^L = -\nu_{\text{phys}} \nabla^\perp \Delta \psi^L, \quad E_{\text{diff}}^L = -\nu_{\text{phys}} \frac{w\Omega}{\lambda^2\kappa} \mathcal{L}^2(Pg). \quad (469)$$

Writing $\mathcal{H} = H : \nabla_a^2$, the last operator is explicitly

$$\begin{aligned} \mathcal{L}^2(Pg) &= \lambda^4 \kappa^2 F''' g + 4\lambda^3 \kappa F'' q \cdot \nabla g + 2\lambda^2 \kappa F' \mathcal{H} g + 4\lambda^2 F' (q \cdot \nabla)^2 g \\ &\quad + 4\lambda F(q \cdot \nabla) \mathcal{H} g + P \mathcal{H}^2 g. \end{aligned} \quad (470)$$

All these forces are supported in $M\bar{B}_\ell$. They vanish in the exact affine region $MB_{\ell/2} \cap \{|s| < s_0\}$.

Proof. The chain rule gives $\partial_{x_i} A = \lambda \zeta_i A_s + \sum_j B_{ji} A_{a_j}$. Applying it a second time and summing in i gives the three terms in (467), since $\sum_i B_{ji} \zeta_i = (B\zeta)_j$ and $\sum_i B_{ji} B_{ki} = (BB^T)_{jk}$. Applying the same identity twice to Pg proves (469). Expanding the square of the commuting constant-in- (s, a) operators in (467), and using $P' = F$, $P'' = F'$, $P''' = F''$, $P'''' = F'''$, proves all six terms of (470). Spatial differentiation does not enlarge the support of a smooth compactly supported function, proving the support statement. On the plateau $g = 1$ all positive derivatives of g vanish. On the linear phase interval $F'' = F''' = 0$. Thus the scalar and curl formulas vanish on their intersection. There $\psi^L = (w\Omega/(\lambda^2\kappa))(P(0) + s^2/2)$ is quadratic in x , so u^L is linear and $\Delta u^L = 0$, proving the vector-force statement as well. Finally $\text{curl}(-\nu_{\text{phys}} \Delta u^L) = -\nu_{\text{phys}} \Delta^2 \psi^L$, so the specified vector force realizes the curl force without any inverse-curl ambiguity. $\square$

For explicit quantitative costs let $C_j = \sup_a \|\nabla^j f(a)\|_{\text{HS}}$, with the Euclidean Hilbert–Schmidt tensor norm, and $C_0 = \|f\|_\infty$. Then $\|\nabla^j g\|_\infty = C_j \ell^{-j}$ by the chain rule. Write $h = \|H\|_{\text{HS}}$ and $q_* = |q|$. Cauchy–Schwarz on the tensor contractions in (468) and (470) proves

$$\|f_{\theta, \text{diff}}^L\|_\infty \leq \kappa_{\text{th}} |w\Theta| [\lambda^2 \kappa \|F''\|_\infty C_0 + 2\lambda q_* \|F'\|_\infty C_1 \ell^{-1} + h \|F\|_\infty C_2 \ell^{-2}], \quad (471)$$

$$\|E_{\text{diff}}^L\|_\infty \leq \frac{\nu_{\text{phys}}|w\Omega|}{\lambda^2\kappa} [\lambda^4\kappa^2\|F'''\|_\infty C_0 + 4\lambda^3\kappa q_*\|F''\|_\infty C_1\ell^{-1} + (2\lambda^2\kappa h + 4\lambda^2 q_*^2)\|F'\|_\infty C_2\ell^{-2} + 4\lambda q_* h\|F\|_\infty C_3\ell^{-3} + h^2\|P\|_\infty C_4\ell^{-4}]. \quad (472)$$

The same reasoning applies to every compact correction: for a specified smooth correction (θ^c, ψ^c) the extra scalar and vector forces are exactly $-\kappa_{\text{th}}\Delta\theta^c$ and $-\nu_{\text{phys}}\nabla^\perp\Delta\psi^c$, with curl $-\nu_{\text{phys}}\Delta^2\psi^c$. This is the exact linear map from an already constructed inviscid forced field to a retained-diffusion forced field:

$$(\theta, u, p, f_\theta, f_u) \mapsto (\theta, u, p, f_\theta - \kappa_{\text{th}}\Delta\theta, f_u - \nu_{\text{phys}}\Delta u). \quad (473)$$

Substitution in the scalar and momentum equations proves the map: adding $\kappa_{\text{th}}\Delta\theta$ or $\nu_{\text{phys}}\Delta u$ to the new right-hand side cancels its respective force decrement. Thus pressure, divergence, initial data and the entire trajectory are retained exactly. On each closed finite smooth stage these are smooth finite forces; where the original fields and their derivatives are compactly supported, the new terms have the same support. This statement supplies no uniform bound as the layer index tends to infinity. The explicit factors λ^2 , ℓ^{-2} , and ℓ^{-4} above must be retained in such an analysis.

Source and scope receipt

The primary source read here is the 76-page PDF *Blowup for the Boussinesq equations with smooth forcing*, Levent Alpöge and Tristan Buckmaster; source equations (3.1)–(3.6) are on pages 8–9 and (4.1)–(4.6) on pages 37–38. The exact file SHA-256 is

895a628d1783bcb039374686f50b895b5f450f53b8ef8aa173523487a7a4a21b.

The companion replay script checks the profile differentiation, full two-dimensional residual algebra, localized Laplacian square, common diffusivity matrix factorization, and eigenfunction obstruction identities. Its checks supplement the proofs above; they do not certify an infinite-stage existence or force-smoothness theorem.